

# Kube-OVN Document

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**v1.15.0**

*Kube-OVN Team*

*2025 Kube-OVN Team*

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# 1. Kube-OVN



## 1.1 What is Kube-OVN?



## 1.2 Why Kube-OVN?



### 1.3.1 eBPF

### 1.3.2 CNI, Ingress, Service Mesh Observability All in One

### 1.3.3 OpenShift

### 1.3.4 Kubernetes EKS/AKS/GKE

### 1.3.5 AI

## 1.4 OVN/ovn-kubernetes/Kube-OVN

### 1.4.1 OVN

### 1.4.2 ovn-kubernetes

### 1.4.3 Kube-OVN

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## 2.

## 2.1

Kube-OVN      CNI      Kubernetes

## 2.1.1

- Kubernetes >= 1.29
- Docker >= 1.12.6, Containerd >= 1.3.4
- : CentOS 7/8, Ubuntu 16.04/18.04/20.04
- Linux `geneve`, `openvswitch`, `ip_tables` `iptables_nat` Kube-OVN

1. 3.10.0-862 `netfilter` bug Kube-OVN CentOS bug [Floating IPs broken after kernel upgrade to Centos/RHEL 7.5 - DNAT not working](#)
2. Rocky Linux 8.6 4.18.0-372.9.1.el8.x86\_64 TCP [TCP connection failed in Rocky Linux 8.6](#) 4.18.0-372.13.1.el8\_6.x86\_64
3. 4.4 `openvswitch` `openvswitch`
4. Geneve IPv6 `cat /proc/cmdline` bug [Geneve tunnels don't work when ipv6 is disabled](#)

## 2.1.2

- IPv6 `ipv6.disable=1` 0
- `kube-proxy` Kube-OVN Service ClusterIP `kube-apiserver`
- `kubelet` CNI , `kubelet` `--network-plugin=cni --cni-bin-dir=/opt/cni/bin --cni-conf-dir=/etc/cni/net.d`
- `/etc/cni/net.d/`

## 2.1.3

|                     |                                               |                     |
|---------------------|-----------------------------------------------|---------------------|
| ovn-central         | 6641/tcp                                      | ovn nb db server    |
| ovn-central         | 6642/tcp                                      | ovn sb db server    |
| ovn-central         | 6643/tcp                                      | ovn northd server   |
| ovn-central         | 6644/tcp                                      | ovn raft server     |
| ovn-ic              | 6645/tcp                                      | ovn ic nb db server |
| ovn-ic              | 6646/tcp                                      | ovn ic sb db server |
| ovs-ovn             | Geneve 6081/udp, STT 7471/tcp, Vxlan 4789/udp |                     |
| kube-ovn-controller | 10660/tcp                                     |                     |
| kube-ovn-daemon     | 10665/tcp                                     |                     |
| kube-ovn-monitor    | 10661/tcp                                     |                     |

firewalld      Packet Forwarding      Masquerade

```
# Packet Forwarding
firewall-cmd --add-forward --permanent
```

```
# IPv4 Masquerade
firewall-cmd --add-masquerade --permanent
# Kube-OVN IPv6/      Masquerade
firewall-cmd --permanent --add-rich-rule 'rule family="ipv6" source address="fd00:10:16::/112" masquerade'

firewall-cmd --reload
```

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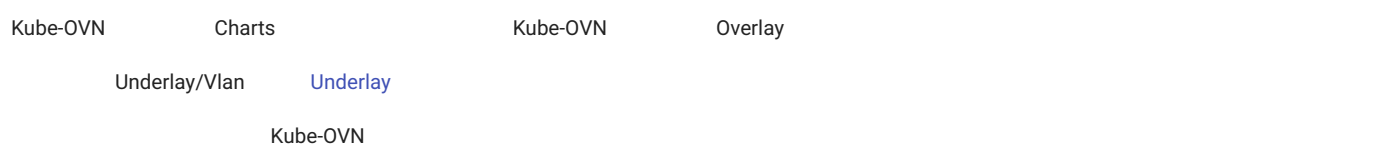
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2.1.4

2.2



2.2.1

release

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/refs/tags/v1.15.21/dist/images/install.sh
```

master

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/images/install.sh
```

```
REGISTRY="kubeovn" #
VERSION="v1.15.21" # /Tag
POD_CIDR="10.16.0.0/16" # CIDR SVC/NODE/JOIN CIDR
SVC_CIDR="10.96.0.0/12" # apiserver service-cluster-ip-range
JOIN_CIDR="100.64.0.0/16" # Pod CIDR SVC/NODE/POD CIDR
LABEL="node-role.kubernetes.io/master" # OVN DB
IFACE="" #
TUNNEL_TYPE="geneve" # geneve, vxlan stt stt ovs
```

```
IFACE=enp6s0f0,eth.*
```

root

```
bash install.sh
```

Kube-OVN

- 1. [Step 4/6] Pod
- 2. Kube-OVN

2.2.2 Helm Chart

| Kube-OVN                   | Helm               | Kube-OVN           |
|----------------------------|--------------------|--------------------|
| IP                         |                    |                    |
| # kubectl get node -o wide |                    |                    |
| NAME                       | STATUS             | ROLES              |
| kube-ovn-control-plane     | NotReady           | control-plane      |
| kube-ovn-worker            | NotReady           | <none>             |
| AGE                        | VERSION            | INTERNAL-IP        |
| 20h                        | v1.26.0            | 172.18.0.3         |
| 20h                        | v1.26.0            | 172.18.0.2         |
| EXTERNAL-IP                | OS-IMAGE           | KERNEL-VERSION     |
| <none>                     | Ubuntu 22.04.1 LTS | 5.10.104-linuxkit  |
| <none>                     | Ubuntu 22.04.1 LTS | 5.10.104-linuxkit  |
|                            |                    | CONTAINER-RUNTIME  |
|                            |                    | containerd://1.6.9 |
|                            |                    | containerd://1.6.9 |

label

```
# kubectl label node -lbeta.kubernetes.io/os=linux kubernetes.io/os=linux --overwrite
node/kube-ovn-control-plane not labeled
node/kube-ovn-worker not labeled

# kubectl label node -lnode-role.kubernetes.io/control-plane kube-ovn/role=master --overwrite
node/kube-ovn-control-plane labeled

# label dpdk dpdk
# kubectl label node -lovn.kubernetes.io/ovs_dp_type!=userspace ovn.kubernetes.io/ovs_dp_type=kernel --overwrite
node/kube-ovn-control-plane labeled
node/kube-ovn-worker labeled
```

Helm Repo

```
# helm repo add kubeovn https://kubeovn.github.io/kube-ovn/
"kubeovn" has been added to your repositories

# helm repo list
NAME      URL
kubeovn   https://kubeovn.github.io/kube-ovn/

# helm repo update kubeovn
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "kubeovn" chart repository
Update Complete. *Happy Helming!*
```

| NAME             | CHART VERSION | APP VERSION | DESCRIPTION             |
|------------------|---------------|-------------|-------------------------|
| kubeovn/kube-ovn | v1.15.21      | v1.15.21    | Helm chart for Kube-OVN |

helm install Kube-OVN

Chart values.yaml

```
# helm install kube-ovn kubeovn/kube-ovn --wait -n kube-system --version v1.15.21
NAME: kube-ovn
LAST DEPLOYED: Thu Apr 24 08:30:13 2025
NAMESPACE: kube-system
STATUS: deployed
REVISION: 1
TEST SUITE: None
```

Helm values.yaml

```
helm upgrade -f values.yaml kube-ovn kubeovn/kube-ovn --wait -n kube-system --version v1.15.21
```

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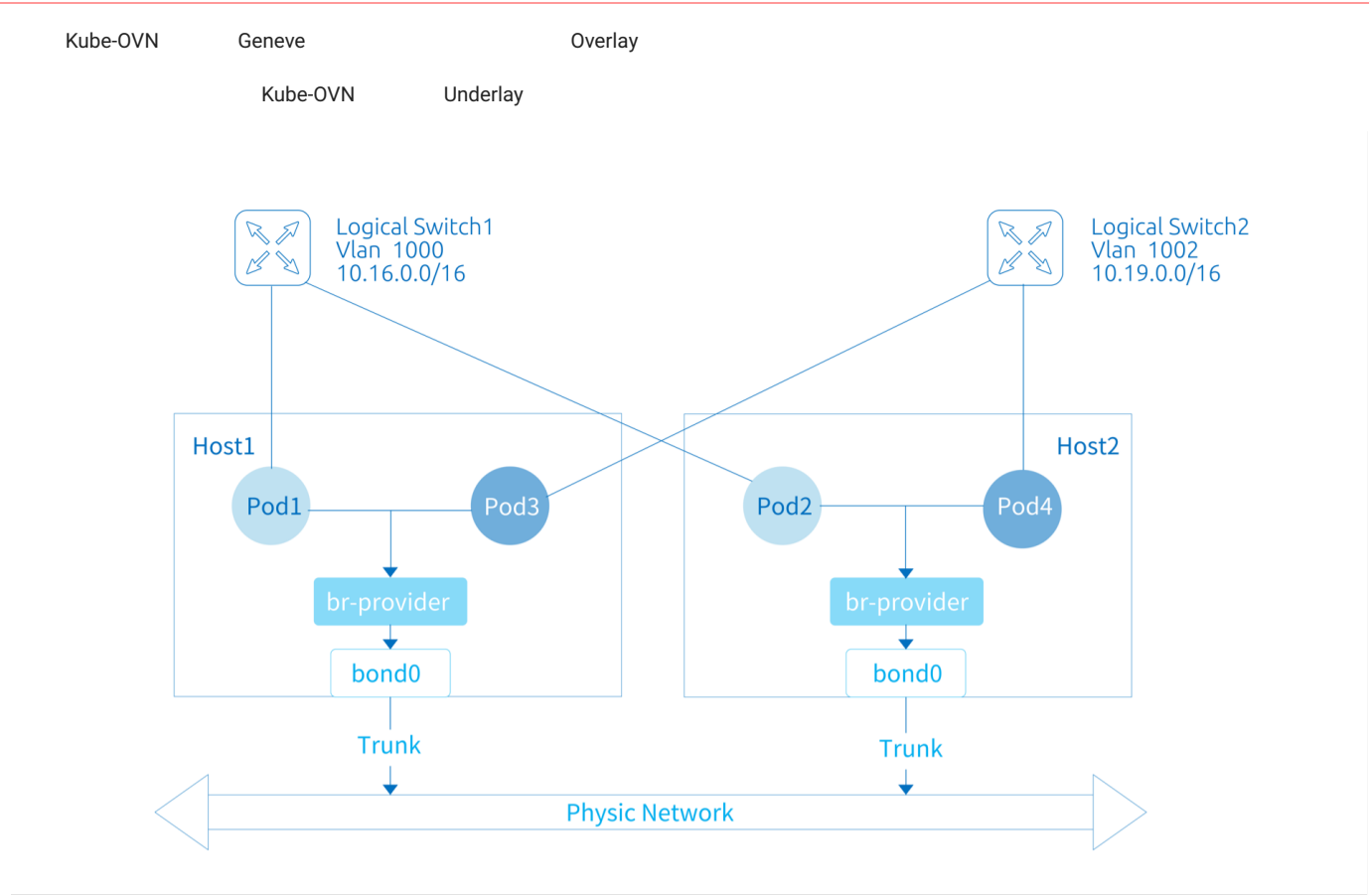


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2.3 Underlay



2.3.1

|         |          |   |    |     |          |
|---------|----------|---|----|-----|----------|
| Overlay | SNAT/EIP | / | L3 | VPC | Underlay |
|---------|----------|---|----|-----|----------|

2.3.2 Macvlan

|          |          |           |         |               |          |     |         |               |  |
|----------|----------|-----------|---------|---------------|----------|-----|---------|---------------|--|
| Kube-OVN | Underlay | Macvlan   |         |               |          |     |         |               |  |
| 1.       | Macvlan  | OVS       | Macvlan |               |          |     |         |               |  |
| 2.       | Kube-OVN | arp-proxy | arp     |               |          |     |         |               |  |
| 3.       | Macvlan  | netfilter | Service | NetworkPolicy | Kube-OVN | OVS | Service | NetworkPolicy |  |
| 4.       | Kube-OVN | Underlay  | Macvlan | IP            | QoS      |     |         |               |  |



2.3.3

| Underlay          | OVS               | OVS                                   | L2/L3                      |          | Vlan       |                                      |
|-------------------|-------------------|---------------------------------------|----------------------------|----------|------------|--------------------------------------|
| 1. OpenStack VM   |                   | PortSecurity                          |                            |          |            |                                      |
| 2. VMware vSwitch |                   | MAC Address Changes, Forged Transmits | Promiscuous Mode Operation |          | allow      |                                      |
| 3. VMware NSX-T   |                   | Underlay                              |                            |          |            |                                      |
| 4. Hyper-V        |                   | MAC Address Spoofing                  |                            |          |            |                                      |
| 5. AWS GCE        |                   | Mac                                   | Underlay                   | Underlay | VPC-CNI    |                                      |
| 6. Linux Bridge   |                   |                                       |                            |          |            |                                      |
|                   | Kube-OVN Underlay |                                       |                            | Underlay |            |                                      |
|                   | Underlay          |                                       |                            |          |            |                                      |
|                   | Kube-OVN          | Mac                                   | IP                         | MTU      | OVS Bridge | Underlay OVS Bridge br-PROVIDER_NAME |
| PROVIDER_NAME     | ProviderNetwork   |                                       | provider                   |          |            |                                      |
| 1. OVS Bridge     |                   | NetworkManager                        |                            | DHCP     |            |                                      |
| 2. OVS Bridge     |                   | OVS                                   | Kube-OVN                   | OVS      |            |                                      |
|                   | Underlay          |                                       |                            |          |            |                                      |

2.3.4

UnderlayPodUnderlay

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.15/dist/images/install.sh
```

```
ENABLE_ARP_DETECT_IP_CONFLICT #          vlan  arp
NETWORK_TYPE                  #          vlan
VLAN_INTERFACE_NAME           #          eth1
VLAN_ID                        #          VLAN Tag  0      VLAN
POD_CIDR                      #          CIDR    192.168.1.0/24
POD_GATEWAY                   #          192.168.1.1
EXCLUDE_IPS                   #          IP      192.168.1.1..192.168.1.100
ENABLE_LB                     #  Underlay  Service true
EXCHANGE_LINK_NAME            #  provider-network OVS      false
LS_DNAT_MOD_DL_DST            # DNAT      MAC      Service  true
```

```
bash install.sh
```

2.3.5 CRD Underlay

UnderlayPodProviderNetworkVlanSubnet

**ProviderNetwork**

ProviderNetwork

ProviderNetwork :

```
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: net1
```

```
spec:
  defaultInterface: eth1
  customInterfaces:
    - interface: eth2
      nodes:
        - node1
  nodeSelector:
    matchLabels:
      kubernetes.io/arch: amd64
      network-type: underlay
  matchExpressions:
    - key: kubernetes.io/hostname
      operator: In
      values:
        - node1
        - node2
```

ProviderNetwork 12

- defaultInterface: Bond Vlan ProviderNetwork excludeNodes br-net1 br-NAME OVS
- customInterfaces:
- nodeSelector: OVS matchLabels matchExpressions
- excludeNodes: net1.provider-network.ovn.kubernetes.io/exclude=true nodeSelector
- autoCreateVlanSubinterfaces: false true VLAN <interface>.<vlanid>

| Key                                               | Value                |
|---------------------------------------------------|----------------------|
| net1.provider-network.ovn.kubernetes.io/ready     | true ProviderNetwork |
| net1.provider-network.ovn.kubernetes.io/interface | eth1                 |
| net1.provider-network.ovn.kubernetes.io/mtu       | 1500 MTU             |

IP IP OVS

VLAN

ProviderNetwork autoCreateVlanSubinterfaces: true VLAN VLAN

```
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: vlan-provider
spec:
  defaultInterface: eth0.341
  autoCreateVlanSubinterfaces: true
  customInterfaces:
    - interface: eth2.341
      nodes:
        - worker-node-1
        - worker-node-2
```

<interface>.<vlanid> VLAN VLAN ProviderNetwork

VLAN

Vlan Vlan Tag ProviderNetwork

VLAN

```
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
```

```
name: vlan1
spec:
  id: 0
  provider: net1
```

- id: VLAN ID/Tag Kube-OVN    Vlan    Vlan    0    vlan    localnet
- provider:        ProviderNetwork        VLAN        ProviderNetwork

Subnet

Vlan

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  vlan: vlan1
  disableGatewayCheck: false
```

- vlan        VLAN        Subnet        VLAN
- disableGatewayCheck    Underlay        true

2.3.6

IP

IP    Pod    IP    Mac

2.3.7

Kube-OVN    Underlay        spec.logicalGateway    true

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  vlan: vlan1
  logicalGateway: true
```

Pod        Kube-OVN        Logical Router

2.3.8 Underlay    Overlay

Underlay    Overlay        Overlay    Pod    NAT    Underlay    Pod IP    Underlay    Pod    Overlay

Overlay

Underlay    Overlay        u2oInterconnection    true        Kube-OVN        Underlay IP    Underlay    ovn-cluster

Kube-OVN    Underlay    Overlay

 **Warning**

Vlan    Underlay        Underlay        u2o

IP

subnet    IP        Underlay Subnet        u2oInterconnectionIP

| Underlay Subnet | VPC |                |     |                    |      |                 |     |
|-----------------|-----|----------------|-----|--------------------|------|-----------------|-----|
| Underlay Subnet | VPC | Overlay Subnet | VPC | u2oInterconnection | true | subnet.spec.vpc | VPC |

## 2.3.9

|                                                                                                                                                                        |         |        |         |          |                |    |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|---------|----------|----------------|----|------|
| IP                                                                                                                                                                     | Netplan | Ubuntu | Netplan | renderer | NetworkManager | IP | DHCP |
| <pre>network:   renderer: NetworkManager   ethernets:     eth0:       dhcp4: no       addresses:         - 172.16.143.129/24   version: 2</pre>                        |         |        |         |          |                |    |      |
| IP                                                                                                                                                                     | netplan |        |         |          |                |    |      |
| <pre>netplan generate  nmcli connection reload netplan-eth0 nmcli device set eth0 managed yes</pre>                                                                    |         |        |         |          |                |    |      |
| Kube-OVN                                                                                                                                                               | IP      | OVS    |         |          |                |    |      |
| NetworkManager                                                                                                                                                         |         | CentOS |         |          |                |    |      |
| <pre>nmcli connection reload eth0 nmcli device set eth0 managed yes nmcli -t -f GENERAL.STATE device show eth0   grep -qw unmanaged    nmcli device reapply eth0</pre> |         |        |         |          |                |    |      |
| IP                                                                                                                                                                     | MAC     |        |         |          |                |    |      |

## 2.3.10

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                     |       |     |          |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------|-----|----------|--|
| hairpin Pod                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                     |       |     |          |  |
| hairpin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Pod                 | Pod   | OVS | MAC      |  |
| hairpin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Kube-OVN            |       |     |          |  |
| Pod                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Pod                 |       |     |          |  |
| Pod                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 300                 | ARP   | OVS | resubmit |  |
| <pre>2022-11-13T08:43:46.782Z 00222 ofproto_dpif_upcall(handler5) WARN Flow: arp,in_port=331,vlan_tci=0x0000,d1_src=00:00:00:25:eb:39,d1_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.213.131.240,arp_tpa=10.213.159.254,arp_op=1,arp_sha=00:00:00:25:eb:39,arp_tha=ff:ff:ff:ff:ff:ff  bridge("br-int") ----- 0. No match.   &gt;&gt;&gt;&gt; received packet on unknown port 331 &lt;&lt;&lt;&lt;   drop  Final flow: unchanged MegafLOW: recirc_id=0,eth,arp,in_port=331,d1_src=00:00:00:25:eb:39 Datapath actions: drop 2022-11-13T08:44:34.077Z 00224 ofproto_dpif_xlate(handler5) WARN over 4096 resubmit actions on bridge br-int while processing arp,in_port=13483,vlan_tci=0x0000,d1_src=00:00:00:59:ef:13,d1_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.213.152.3,arp_tpa=10.213.159.254,arp_op=1,arp_sha=00:00:00:59:ef:13,arp_tha=ff:ff:ff:ff:ff:ff</pre> |                     |       |     |          |  |
| OVN NB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | bcast_arp_req_flood | false |     |          |  |
| <pre>kubectrl ko nbctl set NB_Global . options:bcast_arp_req_flood=false</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |       |     |          |  |

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2.3.11

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## 2.4 CNI

Kube-OVN CNI CNI Cilium Calico Flannel

### 2.4.1

- CNI Kube-OVN CNI Pod CNI eth0
- CNI Kube-OVN VPC
  - 
  - Pod
  - Kube-OVN VPC NAT

### 2.4.2

- CNI Kubernetes CNI Cilium Calico Flannel
- Multus CNI
- Kube-OVN

- CNI
- Multus CNI
- Kube-OVN

### 2.4.3

#### Helm Chart v2

```
# values.yaml
cni:
  nonPrimaryCNI: true
```

#### Helm

```
helm install kube-ovn ./charts/kube-ovn-v2 \
--namespace kube-system \
--set cni.nonPrimaryCNI=true
```

#### Helm Chart v1

```
# values.yaml
cni_conf:
  NON_PRIMARY_CNI: true
```

#### Helm

```
helm install kube-ovn ./charts/kube-ovn \
--namespace kube-system \
--set cni_conf.NON_PRIMARY_CNI=true
```

kube-ovn-controller deployment

```
containers:
- name: kube-ovn-controller
  args:
  - --non-primary-cni-mode=true
```

## 2.4.4

### NADs

#### NAD

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: kube-ovn-vpc-network
  namespace: default
spec:
  config: |
    {
      "cniVersion": "0.3.1",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "kube-ovn-vpc-network.default.ovn"
    }
```

### VPC

#### VPC

```
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  name: vpc-secondary
spec:
  namespaces:
  - default
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet-secondary
spec:
  vpc: vpc-secondary
  cidr: "10.100.0.0/16"
  gateway: "10.100.0.1"
  provider: kube-ovn-vpc-network.default.ovn
```

### Pod

#### Pod

```
apiVersion: v1
kind: Pod
metadata:
  name: multi-network-pod
  annotations:
    k8s.v1.cni.cncf.io/networks: default/kube-ovn-vpc-network
spec:
  containers:
  - name: app
    image: nginx
```

## 2.4.5

### eth0

- CNI Cilium Calico
- 
- Kubernetes

**net1 net2...**

- Kube-OVN
- 
- VPC
- Kube-OVN QoS NAT

## 2.4.6

## 1.

```
# Pod
apiVersion: v1
kind: Pod
metadata:
  name: frontend
  annotations:
    k8s.v1.cni.cncf.io/networks: |
      [
        {"name": "default/public-network"},
        {"name": "default/internal-network"}
      ]
```

## 2.

```
# A Pod
apiVersion: v1
kind: Pod
metadata:
  name: tenant-a-app
  annotations:
    k8s.v1.cni.cncf.io/networks: default/tenant-a-network

---
# B Pod
apiVersion: v1
kind: Pod
metadata:
  name: tenant-b-app
  annotations:
    k8s.v1.cni.cncf.io/networks: default/tenant-b-network
```

## 3. VPC NAT

```
apiVersion: kubeovn.io/v1
kind: VpcNatGateway
metadata:
  name: vpc-nat-gw
spec:
  vpc: vpc-secondary
  subnet: subnet-secondary
  lanIp: "10.100.0.254"
```

## 2.4.7

## Pod

```
metadata:
  annotations:
    k8s.v1.cni.cncf.io/networks: |
      [
        {"name": "default/kube-ovn-network-1", "interface": "net1"},
        {"name": "default/kube-ovn-network-2", "interface": "net2"}
      ]
```



IP

IP

```
# IP
metadata:
  annotations:
    k8s.v1.cni.cncf.io/networks: |
      [{
        "name": "default/kube-ovn-network",
        "ips": ["10.100.0.100/16"]
      }]
```

QoS

QoS

```
metadata:
  annotations:
    ovn.kubernetes.io/ingress_rate: "1000"
    ovn.kubernetes.io/egress_rate: "1000"
```

2.4.8

- 1. Pod - NAD - kube-ovn-cni - Multus
- 2. - VPC - Pod -
- 3. IP - CIDR - IP - IPAM

```
# Pod
kubectl get pod <pod-name> -o yaml | grep -A 10 "networks-status"

# Pod
kubectl exec <pod-name> -- ip addr show

#
kubectl exec <pod-name> -- ip route show

# kube-ovn-cni
kubectl logs -n kube-system daemonset/kube-ovn-cni
```

2.4.9

- 1. Kubernetes
- 2.
- 3. DNS Pod DNS

2.4.10

- 1. IP
- 2.
- 3.
- 4.
- 5.

2.4.11

- Multus CNI

- 
- Kube-OVN

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2.4.12

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## 2.5 Talos

Talos Linux    Kubernetes    Linux

### 2.5.1    Helm Chart    Kube-OVN

Talos Linux    Kube-OVN

```
helm install kube-ovn kubeovn/kube-ovn --wait \
-n kube-system \
--version v1.15.21 \
--set OVN_DIR=/var/lib/ovn \
--set OPENVSWITCH_DIR=/var/lib/openvswitch \
--set DISABLE_MODULES_MANAGEMENT=true \
--set cni_conf.MOUNT_LOCAL_BIN_DIR=false
```

Underlay    Helm    Chart

```
helm install kubeovn kubeovn/kube-ovn --wait \
-n kube-system \
--version v1.15.21 \
--set OVN_DIR=/var/lib/ovn \
--set OPENVSWITCH_DIR=/var/lib/openvswitch \
--set DISABLE_MODULES_MANAGEMENT=true \
--set cni_conf.MOUNT_LOCAL_BIN_DIR=false \
--set networking.NETWORK_TYPE=vlan \
--set networking.vlan.VLAN_INTERFACE_NAME=enp0s5f1 \
--set networking.vlan.VLAN_ID=0 \
--set networking.NET_STACK=ipv4 \
--set-json networking.EXCLUDE_IPS='["172.99.99.11..172.99.99.99"]' \
--set-json ipv4.POD_CIDR='"172.99.99.0/24"' \
--set-json ipv4.POD_GATEWAY='"172.99.99.1"'
```

 **Note**

VLAN Bond Bridge    Underlay    Underlay    Talos    ignore=true

```
machine:
  network:
    interfaces:
      - interface: enp0s5f1
        ignore: true
```

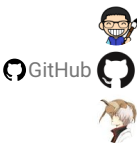
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### 2.5.2

2.6

|          |            |          |
|----------|------------|----------|
| Kube-OVN | pre-v0.0.1 | Kube-OVN |
|----------|------------|----------|

2.6.1

- [Linkerd](#)
- [Elasticsearch](#)
- [EMQX](#)
- [KubeSphere](#)

2.6.2 OpenVswitch/OVN

|                                                                                                             |             |     |
|-------------------------------------------------------------------------------------------------------------|-------------|-----|
| Kube-OVN                                                                                                    | OpenVswitch | OVN |
| <ul style="list-style-type: none"><li>• <a href="#">OVN</a></li><li>• <a href="#">OpenVswitch</a></li></ul> |             |     |
| ovn-architecture                                                                                            |             |     |

2.6.3 Kube-OVN

|          |          |              |
|----------|----------|--------------|
| Kube-OVN |          |              |
| Kube-OVN | Kube-OVN | kubectl bash |

2.6.4

|          |          |              |
|----------|----------|--------------|
| Kube-OVN | Kube-OVN | Kube-OVN E2E |
|----------|----------|--------------|

2.6.5

|          |                               |                          |
|----------|-------------------------------|--------------------------|
| Kube-OVN | <a href="#">OpenTelemetry</a> | <a href="#">DeepFlow</a> |
|----------|-------------------------------|--------------------------|

2.6.6

|                                                                                                                    |                                                                                                                      |                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Kube-OVN                                                                                                           | 7*24                                                                                                                 | Github Issue                                                                                                           |
| Github Issue                                                                                                       | AI                                                                                                                   | AI                                                                                                                     |
| <div> <a href="#">PDF</a></div> | <div> <a href="#">Slack</a></div> | <div> <a href="#">Support</a></div> |

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2.6.7

---

2.7

Kube-OVN

Kube-OVN

OVS

issue

Kube-OVN

2.7.1    Kubernetes

Script Uninstall

Helm Uninstall

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.15/dist/images/cleanup.sh
bash cleanup.sh

helm uninstall kube-ovn -n kube-system
```

2.7.2

ovsdb

openvswitch

```
rm -rf /var/run/openvswitch
rm -rf /var/run/ovn
rm -rf /etc/origin/openvswitch/
rm -rf /etc/origin/ovn/
rm -rf /etc/cni/net.d/00-kube-ovn.conflist
rm -rf /etc/cni/net.d/01-kube-ovn.conflist
rm -rf /var/log/openvswitch
rm -rf /var/log/ovn
rm -fr /var/log/kube-ovn
```

2.7.3

iptables/ipset

reboot

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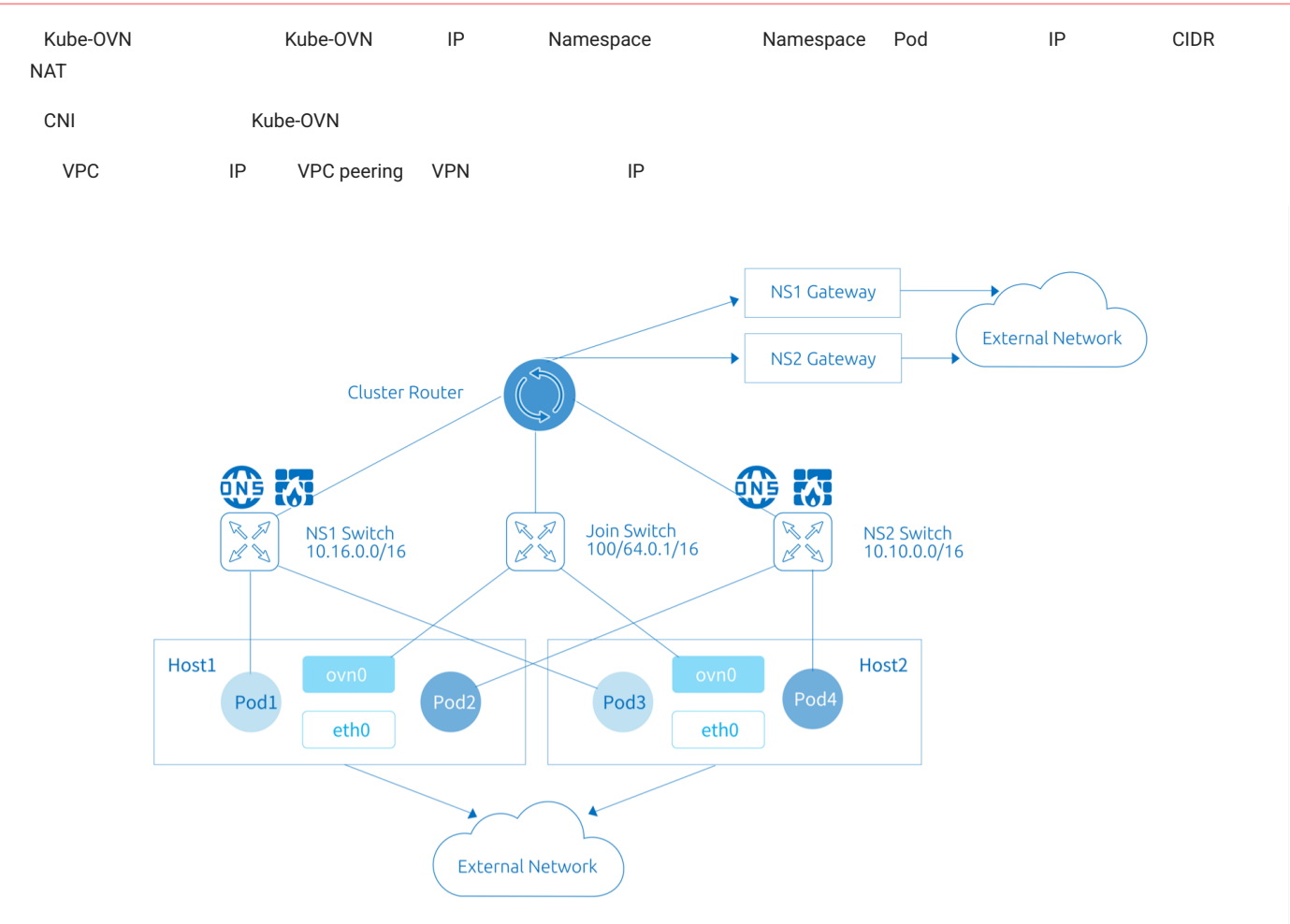
 GitHub



2.7.4

3.

3.1



Overlay Underlay

3.1.1

| Kube-OVN | Namespace | IP      | CIDR |
|----------|-----------|---------|------|
| Overlay  | NAT       | Flannel |      |
| Underlay | arping    |         |      |

|      |         |      |             |
|------|---------|------|-------------|
| spec | default | true | ovn-default |
|------|---------|------|-------------|

```
# kubectl get subnet ovn-default -o yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  creationTimestamp: "2019-08-06T09:33:43Z"
  generation: 1
  name: ovn-default
  resourceVersion: "1571334"
```

```

selfLink: /apis/kubeovn.io/v1/subnets/ovn-default
uid: 7e2451f8-fb44-4f7f-b3e0-cfd27f6fd5d6
spec:
  cidrBlock: 10.16.0.0/16
  default: true
  excludeIps:
    - 10.16.0.1
  gateway: 10.16.0.1
  gatewayType: distributed
  natOutgoing: true
  private: false
  protocol: IPv4

```

### 3.1.2 Join

| Kubernetes | Node  | Pod  | Overlay | Kube-OVN | join | Node | ovn0 | join |
|------------|-------|------|---------|----------|------|------|------|------|
| Pods       | Nodes | ovn0 | Node    | Pod      | ovn0 |      |      |      |
|            |       | join | CIDR    | Join     |      |      |      |      |

#### Join

| Pod  | hostport | externalTrafficPolicy: Local | NodePort | Service |
|------|----------|------------------------------|----------|---------|
| join | CIDR     |                              |          |         |

```

# kubectl get subnet join -o yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  creationTimestamp: "2019-08-06T09:33:43Z"
  generation: 1
  name: join
  resourceVersion: "1571333"
  selfLink: /apis/kubeovn.io/v1/subnets/join
  uid: 9c744810-c678-4d50-8a7d-b8ec12ef91b8
spec:
  cidrBlock: 100.64.0.0/16
  default: false
  excludeIps:
    - 100.64.0.1
  gateway: 100.64.0.1
  gatewayNode: ""
  gatewayType: ""
  natOutgoing: false
  private: false
  protocol: IPv4

```

Node      ovn0

```

# ifconfig ovn0
ovn0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1420
    inet 100.64.0.4 netmask 255.255.0.0 broadcast 100.64.255.255
    inet6 fe80::800:ff:fe40:5 prefixlen 64 scopeid 0x20<link>
    ether 0a:00:00:40:00:05 txqueuelen 1000 (Ethernet)
    RX packets 18 bytes 1428 (1.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 19 bytes 1810 (1.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

### 3.1.3

#### Namespace

```

cat <<EOF | kubectl create -f -
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 10.66.0.0/16
  excludeIps:

```



```
- 10.66.0.1..10.66.0.10
- 10.66.0.101..10.66.0.151
gateway: 10.66.0.1
gatewayType: distributed
natOutgoing: true
routeTable: ""
namespaces:
- ns1
- ns2
EOF
```

- cidrBlock: CIDR VPC Subnet CIDR
- excludeIps: IP Underlay
- gateway Overlay Kube-OVN Underlay
- namespaces: Namespace Namespace Pod
- routeTable:

```
# kubectl create ns ns1
namespace/ns1 created

# kubectl run nginx --image=docker.io/library/nginx:alpine -n ns1
deployment.apps/nginx created

# kubectl get pod -n ns1 -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
nginx-74d5899f46-n8wtg 1/1 Running 0 10s 10.66.0.11 node1 <none> <none>
```

Workload

Pod Namespace IP Namespace Workload Pod Annotation ovn.kubernetes.io/logical\_switch

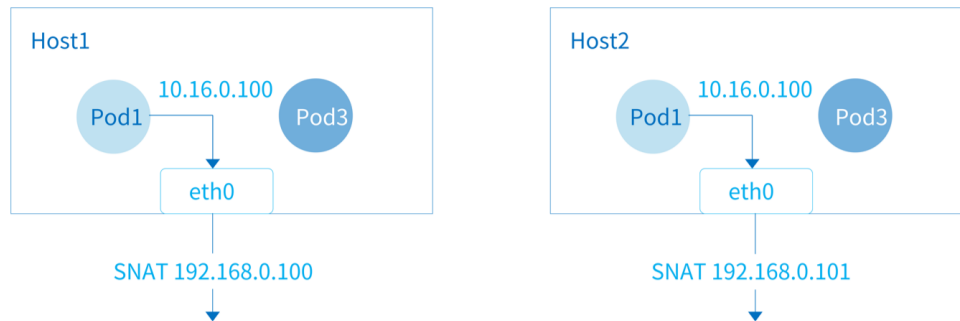
```
apiVersion: v1
kind: Pod
metadata:
  name: another-subnet
  annotations:
    ovn.kubernetes.io/logical_switch: subnet1
spec:
  containers:
  - name: another-subnet
    image: docker.io/library/nginx:alpine
```

Workload Deployment StatefulSet ovn.kubernetes.io/logical\_switch Annotation

spec.template.metadata.annotations

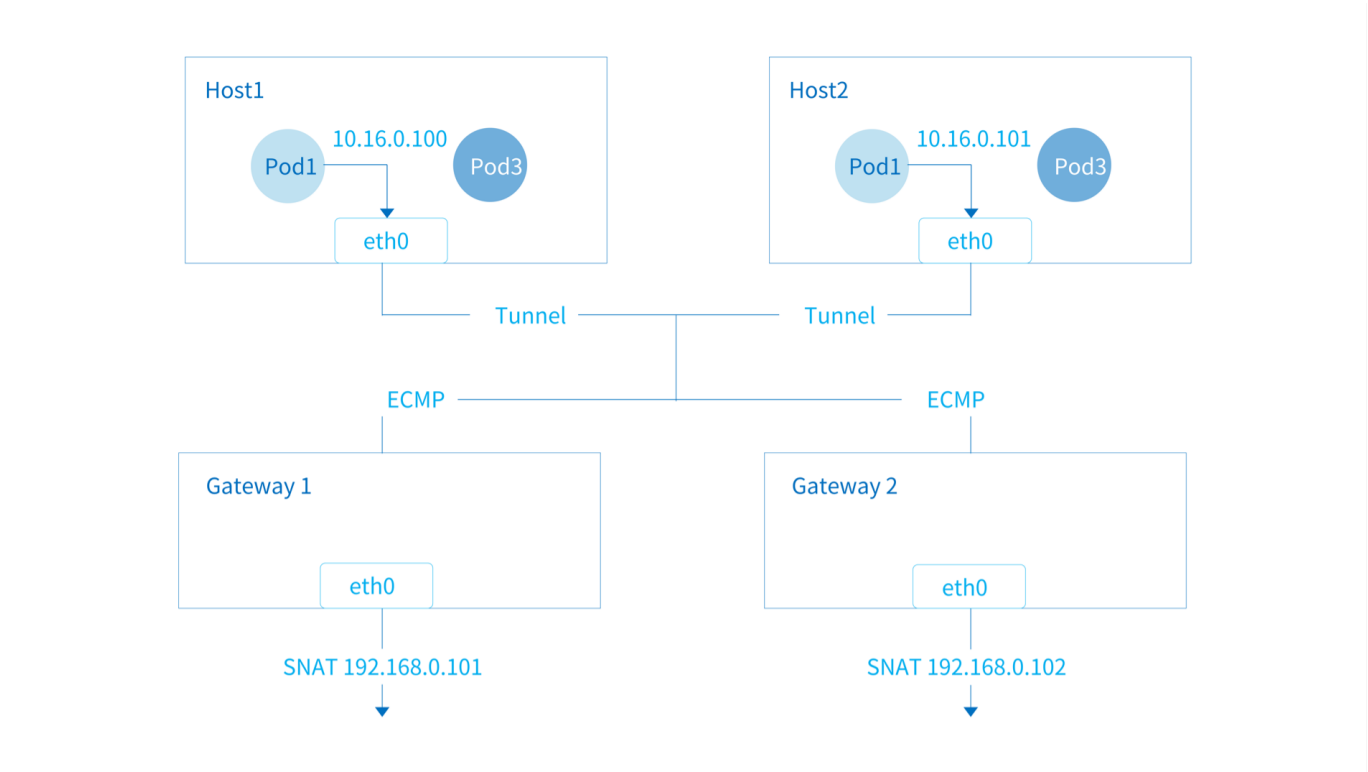
3.1.4 Overlay





gatewayType distributed

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: distributed
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
    - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: distributed
  natOutgoing: true
```



|             |             |             |            |             |             |
|-------------|-------------|-------------|------------|-------------|-------------|
| Pod         | IP          | Pod         | ovn0       | natOutgoing | true        |
|             | IP          |             |            |             |             |
| gatewayType | centralized | gatewayNode | Kubernetes | nodeName    | gatewayNode |

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: centralized
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
    - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: centralized
  gatewayNode: "node1,node2"
  natOutgoing: true
```

- gatewayNode kube-ovn-worker:172.18.0.2, kube-ovn-control-plane:172.18.0.3
- ECMP ECMP
- Kube-OVN v1.12.0 subnet crd spec enableEcmp ECMP ECMP kube-ovn-  
controller Deployment enable-ecmp v1.12.0

|            |                     |      |    |        |
|------------|---------------------|------|----|--------|
| ECMP       | kube-ovn-controller | ping | 5s | 5s-10s |
| Node Ready |                     |      |    |        |

gatewayNodeSelectors

 gatewayNodeSelectors

- gatewayNode gatewayNode gatewayNodeSelectors
- OR
-

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: centralized-selector
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
    - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: centralized
  gatewayNodeSelectors:
    - matchLabels:
        role: gateway
    - matchExpressions:
        - key: node-type
          operator: In
          values: ["gateway", "egress"]
  natOutgoing: true
```

3.1.5 ACL

 Warning

Kube-OVN   NetworkPolicy   Network Policy API   Subnet ACL   Security Group   OVN ACL   NetworkPolicy   NetworkPolicy API

ACL   Kube-OVN   Subnet   ACL

Subnet   ACL   OVN   ACL   ovn-nb ACL Table   match   ovn-sb Logical Flow Table

IP   10.10.0.2   Pod   ACL

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: acl
spec:
  allowEWTraffic: false
  acls:
    - action: drop
      direction: to-lport
      match: ip4.dst == 10.10.0.2 && ip
      priority: 1002
    - action: allow-related
      direction: from-lport
      match: ip4.src == 10.10.0.2 && ip
      priority: 1002
  cidrBlock: 10.10.0.0/24
```

ACL   allowEWTraffic: true

3.1.6

ACL   ACL

Kube-OVN   Pod

CRD   private   true   allowSubnets   allowSubnets

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: private
spec:
  protocol: IPv4
  default: false
  namespaces:
    - ns1
    - ns2
  cidrBlock: 10.69.0.0/16
  private: true
  allowSubnets:
    - 10.16.0.0/16
    - 10.18.0.0/16
```

### 3.1.7 Underlay

```

      Underlay
    • vlan:      Underlay      Subnet   Vlan CR      Underlay
    • logicalGateway: Underlay      OVN      Underlay Overlay false

```

### 3.1.8

|              |     |      |     |          |      |
|--------------|-----|------|-----|----------|------|
| kube-ovn-cni | Pod | ICMP | ARP | Underlay | ICMP |
|--------------|-----|------|-----|----------|------|

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: disable-gw-check
spec:
  disableGatewayCheck: true
```

### 3.1.9 Multicast-Snoop

|        |     |     |     |        |                 |     |                |                 |
|--------|-----|-----|-----|--------|-----------------|-----|----------------|-----------------|
| subnet | Pod | OVN | Pod | subnet | multicast snoop | OVN | South Database | Multicast_Group |
|--------|-----|-----|-----|--------|-----------------|-----|----------------|-----------------|

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: sample1
spec:
  enableMulticastSnoop: true
```

### 3.1.10 Subnet MTU

| Subnet | Pod | MTU | Subnet | Pod |
|--------|-----|-----|--------|-----|
|--------|-----|-----|--------|-----|

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: sample1
spec:
  mtu: 1300
```

### 3.1.11

- IP
- VPC NAT
- QoS
- 
- DHCP
- 
-

• IP

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3.1.12

---

3.2

| Kube-OVN | Pod           | Namespace | IP | Mac | Kube-OVN |
|----------|---------------|-----------|----|-----|----------|
|          | • Pod         | IP/Mac    |    |     |          |
|          | • Workload    | IP Pool   |    |     |          |
|          | • StatefulSet |           |    |     |          |
|          | • KubeVirt VM |           |    |     |          |
|          | • Multus      |           |    |     |          |

3.2.1 Pod IP Mac

| Pod                                                                                                                                                                                                                                                                                             | annotation | Pod        | IP/Mac, kube-ovn-controller |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------|-----------------------------|
| <pre>apiVersion: v1 kind: Pod metadata:   name: static-ip   annotations:     ovn.kubernetes.io/ip_address: 10.16.0.15 # 10.16.0.15, fd00:10:16::15     ovn.kubernetes.io/mac_address: 00:00:00:53:6B:B6 spec:   containers:   - name: static-ip     image: docker.io/library/nginx:alpine</pre> |            |            |                             |
|                                                                                                                                                                                                                                                                                                 | annotation | Pod IP/Mac |                             |
| 1.                                                                                                                                                                                                                                                                                              | IP/Mac     | IP/Mac     |                             |
| 2.                                                                                                                                                                                                                                                                                              | IP         | CIDR       |                             |
| 3.                                                                                                                                                                                                                                                                                              | IP         | Mac        |                             |

3.2.2 Workload IP Pool

| Kube-OVN   | annotation                | ovn.kubernetes.io/ip_pool | Workload   | Deployment/StatefulSet/DaemonSet/Job/CronJob | IP       | kube-ovn- |
|------------|---------------------------|---------------------------|------------|----------------------------------------------|----------|-----------|
| controller | ovn.kubernetes.io/ip_pool | IP                        |            |                                              |          |           |
| IP Pool    | Annotation                | template                  | annotation | Kubernetes                                   | Workload | Workload  |

Deployment IP

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <pre>apiVersion: apps/v1 kind: Deployment metadata:   name: ippool   labels:     app: ippool spec:   replicas: 2   selector:     matchLabels:       app: ippool   template:     metadata:       labels:         app: ippool       annotations:         ovn.kubernetes.io/ip_pool: 10.16.0.15,10.16.0.16,10.16.0.17 # 10.16.0.15, fd00:10:16::000E;10.16.0.16, fd00:10:16::000F;         10.16.0.17, fd00:10:16::0010     spec:       containers:       - name: ippool         image: docker.io/library/nginx:alpine</pre> |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

| Workload                                  | IP |          |      |          |  |                                        |    |
|-------------------------------------------|----|----------|------|----------|--|----------------------------------------|----|
| 1. <code>ovn.kubernetes.io/ip_pool</code> | IP |          | CIDR |          |  |                                        |    |
| 2. <code>ovn.kubernetes.io/ip_pool</code> | IP |          | IP   |          |  |                                        |    |
| 3. <code>ovn.kubernetes.io/ip_pool</code> | IP | replicas | Pod  | Workload |  | <code>ovn.kubernetes.io/ip_pool</code> | IP |

### 3.2.3 StatefulSet

|                |                           |                           |                           |     |           |                           |        |             |        |
|----------------|---------------------------|---------------------------|---------------------------|-----|-----------|---------------------------|--------|-------------|--------|
| StatefulSet    | IP                        | Workload                  | ovn.kubernetes.io/ip_pool | Pod | IP        |                           |        |             |        |
| StatefulSet    | Kube-OVN                  |                           |                           |     |           |                           |        |             |        |
| 1. Pod         | ovn.kubernetes.io/ip_pool | IP                        | StatefulSet               | web | web-0     | ovn.kubernetes.io/ip_pool | IP     | web-1       | IP     |
| 2. StatefulSet | Pod                       | OVN                       | logical_switch_port       | Pod | interface | Pod                       | IP/Mac | StatefulSet | Volume |
| 3.             | 2                         | ovn.kubernetes.io/ip_pool | StatefulSet               | Pod | IP/Mac    | StatefulSet               |        |             |        |

## StatefulSet

```
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: web
spec:
  serviceName: "nginx"
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
  spec:
    containers:
      - name: nginx
        image: docker.io/library/nginx:alpine
        ports:
          - containerPort: 80
            name: web
```

| StatefulSet | Pod | Pod IP |
|-------------|-----|--------|
|-------------|-----|--------|

### StatefulSet Pod IP

| StatefulSet | IP     | Pod Name    | Statefulset | ovn.kubernetes.io/ip_pool | Annotation | Pod         | IP |
|-------------|--------|-------------|-------------|---------------------------|------------|-------------|----|
| StatefulSet | Pod IP | StatefulSet | scale       | 0                         | Annotation | StatefulSet |    |

### 3.2.4 KubeVirt VM

| KubeVirt | VM | kube-ovn-controller | StatefulSet Pod | IP | VM |
|----------|----|---------------------|-----------------|----|----|
| VM IP    |    |                     |                 |    |    |

### 3.2.5 Multus

|        |     |          |            |          |              |      |     |
|--------|-----|----------|------------|----------|--------------|------|-----|
| Multus | Pod | Kube-OVN | annotation | Kube-OVN | CNI Kube-OVN | IPAM | CNI |
|--------|-----|----------|------------|----------|--------------|------|-----|





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3.2.6

---

3.3 IP

| IP | IPPool | Subnet | IPAM | IP | Namespace | Workload |
|----|--------|--------|------|----|-----------|----------|
|----|--------|--------|------|----|-----------|----------|

3.3.1

```
apiVersion: kubeovn.io/v1
kind: IPPool
metadata:
  name: pool-1
spec:
  subnet: ovn-default
  ips:
    - "10.16.0.201"
    - "10.16.0.210/30"
    - "10.16.0.220..10.16.0.230"
  namespaces:
    - ns-1
  enableAddressSet: true
```

Workload

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ippool
  labels:
    app: ippool
spec:
  replicas: 2
  selector:
    matchLabels:
      app: ippool
  template:
    metadata:
      labels:
        app: ippool
      annotations:
        ovn.kubernetes.io/ip_pool: pool-1
    spec:
      containers:
        - name: ippool
          image: docker.io/library/nginx:alpine
```

|                  |            |       |      |     |            |    |
|------------------|------------|-------|------|-----|------------|----|
|                  |            |       |      |     |            |    |
| subnet           |            |       |      |     |            |    |
| ips              | IP         | ..    | IPv6 |     |            |    |
| namespaces       |            |       |      | Pod | IP         | IP |
| enableAddressSet | AddressSet | false | true | ACL | AddressSet |    |

3.3.2

1. [Workload](#) [IP Pool](#) IP IP
2. IP [.spec.ips](#) IP IP [.spec.ips](#) CIDR
3. IP IP
4. IP [.spec.ips](#)
5. IP IP IP IP IP IP
6. IP IP
7. IP Namespace

8. IP `.spec.enableAddressSet` `false` `true` IP OVN NB AddressSet IP IP AddressSet  
NetworkPolicy VPC AddressSet

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3.3.3

---

3.4

Pod Annotations

```
apiVersion: v1
kind: Pod
metadata:
  name: custom-routes
  annotations:
    ovn.kubernetes.io/routes: |
      [{
        "dst": "192.168.0.101/24",
        "gw": "10.16.0.254"
      }, {
        "gw": "10.16.0.254"
      }]
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
```

dst

Deployment DaemonSet StatefulSet Annotation .spec.template.metadata.annotations

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: custom-routes
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
      annotations:
        ovn.kubernetes.io/routes: |
          [{
            "dst": "192.168.0.101/24",
            "gw": "10.16.0.254"
          }, {
            "gw": "10.16.0.254"
          }]
    spec:
      containers:
        - name: nginx
          image: docker.io/library/nginx:alpine
```

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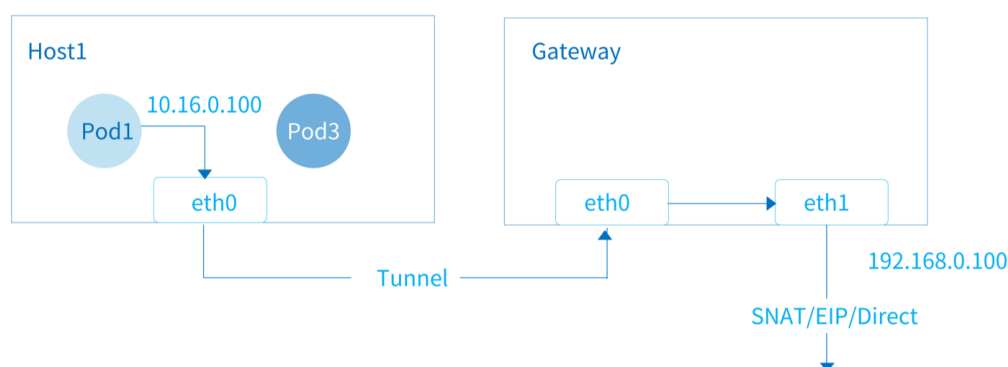
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3.4.1

### 3.5 EIP SNAT

[illegible]

### 3.5.1

- OVN L3 Gateway OVS Overlay Underlay
- NAT Underlay
- EIP SNAT

### 3.5.2

kube-system ConfigMap ovn-external-gw-config

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-external-gw-config
  namespace: kube-system
data:
  enable-external-gw: "true"
  external-gw-nodes: "kube-ovn-worker"
  external-gw-nic: "eth1"
  external-gw-addr: "172.56.0.1/16"
```

```
nic-ip: "172.56.0.254/16"
nic-mac: "16:52:f3:13:6a:25"
```

- enable-external-gw: SNAT EIP
- type: centralized distributed centralized distributed
- external-gw-nodes: centralized
- external-gw-nic:
- external-gw-addr: IP
- nic-ip, nic-mac: IP Mac IP Mac

### 3.5.3 OVN OVS

OVN-NB      ovn-external      ovn-cluster-ovn-external      chassis

```
# kubectl ko nbctl show
switch 3de4cea7-1a71-43f3-8b62-435a57ef16a6 (external)
  port localnet.external
    type: localnet
    addresses: ["unknown"]
  port external-ovn-cluster
    type: router
    router-port: ovn-cluster-external
router e1eb83ad-34be-4ed5-9a02-fcc8b1d357c4 (ovn-cluster)
  port ovn-cluster-external
    mac: "ac:1f:6b:2d:33:f1"
    networks: ["172.56.0.100/16"]
    gateway chassis: [a5682814-2e2c-46dd-9c1c-6803ef0dab66]
```

OVS      br-external

```
# kubectl ko vsctl ${gateway node name} show
e7d81150-7743-4d6e-9e6f-5c688232e130
Bridge br-external
  Port br-external
    Interface br-external
      type: internal
  Port eth1
    Interface eth1
  Port patch-localnet.external-to-br-int
    Interface patch-localnet.external-to-br-int
      type: patch
      options: {peer=patch-br-int-to-localnet.external}
```

### 3.5.4 Pod EIP SNAT

Pod      ovn.kubernetes.io/snat      ovn.kubernetes.io/eip annotation      SNAT EIP

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-snat
  annotations:
    ovn.kubernetes.io/snat: 172.56.0.200
spec:
  containers:
  - name: pod-snat
    image: docker.io/library/nginx:alpine
---
apiVersion: v1
kind: Pod
metadata:
  name: pod-eip
  annotations:
    ovn.kubernetes.io/eip: 172.56.0.233
spec:
  containers:
  - name: pod-eip
    image: docker.io/library/nginx:alpine
```

kubectl      Pod      EIP SNAT      ovn.kubernetes.io/routed annotation

```
kubectl annotate pod pod-gw ovn.kubernetes.io/eip=172.56.0.221 --overwrite
kubectl annotate pod pod-gw ovn.kubernetes.io/routed-
```

EIP SNAT      ovn.kubernetes.io/routed annotation

3.5.5

```
kube-ovn-controller SNAT EIP

• --external-gateway-config-ns: Configmap ovn-external-gw-config Namespace kube-system
• --external-gateway-net: external
• --external-gateway-vlanid: Vlan Tag 0 Vlan
```

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3.5.6

### 3.6 QoS

The diagram illustrates the mapping of Kube-OVN to Pod and QoS, and then to Pod, QoS, Namespace, Subnet, and QoS.

- Kube-OVN maps to Pod and QoS.
- Pod maps to QoS.
- linux-netem maps to QoS.

The bottom row shows the mapping of Pod, QoS, Namespace, Subnet, and QoS.

### 3.6.1 QoS

```
apiVersion: v1
kind: Pod
metadata:
  name: qos
  namespace: ls1
  annotations:
    ovn.kubernetes.io/ingress_rate: "3"
    ovn.kubernetes.io/egress_rate: "1"
spec:
  containers:
    - name: qos
      image: docker.io/library/nginx:alpine
```

annotation      QoS

```
kubectl annotate --overwrite pod nginx-74d5899f46-d7qkn ovn.kubernetes.io/ingress_rate=3
```

## QoS

```
kind: DaemonSet
apiVersion: apps/v1
metadata:
  name: perf
  namespace: ls1
  labels:
    app: perf
spec:
  selector:
    matchLabels:
      app: perf
  template:
    metadata:
      labels:
        app: perf
    spec:
      containers:
        - name: nginx
          image: docker.io/kubeovn/perf
```

Pod      iperf3 server

```
# kubectl exec -it perf-4n4gt -n ls1 sh
# iperf3 -s
-----
Server listening on 5201
-----
```

```
Pod      Pod
# kubectl exec -it perf-d4mqc -n ls1 sh
# iperf3 -c 10.66.0.12
Connecting to host 10.66.0.12, port 5201
[  4] local 10.66.0.14 port 51544 connected to 10.66.0.12 port 5201
[ ID] Interval           Transfer     Bandwidth       Retr   Cwnd
[  4]  0.00-1.00    sec    86.4 MBytes    725 Mbits/sec     3    350 KBytes
[  4]  1.00-2.00    sec    89.9 MBytes    754 Mbits/sec    118   473 KBytes
[  4]  2.00-3.00    sec    101 MBytes    848 Mbits/sec    184   586 KBytes
[  4]  3.00-4.00    sec    104 MBytes    875 Mbits/sec    217   671 KBytes
[  4]  4.00-5.00    sec    111 MBytes    935 Mbits/sec    175   772 KBytes
```



```
[ 4] 5.00-6.00 sec 100 MBytes 840 Mbits/sec 658 598 KBytes
[ 4] 6.00-7.00 sec 106 MBytes 890 Mbits/sec 742 668 KBytes
[ 4] 7.00-8.00 sec 102 MBytes 857 Mbits/sec 764 724 KBytes
[ 4] 8.00-9.00 sec 97.4 MBytes 817 Mbits/sec 1175 764 KBytes
[ 4] 9.00-10.00 sec 111 MBytes 934 Mbits/sec 1083 838 KBytes
-----
[ ID] Interval      Transfer      Bandwidth      Retr
[ 4] 0.00-10.00 sec 1010 MBytes 848 Mbits/sec 5119          sender
[ 4] 0.00-10.00 sec 1008 MBytes 846 Mbits/sec          receiver

iperf Done.
```

Pod QoS

```
kubectl annotate --overwrite pod perf-4n4gt -n ls1 ovn.kubernetes.io/ingress_rate=30
```

Pod Pod

```
# iperf3 -c 10.66.0.12
Connecting to host 10.66.0.12, port 5201
[ 4] local 10.66.0.14 port 52372 connected to 10.66.0.12 port 5201
[ ID] Interval      Transfer      Bandwidth      Retr  Cwnd
[ 4] 0.00-1.00 sec 3.66 MBytes 30.7 Mbits/sec 2 76.1 KBytes
[ 4] 1.00-2.00 sec 3.43 MBytes 28.8 Mbits/sec 0 104 KBytes
[ 4] 2.00-3.00 sec 3.50 MBytes 29.4 Mbits/sec 0 126 KBytes
[ 4] 3.00-4.00 sec 3.50 MBytes 29.3 Mbits/sec 0 144 KBytes
[ 4] 4.00-5.00 sec 3.43 MBytes 28.8 Mbits/sec 0 160 KBytes
[ 4] 5.00-6.00 sec 3.43 MBytes 28.8 Mbits/sec 0 175 KBytes
[ 4] 6.00-7.00 sec 3.50 MBytes 29.3 Mbits/sec 0 212 KBytes
[ 4] 7.00-8.00 sec 3.68 MBytes 30.9 Mbits/sec 0 294 KBytes
[ 4] 8.00-9.00 sec 3.74 MBytes 31.4 Mbits/sec 0 398 KBytes
[ 4] 9.00-10.00 sec 3.80 MBytes 31.9 Mbits/sec 0 526 KBytes
-----
[ ID] Interval      Transfer      Bandwidth      Retr
[ 4] 0.00-10.00 sec 35.7 MBytes 29.9 Mbits/sec 2          sender
[ 4] 0.00-10.00 sec 34.5 MBytes 29.0 Mbits/sec          receiver

iperf Done.
```

3.6.2 linux-netem QoS

```
RHEL netem yum install -y kernel-modules-extra && modprobe sch_netem
```

Pod annotation linux-netem QoS netem QoS Pod Ingress

- ovn.kubernetes.io/latency Pod ms
- ovn.kubernetes.io/jitter Pod ms
- ovn.kubernetes.io/limit qdisc 1000
- ovn.kubernetes.io/loss float 20 20%

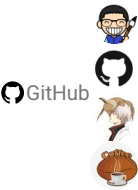
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3.6.3

3.7

|            |            |
|------------|------------|
| Kube-OVN   | IPv4 IPv6  |
| Kubernetes | Kubernetes |

3.7.1

CIDR    cidr=<IPv4 CIDR>,<IPv6 CIDR>    CIDR    IPv4    IPv6

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: ovn-test
spec:
  cidrBlock: 10.16.0.0/16,fd00:10:16::/64
  excludeIps:
    - 10.16.0.1
    - fd00:10:16::1
  gateway: 10.16.0.1,fd00:10:16::1
```

```
POD_CIDR="10.16.0.0/16,fd00:10:16::/64"
JOIN_CIDR="100.64.0.0/16,fd00:100:64::/64"
```

3.7.2 Pod

Pod    IPv4 IPv6    Pod    annotation :

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/allocated: "true"
    ovn.kubernetes.io/cidr: 10.16.0.0/16,fd00:10:16::/64
    ovn.kubernetes.io/gateway: 10.16.0.1,fd00:10:16::1
    ovn.kubernetes.io/ip_address: 10.16.0.9,fd00:10:16::9
    ovn.kubernetes.io/logical_switch: ovn-default
    ovn.kubernetes.io/mac_address: 00:00:00:14:88:09
    ovn.kubernetes.io/network_types: geneve
    ovn.kubernetes.io/routed: "true"
...
podIP: 10.16.0.9
podIPs:
  - ip: 10.16.0.9
  - ip: fd00:10:16::9
```

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3.7.3

## 3.8 Webhook

|         |          |     |          |         |             |
|---------|----------|-----|----------|---------|-------------|
| Webhook | Kube-OVN | CRD | Webhook  | IP      | Subnet CIDR |
| Webhook | Subnet   | Pod | Kube-OVN | Webhook | Pod         |

### 3.8.1 Cert-Manager

|         |              |         |              |
|---------|--------------|---------|--------------|
| Webhook | cert-manager | Webhook | cert-manager |
|---------|--------------|---------|--------------|

cert-manager:

```
kubectl apply -f https://github.com/cert-manager/cert-manager/releases/download/v1.8.0/cert-manager.yaml
```

cert-manager [cert-manager](#)

### 3.8.2 Webhook

Webhook yml :

```
# kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.15/yamls/webhook.yaml
deployment.apps/kube-ovn-webhook created
service/kube-ovn-webhook created
validatingwebhookconfiguration.admissionregistration.k8s.io/kube-ovn-webhook created
certificate.cert-manager.io/kube-ovn-webhook-serving-cert created
issuer.cert-manager.io/kube-ovn-webhook-selfsigned-issuer created
```

### 3.8.3 Webhook

Pod Pod IP 10.16.0.15

```
# kubectl get pod -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
static-7584848b74-fw9dm 1/1 Running 0 2d13h 10.16.0.15 kube-ovn-worker <none>
```

yaml IP Pod

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/ip_address: 10.16.0.15
    ovn.kubernetes.io/mac_address: 00:00:00:53:6B:B6
  labels:
    app: static
managedFields:
  name: staticip-pod
  namespace: default
spec:
  containers:
  - image: docker.io/library/nginx:alpine
    imagePullPolicy: IfNotPresent
    name: qatest
```

yaml Pod IP

```
# kubectl apply -f pod-static.yaml
Error from server (annotation ip address 10.16.0.15 is conflict with ip crd static-7584848b74-fw9dm.default 10.16.0.15): error when creating "pod-static.yaml": admission webhook "pod-ip-validating.kube-ovn.io" denied the request: annotation ip address 10.16.0.15 is conflict with ip crd static-7584848b74-fw9dm.default 10.16.0.15
```



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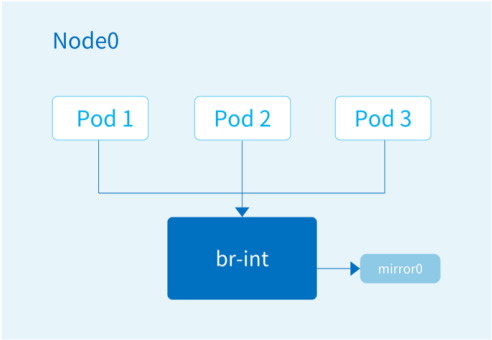
3.8.4

---

3.9

NPM

CPU                      5%~10%                      CPU



3.9.1

kube-ovn-cni DaemonSet

- --enable-mirror=true
- --mirror-iface=mirror0:

br-int

Kube-OVN

tcpdump                      mirror0

```
tcpdump -ni mirror0
```

3.9.2 Pod

Pod                      Pod      `ovn.kubernetes.io/mirror` annotation      Pod

```
apiVersion: v1
kind: Pod
metadata:
  name: mirror-pod
  namespace: ls1
  annotations:
    ovn.kubernetes.io/mirror: "true"
spec:
  containers:
    - name: mirror-pod
      image: docker.io/library/nginx:alpine
```

## 3.9.3

## 1. Pod to Pod in the same Nodes

| Size | TCP Latency | TCP Bandwidth  | UDP Latency | UDP Lost Rate | UDP Bandwidth  |
|------|-------------|----------------|-------------|---------------|----------------|
| 64   | 12.7 us     | 289 Mbits/sec  | 12.6 us     | (1.8%)        | 77.9 Mbits/sec |
| 128  | 15.5 us     | 517 Mbits/sec  | 12.7 us     | (0%)          | 155 Mbits/sec  |
| 512  | 12.2 us     | 1.64 Gbits/sec | 12.4 us     | (0%)          | 624 Mbits/sec  |
| 1k   | 13 us       | 2.96 Gbits/sec | 11.4 us     | (0.53%)       | 1.22 Gbits/sec |
| 4k   | 18 us       | 7.67 Gbits/sec | 25.7 us     | (0.41%)       | 1.50 Gbits/sec |

| Size | TCP Latency | TCP Bandwidth  | UDP Latency | UDP Lost Rate | UDP Bandwidth  |
|------|-------------|----------------|-------------|---------------|----------------|
| 64   | 11.9 us     | 324 Mbits/sec  | 12.2 us     | (0.22%)       | 102 Mbits/sec  |
| 128  | 10.5 us     | 582 Mbits/sec  | 9.5 us      | (0.21%)       | 198 Mbits/sec  |
| 512  | 11.6 us     | 1.84 Gbits/sec | 9.32 us     | (0.091%)      | 827 Mbits/sec  |
| 1k   | 10.5 us     | 3.44 Gbits/sec | 10 us       | (1.2%)        | 1.52 Gbits/sec |
| 4k   | 16.7 us     | 8.52 Gbits/sec | 18.2 us     | (1.3%)        | 2.42 Gbits/sec |

## 2. Pod to Pod in the different Nodes

| Size | TCP Latency | TCP Bandwidth  | UDP Latency | UDP Lost Rate | UDP Bandwidth  |
|------|-------------|----------------|-------------|---------------|----------------|
| 64   | 258 us      | 143 Mbits/sec  | 237 us      | (61%)         | 28.5 Mbits/sec |
| 128  | 240 us      | 252 Mbits/sec  | 231 us      | (64%)         | 54.9 Mbits/sec |
| 512  | 236 us      | 763 Mbits/sec  | 256 us      | (68%)         | 194 Mbits/sec  |
| 1k   | 242 us      | 969 Mbits/sec  | 225 us      | (62%)         | 449 Mbits/sec  |
| 4k   | 352 us      | 1.12 Gbits/sec | 382 us      | (0.71%)       | 21.4 Mbits/sec |

| Size | TCP Latency | TCP Bandwidth  | UDP Latency | UDP Lost Rate | UDP Bandwidth  |
|------|-------------|----------------|-------------|---------------|----------------|
| 64   | 278 us      | 140 Mbits/sec  | 227 us      | (24%)         | 59.6 Mbits/sec |
| 128  | 249 us      | 265 Mbits/sec  | 265 us      | (23%)         | 114 Mbits/sec  |
| 512  | 233 us      | 914 Mbits/sec  | 235 us      | (21%)         | 468 Mbits/sec  |
| 1k   | 238 us      | 1.14 Gbits/sec | 240 us      | (15%)         | 891 Mbits/sec  |
| 4k   | 370 us      | 1.25 Gbits/sec | 361 us      | (0.43%)       | 7.54 Mbits/sec |

### 3. Node to Node

| Size | TCP Latency | TCP Bandwidth  | UDP Latency | UDP Lost Rate | UDP Bandwidth  |
|------|-------------|----------------|-------------|---------------|----------------|
| 64   | 205 us      | 162 Mbits/sec  | 183 us      | (11%)         | 74.2 Mbits/sec |
| 128  | 222 us      | 280 Mbits/sec  | 206 us      | (6.3%)        | 155 Mbits/sec  |
| 512  | 220 us      | 1.04 Gbits/sec | 177 us      | (20%)         | 503 Mbits/sec  |
| 1k   | 213 us      | 2.06 Gbits/sec | 201 us      | (8.6%)        | 1.14 Gbits/sec |
| 4k   | 280 us      | 5.01 Gbits/sec | 315 us      | (37%)         | 1.20 Gbits/sec |

| Size | TCP Latency | TCP Bandwidth  | UDP Latency | UDP Lost Rate | UDP Bandwidth  |
|------|-------------|----------------|-------------|---------------|----------------|
| 64   | 204 us      | 157 Mbits/sec  | 204 us      | (8.8%)        | 81.9 Mbits/sec |
| 128  | 213 us      | 262 Mbits/sec  | 225 us      | (19%)         | 136 Mbits/sec  |
| 512  | 220 us      | 1.02 Gbits/sec | 227 us      | (21%)         | 486 Mbits/sec  |
| 1k   | 217 us      | 1.79 Gbits/sec | 218 us      | (29%)         | 845 Mbits/sec  |
| 4k   | 275 us      | 5.27 Gbits/sec | 336 us      | (34%)         | 1.21 Gbits/sec |

### 4. Pod to the Node where the Pod is located

| Size | TCP Latency | TCP Bandwidth  | UDP Latency | UDP Lost Rate | UDP Bandwidth  |
|------|-------------|----------------|-------------|---------------|----------------|
| 64   | 12.2 us     | 295 Mbits/sec  | 12.7 us     | (0.27%)       | 74.1 Mbits/sec |
| 128  | 14.1 us     | 549 Mbits/sec  | 10.6 us     | (0.41%)       | 153 Mbits/sec  |
| 512  | 13.5 us     | 1.83 Gbits/sec | 12.7 us     | (0.23%)       | 586 Mbits/sec  |
| 1k   | 12 us       | 2.69 Gbits/sec | 13 us       | (1%)          | 1.16 Gbits/sec |
| 4k   | 18.9 us     | 4.51 Gbits/sec | 21.8 us     | (0.42%)       | 1.81 Gbits/sec |

| Size | TCP Latency | TCP Bandwidth  | UDP Latency | UDP Lost Rate | UDP Bandwidth  |
|------|-------------|----------------|-------------|---------------|----------------|
| 64   | 10.4 us     | 335 Mbits/sec  | 12.2 us     | (0.75%)       | 95.4 Mbits/sec |
| 128  | 12.1 us     | 561 Mbits/sec  | 11.3 us     | (0.25%)       | 194 Mbits/sec  |
| 512  | 11.6 us     | 1.87 Gbits/sec | 10.7 us     | (0.66%)       | 745 Mbits/sec  |
| 1k   | 12.7 us     | 3.12 Gbits/sec | 10.9 us     | (1.2%)        | 1.46 Gbits/sec |
| 4k   | 16.5 us     | 8.23 Gbits/sec | 17.9 us     | (1.5%)        | 2.51 Gbits/sec |

## 5. Pod to the Node where the Pod is not located

| Size | TCP Latency | TCP Bandwidth  | UDP Latency | UDP Lost Rate | UDP Bandwidth  |
|------|-------------|----------------|-------------|---------------|----------------|
| 64   | 234 us      | 153 Mbits/sec  | 232 us      | (63%)         | 29.4 Mbits/sec |
| 128  | 237 us      | 261 Mbits/sec  | 238 us      | (49%)         | 76.1 Mbits/sec |
| 512  | 231 us      | 701 Mbits/sec  | 238 us      | (57%)         | 279 Mbits/sec  |
| 1k   | 256 us      | 1.05 Gbits/sec | 228 us      | (56%)         | 524 Mbits/sec  |
| 4k   | 330 us      | 1.08 Gbits/sec | 359 us      | (1.5%)        | 35.7 Mbits/sec |

| Size | TCP Latency | TCP Bandwidth  | UDP Latency | UDP Lost Rate | UDP Bandwidth  |
|------|-------------|----------------|-------------|---------------|----------------|
| 64   | 283 us      | 141 Mbits/sec  | 230 us      | (26%)         | 55.8 Mbits/sec |
| 128  | 234 us      | 255 Mbits/sec  | 234 us      | (25%)         | 113 Mbits/sec  |
| 512  | 246 us      | 760 Mbits/sec  | 234 us      | (22%)         | 458 Mbits/sec  |
| 1k   | 268 us      | 1.23 Gbits/sec | 242 us      | (20%)         | 879 Mbits/sec  |
| 4k   | 326 us      | 1.20 Gbits/sec | 369 us      | (0.5%)        | 7.87 Mbits/sec |

## 6. Pod to the cluster ip service

| Size | TCP Latency | TCP Bandwidth  | UDP Latency | UDP Lost Rate | UDP Bandwidth  |
|------|-------------|----------------|-------------|---------------|----------------|
| 64   | 237 us      | 133 Mbits/sec  | 213 us      | (65%)         | 25.5 Mbits/sec |
| 128  | 232 us      | 271 Mbits/sec  | 222 us      | (62%)         | 54.8 Mbits/sec |
| 512  | 266 us      | 800 Mbits/sec  | 234 us      | (60%)         | 232 Mbits/sec  |
| 1k   | 248 us      | 986 Mbits/sec  | 239 us      | (50%)         | 511 Mbits/sec  |
| 4k   | 314 us      | 1.03 Gbits/sec | 367 us      | (0.6%)        | 13.2 Mbits/sec |

| TCP-Conn-Number | QPS      | Avg-Resp-Time | Stdev-Resp-Time | Max-Resp-Time |
|-----------------|----------|---------------|-----------------|---------------|
| 10              | 14305.17 | 0.87ms        | 1.48ms          | 24.46ms       |
| 100             | 29082.07 | 3.87ms        | 4.35ms          | 102.85ms      |



| Size | TCP Latency | TCP Bandwidth  | UDP Latency | UDP Lost Rate | UDP Bandwidth  |
|------|-------------|----------------|-------------|---------------|----------------|
| 64   | 241 us      | 145 Mbits/sec  | 225 us      | (19%)         | 60.2 Mbits/sec |
| 128  | 245 us      | 261 Mbits/sec  | 212 us      | (15%)         | 123 Mbits/sec  |
| 512  | 252 us      | 821 Mbits/sec  | 219 us      | (14%)         | 499 Mbits/sec  |
| 1k   | 253 us      | 1.08 Gbits/sec | 242 us      | (16%)         | 852 Mbits/sec  |
| 4k   | 320 us      | 1.32 Gbits/sec | 360 us      | (0.47%)       | 6.70 Mbits/sec |

| TCP-Conn-Number | QPS      | Avg-Resp-Time | Stdev-Resp-Time | Max-Resp-Time |
|-----------------|----------|---------------|-----------------|---------------|
| 10              | 13634.07 | 0.96ms        | 1.72ms          | 30.07ms       |
| 100             | 30215.23 | 3.59ms        | 3.20ms          | 77.56ms       |

#### 7. Host to the Node port service where the Pod is not located on the target Node

| TCP-Conn-Number | QPS      | Avg-Resp-Time | Stdev-Resp-Time | Max-Resp-Time |
|-----------------|----------|---------------|-----------------|---------------|
| 10              | 14802.73 | 0.88ms        | 1.66ms          | 31.49ms       |
| 100             | 29809.58 | 3.78ms        | 4.12ms          | 105.34ms      |

| TCP-Conn-Number | QPS      | Avg-Resp-Time | Stdev-Resp-Time | Max-Resp-Time |
|-----------------|----------|---------------|-----------------|---------------|
| 10              | 14273.33 | 0.90ms        | 1.60ms          | 37.16ms       |
| 100             | 30757.81 | 3.62ms        | 3.41ms          | 59.78ms       |

#### 8. Host to the Node port service where the Pod is located on the target Node

| TCP-Conn-Number | QPS      | Avg-Resp-Time | Stdev-Resp-Time | Max-Resp-Time |
|-----------------|----------|---------------|-----------------|---------------|
| 10              | 15402.39 | 802.50us      | 1.42ms          | 30.91ms       |
| 100             | 29424.66 | 4.05ms        | 4.31ms          | 90.60ms       |

| TCP-Conn-Number | QPS      | Avg-Resp-Time | Stdev-Resp-Time | Max-Resp-Time |
|-----------------|----------|---------------|-----------------|---------------|
| 10              | 14649.21 | 0.91ms        | 1.72ms          | 43.92ms       |
| 100             | 32143.61 | 3.66ms        | 3.76ms          | 67.02ms       |


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## 3.10 LoadBalancer Service

Kube-OVN VPC VPC VPC  
 VPC VPC VPC LoadBalancer Service LoadBalancerIP VPC Service

1. multus-cni macvlan cni
2. LoadBalancer Service VPC vpc-nat-gw macvlan
3. VPC VPC LoadBalancer VPC VPC

### 3.10.1 VPC LoadBalancer Service

kube-system namespace deployment kube-ovn-controller args --enable-lb-svc=true false

```
containers:
- args:
  - /kube-ovn/start-controller.sh
  - --default-cidr=10.16.0.0/16
  - --default-gateway=10.16.0.1
  - --default-gateway-check=true
  - --enable-lb-svc=true # true
```

#### NetworkAttachmentDefinition CRD

yaml net-attach-def

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: lb-svc-attachment
  namespace: kube-system
spec:
  # master eth0 / ens192 / bond0
  config: '{
    "cniVersion": "0.3.0",
    "type": "macvlan",
    "master": "eth0",
    "mode": "bridge"
  }'
```

eth0 master

#### Subnet

Subnet LoadBalancer Service LoadBalancerIP Underlay Subnet

yaml

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attach-subnet
spec:
  protocol: IPv4
  provider: lb-svc-attachment.kube-system # provider net-attach-def Name.Namespace
  cidrBlock: 172.18.0.0/16
  gateway: 172.18.0.1
  excludeIps:
  - 172.18.0.0..172.18.0.10
```

Subnet provider ovn .ovn Kube-OVN logical switch

provider ovn .ovn Kube-OVN IPAM IP

LoadBalancer Service

yaml LoadBalancer Service

```
apiVersion: v1
kind: Service
metadata:
  annotations:
    lb-svc-attachment.kube-system.kubernetes.io/logical_switch: attach-subnet #
    ovn.kubernetes.io/attachmentprovider: lb-svc-attachment.kube-system #
  labels:
    app: dynamic
    name: test-service
    namespace: default
spec:
  loadBalancerIP: 172.18.0.18 #
  ports:
    - name: test
      protocol: TCP
      port: 80
      targetPort: 80
  selector:
    app: dynamic
  sessionAffinity: None
  type: LoadBalancer
```

yaml annotation ovn.kubernetes.io/attachmentprovider net-attach-def Name.Namespace net-attach-def Pod

annotation annotation key net-attach-def Name.Namespace.kubernetes.io/logical\_switch LoadBalancerIP LoadBalancerIP LoadBalancerIP spec.loadBalancerIP

yaml Service Service Namespace Pod

```
# kubectl get pod
NAME lb-svc-test-service-6869d98dd8-cjv1l READY STATUS RESTARTS AGE
lb-svc-test-service-6869d98dd8-cjv1l 1/1 Running 0 107m

# kubectl get svc
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
test-service LoadBalancer 10.109.201.193 172.18.0.18 80:30056/TCP 107m
```

service.spec.loadBalancerIP service external-ip

Pod yaml

```
# kubectl get pod -o yaml lb-svc-test-service-6869d98dd8-cjv1l
apiVersion: v1
kind: Pod
metadata:
  annotations:
    k8s.v1.cni.cncf.io/network-status: |-
      [{"name": "kube-ovn", "ips": [{"ip": "10.16.0.2"}], "default": true, "dns": {}}, {"name": "default/test-service", "interface": "net1", "mac": "ba:85:f7:02:9f:42", "dns": {}}]
    k8s.v1.cni.cncf.io/networks: default/test-service
    k8s.v1.cni.cncf.io/networks-status: |-
      [{"name": "kube-ovn", "ips": [{"ip": "10.16.0.2"}], "default": true, "dns": {}}, {"name": "default/test-service", "interface": "net1", "mac": "ba:85:f7:02:9f:42", "dns": {}}]
  labels:
    app: dynamic
    name: test-service
    namespace: default
spec:
  containers:
    - name: test-service
      image: nginx
      ports:
        - containerPort: 80
  dnsPolicy: ClusterFirst
  nodeName: kube-ovn
  restartPolicy: Always
  schedulerName: default-scheduler
  securityContext: {}
  serviceAccountName: default
  terminationGracePeriodSeconds: 30
  volumes: {}
```

```

ovn.kubernetes.io/ip_address: 10.16.0.2
ovn.kubernetes.io/logical_router: ovn-cluster
ovn.kubernetes.io/logical_switch: ovn-default
ovn.kubernetes.io/mac_address: 00:00:00:45:F4:29
ovn.kubernetes.io/pod_nic_type: veth-pair
ovn.kubernetes.io/routed: "true"
test-service.default.kubernetes.io/allocated: "true"
test-service.default.kubernetes.io/cidr: 172.18.0.0/16
test-service.default.kubernetes.io/gateway: 172.18.0.1
test-service.default.kubernetes.io/ip_address: 172.18.0.18
test-service.default.kubernetes.io/logical_switch: attach-subnet
test-service.default.kubernetes.io/mac_address: 00:00:00:AF:AA:BF
test-service.default.kubernetes.io/pod_nic_type: veth-pair

```

## Service

```

# kubectl get svc -o yaml test-service
apiVersion: v1
kind: Service
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"v1","kind":"Service","metadata":{"annotations":{"test-service.default.kubernetes.io/logical_switch":"attach-subnet"},"labels":{"app":"dynamic"},"name":"test-service","namespace":"default"},"spec":{"ports":[{"name":"test","port":80,"protocol":"TCP","targetPort":80}],"selector":{"app":"dynamic"},"sessionAffinity":"None","type":"LoadBalancer"}}
    ovn.kubernetes.io/vpc: ovn-cluster
    test-service.default.kubernetes.io/logical_switch: attach-subnet
  creationTimestamp: "2022-06-15T09:01:58Z"
  labels:
    app: dynamic
  name: test-service
  namespace: default
  resourceVersion: "38485"
  uid: 161ede1-7f6e-40f5-9e09-5a52c44267d0
spec:
  allocateLoadBalancerNodePorts: true
  clusterIP: 10.109.201.193
  clusterIPs:
    - 10.109.201.193
  externalTrafficPolicy: Cluster
  internalTrafficPolicy: Cluster
  ipFamilies:
    - IPv4
  ipFamilyPolicy: SingleStack
  ports:
    - name: test
      nodePort: 30056
      port: 80
      protocol: TCP
      targetPort: 80
  selector:
    app: dynamic
  sessionAffinity: None
  type: LoadBalancer
status:
  loadBalancer:
    ingress:
      - ip: 172.18.0.18

```

## 3.10.2 LoadBalancerIP

yaml, Pod Service Endpoints

```

apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: dynamic
  name: dynamic
  namespace: default
spec:
  replicas: 2
  selector:
    matchLabels:
      app: dynamic
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: dynamic
    spec:
      containers:
        - image: docker.io/library/nginx:alpine

```

```

imagePullPolicy: IfNotPresent
name: nginx
dnsPolicy: ClusterFirst
restartPolicy: Always

```

### Service LoadBalancerIP:Port

```

# curl 172.18.0.18:80
<html>
<head>
  <title>Hello World!</title>
  <link href="//fonts.googleapis.com/css?family=Open+Sans:400,700" rel="stylesheet" type="text/css">
  <style>
    body {
      background-color: white;
      text-align: center;
      padding: 50px;
      font-family: "Open Sans", "Helvetica Neue", Helvetica, Arial, sans-serif;
    }
    #logo {
      margin-bottom: 40px;
    }
  </style>
</head>
<body>
  <h1>Hello World!</h1>
  <h3>Links found</h3>
  <h3>I am on dynamic-7d8d7874f5-hsgc4</h3>
  <h3>Cookie =</h3>
  <b>KUBERNETES</b> listening in 443 available at tcp://10.96.0.1:443<br />
  <h3>my name is hanhouchao!</h3>
  <h3> RequestURI='/'</h3>
</body>
</html>

```

### Service Pod

```

# ip a
4: net1@if62: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether ba:85:f7:02:9f:42 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.18.0.18/16 scope global net1
        valid_lft forever preferred_lft forever
    inet6 fe80::b885:f7ff:fe02:9f42/64 scope link
        valid_lft forever preferred_lft forever
36: eth0@if37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP group default
    link/ether 00:00:00:45:f4:29 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.16.0.2/16 brd 10.16.255.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe45:f429/64 scope link
        valid_lft forever preferred_lft forever

# ip rule
0: from all lookup local
32764: from all iif eth0 lookup 100
32765: from all iif net1 lookup 100
32766: from all lookup main
32767: from all lookup default

# ip route show table 100
default via 172.18.0.1 dev net1
10.109.201.193 via 10.16.0.1 dev eth0
172.18.0.0/16 dev net1 scope link

# iptables -t nat -L -n -v
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source    destination
    0     0 DNAT      tcp -- *      *        0.0.0.0/0 172.18.0.18      tcp dpt:80 to:10.109.201.193:80

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source    destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source    destination

Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source    destination
    0     0 MASQUERADE all -- *      *        0.0.0.0/0 10.109.201.193

```

### lb service Pod nodeSelector

```

ovn-vpc-nat-config ConfigMap nodeSelector LoadBalancer service Pod

```

```

apiVersion: v1
data:
  image: docker.io/kubeovn/vpc-nat-gateway:v1.14.0
  nodeSelector: |
    kubernetes.io/hostname: kube-ovn-control-plane

```

```
kubernetes.io/os: linux
kind: ConfigMap
metadata:
  name: ovn-vpc-nat-config
  namespace: kube-system
```

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3.10.3

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3.11

|                                 |            |            |                                 |     |          |                          |
|---------------------------------|------------|------------|---------------------------------|-----|----------|--------------------------|
| Kube-OVN                        | Prometheus |            |                                 |     |          |                          |
| <a href="#">kube-prometheus</a> | CRD        | Prometheus | <a href="#">kube-prometheus</a> | CRD | Kube-OVN | <a href="#">Kube-OVN</a> |

3.11.1 Prometheus Monitor

Kube-OVN Prometheus Monitor CRD

```
#
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/pinger-monitor.yaml
# kube-ovn-controller
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/controller-monitor.yaml
# kube-ovn-cni
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/cni-monitor.yaml
# ovn
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/ovn-monitor.yaml
```

Prometheus 15s yml interval

3.11.2 Grafana

Kube-OVN Grafana Dashboard

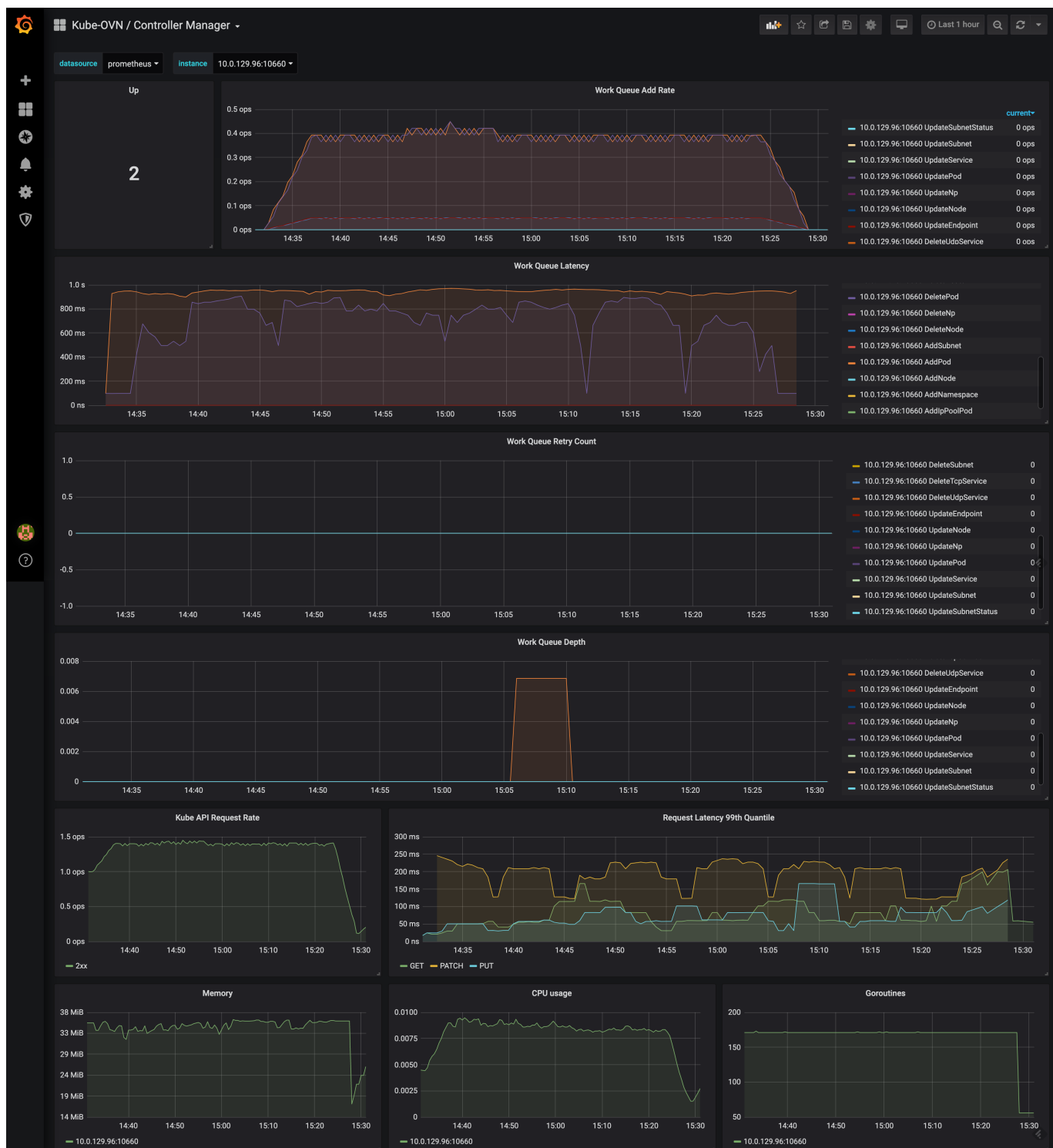
Dashboard

```
#
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/pinger-grafana.json
# kube-ovn-controller
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/controller-grafana.json
# kube-ovn-cni
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/cni-grafana.json
# ovn
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/ovn-grafana.json
# ovs
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/ovs-grafana.json
```

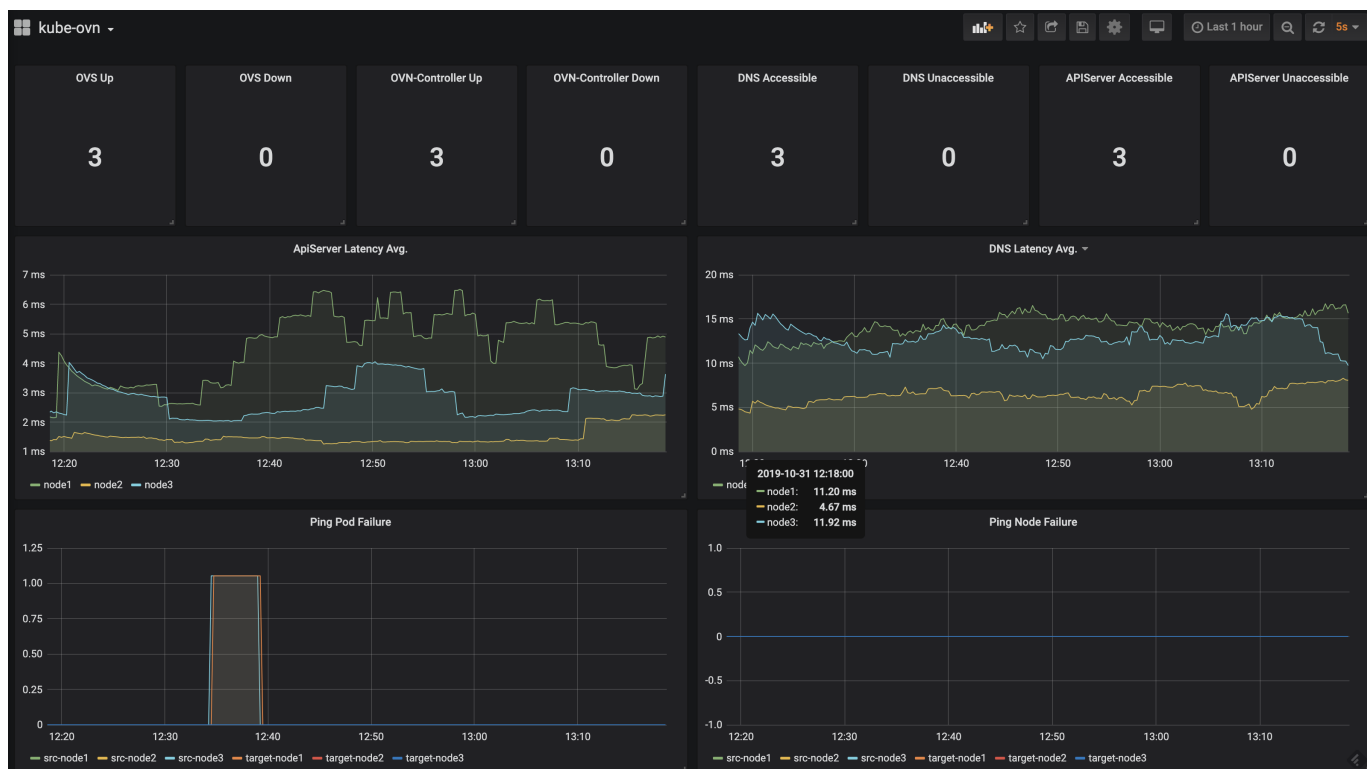
Grafana Prometheus Dashboard

kube-ovn-controller





kube-ovn-pinger



## kube-ovn-cni



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3.11.3

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3.12

3.12.1 NetworkPolicy

|               |               |     |     |          |     |     |                          |
|---------------|---------------|-----|-----|----------|-----|-----|--------------------------|
| NetworkPolicy | Kubernetes    | Pod | Pod | Kube-OVN | OVN | ACL | Kubernetes NetworkPolicy |
|               | NetworkPolicy |     |     |          |     |     |                          |

Kube-OVN

|          |     |                      |               |     |
|----------|-----|----------------------|---------------|-----|
| Kube-OVN | OVN | Open Virtual Network | NetworkPolicy | OVN |
|----------|-----|----------------------|---------------|-----|

PORT GROUP

|               |          |            |             |     |            |   |   |   |   |   |   |
|---------------|----------|------------|-------------|-----|------------|---|---|---|---|---|---|
| NetworkPolicy | Kube-OVN | Port Group | podSelector | Pod | Port Group | < | > | < | > | - | . |
|---------------|----------|------------|-------------|-----|------------|---|---|---|---|---|---|

ADDRESS SET

|             |               |    |               |          |             |                   |         |
|-------------|---------------|----|---------------|----------|-------------|-------------------|---------|
| Address Set | NetworkPolicy | IP | NetworkPolicy | Kube-OVN | podSelector | namespaceSelector | ipBlock |
| IP          | Address Set   |    |               |          |             |                   |         |

|                                |         |        |       |        |             |
|--------------------------------|---------|--------|-------|--------|-------------|
| NetworkPolicy                  | Ingress | Egress | Allow | Except | Address Set |
| • < > . < > . ingress . allow  |         |        | IP    |        |             |
| • < > . < > . ingress . except |         |        | IP    |        |             |
| • < > . < > . egress . allow   |         |        | IP    |        |             |
| • < > . < > . egress . except  |         |        | IP    |        |             |

ACL

|     |     |          |               |         |            |     |
|-----|-----|----------|---------------|---------|------------|-----|
| ACL | OVN | Kube-OVN | NetworkPolicy | OVN ACL | Port Group | ACL |
|-----|-----|----------|---------------|---------|------------|-----|

|          |               |     |
|----------|---------------|-----|
| Kube-OVN | NetworkPolicy | ACL |
|----------|---------------|-----|

|                 |                          |     |     |  |
|-----------------|--------------------------|-----|-----|--|
| • Ingress Allow | 2001                     |     |     |  |
| • Egress Allow  | 2001                     |     |     |  |
| • Default Deny  | 1000                     |     |     |  |
| Kube-OVN        | Kubernetes NetworkPolicy | OVN | ACL |  |

Kube-OVN

|                             |                    |                            |
|-----------------------------|--------------------|----------------------------|
| • <b>NetworkPolicy</b>      | Kubernetes         |                            |
| • <b>Network Policy API</b> | AdminNetworkPolicy | BaselineAdminNetworkPolicy |
| • <b>Subnet ACL</b>         |                    |                            |
| • <b>Security Group</b>     |                    |                            |
| OVN ACL                     | NetworkPolicy      | Network Policy API         |

NAMED PORT

|               |            |
|---------------|------------|
| NetworkPolicy | Named Port |
|---------------|------------|

```
ports:
- protocol: TCP
  port: http
```

| Kube-OVN | Named Port |               | Named Port    |  |       |      |      |       |      |      |
|----------|------------|---------------|---------------|--|-------|------|------|-------|------|------|
|          | Pod        | Named Port    | NetworkPolicy |  | Pod A | http | 8080 | Pod B | http | 8081 |
|          | port: http | NetworkPolicy |               |  |       |      |      |       |      |      |
|          | Named Port | Pod           |               |  |       |      |      |       |      |      |

IPBLOCK EXCEPT

NetworkPolicy ipBlock except IP

```
egress:
- to:
  - ipBlock:
      cidr: 10.0.0.0/8
      except:
        - 10.0.1.0/24
        - 10.0.2.0/24
```

OVN except except ACL OVN

except IP

- CIDR
- podSelector namespaceSelector IP

## NetworkPolicy

Kube-OVN NetworkPolicy

### Warning

NetworkPolicy CPU

ovn.kubernetes.io/acl\_log\_meter\_rate annotation ACL

NetworkPolicy annotation ovn.kubernetes.io/enable\_log

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: default-deny-ingress
  namespace: default
  annotations:
    ovn.kubernetes.io/enable_log: "true"
spec:
  podSelector: {}
  policyTypes:
  - Ingress
```

Drop

Pod /var/log/ovn/ovn-controller.log

```
# tail -f /var/log/ovn/ovn-controller.log
2022-07-20T05:55:03.229Z|00394|acl_log(ovn_pinctrl0)|INFO|name="np/default-deny-ingress.default/IPv4/0", verdict=drop, severity=warning, direction=to-lport:
udp,vlan_tci=0x0000,dl_src=00:00:00:21:b7:d1,dl_dst=00:00:00:8d:0b:86,nw_src=10.16.0.10,nw_dst=10.16.0.7,nw_tos=0,nw_ecn=0,nw_ttl=63,tp_src=54343,tp_dst=53
```

IP IP

Kube-OVN v1.13.0 ovn.kubernetes.io/log\_acl\_actions annotation Allow

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-from-client
  namespace: default
  annotations:
    ovn.kubernetes.io/enable_log: "true"
    ovn.kubernetes.io/log_acl_actions: "allow"
spec:
  podSelector:
    matchLabels:
      app: web
  policyTypes:
    - Ingress
  ingress:
    - from:
      - podSelector:
          matchLabels:
            app: client
```

ovn.kubernetes.io/log\_acl\_actions

- drop
- allow
- allow, drop

```
# tail -f /var/log/ovn/ovn-controller.log
2024-08-14T09:27:49.590Z|00004|acl_log(ovn_pinctrl0)|INFO|name="np/allow-from-client.default/ingress/IPv4/0", verdict=allow, severity=info, direction=to-
lport: icmp, vlan_tci=0x0000, dl_src=96:7b:b0:2f:a0:1a, dl_dst=a6:e5:1b:c2:1b:f8, nw_src=10.16.0.7, nw_dst=10.
16.0.12, nw_tos=0, nw_ecn=0, nw_ttl=64, nw_frag=no, icmp_type=8, icmp_code=0
```

ovn.kubernetes.io/acl\_log\_meter\_rate annotation

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-from-client
  namespace: default
  annotations:
    ovn.kubernetes.io/enable_log: "true"
    ovn.kubernetes.io/log_acl_actions: "allow"
    ovn.kubernetes.io/acl_log_meter_rate: "100"
spec:
  podSelector:
    matchLabels:
      app: web
  policyTypes:
    - Ingress
  ingress:
    - from:
      - podSelector:
          matchLabels:
            app: client
```

ovn.kubernetes.io/acl\_log\_meter\_rate / 100 100

annotation ovn.kubernetes.io/enable\_log false

```
kubectctl annotate networkpolicy -n default allow-from-client ovn.kubernetes.io/enable_log=false --overwrite
```

Kube-OVN

|            |               |              |      |    |    |      |     |  |
|------------|---------------|--------------|------|----|----|------|-----|--|
| • standard | NetworkPolicy | IP           |      |    |    |      |     |  |
| • lax      |               | TCP/UDP/SCTP | ICMP | L4 | IP | DHCP | UDP |  |
|            | NetworkPolicy | annotation   |      |    |    |      |     |  |

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: example-policy
  namespace: default
  annotations:
    ovn.kubernetes.io/enforcement: "lax"
spec:
  podSelector: {}
  policyTypes:
  - Ingress
```

Kube-OVN

`--network-policy-enforcement`

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|                        |               |                    |            |                          |                        |       |            |      |     |
|------------------------|---------------|--------------------|------------|--------------------------|------------------------|-------|------------|------|-----|
| Kubernetes             | NetworkPolicy | L3                 | L4         | AdminNetworkPolicy (ANP) |                        |       |            |      | Pod |
| DNSNameResolver        |               | CoreDNS            |            |                          |                        |       |            |      |     |
| NetworkPolicy          |               | OVN                | AddressSet | IP                       | IP                     | OVN   | AddressSet | DNS  |     |
| 1. kube-ovn-controller |               | AdminNetworkPolicy |            | DNSNameResolver CR       |                        |       |            |      |     |
| 2. CoreDNS             |               | DNSNameResolver CR |            | IP                       | DNSNameResolver status |       |            |      |     |
| 3. kube-ovn-controller |               | DNSNameResolver CR |            | status                   | AddressSet             |       |            |      |     |
| IP                     |               | Deny               |            | Allow                    |                        | Allow |            | Deny |     |

ANP BANP CRD

AdminNetworkPolicyCRD

```
kubectl apply -f https://raw.githubusercontent.com/kubernetes-sigs/network-policy-api/refs/heads/main/config/crd/experimental/policy.networking.k8s.io_adminnetworkpolicies.yaml
kubectl apply -f https://raw.githubusercontent.com/kubernetes-sigs/network-policy-api/refs/heads/main/config/crd/experimental/policy.networking.k8s.io_baselineadminnetworkpolicies.yaml
```

DNSNAMERESOLVER

DNSNameResolver

```
kubectl apply -f https://raw.githubusercontent.com/kubeovn/dnsresolver/refs/heads/main/manifest/crd.yaml
kubectl apply -f https://raw.githubusercontent.com/kubeovn/dnsresolver/refs/heads/main/manifest/rbac.yaml
kubectl apply -f https://raw.githubusercontent.com/kubeovn/dnsresolver/refs/heads/main/manifest/cm.yaml
```

COREDNS

DNSNameResolverCoreDNS

```
kubectl set image deployment/coredns coredns=kubeovn/dnsresolver:dev -n kube-system
```

CoreDNS

```
kubectl get pod -n kube-system -l k8s-app=kube-dns
```

ANP

kube-ovn-controller

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: kube-ovn-controller
spec:
  template:
    spec:
      containers:
        - name: kube-ovn-controller
          args:
            - --enable-anp=true
            - --enable-dns-name-resolver=true
            # ...
```



```
apiVersion: policy.networking.k8s.io/v1alpha1
kind: AdminNetworkPolicy
metadata:
  name: deny-external-domains
spec:
  priority: 55
  subject:
    namespaces:
      matchLabels:
        kubernetes.io/metadata.name: kube-system
  egress:
    - action: Deny
      name: deny-baidu-google
      to:
        - domainNames:
            - '*.baidu.com.'
            - '*.google.com.'
```

|             |                 |  |
|-------------|-----------------|--|
|             |                 |  |
| priority    |                 |  |
| subject     | Pod             |  |
| egress      |                 |  |
| action      | Allow Deny Pass |  |
| domainNames | .               |  |

kube-ovn-pinger

```
#
kubectl exec -it -n kube-system kube-ovn-pinger-xxxxx -- ping baidu.com
```

DNS    ACL

DNSNameResolver

# kubectl get dnsnameresolver

| NAME                               | DNS NAME      | RESOLVED IPS  |
|------------------------------------|---------------|---------------|
| anp-deny-external-domains-88dc32ab | *.google.com. |               |
| anp-deny-external-domains-fb3029ce | *.baidu.com.  | 220.181.7.203 |

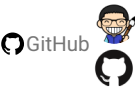
PDF

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- OVN
- OVN
- OVN
- OVN

```
apiVersion: v1
kind: Pod
metadata:
  name: secure-pod
  annotations:
    ovn.kubernetes.io/port_security: "true"
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
```



# 4. KubeVirt

## 4.1 VM IP

| IP                                                               | VM | VM | IP |
|------------------------------------------------------------------|----|----|----|
| CNI                                                              |    |    |    |
| • IP VM VM IP                                                    |    |    |    |
| • IP Node VM IP                                                  |    |    |    |
| • IP VM IP                                                       |    |    |    |
| KubeVirt masquerade iptables VM VM IP masquerade bridge          |    |    |    |
| • Pod IP VM IP Pod IP                                            |    |    |    |
| • masquerade iptables bridge                                     |    |    |    |
| • masquerade                                                     |    |    |    |
| • masquerade conntrack                                           |    |    |    |
| Kube-OVN KubeVirt bridge managedTap IP VM IP VM annotation VM IP |    |    |    |
| Kube-OVN Multus                                                  |    |    |    |

### 4.1.1 IP VM

| VM     | IP | IP | VM | Kube-OVN | IPAM | VM | VM | IP |
|--------|----|----|----|----------|------|----|----|----|
| bridge |    | VM | IP |          |      |    |    |    |

#### 1. VM

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
      annotations:
        kubevirt.io/allow-pod-bridge-network-live-migration: "true"
    spec:
      domain:
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
            - name: cloudinitdisk
              disk:
                bus: virtio
          interfaces:
            - name: default
              bridge: {}
        resources:
          requests:
            memory: 64M
        networks:
          - name: default
            pod: {}
        volumes:
          - name: containerdisk
            containerDisk:
              image: quay.io/kubevirt/cirros-container-disk-demo
          - name: cloudinitdisk
```

```
cloudInitNoCloud:
  userDataBase64: SGkuXG4=
```

1. VM

```
kubectl get vmi testvm
```

1. VM

```
virtctl restart testvm
```

1. VM

```
virtctl migrate testvm
```

bridge      VM      IP

4.1.2 IP/Mac

VM   IP/Mac                  VM   annotation      VM   IP                  KubeVirt

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
    annotations:
      ovn.kubernetes.io/ip_address: 10.16.0.15 #(1)
      ovn.kubernetes.io/mac_address: 00:00:00:53:6B:B6 #(2)
      kubevirt.io/allow-pod-bridge-network-live-migration: "true"
  spec:
    domain:
      devices:
        disks:
          - name: containerdisk
            disk:
              bus: virtio
          - name: cloudinitdisk
            disk:
              bus: virtio
        interfaces:
          - name: default
            bridge: {}
        resources:
          requests:
            memory: 64M
    networks:
      - name: default
        pod: {}
    volumes:
      - name: containerdisk
        containerDisk:
          image: quay.io/kubevirt/cirros-container-disk-demo
      - name: cloudinitdisk
        cloudInitNoCloud:
          userDataBase64: SGkuXG4=
```

- 1. 🧑 IP
- 2. 🧑 Mac

 **Warning**

|                 |                                                         |     |     |          |
|-----------------|---------------------------------------------------------|-----|-----|----------|
| KubeVirt VM API | spec.template.spec.domain.devices.interfaces.macAddress | Mac | Mac | Kube-OVN |
|                 | annotation                                              | Mac | Mac |          |

4.1.3 VM IP

Kube-OVN      VM IP      IP      VM

VM IP

- 1. VM Annotation IP
- 2. `virtctl restart <vm name>` VM IP

4.1.4

VM IP VM `ovn.kubernetes.io/logical_switch`

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
      annotations:
        ovn.kubernetes.io/logical_switch: subnet1 #(1)
        kubevirt.io/allow-pod-bridge-network-live-migration: "true"
    spec:
      domain:
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
            - name: cloudinitdisk
              disk:
                bus: virtio
          interfaces:
            - name: default
              bridge: {}
          resources:
            requests:
              memory: 64M
      networks:
        - name: default
          pod: {}
      volumes:
        - name: containerdisk
          containerDisk:
            image: quay.io/kubevirt/cirros-container-disk-demo
        - name: cloudinitdisk
          cloudInitNoCloud:
            userDataBase64: SGkuXG4=
```

1. 🧑 VM

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4.1.5

4.2

|          |       |       |            |          |            |    |     |
|----------|-------|-------|------------|----------|------------|----|-----|
| KubeVirt | Clone | Label | Annotation | Kube-OVN | Annotation | IP | MAC |
|----------|-------|-------|------------|----------|------------|----|-----|

4.2.1 Annotation

|                     |          |            |
|---------------------|----------|------------|
| VirtualMachineClone | Kube-OVN | Annotation |
|---------------------|----------|------------|

```
kind: VirtualMachineClone
apiVersion: "clone.kubevirt.io/v1beta1"
metadata:
  name: testclone
spec:
  source:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-source
  target:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-target
  template:
    annotationFilters:
      - "ovn.kubernetes.io/*"
```

4.2.2

 **Note**

patches    KubeVirt 1.6

IP

```
kind: VirtualMachineClone
apiVersion: "clone.kubevirt.io/v1beta1"
metadata:
  name: testclone
spec:
  source:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-source
  target:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-target
  patches:
    - '{"op": "replace", "path": "/spec/template/metadata/annotations/ovn.kubernetes.io~1ip_address", "value": "10.16.0.15"}'
```

```
kind: VirtualMachineClone
apiVersion: "clone.kubevirt.io/v1beta1"
metadata:
  name: testclone
spec:
  source:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-source
  target:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-target
  patches:
    - '{"op": "remove", "path": "/spec/template/metadata/annotations/ovn.kubernetes.io~1ip_address"}'
```

Annotation      KubeVirt Clone API

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4.2.3

---

4.3

|                |         |                        |               |            |      |          |        |             |    |          |       |
|----------------|---------|------------------------|---------------|------------|------|----------|--------|-------------|----|----------|-------|
| KubeVirt       | Bridge  | DHCP                   | virt-launcher | KubeVirt   | IPv4 | DHCP     | Bridge | KubeVirt VM | RA | IPv6     | Kube- |
| OVN            | DHCP RA | KubeVirt               | DHCP/RA       |            |      |          |        |             |    |          |       |
| KubeVirt 1.4.0 |         | Network Binding Plugin | Bridge        | managedTap |      | KubeVirt | DHCP   | managedTap  |    | Kube-OVN |       |
| DHCP/RA        | VM      |                        |               |            |      |          |        |             |    |          |       |

4.3.1 DHCP

|          |        |      |         |      |
|----------|--------|------|---------|------|
| Kube-OVN | Subnet | DHCP | IPv6 RA | YAML |
|----------|--------|------|---------|------|

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: dual-stack-subnet
spec:
  cidrBlock: "10.244.0.0/16,fd00:10:244::/64"
  enableDHCP: true
  enableIPv6RA: true
```

4.3.2 managedTap

|          |            |                         |
|----------|------------|-------------------------|
| KubeVirt | managedTap | Network Binding Plugin: |
|----------|------------|-------------------------|

```
# kubectl patch kubevirt -n kubevirt kubevirt --type=json -p=\
' [{"op": "add", "path": "/spec/configuration/network", "value": {
  "binding": {
    "managedtap": {
      "domainAttachmentType": "managedTap"
    }
  }
} ]'
}}
```

4.3.3 managedTap

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: dual-stack-vm
  namespace: default
spec:
  running: false
  template:
    spec:
      domain:
        devices:
          interfaces:
            - name: default
              binding:
                name: managedtap
      networks:
        - name: default
      pod: {}
```

VM DHCP IPv6 RA IPv4/IPv6

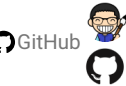
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#### 4.3.4

---

## 4.4

### KubeVirt

- KubeVirt Bridge
- KubeVirt
- IP
- 

### Kube-OVN

0.5

TCP

### 4.4.1

VM Spec    `kubevirt.io/allow-pod-bridge-network-live-migration: "true"`    annotation Kube-OVN

#### 1. VM

```
kubectl apply -f - <<EOF
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
      annotations:
        kubevirt.io/allow-pod-bridge-network-live-migration: "true"
    spec:
      domain:
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
            - name: cloudinitdisk
              disk:
                bus: virtio
          interfaces:
            - name: default
              bridge: {}
          resources:
            requests:
              memory: 64M
        networks:
          - name: default
            pod: {}
        volumes:
          - name: containerdisk
            containerDisk:
              image: quay.io/kubevirt/cirros-container-disk-demo
          - name: cloudinitdisk
            cloudInitNoCloud:
              userDataBase64: SGkuXG4=
EOF
```

#### 1. SSH

```
# password: gocubsgo
virtctl ssh cirros@testvm
ping 8.8.8.8
```

#### 1.

```
virtctl migrate testvm
```

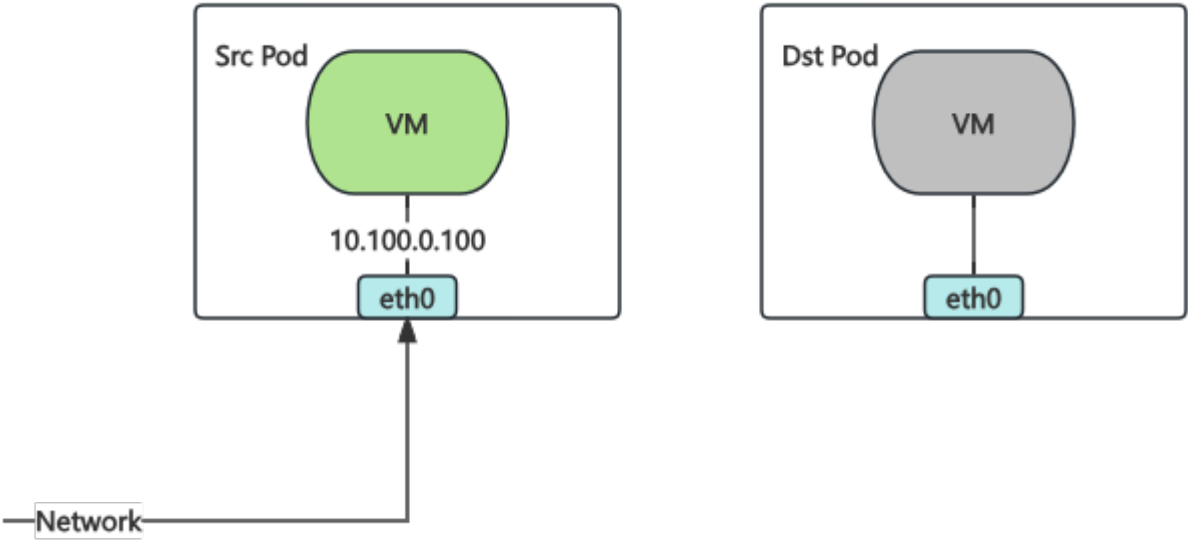
VM      SSH      ping

4.4.2

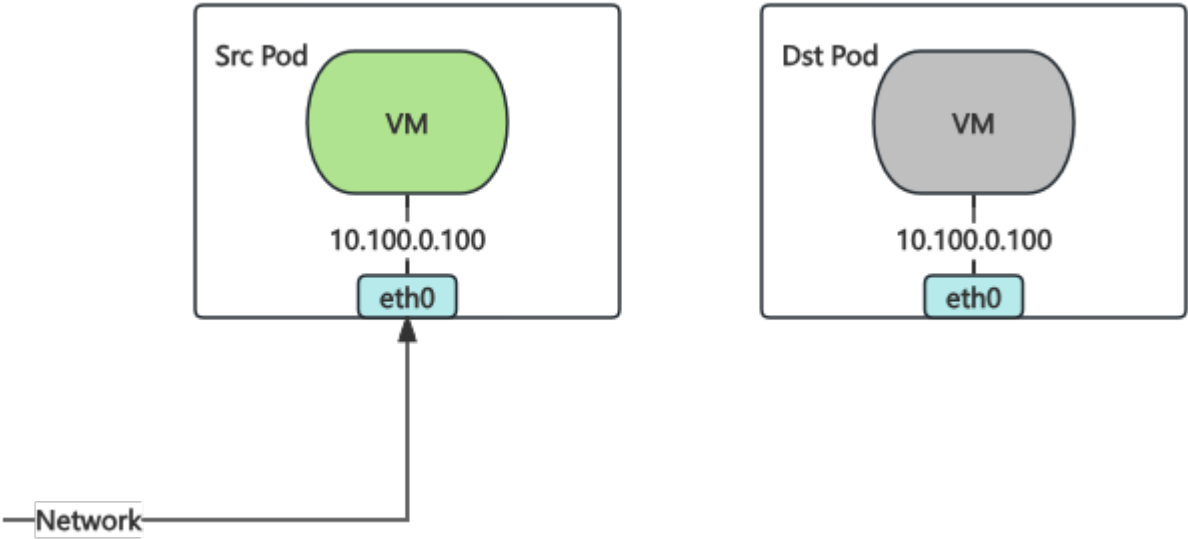
Kube-OVN [Live migration - Reducing downtime with multichassis port bindings](#)

IP

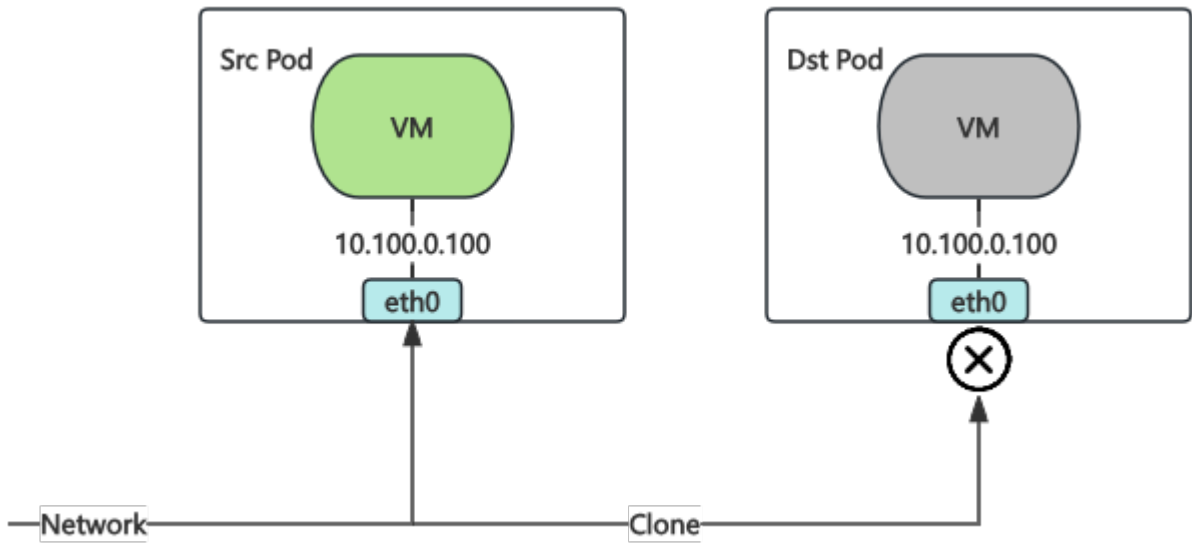
1. KubeVirt Pod



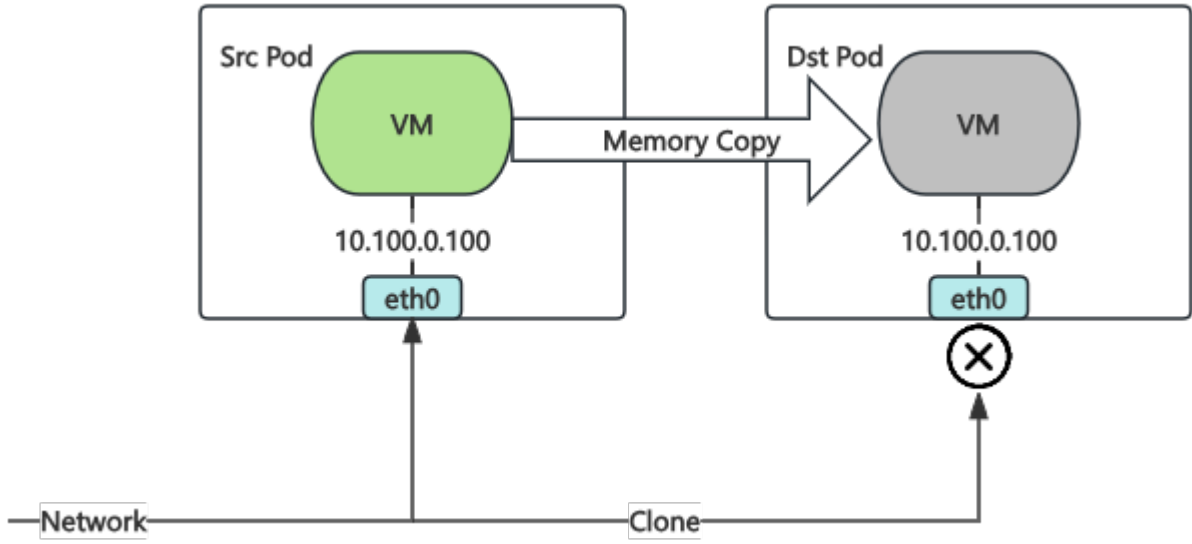
1. Kube-OVN Pod Pod Pod



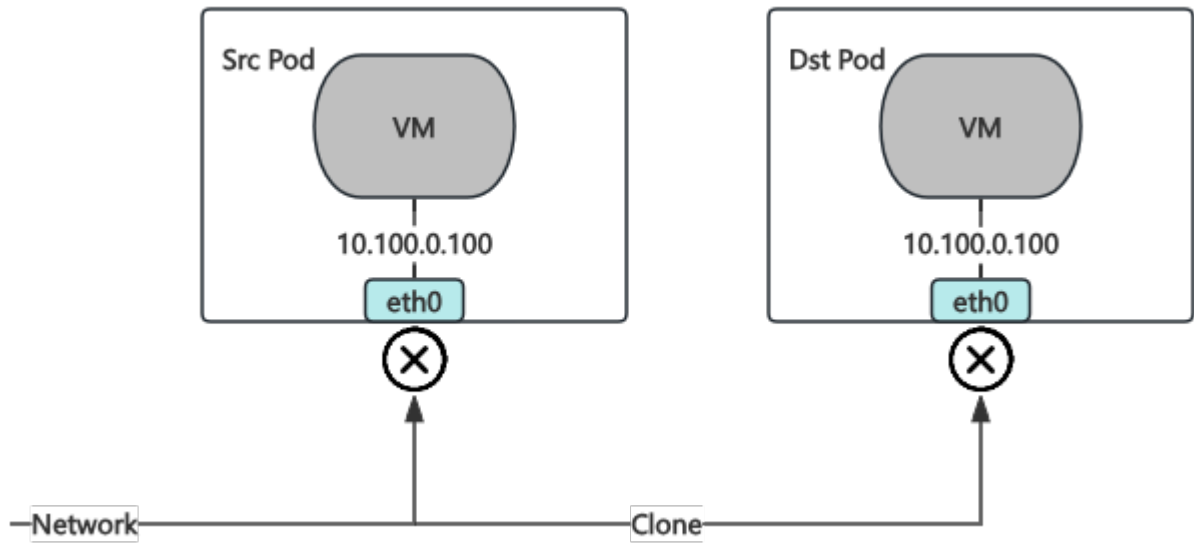
1. Kube-OVN Pod Pod Pod Pod Pod



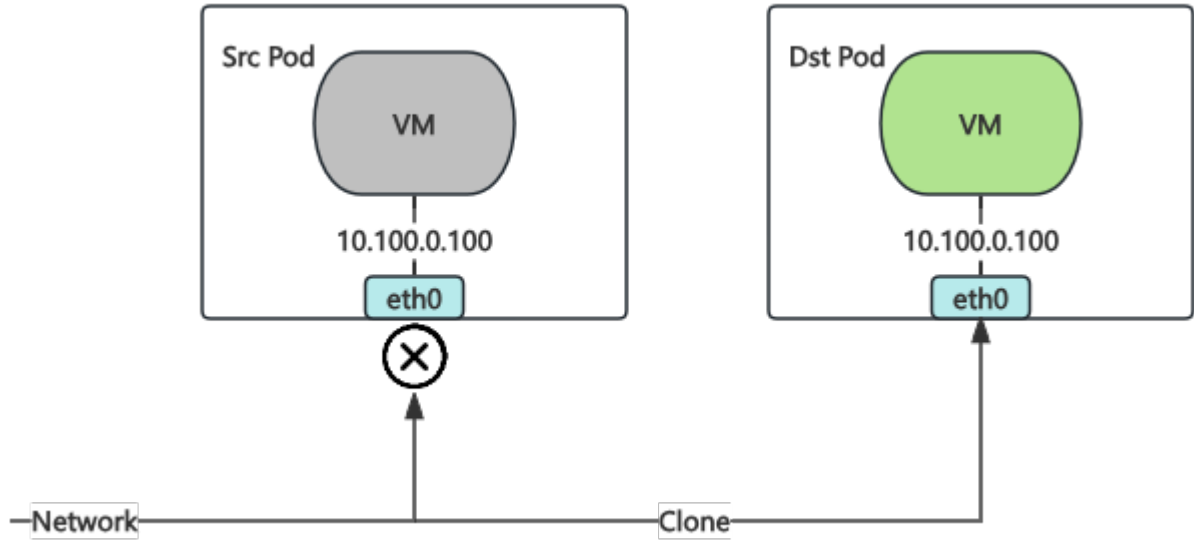
1. KubeVirt VM



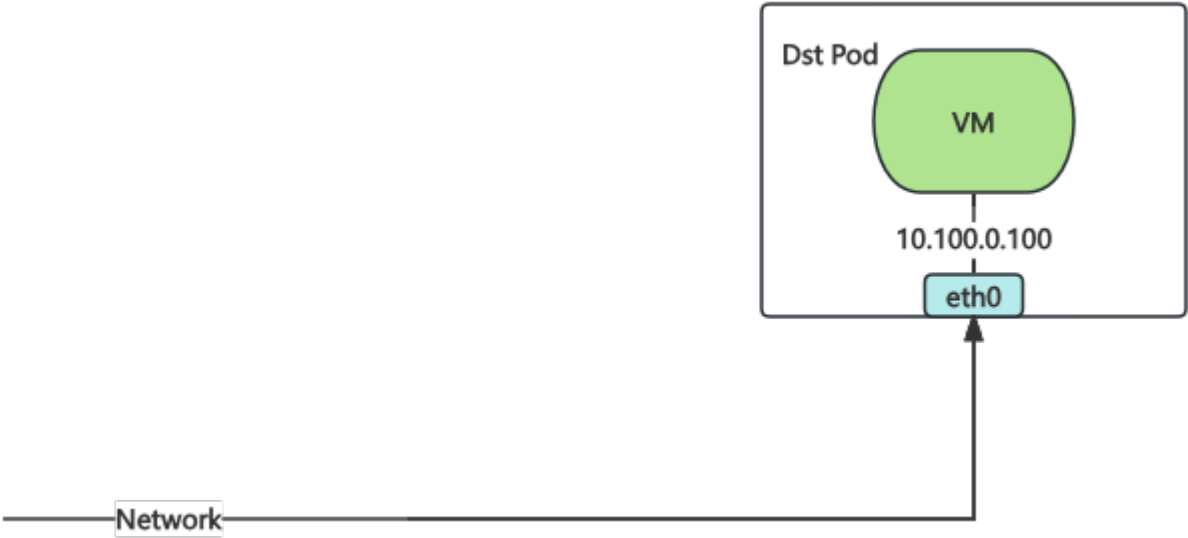
1. KubeVirt Pod Pod



1. KubeVirt Pod libvirt RARP Pod Pod



1. KubeVirt Pod Kube-OVN Watch Migration CR



5 6 libvirt RARP 0.5 TCP

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4.4.3

4.5

Kube-OVN

Multus Dynamic Networks Controller

KubeVirt

v1.4.0

VM

4.5.1

Thick

Multus

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/refs/heads/master/deployments/multus-daemonset-thick.yaml
```

Multus Dynamic Networks Controller

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-dynamic-networks-controller/refs/heads/main/manifests/dynamic-networks-controller.yaml
```

4.5.2

NetworkAttachmentDefinition

provider

ovn

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: attachnet
  namespace: default
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "kube-ovn",
    "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
    "provider": "attachnet.default.ovn"
  }'
```

- spec.config.type: kube-ovn CNI Kube-OVN
- server\_socket: Kube-OVN socket /run/openvswitch/kube-ovn-daemon.sock
- provider: NetworkAttachmentDefinition <name>.<namespace>.ovn , Kube-OVN Subnet ovn

Kube-OVN Subnet

Kube-OVN

provider

NetworkAttachmentDefinition

<name>.<namespace>.ovn

ovn

Kube-OVN

Subnet

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attachnet
spec:
  protocol: IPv4
  provider: attachnet.default.ovn
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
    - 172.17.0.0..172.17.0.10
```

4.5.3 VM

yaml

VM

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-fedora
spec:
  runStrategy: Always
  template:
    spec:
      domain:
```

```

    devices:
      disks:
        - disk:
            bus: virtio
            name: containerdisk
      interfaces:
        - masquerade: {}
          name: defaultnetwork
      rng: {}
    resources:
      requests:
        memory: 1024M
    networks:
      - name: defaultnetwork
        pod: {}
    terminationGracePeriodSeconds: 0
    volumes:
      - containerDisk:
          image: quay.io/kubevirt/fedora-with-test-tooling-container-disk:devel
          name: containerdisk

```

## VM Spec

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-fedora
template:
  spec:
    domain:
      devices:
        interfaces:
          - name: defaultnetwork
            masquerade: {}
          # new interface
          - name: dyniface1
            bridge: {}
        networks:
          - name: defaultnetwork
            pod: {}
          # new network
          - name: dyniface1
        multus:
          networkName: attachnet

```

interface state absent

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-fedora
template:
  spec:
    domain:
      devices:
        interfaces:
          - name: defaultnetwork
            masquerade: {}
          # set the interface state to absent
          - name: dyniface1
            state: absent
            bridge: {}
        networks:
          - name: defaultnetwork
            pod: {}
          - name: dyniface1
        multus:
          networkName: attachnet

```


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4.5.4

---

4.6 DHCP

managedTapSR-IOVDPDKKubeVirtDHCPKube-OVNOVNDHCPDHCPKubeVirt

DHCPIPKube-OVNDHCPDHCPv6, IPv6RA, DNSTFTPDHCPDHCPDHCP

Warning

1. bridgeKubeVirtDHCPKube-OVNDHCPKube-OVNDHCPKube-OVNDHCPmanagedTap

bridgemanagedTapmanagedTap

2. DHCPPodDHCP

DHCP

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: sn-dual
spec:
  cidrBlock: "10.0.0.0/24,240e::a00/120"
  default: false
  disableGatewayCheck: true
  disableInterConnection: false
  excludeIps:
    - 10.0.0.1
    - 240e::a01
  gateway: 10.0.0.1,240e::a01
  gatewayNode: ''
  gatewayType: distributed
  natOutgoing: false
  private: false
  protocol: Dual
  provider: ovn
  vpc: vpc-test
  enableDHCP: true
  dhcpV4Options: "lease_time=3600,router=10.0.0.1,server_id=169.254.0.254,server_mac=00:00:00:2E:2F:B8"
  dhcpV6Options: "server_id=00:00:00:2E:2F:C5"
  enableIPv6RA: true
  ipv6RAConfigs: "address_mode=dhcpv6_stateful,max_interval=30,min_interval=5,send_periodic=true"
```

- enableDHCP: DHCP
- dhcpV4Options, dhcpV6Options: ovn-nb DHCP DHCP Options "lease\_time=3600, router=\$ipv4\_gateway, server\_id=169.254.0.254, server\_mac=\$random\_mac" server\_id=\$random\_mac
- enableIPv6RA: DHCPv6
- ipv6RAConfigs ovn-nb Logical\_Router\_Port Logical Router Port address\_mode=dhcpv6\_stateful, max\_interval=30, min\_interval=5, send\_periodic=true

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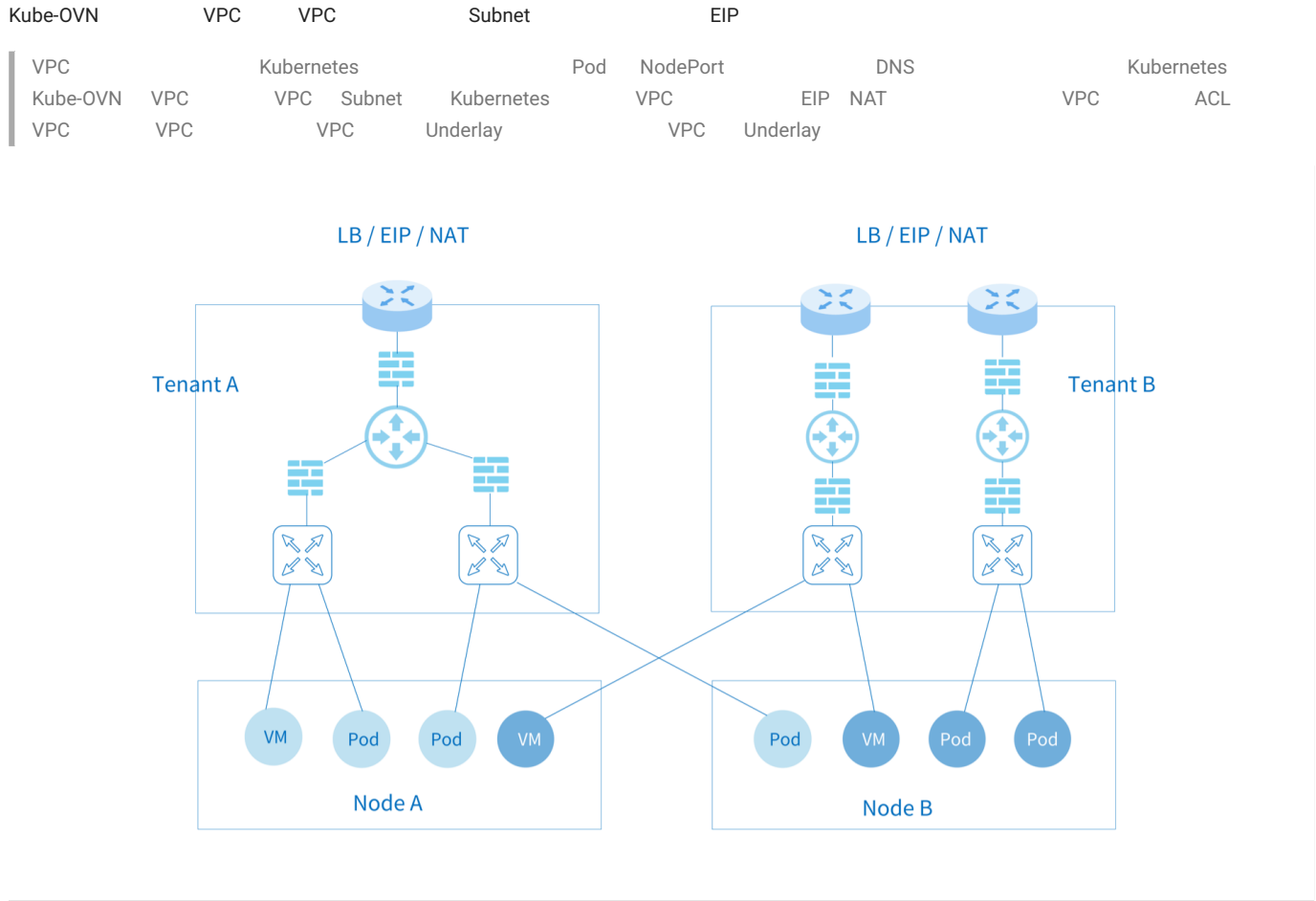
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4.6.1

# 5. VPC

## 5.1 VPC



### 5.1.1

| Kube-OVN | VPC         | OVN | IP | VPC | IP | OVN | Datapath ID |
|----------|-------------|-----|----|-----|----|-----|-------------|
|          | Datapath ID |     |    |     |    |     |             |

### 5.1.2 VPC

VPC

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  namespaces:
    - ns1
---
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-2
spec:
  namespaces:
    - ns2
```

namespaces

Namespace

VPC

VPC      CIDR:

```
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net1
spec:
  vpc: test-vpc-1
  cidrBlock: 10.0.1.0/24
  protocol: IPv4
  namespaces:
    - ns1
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net2
spec:
  vpc: test-vpc-2
  cidrBlock: 10.0.1.0/24
  protocol: IPv4
  namespaces:
    - ns2
```

Namespace      Pod:

```
apiVersion: v1
kind: Pod
metadata:
  namespace: ns1
  name: vpc1-pod
spec:
  containers:
    - name: vpc1-pod
      image: docker.io/library/nginx:alpine
---
apiVersion: v1
kind: Pod
metadata:
  namespace: ns2
  name: vpc2-pod
spec:
  containers:
    - name: vpc2-pod
      image: docker.io/library/nginx:alpine
```

Pod                  CIDR                  VPC      Pod

5.1.3 VPC

|     |            |            |    |      |      |
|-----|------------|------------|----|------|------|
| VPC | VPC        | VPC        | IP | SNAT | DNAT |
| VPC | Multus-CNI | multus-cni |    |      |      |

 **Note**

|                |         |          |                |         |          |         |     |     |  |
|----------------|---------|----------|----------------|---------|----------|---------|-----|-----|--|
| VPC            | VPC     |          |                |         |          |         |     |     |  |
| VPC            | VPC NAT | OVN      | Egress Gateway | VPC NAT | Kube-OVN | VPC NAT | Pod | VPC |  |
| Macvlan        | Pod     | iptables |                |         |          |         |     |     |  |
| OVN            | OVN     | NAT      | OVN            | BFD     | OVN      | OVN     |     |     |  |
| Egress Gateway | VPC NAT |          |                |         |          |         |     |     |  |

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: ovn-vpc-external-network
spec:
  protocol: IPv4
  provider: ovn-vpc-external-network.kube-system
  cidrBlock: 192.168.0.0/24
  gateway: 192.168.0.1 # IP address of the physical gateway
  excludeIps:
```

```
- 192.168.0.1..192.168.0.10
---
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-vpc-external-network
  namespace: kube-system
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "macvlan",
    "master": "eth1",
    "mode": "bridge",
    "ipam": {
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "ovn-vpc-external-network.kube-system"
    }
  }'

      'ipam'      VPC      net1      Kube-OVN      IP
---
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-vpc-external-network
  namespace: kube-system
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "macvlan",
    "master": "eth1",
    "mode": "bridge"
  }'
```

- Subnet                      Macvlan    VPC                      IP
  - VPC      Macvlan           NetworkAttachmentDefinition    master
  - name
- | Macvlan                 | L2/L3                                 | Vlan                                |
|-------------------------|---------------------------------------|-------------------------------------|
| 1. OpenStack VM         | PortSecurity                          |                                     |
| 2. VMware vSwitch       | MAC Address Changes, Forged Transmits | Promiscuous Mode Operation    allow |
| 3. Hyper-V              | MAC Address Spoofing                  |                                     |
| 4. AWS GCE              | Mac      Macvlan                      |                                     |
| 5. Macvlan      Macvlan |                                       | VpcNATGateway Pod      Pod          |
| 6.              Trunk   | Macvlan                               |                                     |

VPC

VPC      kube-system      ovn-vpc-nat-gw-config      nodeSelector

```
---
kind: ConfigMap
apiVersion: v1
metadata:
  name: ovn-vpc-nat-config
  namespace: kube-system
data:
  image: 'docker.io/kubeovn/vpc-nat-gateway:v1.15.21'
  nodeSelector: |
    kubernetes.io/hostname: kube-ovn-control-plane
---
kind: ConfigMap
apiVersion: v1
metadata:
  name: ovn-vpc-nat-gw-config
  namespace: kube-system
data:
  enable-vpc-nat-gw: 'true'
```

- image:    Pod
- enable-vpc-nat-gw      VPC

VPC

```
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: gw1
spec:
  vpc: test-vpc-1
  subnet: net1
  lanIp: 10.0.1.254
  selector:
    - "kubernetes.io/hostname: kube-ovn-worker"
    - "kubernetes.io/os: linux"
  externalSubnets:
    - ovn-vpc-external-network
  noDefaultEIP: false
```

- vpc VpcNatGateway VPC
- subnet VPC Subnet VPC Pod lanIp
- lanIp subnet IP VPC Pod IP VPC VpcNatGateway nextHopIP lanIp
- selector VpcNatGateway Pod Kubernetes NodeSelector
- externalSubnets VPC ovn-vpc-external-network
- noDefaultEIP VPC EIP false v1.15 BGP true Underlay
- tolerations VPC
- affinity VPC Pod

VPC-NAT-GW

1. nat gw pod net1 arp ping eip arp ping

EIP

EIP IP VPC DNAT SNAT IP

EIP

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-random
spec:
  natGwDp: gw1
```

EIP

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  natGwDp: gw1
  v4ip: 192.168.0.100
```

EIP

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-random
spec:
  natGwDp: gw1
  externalSubnet: ovn-vpc-external-network
```

- externalSubnet EIP ovn-vpc-external-network VPC externalSubnets

DNAT

DNATEIPVPCIP

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eipd01
spec:
  natGwDp: gw1

---
kind: IptablesDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: dnat01
spec:
  eip: eipd01
  externalPort: '8888'
  internalIp: 10.0.1.10
  internalPort: '80'
  protocol: tcp
```

SNAT

SNATVPCPodEIPSNAT

```
---
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eips01
spec:
  natGwDp: gw1
---
kind: IptablesSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat01
spec:
  eip: eips01
  internalCIDR: 10.0.1.0/24
```

IP

IPVPCIP EIP EIPVPCIP VPCIPVPCIP SNATEIP

```
---
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eipf01
spec:
  natGwDp: gw1

---
kind: IptablesFIPRule
apiVersion: kubeovn.io/v1
metadata:
  name: fip01
spec:
  eip: eipf01
  internalIp: 10.0.1.5
```

5.1.4

VCKube-OVN

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  staticRoutes:
    - cidr: 0.0.0.0/0
      nextHopIP: 10.0.1.254
      policy: policyDst
```

```
- cidr: 172.31.0.0/24
  nextHopIP: 10.0.1.253
  policy: policySrc
  routeTable: "rtb1"
```

- policy: policyDst policySrc
- CIDR
- routeTable:

OVN Logical Router Policy

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  policyRoutes:
    - action: drop
      match: ip4.src==10.0.1.0/24 && ip4.dst==10.0.1.250
      priority: 11
    - action: reroute
      match: ip4.src==10.0.1.0/24
      nextHopIP: 10.0.1.252
      priority: 10
```

5.1.5

|            |         |            |            |     |     |
|------------|---------|------------|------------|-----|-----|
| Kubernetes | Service | Kubernetes | Service IP | VPC | VPC |
| Kubernetes | Service |            |            |     |     |

Kube-OVN SwitchLBRule

SwitchLBRule

```
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name: cjh-slr-nginx
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  selector:
    - app: nginx
  ports:
    - name: dns
      port: 8888
      targetPort: 80
      protocol: TCP
```

- vip
- namespace Pod Namespace
- sessionAffinity Service sessionAffinity
- selector Service selector
- ports Service port

```
# kubectl get slr
NAME          VIP      PORT(S)          SERVICE          AGE
vpc-dns-test-cjh2 10.96.0.3 53/UDP, 53/TCP, 9153/TCP kube-system/slr-vpc-dns-test-cjh2 88m
```

5.1.6 vpc-dns

|     |     |     |     |         |     |         |         |          |         |
|-----|-----|-----|-----|---------|-----|---------|---------|----------|---------|
| VPC | VPC | VPC | Pod | coredns | VPC | CoreDNS | Service | Kube-OVN | vpc-dns |
|-----|-----|-----|-----|---------|-----|---------|---------|----------|---------|



```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-nad
  namespace: default
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "kube-ovn",
    "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
    "provider": "ovn-nad.default.ovn"
  }'
```

| ovn-default | provider                                  |
|-------------|-------------------------------------------|
| ovn-default | provider nad provider ovn-nad.default.ovn |

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: ovn-default
spec:
  cidrBlock: 10.16.0.0/16
  default: true
  disableGatewayCheck: false
  disableInterConnection: false
  enableDHCP: false
  enableIPv6RA: false
  excludeIps:
    - 10.16.0.1
  gateway: 10.16.0.1
  gatewayType: distributed
  logicalGateway: false
  natOutgoing: true
  private: false
  protocol: IPv4
  provider: ovn-nad.default.ovn
  vpc: ovn-cluster
```

vpc-dns ConfigMap

|             |           |         |         |
|-------------|-----------|---------|---------|
| kube-system | configmap | vpc-dns | vpc-dns |
|-------------|-----------|---------|---------|

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-config
  namespace: kube-system
data:
  coredns-vip: 10.96.0.3
  enable-vpc-dns: "true"
  nad-name: ovn-nad
  nad-provider: ovn-nad.default.ovn
```

- enable-vpc-dns true false true
- coredns-image dns coredns
- coredns-template dns URL yamls/coredns-template.yaml
- coredns-vip coredns lb vip
- nad-name network-attachment-definitions
- nad-provider provider
- k8s-service-host coredns k8s apiserver ip
- k8s-service-port coredns k8s apiserver port

vpc-dns

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: system:vpc-dns
```

```

rules:
- apiGroups:
  - ""
  resources:
  - endpoints
  - services
  - pods
  - namespaces
  verbs:
  - list
  - watch
- apiGroups:
  - discovery.k8s.io
  resources:
  - endpointslices
  verbs:
  - list
  - watch
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  annotations:
    rbac.authorization.kubernetes.io/autoupdate: "true"
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: vpc-dns
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: system:vpc-dns
subjects:
- kind: ServiceAccount
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-corefile
  namespace: kube-system
data:
  Corefile: |
    .:53 {
      errors
      health {
        lameduck 5s
      }
      ready
      kubernetes cluster.local in-addr.arpa ip6.arpa {
        pods insecure
        fallthrough in-addr.arpa ip6.arpa
      }
      prometheus :9153
      forward . /etc/resolv.conf {
        prefer_udp
      }
      cache 30
      loop
      reload
      loadbalance
    }

```

## vpc-dns

```

kind: VpcDns
apiVersion: kubeovn.io/v1
metadata:
  name: test-cjh1
spec:
  vpc: cjh-vpc-1
  subnet: cjh-subnet-1

```

- vpc      dns      vpc
- subnet      dns

```

[root@hci-dev-mst-1 kubeovn]# kubectl get vpc-dns
NAME          ACTIVE    VPC          SUBNET

```

|           |       |           |              |
|-----------|-------|-----------|--------------|
| test-cjh1 | false | cjh-vpc-1 | cjh-subnet-1 |
| test-cjh2 | true  | cjh-vpc-1 | cjh-subnet-2 |

• ACTIVE: true          dns    false

- VPC                  DNS
- VPC          vpc-dns          VPC          subnet          vpc-dns          true          false
- true    vpc-dns          false    vpc-dns

5.1.7

VPC                  VPC

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  namespaces:
  - ns1
  defaultSubnet: test
```

• defaultSubnet VPC

Namespace    ovn.kubernetes.io/logical\_switch          ovn.kubernetes.io/logical\_switch          Pod

VPC Pod    livenessProbe    readinessProbe

VPC    Pod                  kubelet                  VPC    Pod Kube-OVN    TProxy    kubelet                  VPC    Pod

DaemonSet kube-ovn-cni          --enable-tproxy=true

```
spec:
  template:
    spec:
      containers:
      - args:
        - --enable-tproxy=true
```

- 1.                  VPC    Pod    IP
- 2.          tcpSocket    httpGet

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5.1.8

5.2 VPC Egress Gateway

 **Note**

|                |         |         |          |                |         |          |         |     |     |
|----------------|---------|---------|----------|----------------|---------|----------|---------|-----|-----|
| VPC            | VPC     |         |          |                |         |          |         |     |     |
| VPC            |         | VPC NAT | OVN      | Egress Gateway | VPC NAT | Kube-OVN | VPC NAT | Pod | VPC |
| Macvlan        |         | Pod     | iptables |                |         |          |         |     |     |
| OVN            | OVN     | NAT     |          | OVN            | BFD     | OVN      | OVN     |     |     |
| Egress Gateway | VPC NAT |         |          |                |         |          |         |     |     |

VPC Egress Gateway      VPC      VPC    Pod

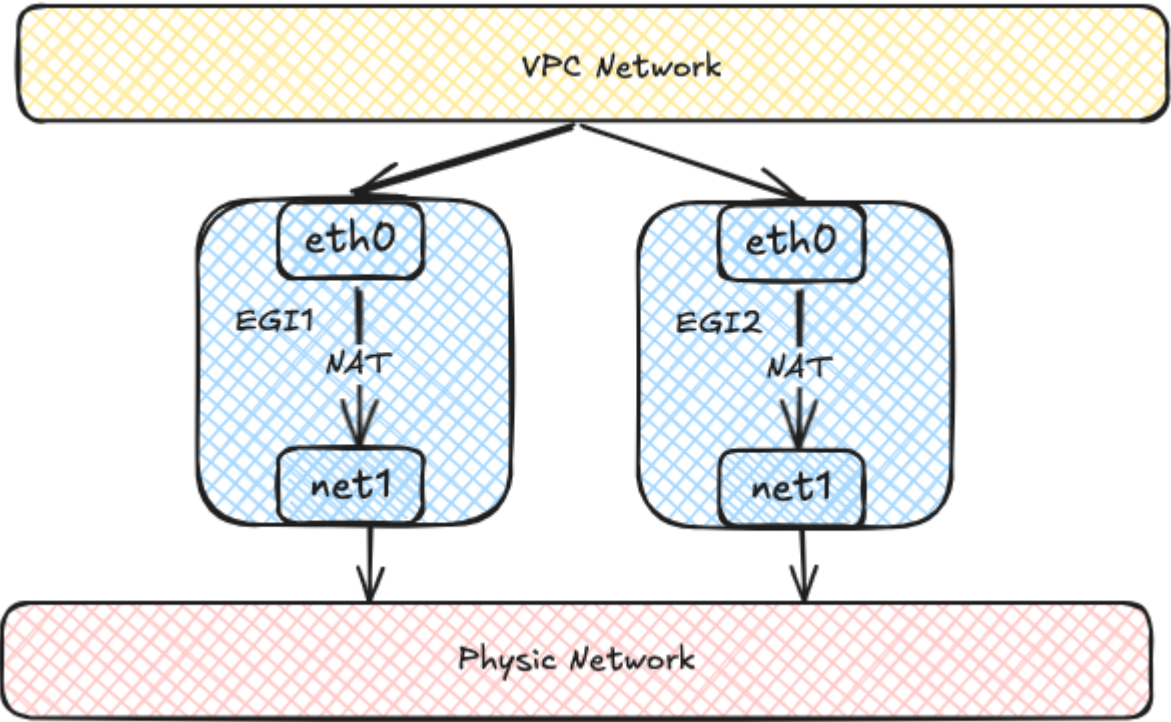
- ECMP    Active-Active
- BFD    <1s
- IPv6
- Namespace    Pod
- Node    Egress Gateway

VPC Egress Gateway

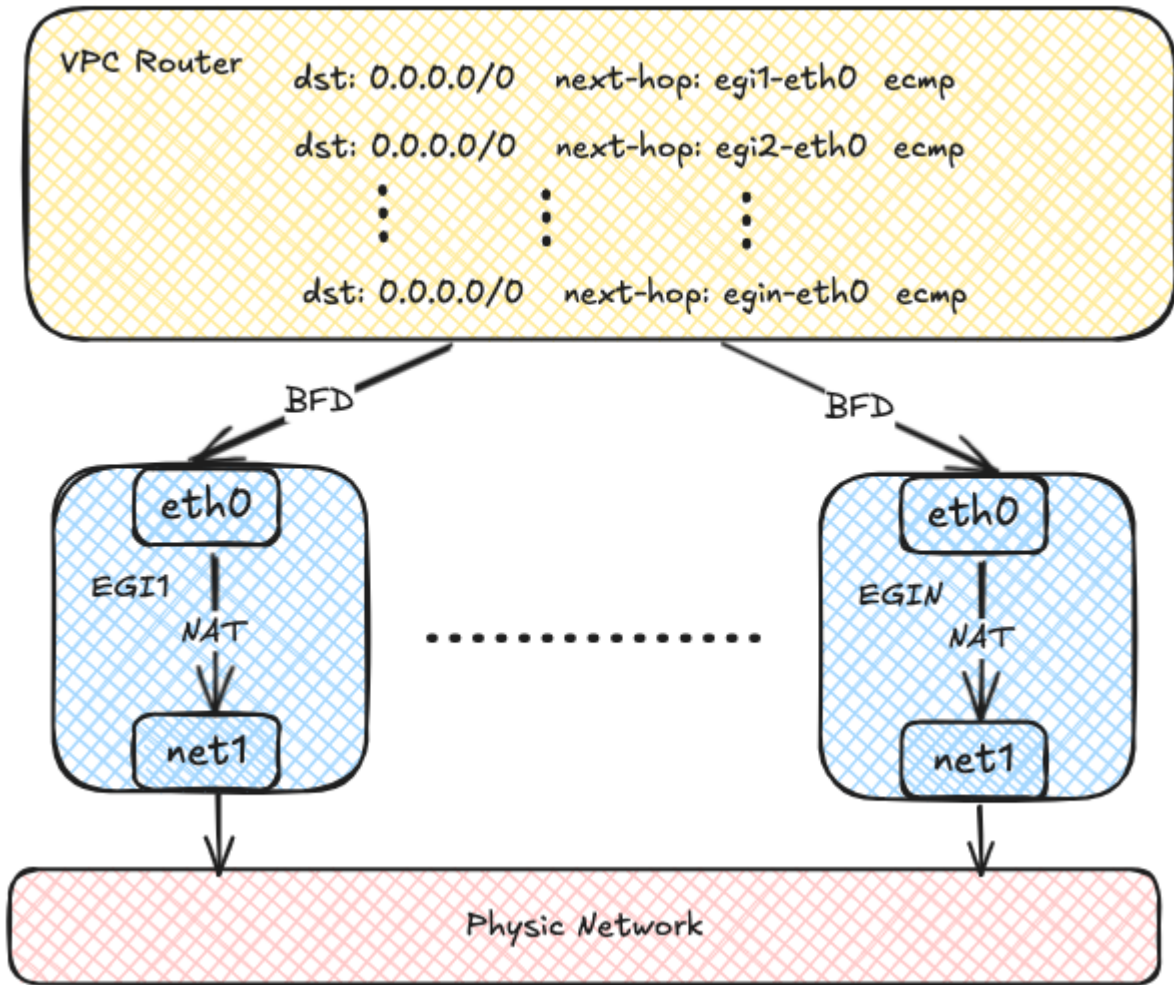
- Macvlan                      [Underlay](#)
- Gateway                      Egress IP
- SNAT      EIP    DNAT
- 

5.2.1

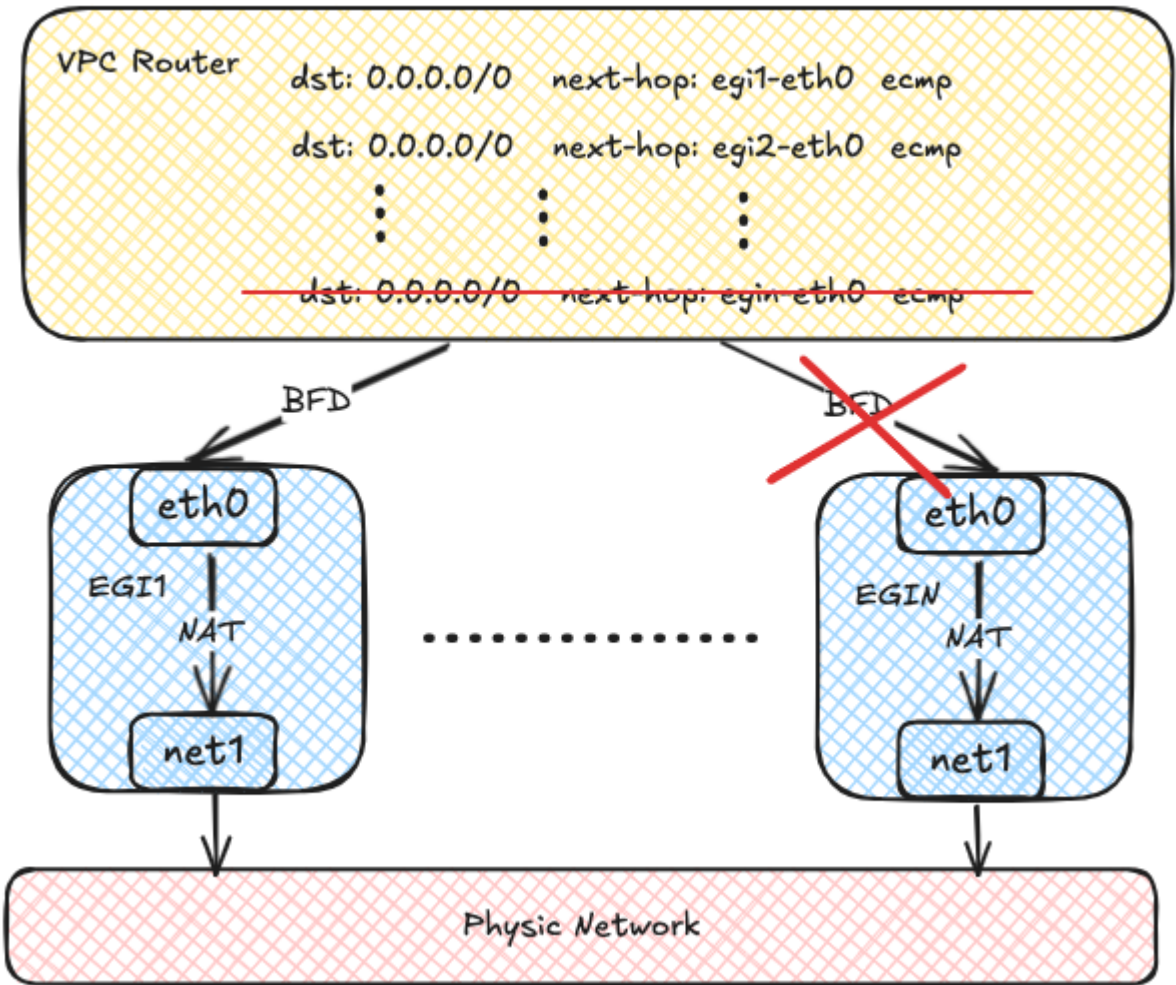
Egress Gateway      Pod      Pod                      VPC                      Macvlan                      Egress Gateway      NAT



Egress Gateway      OVN      VPC Pod      OVN      Egress Gateway      Egress Gateway



OVN    BFD    Egress Gateway    Egress Gateway    OVN



5.2.2

VPC Egress Gateway   VPC NAT Gateway   [Multus-CNI](#)

VPC Egress Gateway   ConfigMap

5.2.3

NetworkAttachmentDefinition

VPC Egress Gateway   VPC   NetworkAttachmentDefinition   `macvlan`   Kube-OVN   IPAM

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: eth1
  namespace: default
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "macvlan",
    "master": "eth1",
    "mode": "bridge",
    "ipam": {
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "eth1.default"
    }
  }'
```

---

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: macvlan1
```

```
spec:
  protocol: IPv4
  provider: eth1.default
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
    - 172.17.0.0..172.17.0.10
```

CNI      NetworkAttachmentDefinition    VPC Egress Gateway

VPC Egress Gateway

VPC Egress Gateway

```
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway1
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 1
  externalSubnet: macvlan1
  policies:
    - snat: true
      subnets:
        - ovn-default
```

default      VPC ovn-cluster      gateway1      VPC Egress Gateway    ovn-cluster    ovn-default    10.16.0.0/16      Pod  
macvlan1      SNAT

VPC Egress Gateway

```
$ kubectl get veg gateway1
NAME      VPC      REPLICAS  BFD ENABLED  EXTERNAL SUBNET  PHASE      READY  AGE
gateway1  ovn-cluster  1          false        macvlan1         Completed  true   13s
```

```
kubectl get veg gateway1 -o wide
NAME      VPC      REPLICAS  BFD ENABLED  EXTERNAL SUBNET  PHASE      READY  INTERNAL IPS  EXTERNAL IPS  WORKING NODES  AGE
gateway1  ovn-cluster  1          false        macvlan1         Completed  true   ["10.16.0.12"]  ["172.17.0.11"]  ["kube-ovn-worker"]  82s
```

```
$ kubectl get deployment -l ovn.kubernetes.io/vpc-egress-gateway=gateway1
NAME      READY  UP-TO-DATE  AVAILABLE  AGE
gateway1  1/1    1            1          4m40s

$ kubectl get pod -l ovn.kubernetes.io/vpc-egress-gateway=gateway1 -o wide
NAME                        READY  STATUS    RESTARTS  AGE  IP           NODE           NOMINATED NODE  READINESS GATES
gateway1-b9f8b4448-76lhm    1/1    Running   0          4m48s  10.16.0.12  kube-ovn-worker  <none>          <none>
```

Pod    IP      iptables

```
$ kubectl exec gateway1-b9f8b4448-76lhm -c gateway -- ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: net1@if13: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 62:d8:71:90:7b:86 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.17.0.11/16 brd 172.17.255.255 scope global net1
        valid_lft forever preferred_lft forever
    inet6 fe80::60d8:71ff:fe90:7b86/64 scope link
        valid_lft forever preferred_lft forever
17: eth0@if18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP group default
    link/ether 36:7c:6b:c7:82:6b brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.16.0.12/16 brd 10.16.255.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::347c:6bff:fec7:826b/64 scope link
        valid_lft forever preferred_lft forever

$ kubectl exec gateway1-b9f8b4448-76lhm -c gateway -- ip route show
default via 172.17.0.1 dev net1
10.16.0.0/16 dev eth0 proto kernel scope link src 10.16.0.12
```



```
172.17.0.0/16 dev net1 proto kernel scope link src 172.17.0.11

$ kubectl exec gateway1-b9f8b4448-76lhm -c gateway -- iptables -t nat -S
-P PREROUTING ACCEPT
-P INPUT ACCEPT
-P OUTPUT ACCEPT
-P POSTROUTING ACCEPT
-A POSTROUTING -s 10.16.0.0/16 -j MASQUERADE --random-fully
```

## Gateway Pod

```
$ kubectl exec -ti gateway1-b9f8b4448-76lhm -c gateway -- bash
nobody@gateway1-b9f8b4448-76lhm:/kube-ovn$ tcpdump -i any -nnve icmp and host 172.17.0.1
tcpdump: data link type LINUX_SLL2
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
06:50:58.936528 eth0 In ifindex 17 92:26:b8:9e:f2:1c ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 63, id 30481, offset 0, flags [DF], proto ICMP (1), length 84)
    10.16.0.9 > 172.17.0.1: ICMP echo request, id 37989, seq 0, length 64
06:50:58.936574 net1 Out ifindex 2 62:d8:71:90:7b:86 ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 62, id 30481, offset 0, flags [DF], proto ICMP (1), length 84)
    172.17.0.11 > 172.17.0.1: ICMP echo request, id 39449, seq 0, length 64
06:50:58.936613 net1 In ifindex 2 02:42:39:79:7f:08 ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 64, id 26701, offset 0, flags [none], proto ICMP (1), length 84)
    172.17.0.1 > 172.17.0.11: ICMP echo reply, id 39449, seq 0, length 64
06:50:58.936621 eth0 Out ifindex 17 36:7c:6b:c7:82:6b ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 63, id 26701, offset 0, flags [none], proto ICMP (1), length 84)
    172.17.0.1 > 10.16.0.9: ICMP echo reply, id 37989, seq 0, length 64
```

## OVN Logical Router

## VPC

```
$ kubectl ko nbctl lr-policy-list ovn-cluster
Routing Policies
31000 ip4.dst == 10.16.0.0/16 allow
31000 ip4.dst == 100.64.0.0/16 allow
30000 ip4.dst == 172.18.0.2 reroute 100.64.0.3
30000 ip4.dst == 172.18.0.3 reroute 100.64.0.2
30000 ip4.dst == 172.18.0.4 reroute 100.64.0.4
29100 ip4.src == 10.16.0.0/16 reroute 10.16.0.12
29000 ip4.src == $ovn.default.kube.ovn.control.plane_ip4 reroute 100.64.0.2
29000 ip4.src == $ovn.default.kube.ovn.worker2_ip4 reroute 100.64.0.4
29000 ip4.src == $ovn.default.kube.ovn.worker_ip4 reroute 100.64.0.3
```

.spec.replicas

```
$ kubectl scale veg gateway1 --replicas=2
vpcgressgateway.kubeovn.io/gateway1 scaled

$ kubectl get veg gateway1
NAME VPC REPLICAS BFD ENABLED EXTERNAL SUBNET PHASE READY AGE
gateway1 ovn-cluster 2 false macvlan Completed true 39m

$ kubectl get pod -l ovn.kubernetes.io/vpc-egress-gateway=gateway1 -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
gateway1-b9f8b4448-76lhm 1/1 Running 0 40m 10.16.0.12 kube-ovn-worker <none> <none>
gateway1-b9f8b4448-zd4dl 1/1 Running 0 64s 10.16.0.13 kube-ovn-worker2 <none> <none>

$ kubectl ko nbctl lr-policy-list ovn-cluster
Routing Policies
31000 ip4.dst == 10.16.0.0/16 allow
31000 ip4.dst == 100.64.0.0/16 allow
30000 ip4.dst == 172.18.0.2 reroute 100.64.0.3
30000 ip4.dst == 172.18.0.3 reroute 100.64.0.2
30000 ip4.dst == 172.18.0.4 reroute 100.64.0.4
29100 ip4.src == 10.16.0.0/16 reroute 10.16.0.12, 10.16.0.13
29000 ip4.src == $ovn.default.kube.ovn.control.plane_ip4 reroute 100.64.0.2
29000 ip4.src == $ovn.default.kube.ovn.worker2_ip4 reroute 100.64.0.4
29000 ip4.src == $ovn.default.kube.ovn.worker_ip4 reroute 100.64.0.3
```

## Egress Gateway IP

externalIPs nodeSelector Egress Gateway Pods Egress IP

```
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway1
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 2
  externalSubnet: macvlan1
  policies:
    - snat: true
      subnets:
        - ovn-default
```

```
externalIPs:
  - 172.17.0.10
  - 172.17.0.11
nodeSelector:
  - matchLabels:
      kubernetes.io/hostname: kube-ovn-worker
```

BFD

BFD      VPC    BFD LRP                  VPC                  BFD Port

```
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  name: vpc1
spec:
  bfdPort:
    enabled: true
    ip: 10.255.255.255
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  vpc: vpc1
  protocol: IPv4
  cidrBlock: 192.168.0.0/24
```

BFD Port                  OVN LR                  BFD    LRP

```
$ kubectl ko nbctl show vpc1
router 0c1d1e8f-4c86-4d96-88b2-c4171c7ff824 (vpc1)
  port bfd@vpc1
    mac: "8e:51:4b:16:3c:90"
    networks: ["10.255.255.255"]
  port vpc1-subnet1
    mac: "de:c9:5c:38:7a:61"
    networks: ["192.168.0.1/24"]
```

VPC Egress Gateway    .spec.bfd.enabled    true

```
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway2
  namespace: default
spec:
  vpc: vpc1
  replicas: 2
  internalSubnet: subnet1
  externalSubnet: macvlan
  bfd:
    enabled: true
  policies:
    - snat: true
      ipBlocks:
        - 192.168.0.0/24
```

VPC Egress Gateway

```
$ kubectl get veg gateway2 -o wide
NAME      VPC  REPLICAS  BFD  ENABLED  EXTERNAL SUBNET  PHASE  READY  INTERNAL IPS  EXTERNAL IPS  WORKING
NODES
gateway2  vpc1  2         true  macvlan  Completed  true   ["192.168.0.2","192.168.0.3"]  ["172.17.0.13","172.17.0.14"]  ["kube-ovn-
worker","kube-ovn-worker2"]  58s

$ kubectl get pod -l ovn.kubernetes.io/vpc-egress-gateway=gateway2 -o wide
NAME      READY  STATUS  RESTARTS  AGE  IP  NODE  NOMINATED NODE  READINESS GATES
gateway2-fcc6b8b87-8lgvx  1/1    Running  0         2m18s  192.168.0.3  kube-ovn-worker2  <none>  <none>
gateway2-fcc6b8b87-wmmw6  1/1    Running  0         2m18s  192.168.0.2  kube-ovn-worker  <none>  <none>

$ kubectl ko nbctl lr-route-list vpc1
IPv4 Routes
Route Table <main>:
      192.168.0.0/24      192.168.0.2 src-ip ecmp ecmp-symmetric-reply bfd
      192.168.0.0/24      192.168.0.3 src-ip ecmp ecmp-symmetric-reply bfd

$ kubectl ko nbctl list bfd
_uuid      : 223ede10-9169-4c7d-9524-a546e24bfab5
detect_mult : 3
dst_ip      : "192.168.0.2"
external_ids : {af="4", vendor=kube-ovn, vpc-egress-gateway="default/gateway2"}
logical_port : "bfd@vpc1"
min_rx      : 1000
```

```
min_tx      : 1000
options     : {}
status      : up

_uuid       : b050c75e-2462-470b-b89c-7bd38889b758
detect_mult : 3
dst_ip      : "192.168.0.3"
external_ids : {af="4", vendor=kube-ovn, vpc-egress-gateway="default/gateway2"}
logical_port : "bfd@vpc1"
min_rx      : 1000
min_tx      : 1000
options     : {}
status      : up
```

Pod BFD

```
$ kubectl exec gateway2-fcc6b8b87-8lgvx -c bfd -- bfd-control status
There are 1 sessions:
Session 1
id=1 local=192.168.0.3 (p) remote=10.255.255.255 state=Up

$ kubectl exec gateway2-fcc6b8b87-wmww6 -c bfd -- bfd-control status
There are 1 sessions:
Session 1
id=1 local=192.168.0.2 (p) remote=10.255.255.255 state=Up
```

VPC BFD PORT

|                               |              |       |                                                                                                                                               |                                 |
|-------------------------------|--------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| enabled                       | boolean      | false | BFD Port                                                                                                                                      | true                            |
| ip                            | string       | -     | BFD Port IP IPv6                                                                                                                              | 169.255.255.255 169.255.255.255 |
| nodeSelector                  | object       | -     | BFD Port BFD Port OVN HA Chassis Group Active/Backup Active nodeSelector Kube-OVN kubectl ko nbctl list ha_chassis_group OVN HA Chassis Group | -                               |
| nodeSelector.matchLabels      | dict/map     | -     |                                                                                                                                               | -                               |
| nodeSelector.matchExpressions | object array | -     |                                                                                                                                               | -                               |

VPC EGRESS GATEWAY

Spec

|                   |                   |             |      |                                                                                                    |                         |
|-------------------|-------------------|-------------|------|----------------------------------------------------------------------------------------------------|-------------------------|
| vpc               | string            | VPC cluster | ovn- | VPC                                                                                                | vpc1                    |
| replicas          | integer/int32     | 1           |      |                                                                                                    | 2                       |
| prefix            | string            | -           |      | Deployment                                                                                         | veg-                    |
| image             | string            | -           |      | Deployment                                                                                         | docker.io/kubeovn/kube- |
| internalSubnet    | string            | VPC         |      | VPC                                                                                                | subnet1                 |
| externalSubnet    | string            | -           |      |                                                                                                    | ext1                    |
| internalIPs       | string array      | -           |      | VPC IP<br>IPv6 IP<br><br><replicas> + 1<br>Pod                                                     | 10.16.0.101 / fd00::11  |
| externalIPs       | string array      | -           |      | IP<br>IPv6 IP<br><br><replicas> + 1<br>Pod                                                         | 10.16.0.101 / fd00::11  |
| bfd               | object            | -           |      | BFD                                                                                                | -                       |
| policies          | object array      | -           |      | Egress<br>selectors                                                                                | -                       |
| selectors         | object array      | -           |      | Namespace<br>Selector Pod<br>Selector Egress<br>Pod<br>SNAT/<br>MASQUERADE<br>policies             | -                       |
| nodeSelector      | object array      | -           |      | Deployment/Pod                                                                                     | -                       |
| trafficPolicy     | string            | Cluster     |      | Cluster /<br>Local BFD<br>Local Egress<br>VPC<br>Egress Gateway<br>VPC Egress<br>Gateway<br>Egress | Local                   |
| bandwidth         | object            | -           |      |                                                                                                    | -                       |
| bandwidth.ingress | integer<br>string | -           |      | VPC                                                                                                | 1024 / 1Gi              |
| bandwidth.egress  | integer<br>string | -           |      | VPC                                                                                                | 1024 / 1Gi              |

MbpsM Mi G Gi

quantitybit/s

Mbps100M100Mi1G1Gi

10010510001074 Mbps

9223372036854 Mbps

ingress

egress

spec:

bandwidth:

ingress: 1024

egress: 1Gi

09223372036854MbpsKube-OVNvpc-egress-gateways.kubeovn.ioCRDx-kubernetes-int-or-

string: true

HelmChartcrds/CRDkube-ovn-v2ChartKUBE\_OVN\_VERSIONKube-OVNCRD bundle

kubectl apply -f "https://raw.githubusercontent.com/kubeovn/kube-ovn/\${KUBE\_OVN\_VERSION}/charts/kube-ovn-v2/crds/kube-ovn-crd.yaml"

quantityMbps

BFD

|            |               |       |                |      |
|------------|---------------|-------|----------------|------|
| enabled    | boolean       | false | BFD            | true |
| minRX      | integer/int32 | 1000  | BFD minRX ms   | 500  |
| minTX      | integer/int32 | 1000  | BFD minTX ms   | 500  |
| multiplier | integer/int32 | 3     | BFD multiplier | 1    |

Egress

|          |              |       |                  |                              |
|----------|--------------|-------|------------------|------------------------------|
| snat     | boolean      | false | SNAT/MASQUERADE  | true                         |
| ipBlocks | string array | -     | Gateway IP IPv6  | 192.168.0.1 / 192.168.0.0/24 |
| subnets  | string array | -     | Gateway VPC IPv6 | subnet1                      |

Selectors

|                                    |              |   |            |   |
|------------------------------------|--------------|---|------------|---|
| namespaceSelector                  | object       | - | Namespace  | - |
|                                    |              |   | Namespace  |   |
| namespaceSelector.matchLabels      | dict/map     | - |            | - |
| namespaceSelector.matchExpressions | object array | - |            | - |
| podSelector                        | object       | - | Pod<br>Pod | - |
| podSelector.matchLabels            | dict/map     | - |            | - |
| podSelector.matchExpressions       | object array | - |            | - |

|                  |              |   |   |
|------------------|--------------|---|---|
| matchLabels      | dict/map     | - | - |
| matchExpressions | object array | - | - |
| matchFields      | object array | - | - |

Status

|                     |              |         |                                  |
|---------------------|--------------|---------|----------------------------------|
| ready               | boolean      | Gateway | true                             |
| phase               | string       | Gateway | Pending / Processing / Completed |
| internalIPs         | string array | VPC IP  | -                                |
| externalIPs         | string array | IP      | -                                |
| workload            | object       |         | -                                |
| workload.apiVersion | string       | API     | apps/v1                          |
| workload.kind       | string       |         | Deployment                       |
| workload.name       | string       |         | gateway1                         |
| workload.nodes      | string array |         | -                                |
| conditions          | object array | -       | -                                |

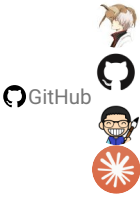
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5.2.4

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### 5.3 VPC QoS

Kube-OVN      QoSPolicy CRD      VPC

#### 5.3.1 EIP QoS

EIP      1Mbps      1      shared=false      QoSPolicy      EIP      QoSPolicy      QoS

QoSPolicy

```
apiVersion: kubeovn.io/v1
kind: QoSPolicy
metadata:
  name: qos-eip-example
spec:
  shared: false
  bindingType: EIP
  bandwidthLimitRules:
    - name: eip-ingress
      rateMax: "1" # Mbps
      burstMax: "1" # Mbps
      priority: 1
      direction: ingress
    - name: eip-egress
      rateMax: "1" # Mbps
      burstMax: "1" # Mbps
      priority: 1
      direction: egress
```

IptablesEIP

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-1
spec:
  natGwDp: gw1
  qosPolicy: qos-eip-example
```

`.spec.qosPolicy`

#### 5.3.2      QoS    EIP

label      qos    eip

```
# kubectl get eip -l ovn.kubernetes.io/qos=qos-eip-example
NAME  IP          MAC           NAT  NATGWDP  READY
eip-1  172.18.11.24  00:00:00:34:41:0B  fip  gw1      true
```

#### 5.3.3 VPC NATGW net1      QoS

VPC NATGW    net1      10Mbps      3      shared=true      QoSPolicy      QoSPolicy

QoSPolicy

```
apiVersion: kubeovn.io/v1
kind: QoSPolicy
metadata:
  name: qos-natgw-example
spec:
  shared: true
  bindingType: NATGW
  bandwidthLimitRules:
    - name: net1-ingress
      interface: net1
      rateMax: "10" # Mbps
      burstMax: "10" # Mbps
      priority: 3
      direction: ingress
    - name: net1-egress
      interface: net1
      rateMax: "10" # Mbps
      burstMax: "10" # Mbps
      direction: egress
```

```
priority: 3
direction: egress
```

VpcNatGateway

```
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: gw1
spec:
  vpc: test-vpc-1
  subnet: net1
  lanIp: 10.0.1.254
  qosPolicy: qos-natgw-example
  selector:
    - "kubernetes.io/hostname: kube-ovn-worker"
    - "kubernetes.io/os: linux"
```

```
.spec.qosPolicy
```

5.3.4 net1 QoS

|      |       |   |             |           |           |
|------|-------|---|-------------|-----------|-----------|
| net1 | 5Mbps | 2 | shared=true | QoSPolicy | QoSPolicy |
|------|-------|---|-------------|-----------|-----------|

QoSPolicy

```
apiVersion: kubeovn.io/v1
kind: QoSPolicy
metadata:
  name: qos-natgw-example
spec:
  shared: true
  bindingType: NATGW
  bandwidthLimitRules:
    - name: net1-extip-ingress
      interface: net1
      rateMax: "5" # Mbps
      burstMax: "5" # Mbps
      priority: 2
      direction: ingress
      matchType: ip
      matchValue: src 172.18.11.22/32
    - name: net1-extip-egress
      interface: net1
      rateMax: "5" # Mbps
      burstMax: "5" # Mbps
      priority: 2
      direction: egress
      matchType: ip
      matchValue: dst 172.18.11.23/32
```

VpcNatGateway

```
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: gw1
spec:
  vpc: test-vpc-1
  subnet: net1
  lanIp: 10.0.1.254
  qosPolicy: qos-natgw-example
  selector:
    - "kubernetes.io/hostname: kube-ovn-worker"
    - "kubernetes.io/os: linux"
```

5.3.5 QoS NATGW

|       |     |     |
|-------|-----|-----|
| label | qos | eip |
|-------|-----|-----|

```
# kubectl get vpc-nat-gw -l ovn.kubernetes.io/qos=qos-natgw-example
NAME  VPC      SUBNET  LANIP
gw1   test-vpc-1  net1    10.0.1.254
```

5.3.6 qos

```
# kubectl get qos -A
NAME      SHARED  BINDINGTYPE
```

|                   |       |       |
|-------------------|-------|-------|
| qos-eip-example   | false | EIP   |
| qos-natgw-example | true  | NATGW |

5.3.7

.

QoS

QoS

QoS

EIP

NATGW

spec.qosPolicy

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5.3.8

## 5.4 VPC

| Kubernetes    | Service | VPC              | Service |
|---------------|---------|------------------|---------|
| 1. Service IP | VPC     |                  |         |
| 2.            | IP      |                  |         |
| Kube-OVN      | 1.11    | SwitchLBRule CRD | VPC     |
| SwitchLBRule  | VPC     |                  |         |

### 5.4.1 Selector

| selector                                                                                                                                                                                                                                                                     | label        | pod                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------------|
| SwitchLBRule                                                                                                                                                                                                                                                                 |              |                    |
| <pre>apiVersion: kubeovn.io/v1 kind: SwitchLBRule metadata:   name: cjh-slr-nginx spec:   vip: 1.1.1.1   sessionAffinity: ClientIP   namespace: default   selector:     - app:nginx   ports:     - name: dns       port: 8888       targetPort: 80       protocol: TCP</pre> |              |                    |
| • selector, sessionAffinity                                                                                                                                                                                                                                                  | port         | Kubernetes Service |
| • vip                                                                                                                                                                                                                                                                        | IP           |                    |
| • namespace                                                                                                                                                                                                                                                                  | selector     | Pod                |
| Kube-OVN                                                                                                                                                                                                                                                                     | SwitchLBRule | Pod Pod VPC L2 LB  |

### 5.4.2 Endpoints

| endpoints                                                                                                                                                                                                                                                                                                                 | selector | kubevirt           | vm |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------|----|
| SwitchLBRule                                                                                                                                                                                                                                                                                                              |          |                    |    |
| <pre>apiVersion: kubeovn.io/v1 kind: SwitchLBRule metadata:   name: cjh-slr-nginx spec:   vip: 1.1.1.1   sessionAffinity: ClientIP   namespace: default   endpoints:     - 192.168.0.101     - 192.168.0.102     - 192.168.0.103   ports:     - name: dns       port: 8888       targetPort: 80       protocol: TCP</pre> |          |                    |    |
| • sessionAffinity                                                                                                                                                                                                                                                                                                         | port     | Kubernetes Service |    |
| • vip                                                                                                                                                                                                                                                                                                                     | IP       |                    |    |
| • namespace                                                                                                                                                                                                                                                                                                               | selector | Pod                |    |
| • endpoints                                                                                                                                                                                                                                                                                                               | IP       |                    |    |

selector endpoints, selector

## 5.4.3

OVN IPv4

[Health Checks](https://www.ovn.org/support/dist-docs/ovn-nb.5.html)

ovn SwitchLBRule SwitchLBRule VPC subnet vip ip\_port\_mappings

load\_balancer\_health\_check

- vip subnet subnet SwitchLBRule
- Selector

```
root@server:~# kubectl get po -o wide -n vulpecula
NAME          READY   STATUS    RESTARTS   AGE   IP            NODE   NOMINATED NODE   READINESS GATES
nginx-78d9578975-f4qn4  1/1     Running   3           4d16h  10.16.0.4     worker  <none>           <none>
nginx-78d9578975-t8tm5  1/1     Running   3           4d16h  10.16.0.6     worker  <none>           <none>
```

```
# slr
root@server:~# cat << END > slr.yaml
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name: nginx
  namespace: vulpecula
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  selector:
    - app:nginx
  ports:
    - name: dns
      port: 8888
      targetPort: 80
      protocol: TCP
END
root@server:~# kubectl apply -f slr.yaml
root@server:~# kubectl get slr
NAME          VIP      PORT(S)   SERVICE                     AGE
vulpecula-nginx  1.1.1.1  8888/TCP  default/slr-vulpecula-nginx  3d21h
```

subnet vip

```
# vip
root@server:~# kubectl get vip
NAME          NS      V4IP      MAC              V6IP      PMAC      SUBNET      READY   TYPE
vulpecula-subnet  10.16.0.2  00:00:00:39:95:C1 <nil>      vulpecula-subnet  true
```

Load\_Balancer\_Health\_Check Service\_Monitor

```
root@server:~# kubectl ko nbctl list Load_Balancer
 _uuid      : 3cbb6d43-44aa-4028-962f-30d2dba9f0b8
external_ids : {}
health_check : [5bee3f12-6b54-411c-9cc8-c9def8f67356]
ip_port_mappings : {"10.16.0.4"="nginx-78d9578975-f4qn4.default:10.16.0.2", "10.16.0.6"="nginx-78d9578975-t8tm5.default:10.16.0.2"}
name        : cluster-tcp-session-loadbalancer
options      : {affinity_timeout="10800"}
protocol     : tcp
selection_fields : [ip_src]
vips         : {"1.1.1.1:8888"="10.16.0.4:80,10.16.0.6:80"}
```

```
root@server:~# kubectl ko nbctl list Load_Balancer_Health_Check
 _uuid      : 5bee3f12-6b54-411c-9cc8-c9def8f67356
external_ids : {switch_lb_subnet=vulpecula-subnet}
options      : {failure_count="3", interval="5", success_count="3", timeout="20"}
vip          : "1.1.1.1:8888"
```

```
root@server:~# kubectl ko sbctl list Service_Monitor
 _uuid      : 1bddc541-cc49-44ea-9935-a4208f627a91
external_ids : {}
ip          : "10.16.0.4"
logical_port : nginx-78d9578975-f4qn4.default
options      : {failure_count="3", interval="5", success_count="3", timeout="20"}
port        : 80
protocol     : tcp
```

```

src_ip      : "10.16.0.2"
src_mac     : "c6:d4:b8:08:54:e7"
status      : online

_uuid       : 84dd24c5-e1b4-4e97-9daa-13687ed59785
external_ids : {}
ip          : "10.16.0.6"
logical_port : nginx-78d9578975-t8tm5.default
options     : {failure_count="3", interval="5", success_count="3", timeout="20"}
port        : 80
protocol    : tcp
src_ip      : "10.16.0.2"
src_mac     : "c6:d4:b8:08:54:e7"
status      : online

```

vip

```

root@server:~# kubectl exec -it -n vulpecula nginx-78d9578975-t8tm5 -- curl 1.1.1.1:8888
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<p><em>Thank you for using nginx.</em></p>
</body>
</html>

```

pod

```

kubectl delete po nginx-78d9578975-f4qn4
kubectl get po -o wide -n vulpecula

```

| NAME                   | READY | STATUS  | RESTARTS | AGE   | IP        | NODE   | NOMINATED NODE | READINESS GATES |
|------------------------|-------|---------|----------|-------|-----------|--------|----------------|-----------------|
| nginx-78d9578975-lxmvh | 1/1   | Running | 0        | 31s   | 10.16.0.8 | worker | <none>         | <none>          |
| nginx-78d9578975-t8tm5 | 1/1   | Running | 3        | 4d16h | 10.16.0.6 | worker | <none>         | <none>          |

Load\_Balancer\_Health\_Check Service\_Monitor

```

root@server:~# kubectl ko nbctl list Load_Balancer
_uuid       : 3cbb6d43-44aa-4028-962f-30d2dba9f0b8
external_ids : {}
health_check : [5bee3f12-6b54-411c-9cc8-c9def8f67356]
ip_port_mappings : {"10.16.0.4"="nginx-78d9578975-f4qn4.default:10.16.0.2", "10.16.0.6"="nginx-78d9578975-t8tm5.default:10.16.0.2", "10.16.0.8"="nginx-78d9578975-lxmvh.default:10.16.0.2"}
name        : cluster-tcp-session-loadbalancer
options     : {affinity_timeout="10800"}
protocol    : tcp
selection_fields : [ip_src]
vips        : {"1.1.1.1:8888"="10.16.0.6:80,10.16.0.8:80"}

root@server:~# kubectl ko nbctl list Load_Balancer_Health_Check
_uuid       : 5bee3f12-6b54-411c-9cc8-c9def8f67356
external_ids : {switch_lb_subnet=vulpecula-subnet}
options     : {failure_count="3", interval="5", success_count="3", timeout="20"}
vip         : "1.1.1.1:8888"

root@server:~# kubectl ko sbctl list Service_Monitor
_uuid       : 84dd24c5-e1b4-4e97-9daa-13687ed59785
external_ids : {}
ip          : "10.16.0.6"
logical_port : nginx-78d9578975-t8tm5.default
options     : {failure_count="3", interval="5", success_count="3", timeout="20"}
port        : 80
protocol    : tcp
src_ip      : "10.16.0.2"
src_mac     : "c6:d4:b8:08:54:e7"
status      : online

_uuid       : 5917b7b7-a999-49f2-a42d-da81f1eeb28f
external_ids : {}
ip          : "10.16.0.8"
logical_port : nginx-78d9578975-lxmvh.default
options     : {failure_count="3", interval="5", success_count="3", timeout="20"}
port        : 80
protocol    : tcp
src_ip      : "10.16.0.2"
src_mac     : "c6:d4:b8:08:54:e7"
status      : online

```

SwitchLBRule

Load\_Balancer\_Health\_Check

Service\_Monitor

vip

```

root@server:~# kubectl delete -f slr.yaml
switchlbrule.kubeovn.io "vulpecula-nginx" deleted
root@server:~# kubectl get vip
No resources found

```

```
root@server:~# kubectl ko sbctl list Service_Monitor
root@server:~#
root@server:~# kubectl ko nbctl list Load_Balancer_Health_Check
root@server:~#
```



PDF



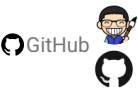
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5.4.4

---

## 5.5 VPC DNS

|                                                                                                                                                                |         |     |     |         |     |            |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----|-----|---------|-----|------------|-------------|
| VPC                                                                                                                                                            | VPC     | VPC | VPC | coredns | VPC | Kubernetes | vpc-dns CRD |
| CRD                                                                                                                                                            | coredns | Pod | VPC | VPC     | VPC | VPC        |             |
| <div><div> <b>Note</b></div><div>DNSPod VMWebhook/etc/resolv.conf</div></div> |         |     |     |         |     |            |             |

### 5.5.1 vpc-dns

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: system:vpc-dns
rules:
- apiGroups:
  - ""
  resources:
  - endpoints
  - services
  - pods
  - namespaces
  verbs:
  - list
  - watch
- apiGroups:
  - discovery.k8s.io
  resources:
  - endpointslices
  verbs:
  - list
  - watch
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  annotations:
    rbac.authorization.kubernetes.io/autoupdate: "true"
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: vpc-dns
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: system:vpc-dns
subjects:
- kind: ServiceAccount
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-corefile
  namespace: kube-system
data:
  Corefile: |
    .:53 {
      errors
      health {
        lameduck 5s
      }
      ready
      kubernetes cluster.local in-addr.arpa ip6.arpa {
        pods insecure
        fallthrough in-addr.arpa ip6.arpa
      }
      prometheus :9153
      forward . /etc/resolv.conf {
        prefer_udp
      }
      cache 30
```



```

    loop
    reload
    loadbalance
  }

```

nat-gw-pod

## 5.5.2

```

apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-nad
  namespace: default
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "kube-ovn",
    "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
    "provider": "ovn-nad.default.ovn"
  }'

```

## 5.5.3 vpc-dns Configmap

kube-system      configmap    vpc-dns      vpc-dns

```

apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-config
  namespace: kube-system
data:
  coredns-vip: 10.96.0.3
  enable-vpc-dns: "true"
  nad-name: ovn-nad
  nad-provider: ovn-nad.default.ovn

```

- enable-vpc-dns      true
- coredns-image    dns      coredns
- coredns-vip    coredns    lb    vip
- coredns-template    coredns      URL      ovn      coredns-template.yaml      <https://raw.githubusercontent.com/kubeovn/kube-ovn/master/yamls/coredns-template.yaml>
- nad-name      network-attachment-definitions
- nad-provider      provider
- k8s-service-host    coredns    k8s apiserver    ip      apiserver
- k8s-service-port    coredns    k8s apiserver    port      apiserver

## 5.5.4 vpc-dns

vpc-dns yaml

```

kind: VpcDns
apiVersion: kubeovn.io/v1
metadata:
  name: test-cjh1
spec:
  vpc: cjh-vpc-1
  subnet: cjh-subnet-1
  replicas: 2

```

- vpc      dns      vpc
- subnet      dns
- replicas: vpc dns deployment replicas

```
# kubectl get vpc-dns
NAME      ACTIVE  VPC      SUBNET
test-cjh1 false    cjh-vpc-1 cjh-subnet-1
test-cjh2 true     cjh-vpc-1 cjh-subnet-2
```

ACTIVE : true          dns    false

VPC                  DNS

- VPC      vpc-dns          VPC    subnet      vpc-dns    true    false
- true   vpc-dns            false   vpc-dns

5.5.5

vpc-dns Pod          label app=vpc-dns          vpc-dns Pod

# kubectl -n kube-system get pods -l app=vpc-dns

| NAME                               | READY | STATUS  | RESTARTS | AGE |
|------------------------------------|-------|---------|----------|-----|
| vpc-dns-test-cjh1-7b878d96b4-g5979 | 1/1   | Running | 0        | 28s |
| vpc-dns-test-cjh1-7b878d96b4-ltmf9 | 1/1   | Running | 0        | 28s |

slr

# kubectl -n kube-system get slr

| NAME              | VIP       | PORT(S)                  | SERVICE                           | AGE  |
|-------------------|-----------|--------------------------|-----------------------------------|------|
| vpc-dns-test-cjh1 | 10.96.0.3 | 53/UDP, 53/TCP, 9153/TCP | kube-system/slr-vpc-dns-test-cjh1 | 113s |

VPC   Pod   dns   :

nslookup kubernetes.default.svc.cluster.local 10.96.0.3

VPC   switch lb rule          VPC          Pod

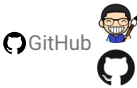
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5.5.6

5.6 SecurityGroup

Kube-OVN Pod

 **Warning**

Kube-OVN

[NetworkPolicy](#) [Network Policy API](#) [Subnet ACL](#) [Security Group](#) [OVN ACL](#) [NetworkPolicy](#) [NetworkPolicy API](#)

5.6.1

```
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-example
spec:
  allowSameGroupTraffic: true
  egressRules:
    - ipVersion: ipv4
      policy: allow
      priority: 1
      protocol: all
      remoteAddress: 10.16.0.13 # 10.16.0.0/16
      remoteType: address
  ingressRules:
    - ipVersion: ipv4
      policy: deny
      priority: 1
      protocol: icmp
      remoteAddress: 10.16.0.14
      remoteType: address
```

[Kube-OVN](#)

Pod `ovn.kubernetes.io/security_groups` annotation

```
ovn.kubernetes.io/security_groups: sg-example
```

Port Security

5.6.2

|   |          |          |     |     |     |          |                   |         |  |
|---|----------|----------|-----|-----|-----|----------|-------------------|---------|--|
| • | ACL      | OVN      | ACL | ACL |     |          |                   |         |  |
| • | priority | 1-200    | ACL | ACL | ACL | = 2300 - |                   | ACL     |  |
| • |          | Kube-OVN | CNI | Pod | Pod | Pod      | ContainerCreating | Running |  |

5.6.3

YAML Pod annotation

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: static
  annotations:
    ovn.kubernetes.io/security_groups: 'sg-example'
  name: sg-test-pod
  namespace: default
spec:
  nodeName: kube-ovn-worker
  containers:
    - image: docker.io/library/nginx:alpine
      imagePullPolicy: IfNotPresent
      name: qatest
```

```
# kubectl get pod -o wide
NAME                READY   STATUS             RESTARTS   AGE   IP            NODE                NOMINATED NODE   READINESS GATES
sg-test-pod          0/1     ContainerCreating   0          5h32m   <none>         kube-ovn-worker     <none>           <none>
test-99fff7f86-52h9r 1/1     Running             0          5h41m   10.16.0.14    kube-ovn-control-plane <none>           <none>
test-99fff7f86-qcgjw 1/1     Running             0          5h43m   10.16.0.13    kube-ovn-worker     <none>           <none>
```

## kubectl describe pod Pod

```
# kubectl describe pod sg-test-pod
Name:          sg-test-pod
Namespace:     default
Priority:       0
Node:          kube-ovn-worker/172.18.0.2
Start Time:    Tue, 28 Feb 2023 10:29:36 +0800
Labels:        app=static
Annotations:   ovn.kubernetes.io/allocated: true
               ovn.kubernetes.io/cidr: 10.16.0.16
               ovn.kubernetes.io/gateway: 10.16.0.1
               ovn.kubernetes.io/ip_address: 10.16.0.15
               ovn.kubernetes.io/logical_router: ovn-cluster
               ovn.kubernetes.io/logical_switch: ovn-default
               ovn.kubernetes.io/mac_address: 00:00:00:FA:17:97
               ovn.kubernetes.io/pod_nic_type: veth-pair
               ovn.kubernetes.io/port_security: true
               ovn.kubernetes.io/routed: true
               ovn.kubernetes.io/security_groups: sg-allow-reject
Status:        Pending
IP:
IPs:           <none>
.
.
.
Events:
  Type     Reason          Age          From          Message
  ----     -
Warning   FailedCreatePodSandbox 5m3s (x70 over 4h59m) kubelet (combined from similar events): Failed to create pod sandbox: rpc error: code = Unknown desc = failed to setup network for sandbox "40636e0c7f1ade5500fa958486163d74f2e2300051a71522a9afd7ba0538afb6": plugin type="kube-ovn" failed (add): RPC failed; request ip return 500 configure nic failed 10.16.0.15 network not ready after 200 ping 10.16.0.1
```

```
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-gw-both
spec:
  allowSameGroupTraffic: true
  egressRules:
  - ipVersion: ipv4
    policy: allow
    priority: 2
    protocol: all
    remoteAddress: 10.16.0.13
    remoteType: address
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: all
    remoteAddress: 10.16.0.1
    remoteType: address
  ingressRules:
  - ipVersion: ipv4
    policy: deny
    priority: 2
    protocol: icmp
    remoteAddress: 10.16.0.14
    remoteType: address
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: icmp
    remoteAddress: 10.16.0.1
    remoteType: address
```

## yaml Pod Pod

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: static
  annotations:
    ovn.kubernetes.io/security_groups: 'sg-gw-both'
  name: sg-gw-both
  namespace: default
```

```
spec:
  nodeName: kube-ovn-worker
  containers:
  - image: docker.io/library/nginx:alpine
    imagePullPolicy: IfNotPresent
    name: qatest
```

Pod

# kubectl get pod -o wide

| NAME        | READY | STATUS            | RESTARTS | AGE   | IP         | NODE            | NOMINATED | NODE | READINESS | GATES |
|-------------|-------|-------------------|----------|-------|------------|-----------------|-----------|------|-----------|-------|
| sg-test-pod | 0/1   | ContainerCreating | 0        | 5h41m | <none>     | kube-ovn-worker | <none>    |      | <none>    |       |
| sg-gw-both  | 1/1   | Running           | 0        | 5h37m | 10.16.0.19 | kube-ovn-worker | <none>    |      | <none>    |       |

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5.7 OVN EIP FIP SNAT DNAT

 **Note**

|                |         |          |                |         |          |     |         |     |     |
|----------------|---------|----------|----------------|---------|----------|-----|---------|-----|-----|
| VPC            | VPC     |          |                |         |          |     |         |     |     |
| VPC            | VPC NAT | OVN      | Egress Gateway | VPC NAT | Kube-OVN |     | VPC NAT | Pod | VPC |
| Macvlan        | Pod     | iptables |                |         |          |     |         |     |     |
| OVN            | OVN     | NAT      |                | OVN     | BFD      | OVN | OVN     |     |     |
| Egress Gateway | VPC NAT |          |                |         |          |     |         |     |     |

VPC OVN NAT provider-network vlan (external) subnet VPC EIP/SNAT

5.7.1

- kube-ovn-controller kube-ovn-cni ovn-external-gw-config VPC spec enableExternal
- CRD provider-network vlan subnet VPC spec extraExternalSubnets ovn-eip ovn-dnat ovn-fip ovn-snat CRD

```
graph LR
pod-->subnet-->vpc-->lrp--bind-->gw-chassis-->snat-->lsp-->external-subnet
lrp-.peer-.lsp
```

Pod SNAT Pod Fip

```
graph LR
pod-->subnet-->vpc-->lrp--bind-->local-chassis-->snat-->lsp-->external-subnet
lrp-.peer-.lsp
```

Pod FIP (dnat\_and\_snat)

- CRD iptables nat gw
- ovn eip: ip underlay provider network vlan subnet
  - ovn fip dnat snat VPC ip vip
  - ovn snat VPC ip snat
  - ovn dnat router lb , ip + VPC endpoints

5.7.2 1.

OpenStack Neutron ovn provider network VPC EIP/SNAT

vlan vlan 0 vlan id

```
#
# 1. kube-ovn-controller
- --external-gateway-vlanid=204
- --external-gateway-switch=external204
# 2. kube-ovn-cni :
- --external-gateway-switch=external204
```

```
###          vlan id          underlay
```

- provider network vlan subnet
- VPC enable\_eip\_snat vlan subnet ip ipam
- VPC enable\_eip\_snat , pod annotation fip snat
- VPC enable\_eip\_snat vlan subnet VPC eip snat

## 1.1 underlay

```
# provider-network vlan subnet
# cat 01-provider-network.yaml

apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: external204
spec:
  defaultInterface: vlan

# cat 02-vlan.yaml

apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan204
spec:
  id: 204
  provider: external204

# cat 03-vlan-subnet.yaml

apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: external204
spec:
  protocol: IPv4
  cidrBlock: 10.5.204.0/24
  gateway: 10.5.204.254
  vlan: vlan204
  excludeIps:
    - 10.5.204.1..10.5.204.100
```

## 1.2 VPC eip\_snat

```
# VPC underlay provider subnet
# cat 00-centralized-external-gw-no-ip.yaml

apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-external-gw-config
  namespace: kube-system
data:
  enable-external-gw: "true"
  external-gw-nodes: "pc-node-1,pc-node-2,pc-node-3"
  type: "centralized"
  external-gw-nic: "vlan" # ovs
  external-gw-addr: "10.5.204.254/24" # underlay ip
```

```
logical router port (lrp) ip mac          underlay          lrp          ovn eip
ip          lrp          ovn-eip          lrp          ovn eip
```

## 1.3 VPC eip snat fip

node

```
# external-gw-nodes
kubectl label nodes pc-node-1 pc-node-2 pc-node-3 ovn.kubernetes.io/external-gw=true

# cat 00-ns.yaml

apiVersion: v1
```

```
kind: Namespace
metadata:
  name: vpc1

# cat 01-vpc-ecmp-enable-external-bfd.yml
```

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc1
spec:
  namespaces:
  - vpc1
  enableExternal: true
  staticRoutes:
  - cidr: 0.0.0.0/0
    nextHopIP: 10.5.204.254
    policy: policyDst
```

```
# VPC      enableExternal      lrp
```

```
# cat 02-subnet.yml
```

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vpc1-subnet1
spec:
  cidrBlock: 192.168.0.0/24
  default: false
  disableGatewayCheck: false
  disableInterConnection: true
  enableEcmp: true
  gatewayNode: ""
  gatewayType: distributed
  #gatewayType: centralized
  natOutgoing: false
  private: false
  protocol: IPv4
  provider: ovn
  vpc: vpc1
  namespaces:
  - vpc1
```

```
# subnet
```

```
# kubectl ko nbctl show vpc1
```

```
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
  port vpc1-vpc1-subnet1
    mac: "00:00:00:ED:8E:C7"
    networks: ["192.168.0.1/24"]
  port vpc1-external204
    mac: "00:00:00:EF:05:C7"
    networks: ["10.5.204.105/24"]
  gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
  nat 21d853b0-f7b4-40bd-9a53-31d2e2745739
    external ip: "10.5.204.115"
    logical ip: "192.168.0.0/24"
    type: "snat"
```

```
# kubectl ko nbctl lr-route-list vpc1
```

```
IPv4 Routes
Route Table <main>:
      0.0.0.0/0      10.5.204.254 dst-ip
# VPC CRD
```

enableExternal

VPC CRD

1.4

1.4.1 UNDERLAY

eip snat fip 1 eip snat fip

```
# provider-network vlan subnet
# cat 01-extra-provider-network.yaml
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: extra
spec:
  defaultInterface: vlan
# cat 02-extra-vlan.yaml
```



```
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan0
spec:
  id: 0
  provider: extra
# cat 03-extra-vlan-subnet.yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: extra
spec:
  protocol: IPv4
  cidrBlock: 10.10.204.0/24
  gateway: 10.10.204.254
  vlan: vlan0
  excludeIps:
    - 10.10.204.1..10.10.204.100
```

1.4.2 VPC

```
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  name: vpc1
spec:
  namespaces:
    - vpc1
  enableExternal: true # enableExternal VPC external ls
  extraExternalSubnets: # extraExternalSubnets
    - extra

# kubectl ko nbctl show vpc1
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
  port vpc1-vpc1-subnet1
    mac: "00:00:00:ED:8E:C7"
    networks: ["192.168.0.1/24"]
  port vpc1-external204
    mac: "00:00:00:EF:05:C7"
    networks: ["10.5.204.105/24"]
  gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
  port vpc1-extra
    mac: "00:00:00:EF:6A:C7"
    networks: ["10.10.204.105/24"]
  gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
```

5.7.3 2. ovn-eip

| iptables-eip |                    | ovn-eip |    | type              |      |
|--------------|--------------------|---------|----|-------------------|------|
| • nat:       | ovn dn timer, snat | nat     |    |                   |      |
| • lrp:       | underlay           | lrp     | ip | dnat snat         |      |
| • lsp:       | ovn bfd ecmp       |         |    | ovs internal port | ecmp |

```
---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  externalSubnet: external204
  type: nat

# eip fip
```

|                |             |
|----------------|-------------|
| externalSubnet | external204 |
| externalSubnet | extra       |

2.1 ovn-fip pod fip

```
# kubectl get po -o wide -n vpc1 vpc-1-busybox01
NAME READY STATUS RESTARTS AGE IP NODE
vpc-1-busybox01 1/1 Running 0 3d15h 192.168.0.2 pc-node-2

# kubectl get ip vpc-1-busybox01.vpc1
```

```

NAME          V4IP          V6IP          MAC          NODE          SUBNET
vpc-1-busybox01.vpc1  192.168.0.2          00:00:00:0A:DD:27  pc-node-2  vpc1-subnet1

```

```
---
```

```

kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  externalSubnet: external204
  type: nat

```

```
---
```

```

kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  ovnEip: eip-static
  ipName: vpc-1-busybox01.vpc1 # ip crd
  type: "centralized" # centralized distributed

```

```
--
```

```

#          VPC          ip

kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  ovnEip: eip-static
  vpc: vpc1
  v4Ip: 192.168.0.2
  type: "centralized" # centralized distributed

```

```

# kubectl get ofip
NAME      VPC      V4EIP          V4IP          READY  IPTYPE  IPNAME
eip-for-vip  vpc1    10.5.204.106  192.168.0.3  true   vip     test-fip-vip
eip-static  vpc1    10.5.204.101  192.168.0.2  true
# kubectl get ofip eip-static
NAME      VPC      V4EIP          V4IP          READY  IPTYPE  IPNAME
eip-static  vpc1    10.5.204.101  192.168.0.2  true   vip     vpc-1-busybox01.vpc1

```

```

[root@pc-node-1 03-cust-vpc]# ping 10.5.204.101
PING 10.5.204.101 (10.5.204.101) 56(84) bytes of data.
64 bytes from 10.5.204.101: icmp_seq=2 ttl=62 time=1.21 ms
64 bytes from 10.5.204.101: icmp_seq=3 ttl=62 time=0.624 ms
64 bytes from 10.5.204.101: icmp_seq=4 ttl=62 time=0.368 ms
^C
--- 10.5.204.101 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3049ms
rtt min/avg/max/mdev = 0.368/0.734/1.210/0.352 ms
[root@pc-node-1 03-cust-vpc]#

```

```
#          node ping      VPC      pod      ip
```

```

#          ip
# kubectl ko nbctl show vpc1
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
  port vpc1-vpc1-subnet1
    mac: "00:00:00:ED:8E:C7"
    networks: ["192.168.0.1/24"]
  port vpc1-external204
    mac: "00:00:00:EF:05:C7"
    networks: ["10.5.204.105/24"]
  gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
  nat 813523e7-c68c-408f-bd8c-cba30cb2e4f4
    external ip: "10.5.204.101"
    logical ip: "192.168.0.2"
    type: "dnat_and_snat"

```

## 2.2 ovn-fip vip fip

```
vip          kubevirt          vip          keepalived kube-vip
```

```
fip  VPC  vip          vip
```

```

#          vip eip      eip      vip
# cat vip.yaml

apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: test-fip-vip
spec:
  subnet: vpc1-subnet1

```

```
# cat 04-fip.yaml

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-for-vip
spec:
  externalSubnet: external204
  type: nat
```

```
---
kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-for-vip
spec:
  ovnEip: eip-for-vip
  ipType: vip # fip pod ip vip
  ipName: test-fip-vip
```

```
---
# VPC ip

kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-for-vip
spec:
  ovnEip: eip-for-vip
  ipType: vip # fip pod ip vip
  vpc: vpc1
  v4Ip: 192.168.0.3
```

```
# kubectl get ofip
NAME      VPC    V4EIP      V4IP      READY  IPTYPE  IPNAME
eip-for-vip vpc1    10.5.204.106 192.168.0.3 true    vip     test-fip-vip
```

```
[root@pc-node-1 fip-vip]# ping 10.5.204.106
PING 10.5.204.106 (10.5.204.106) 56(84) bytes of data.
64 bytes from 10.5.204.106: icmp_seq=1 ttl=62 time=0.694 ms
64 bytes from 10.5.204.106: icmp_seq=2 ttl=62 time=0.436 ms
```

```
# node ping
```

```
# pod ip
```

```
[root@pc-node-1 fip-vip]# kubectl -n vpc1 exec -it vpc-1-busybox03 -- bash
[root@vpc-1-busybox03 /]#
[root@vpc-1-busybox03 /]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
1568: eth0@if1569: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 00:00:00:56:40:e5 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.0.5/24 brd 192.168.0.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet 192.168.0.3/24 scope global secondary eth0 # vip
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe56:40e5/64 scope link
        valid_lft forever preferred_lft forever
```

```
[root@vpc-1-busybox03 /]# tcpdump -i eth0 host 192.168.0.3 -netvv
tcpdump: listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
00:00:00:ed:8e:c7 > 00:00:00:56:40:e5, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 62, id 44830, offset 0, flags [DF], proto ICMP (1), length 84)
10.5.32.51 > 192.168.0.3: ICMP echo request, id 177, seq 1, length 64
00:00:00:56:40:e5 > 00:00:00:ed:8e:c7, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 64, id 43962, offset 0, flags [none], proto ICMP (1), length 84)
192.168.0.3 > 10.5.32.51: ICMP echo reply, id 177, seq 1, length 64
```

```
# pod fip icmp
```

## 5.7.4 3. ovn-snat

### 3.1 ovn-snat subnet cidr

#### iptables-snat

```
# cat 03-subnet-snat.yaml

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
```

```

name: snat-for-subnet-in-vpc
spec:
  externalSubnet: external204
  type: nat
---
kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  ovnEip: snat-for-subnet-in-vpc
  vpcSubnet: vpc1-subnet1 # eip
---
#           VPC      subnet cidr

kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  ovnEip: snat-for-subnet-in-vpc
  vpc: vpc1
  v4IpCidr: 192.168.0.0/24 #      cidr      ip

```

externalSubnet

extra

### 3.2 ovn-snat pod ip

#### iptables-snat

```

# cat 03-pod-snat.yaml
---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-pod-vpc-ip
spec:
  externalSubnet: external204
  type: nat
---
kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat01
spec:
  ovnEip: snat-for-pod-vpc-ip
  ipName: vpc-1-busybox02.vpc1 # eip      pod ip
---
#           VPC      ip

kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  ovnEip: snat-for-subnet-in-vpc
  vpc: vpc1
  v4IpCidr: 192.168.0.4

```

externalSubnet

extra

#### snat

```

# kubectl ko nbctl show vpc1
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
  port vpc1-vpc1-subnet1
    mac: "00:00:00:ED:8E:C7"
    networks: ["192.168.0.1/24"]
  port vpc1-external204
    mac: "00:00:00:EF:05:C7"
    networks: ["10.5.204.105/24"]
    gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
  nat 21d853b0-f7b4-40bd-9a53-31d2e2745739
    external ip: "10.5.204.115"
    logical ip: "192.168.0.0/24"
    type: "snat"
  nat da77a11f-c523-439c-b1d1-72c664196a0f
    external ip: "10.5.204.116"
    logical ip: "192.168.0.4"
    type: "snat"

```

```
[root@pc-node-1 03-cust-vpc]# kubectl get po -A -o wide | grep busy
vpc1          vpc-1-busybox01          1/1    Running    0          3d15h    192.168.0.2    pc-node-2    <none>      <none>
vpc1          vpc-1-busybox02          1/1    Running    0          17h      192.168.0.4    pc-node-1    <none>      <none>
vpc1          vpc-1-busybox03          1/1    Running    0          17h      192.168.0.5    pc-node-1    <none>      <none>
vpc1          vpc-1-busybox04          1/1    Running    0          17h      192.168.0.6    pc-node-3    <none>      <none>
vpc1          vpc-1-busybox05          1/1    Running    0          17h      192.168.0.7    pc-node-1    <none>      <none>

# kubectl exec -it -n vpc1          vpc-1-busybox04    bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl exec [POD] -- [COMMAND] instead.
[root@vpc-1-busybox04 /]#
[root@vpc-1-busybox04 /]#
[root@vpc-1-busybox04 /]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
17895: eth0@if17895: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 00:00:00:76:94:55 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.0.6/24 brd 192.168.0.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe76:9455/64 scope link
        valid_lft forever preferred_lft forever
[root@vpc-1-busybox04 /]# ping 223.5.5.5
PING 223.5.5.5 (223.5.5.5) 56(84) bytes of data.
64 bytes from 223.5.5.5: icmp_seq=1 ttl=114 time=22.2 ms
64 bytes from 223.5.5.5: icmp_seq=2 ttl=114 time=21.8 ms

[root@pc-node-1 03-cust-vpc]# kubectl exec -it -n vpc1          vpc-1-busybox02    bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl exec [POD] -- [COMMAND] instead.
[root@vpc-1-busybox02 /]#
[root@vpc-1-busybox02 /]#
[root@vpc-1-busybox02 /]#
[root@vpc-1-busybox02 /]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
1566: eth0@if1567: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 00:00:00:b:e9:d0 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.0.4/24 brd 192.168.0.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe0b:e9d0/64 scope link
        valid_lft forever preferred_lft forever
[root@vpc-1-busybox02 /]# ping 223.5.5.5
PING 223.5.5.5 (223.5.5.5) 56(84) bytes of data.
64 bytes from 223.5.5.5: icmp_seq=2 ttl=114 time=22.7 ms
64 bytes from 223.5.5.5: icmp_seq=3 ttl=114 time=22.6 ms
64 bytes from 223.5.5.5: icmp_seq=4 ttl=114 time=22.1 ms
^C
--- 223.5.5.5 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3064ms
rtt min/avg/max/mdev = 22.126/22.518/22.741/0.278 ms

#          pod          snat
```

## 5.7.5 4. ovn-dnat

### 4.1 ovn-dnat pod dnat

```
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  externalSubnet: underlay

---
kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ovnEip: eip-dnat
  ipName: vpc-1-busybox01.vpc1 #          pod ip crd
  protocol: tcp
  internalPort: "22"
  externalPort: "22"

---
#          VPC          ip

kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
```

```
spec:
  ovnEip: eip-dnat
  protocol: tcp
  internalPort: "22"
  externalPort: "22"
  vpc: vpc1
  v4Ip: 192.168.0.3
```

externalSubnet

extra

OvnDnatRule      IptablesDnatRule

```
# kubectl get oep eip-dnat
NAME          V4IP          MAC          TYPE          READY
eip-dnat      10.5.49.4     00:00:00:4D:CE:49  dnat          true

# kubectl get onat
NAME          EIP          PROTOCOL    V4EIP          V4IP          INTERNALPORT    EXTERNALPORT    IPNAME          READY
eip-dnat      eip-dnat     tcp         10.5.49.4     192.168.0.3   22             22             vpc-1-busybox01.vpc1  true
```

## 4.2 ovn-dnat vip dn timer

```
kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ipType: vip # dnat pod ip vip
  ovnEip: eip-dnat
  ipName: test-dnat-vip
  protocol: tcp
  internalPort: "22"
  externalPort: "22"

---
# VPC ip

kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ipType: vip # dnat pod ip vip
  ovnEip: eip-dnat
  ipName: test-dnat-vip
  protocol: tcp
  internalPort: "22"
  externalPort: "22"
  vpc: vpc1
  v4Ip: 192.168.0.4
```

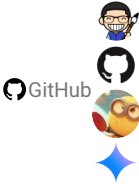
OvnDnatRule      IptablesDnatRule

|                                 |             |          |                   |             |              |              |               |       |  |
|---------------------------------|-------------|----------|-------------------|-------------|--------------|--------------|---------------|-------|--|
| # kubectl get vip test-dnat-vip |             |          |                   |             |              |              |               |       |  |
| NAME                            | V4IP        | PV4IP    | MAC               | PMAC        | V6IP         | PV6IP        | SUBNET        | READY |  |
| test-dnat-vip                   | 192.168.0.4 |          | 00:00:00:D0:C0:B5 |             |              |              | vpc1-subnet1  | true  |  |
|                                 |             |          |                   |             |              |              |               |       |  |
| # kubectl get oep eip eip-dnat  |             |          |                   |             |              |              |               |       |  |
| NAME                            | V4IP        | V6IP     | MAC               | TYPE        | READY        |              |               |       |  |
| eip-dnat                        | 10.5.49.4   |          | 00:00:00:4D:CE:49 | dnat        | true         |              |               |       |  |
|                                 |             |          |                   |             |              |              |               |       |  |
| # kubectl get ondat eip-dnat    |             |          |                   |             |              |              |               |       |  |
| NAME                            | EIP         | PROTOCOL | V4EIP             | V4IP        | INTERNALPORT | EXTERNALPORT | IPNAME        | READY |  |
| eip-dnat                        | eip-dnat    | tcp      | 10.5.49.4         | 192.168.0.4 | 22           | 22           | test-dnat-vip | true  |  |



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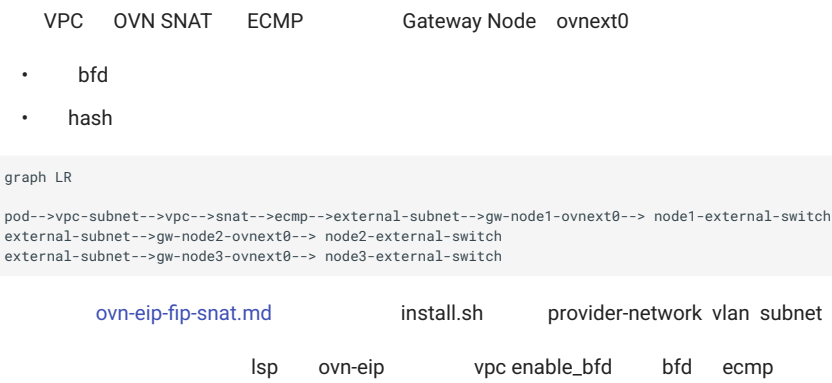
🕒 2023 3 3



5.7.6

---

## 5.8 OVN SNAT    ECMP BFD    L3 HA



### 5.8.1 1.

#### 1.1    underlay

#### 1.2    vpc    eip\_snat

#### 1.3    VPC    eip snat fip



### 5.8.2 2.    vpc    ecmp bfd L3 HA

```
# cat 01-vpc-ecmp-enable-external-bfd.yml
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc1
spec:
  namespaces:
    - vpc1
  enableExternal: true
  enableBfd: true # bfd
  #enableBfd: false
```



```
# cat 02-subnet.yml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vpc1-subnet1
spec:
  cidrBlock: 192.168.0.0/24
  default: false
  disableGatewayCheck: false
  disableInterConnection: true
  enableEcmp: true # ecmp
  gatewayNode: ""
  gatewayType: distributed
  #gatewayType: centralized
  natOutgoing: false
  private: false
  protocol: IPv4
  provider: ovn
  vpc: vpc1
  namespaces:
  - vpc1
```

:

1. vpc ecmp ecmp bfd vpc enableBfd subnet enableEcmp ecmp bfd
- 2.
3. VPC VPC VPC snat
4. vpc subnet enableEcmp gatewayType
5. EnableExternal vpc
6. EnableExternal EnableBfd

```
# ovn
# vpc
# k get vpc
NAME      ENABLEEXTERNAL  ENABLEBFD  STANDBY  SUBNETS                                NAMESPACES
ovn-cluster true            true       true     ["external204","join","ovn-default"]
vpc1      true            true       true     ["vpc1-subnet1"]                      ["vpc1"]

# vpc ENABLEBFD
# vpc

# 1. bfd
# k ko nbctl list bfd
_uuid      : be7df545-2c4c-4751-878f-b3507987f050
detect_mult : 3
dst_ip      : "10.5.204.121"
external_ids : {}
logical_port : vpc1-external204
min_rx      : 100
min_tx      : 100
options     : {}
status      : up

_uuid      : 684c4489-5b59-4693-8d8c-3beab93f8093
detect_mult : 3
dst_ip      : "10.5.204.109"
external_ids : {}
logical_port : vpc1-external204
min_rx      : 100
min_tx      : 100
options     : {}
status      : up

_uuid      : f0f62077-2ae9-4e79-b4f8-a446ec6e784c
detect_mult : 3
dst_ip      : "10.5.204.108"
external_ids : {}
logical_port : vpc1-external204
min_rx      : 100
min_tx      : 100
options     : {}
status      : up

###      status      up

# 2. bfd
# k ko nbctl lr-route-list vpc1
IPv4 Routes
Route Table <main>:
      192.168.0.0/24      10.5.204.108 src-ip ecmp ecmp-symmetric-reply bfd
      192.168.0.0/24      10.5.204.109 src-ip ecmp ecmp-symmetric-reply bfd
      192.168.0.0/24      10.5.204.121 src-ip ecmp ecmp-symmetric-reply bfd

# 3.
```

```
# k ko nbctl find Logical_Router_Static_Route policy=src-ip options=ecmp_symmetric_reply="true"
_uuid      : 3aacb384-d5ee-4b14-aebf-59e8c11717ba
bfd        : 684c4489-5b59-4693-8d8c-3beab93f8093
external_ids : {}
ip_prefix   : "192.168.0.0/24"
nexthop     : "10.5.204.109"
options     : {ecmp_symmetric_reply="true"}
output_port : []
policy      : src-ip
route_table : ""

_uuid      : 18bcc585-bc05-430b-925b-ef673c8e1aef
bfd        : f0f62077-2ae9-4e79-b4f8-a446ec6e784c
external_ids : {}
ip_prefix   : "192.168.0.0/24"
nexthop     : "10.5.204.108"
options     : {ecmp_symmetric_reply="true"}
output_port : []
policy      : src-ip
route_table : ""

_uuid      : 7d0a4e6b-cde0-4110-8176-fbaf19738498
bfd        : be7df545-2c4c-4751-878f-b3507987f050
external_ids : {}
ip_prefix   : "192.168.0.0/24"
nexthop     : "10.5.204.121"
options     : {ecmp_symmetric_reply="true"}
output_port : []
policy      : src-ip
route_table : ""
```

```
#
[root@pc-node-1 ~]# ip netns exec ovnext ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
1541: ovnext0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UNKNOWN group default qlen 1000
    link/ether 00:00:00:ab:bd:87 brd ff:ff:ff:ff:ff:ff
    inet 10.5.204.108/24 brd 10.5.204.255 scope global ovnext0
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:feab:bd87/64 scope link
        valid_lft forever preferred_lft forever
[root@pc-node-1 ~]#
[root@pc-node-1 ~]# ip netns exec ovnext route -n
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
0.0.0.0 10.5.204.254 0.0.0.0 UG 0 0 0 ovnext0
10.5.204.0 0.0.0.0 255.255.255.0 U 0 0 0 ovnext0

## internal port unerlay pod ns ns
[root@pc-node-1 ~]# ip netns exec ovnext bfd-control status
There are 1 sessions:
Session 1
id=1 local=10.5.204.108 (p) remote=10.5.204.122 state=Up

## lrp bfd lrp ecmp
[root@pc-node-1 ~]# ip netns exec ovnext ping -c1 223.5.5.5
PING 223.5.5.5 (223.5.5.5) 56(84) bytes of data.
64 bytes from 223.5.5.5: icmp_seq=1 ttl=115 time=21.6 ms

#
```

### ovnext ns

```
[root@pc-node-3 ~]# ip netns exec ovnext bash
[root@pc-node-3 ~]# tcpdump -i ovnext0 host 223.5.5.5 -netvv
dropped privs to tcpdump
tcpdump: listening on ovnext0, link-type EN10MB (Ethernet), capture size 262144 bytes
00:00:00:2d:f8:ce > 00:00:00:fd:b2:a4, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 63, id 57978, offset 0, flags [DF], proto ICMP (1), length 84)
    10.5.204.102 > 223.5.5.5: ICMP echo request, id 22, seq 71, length 64
00:00:00:fd:b2:a4 > dc:ef:80:5a:44:1a, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 62, id 57978, offset 0, flags [DF], proto ICMP (1), length 84)
    10.5.204.102 > 223.5.5.5: ICMP echo request, id 22, seq 71, length 64
^C
2 packets captured
2 packets received by filter
0 packets dropped by kernel
[root@pc-node-3 ~]#

# down pod
# 3
```

## 5.8.3.3. bfd

vpc enable\_eip\_snat

```
# cat 01-vpc-ecmp-enable-external-bfd.yml
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc2
spec:
  namespaces:
  - vpc2
  enableExternal: true
  #enableBfd: true
  enableBfd: false

##  bfd

# k ko nbctl lr-route-list vpc2
IPv4 Routes
Route Table <main>:
      0.0.0.0/0          10.5.204.254 dst-ip

#
#  nbctl list bfd      lrp      bfd
#  ovnext ns      bfd
#      vpc subnet      ping      ( )
#      ( )
```

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## 5.9 VPC

|     |     |     |     |
|-----|-----|-----|-----|
| VPC | VPC | VPC | NAT |
|-----|-----|-----|-----|

### 5.9.1

|    |     |      |
|----|-----|------|
| 1. | VPC |      |
| 2. | VPC | CIDR |
| 3. | VPC | VPC  |

### 5.9.2

|     |     |        |        |      |
|-----|-----|--------|--------|------|
| VPC | VPC | Subnet | Subnet | CIDR |
|-----|-----|--------|--------|------|

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-1
spec: {}
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net1
spec:
  vpc: vpc-1
  cidrBlock: 10.0.0.0/16
---
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-2
spec: {}
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net2
spec:
  vpc: vpc-2
  cidrBlock: 172.31.0.0/16
```

|     |             |
|-----|-------------|
| VPC | vpcPeerings |
|-----|-------------|

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-1
spec:
  vpcPeerings:
    - remoteVpc: vpc-2
      localConnectIP: 169.254.0.1/30
  staticRoutes:
    - cidr: 172.31.0.0/16
      nextHopIP: 169.254.0.2
      policy: policyDst
---
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-2
spec:
  vpcPeerings:
    - remoteVpc: vpc-1
      localConnectIP: 169.254.0.2/30
  staticRoutes:
    - cidr: 10.0.0.0/16
      nextHopIP: 169.254.0.1
      policy: policyDst
```

|                   |        |                |    |      |
|-------------------|--------|----------------|----|------|
| • remoteVpc:      | VPC    |                |    |      |
| • localConnectIP: | IP     | CIDR           | IP | CIDR |
| • cidr            | Subnet | CIDR           |    |      |
| • nextHopIP       | VPC    | localConnectIP |    |      |

Subnet Pod

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/logical_switch: net1
  name: vpc-1-pod
spec:
  containers:
    - name: vpc-1-pod
      image: docker.io/library/nginx:alpine
---
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/logical_switch: net2
  name: vpc-2-pod
spec:
  containers:
    - name: vpc-2-pod
      image: docker.io/library/nginx:alpine
```

```
# kubectl exec -it vpc-1-pod -- ping $(kubectl get pod vpc-2-pod -o jsonpath='{.status.podIP}')
PING 172.31.0.2 (172.31.0.2): 56 data bytes
64 bytes from 172.31.0.2: seq=0 ttl=62 time=0.655 ms
64 bytes from 172.31.0.2: seq=1 ttl=62 time=0.086 ms
64 bytes from 172.31.0.2: seq=2 ttl=62 time=0.098 ms
^C
--- 172.31.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.086/0.279/0.655 ms
# kubectl exec -it vpc-2-pod -- ping $(kubectl get pod vpc-1-pod -o jsonpath='{.status.podIP}')
PING 10.0.0.2 (10.0.0.2): 56 data bytes
64 bytes from 10.0.0.2: seq=0 ttl=62 time=0.594 ms
64 bytes from 10.0.0.2: seq=1 ttl=62 time=0.093 ms
64 bytes from 10.0.0.2: seq=2 ttl=62 time=0.088 ms
^C
--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.088/0.258/0.594 ms
```

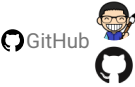
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6.

6.1 kubectl

Kube-OVN      kubectl                      OVN              OVN              OVS              tcpdump

6.1.1

Kube-OVN                      kubectl

kubectl-ko

wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.15/dist/images/kubectl-ko

\$PATH

mv kubectl-ko /usr/local/bin/kubectl-ko

chmod +x /usr/local/bin/kubectl-ko

# kubectl plugin list  
The following compatible plugins are available:  
  
/usr/local/bin/kubectl-ko

6.1.2

kubectl ko

```
# kubectl ko
kubectl ko {subcommand} [option...]
Available Subcommands:
[nb|sb] [status|kick|backup|dbstatus|restore]      ovn-db operations show cluster status, kick stale server, backup database, get db consistency status or
restore ovn nb db when met 'inconsistent data' error
nbctl [ovn-nbctl options ...]      invoke ovn-nbctl
sbctl [ovn-sbctl options ...]      invoke ovn-sbctl
vsctl {nodeName} [ovs-vsctl options ...]      invoke ovs-vsctl on the specified node
ofctl {nodeName} [ovs-ofctl options ...]      invoke ovs-ofctl on the specified node
dpctl {nodeName} [ovs-dpctl options ...]      invoke ovs-dpctl on the specified node
appctl {nodeName} [ovs-appctl options ...]      invoke ovs-appctl on the specified node
tcpdump {namespace/podname} [tcpdump options ...]      capture pod traffic
{trace|ovn-trace} ...      trace ovn microflow of specific packet
{trace|ovn-trace} {namespace/podname} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]      trace ICMP/TCP/UDP
{trace|ovn-trace} {namespace/podname} {target ip address} [target mac address] arp {request|reply}      trace ARP request/reply
{trace|ovn-trace} {node//nodename} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]      trace ICMP/TCP/UDP
{trace|ovn-trace} {node//nodename} {target ip address} [target mac address] arp {request|reply}      trace ARP request/reply
diagnose {all|node|subnet|IPPorts} [nodename|subnetName|{proto1}-{IP1}-{Port1},{proto2}-{IP2}-{Port2}]      diagnose connectivity of all nodes or a specific
node or specify subnet's ds pod or IPPorts like 'tcp-172.18.0.2-53,udp-172.18.0.3-53'
env-check      check the environment configuration
reload      restart all kube-ovn components
log {kube-ovn|ovn|ovs|linux|all}      save log to ./kubectl-ko-log/
perf [image] performance test default image is docker.io/kubeovn/test:v1.13.0
icnbctl [ovn-nbctl options ...]      invoke ovn-ic-nbctl
icsbctl [ovn-sbctl options ...]      invoke ovn-ic-sbctl
```

[nb | sb] [status | kick | backup | dbstatus | restore]

OVN

OVN leader ovs-appctl cluster/status :

```
# kubect1 ko nb status
306b
Name: OVN_Northbound
Cluster ID: 9a87 (9a872522-3e7d-47ca-83a3-d74333e1a7ca)
Server ID: 306b (306b256b-b5e1-4eb0-be91-4ca96adf6bad)
Address: tcp:[172.18.0.2]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 280309 ms ago, reason: timeout
Last Election won: 280309 ms ago
Election timer: 5000
Log: [139, 139]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-8723 ->8723 <-85d6 ->85d6
Disconnections: 0
Servers:
  85d6 (85d6 at tcp:[172.18.0.4]:6643) next_index=139 match_index=138 last msg 763 ms ago
  8723 (8723 at tcp:[172.18.0.3]:6643) next_index=139 match_index=138 last msg 763 ms ago
  306b (306b at tcp:[172.18.0.2]:6643) (self) next_index=2 match_index=138
status: ok
```

| Server | match_index | last msg | Server |
|--------|-------------|----------|--------|
|--------|-------------|----------|--------|

OVN 172.18.0.3 :

```
# kubect1 ko nb kick 8723
started removal
```

```
# kubect1 ko nb status
306b
Name: OVN_Northbound
Cluster ID: 9a87 (9a872522-3e7d-47ca-83a3-d74333e1a7ca)
Server ID: 306b (306b256b-b5e1-4eb0-be91-4ca96adf6bad)
Address: tcp:[172.18.0.2]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 324356 ms ago, reason: timeout
Last Election won: 324356 ms ago
Election timer: 5000
Log: [140, 140]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-85d6 ->85d6
Disconnections: 2
Servers:
  85d6 (85d6 at tcp:[172.18.0.4]:6643) next_index=140 match_index=139 last msg 848 ms ago
  306b (306b at tcp:[172.18.0.2]:6643) (self) next_index=2 match_index=139
status: ok
```

OVN

```
# kubect1 ko nb backup
tar: Removing leading '/' from member names
backup ovn-nb db to /root/ovnnb_db.060223191654183154.backup
```

```
# kubect1 ko nb dbstatus
status: ok
```

inconsistent data

inconsistent data

```
# kubectl ko nb restore
deployment.apps/ovn-central scaled
ovn-central original replicas is 3
first nodeIP is 172.18.0.5
ovs-ovn pod on node 172.18.0.5 is ovs-ovn-8jxv9
ovs-ovn pod on node 172.18.0.3 is ovs-ovn-sjzb6
ovs-ovn pod on node 172.18.0.4 is ovs-ovn-t87zk
backup nb db file
restore nb db file, operate in pod ovs-ovn-8jxv9
deployment.apps/ovn-central scaled
finish restore nb db file and ovn-central replicas
recreate ovs-ovn pods
pod "ovs-ovn-8jxv9" deleted
pod "ovs-ovn-sjzb6" deleted
pod "ovs-ovn-t87zk" deleted
```

[nbctl | sbctl] [options ...]

OVN

leader

ovn-nbctl

ovn-sbctl

OVN

ovn-nbctl(8)

ovn-sbctl(8)

```
# kubectl ko nbctl show
switch c7cd17e8-ceee-4a91-9bb3-e5a313fe1ece (snat)
  port snat-ovn-cluster
    type: router
    router-port: ovn-cluster-snat
switch 20e0c6d0-023a-4756-aec5-200e0c60f95d (join)
  port node-liumengxin-ovn3-192.168.137.178
    addresses: ["00:00:00:64:FF:A8 100.64.0.4"]
  port node-liumengxin-ovn1-192.168.137.176
    addresses: ["00:00:00:AF:98:62 100.64.0.2"]
  port node-liumengxin-ovn2-192.168.137.177
    addresses: ["00:00:00:D9:58:B8 100.64.0.3"]
  port join-ovn-cluster
    type: router
    router-port: ovn-cluster-join
switch 0191705c-f827-427b-9de3-3c3b7d971ba5 (central)
  port central-ovn-cluster
    type: router
    router-port: ovn-cluster-central
switch 2a45ff05-388d-4f85-9daf-e6fccd5833dc (ovn-default)
  port alertmanager-main-0.monitoring
    addresses: ["00:00:00:6C:DF:A3 10.16.0.19"]
  port kube-state-metrics-5d6885d89-4nf8h.monitoring
    addresses: ["00:00:00:6F:02:1C 10.16.0.15"]
  port fake-kubelet-67c55dfd89-pv86k.kube-system
    addresses: ["00:00:00:5C:12:E8 10.16.19.177"]
  port ovn-default-ovn-cluster
    type: router
    router-port: ovn-cluster-ovn-default
router 212f73dd-d63d-4d72-864b-a537e9afbee1 (ovn-cluster)
  port ovn-cluster-snat
    mac: "00:00:00:7A:82:8F"
    networks: ["172.22.0.1/16"]
  port ovn-cluster-join
    mac: "00:00:00:F8:18:5A"
    networks: ["100.64.0.1/16"]
  port ovn-cluster-central
    mac: "00:00:00:4D:8C:F5"
    networks: ["192.101.0.1/16"]
  port ovn-cluster-ovn-default
    mac: "00:00:00:A3:F8:18"
    networks: ["10.16.0.1/16"]
```

vsctl {nodeName} [options ...]

nodeName

ovs-ovn

ovs-vsctl

vswitchd

OVS

ovs-vsctl(8)

```
# kubectl ko vsctl kube-ovn-01 show
0d4c4675-c9cc-440a-8c1a-878e17f81b88
  Bridge br-int
    fail_mode: secure
    datapath_type: system
    Port a2c1a8a8b83a_h
      Interface a2c1a8a8b83a_h
      Port "4fa5c4cbb1a5_h"
      Interface "4fa5c4cbb1a5_h"
    Port ovn-eef07d-0
      Interface ovn-eef07d-0
      type: stt
      options: {csum="true", key=flow, remote_ip="192.168.137.178"}
    Port ovn0
```



```

    Interface ovn0
      type: internal
    Port mirror0
      Interface mirror0
        type: internal
    Port ovn-efa253-0
      Interface ovn-efa253-0
        type: stt
        options: {csum="true", key=flow, remote_ip="192.168.137.177"}
    Port br-int
      Interface br-int
        type: internal
    ovs_version: "2.17.2"

```

### ofctl {nodeName} [options ...]

nodeName    ovs-ovn    ovs-ofctl    OpenFlow    OVS    ovs-ofctl(8)

```

# kubectl ko ofctl kube-ovn-01 dump-flows br-int
NXST_FLOW reply (xid=0x4): flags=[more]
cookie=0xcfc3429e6, duration=671791.432s, table=0, n_packets=0, n_bytes=0, idle_age=65534, hard_age=65534, priority=100,in_port=2 actions=load:0x4->NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x1->NXM_NX_REG14[],resubmit(,8)
cookie=0xc91413c6, duration=671791.431s, table=0, n_packets=907489, n_bytes=99978275, idle_age=0, hard_age=65534, priority=100,in_port=7 actions=load:0x1->NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x4->NXM_NX_REG14[],resubmit(,8)
cookie=0xf180459, duration=671791.431s, table=0, n_packets=17348582, n_bytes=2667811214, idle_age=0, hard_age=65534, priority=100,in_port=6317 actions=load:0xa->NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x9->NXM_NX_REG14[],resubmit(,8)
cookie=0x7806dd90, duration=671791.431s, table=0, n_packets=3235428, n_bytes=833821312, idle_age=0, hard_age=65534, priority=100,in_port=1 actions=load:0xd->NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x3->NXM_NX_REG14[],resubmit(,8)
...

```

### dpctl {nodeName} [options ...]

nodeName    ovs-ovn    ovs-dpctl    OVS datapath    OVS    ovs-dpctl(8)

```

# kubectl ko dpctl kube-ovn-01 show
system@ovs-system:
  lookups: hit:350805055 missed:21983648 lost:73
  flows: 105
  masks: hit:1970748791 total:22 hit/pkt:5.29
  port 0: ovs-system (internal)
  port 1: ovn0 (internal)
  port 2: mirror0 (internal)
  port 3: br-int (internal)
  port 4: stt_sys_7471 (stt: packet_type=ptap)
  port 5: eeb4d9e51b5d_h
  port 6: a2c1a8a8b83a_h
  port 7: 4fa5c4cbb1a5_h

```

### apctl {nodeName} [options ...]

nodeName    ovs-ovn    ovs-apctl    daemon    OVS    ovs-apctl(8)

```

# kubectl ko apctl kube-ovn-01 vlog/list
      console      syslog      file
      -----
backtrace      OFF      ERR      INFO
bfd            OFF      ERR      INFO
bond           OFF      ERR      INFO
bridge         OFF      ERR      INFO
bundle         OFF      ERR      INFO
bundles        OFF      ERR      INFO
...

```

### tcpdump {namespace/podname} [tcpdump options ...]

namespace/podname    kube-ovn-cni    tcpdump    veth

```

# kubectl ko tcpdump default/ds1-l6n7p icmp
+ kubectl exec -it kube-ovn-cni-wlg4s -n kube-ovn -- tcpdump -nn -i d7176fe7b4e0_h icmp
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on d7176fe7b4e0_h, link-type EN10MB (Ethernet), capture size 262144 bytes
06:52:36.619688 IP 100.64.0.3 > 10.16.0.4: ICMP echo request, id 2, seq 1, length 64
06:52:36.619746 IP 10.16.0.4 > 100.64.0.3: ICMP echo reply, id 2, seq 1, length 64
06:52:37.619588 IP 100.64.0.3 > 10.16.0.4: ICMP echo request, id 2, seq 2, length 64
06:52:37.619630 IP 10.16.0.4 > 100.64.0.3: ICMP echo reply, id 2, seq 2, length 64
06:52:38.619933 IP 100.64.0.3 > 10.16.0.4: ICMP echo request, id 2, seq 3, length 64
06:52:38.619973 IP 10.16.0.4 > 100.64.0.3: ICMP echo reply, id 2, seq 3, length 64

```

**trace [arguments ...]**

|     |     |          |
|-----|-----|----------|
| Pod | OVN | Openflow |
|-----|-----|----------|

```
kubectl ko trace {namespace/podname} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]
kubectl ko trace {namespace/podname} {target ip address} [target mac address] arp {request|reply}
kubectl ko trace {node//nodename} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]
kubectl ko trace {node//nodename} {target ip address} [target mac address] arp {request|reply}
```

```
# kubectl ko trace default/ds1-l6n7p 8.8.8.8 icmp
+ kubectl exec ovn-central-5bc494cb5-np9hm -n kube-ovn -- ovn-trace --ct=new ovn-default 'inport == "ds1-l6n7p.default" && ip.ttl == 64 && icmp && eth.src ==
0a:00:00:10:00:05 && ip4.src == 10.16.0.4 && eth.dst == 00:00:00:b8:ca:43 && ip4.dst == 8.8.8.8'
# icmp, reg14=0xf, vlan_tci=0x0000, dl_src=0a:00:00:10:00:05, dl_dst=00:00:00:b8:ca:
43, nw_src=10.16.0.4, nw_dst=8.8.8.8, nw_tos=0, nw_ecn=0, nw_ttl=64, icmp_type=0, icmp_code=0

ingress(dp="ovn-default", inport="ds1-l6n7p.default")
-----
0. ls_in_port_sec_l2 (ovn-northd.c:4143): inport == "ds1-l6n7p.default" && eth.src == {0a:00:00:10:00:05}, priority 50, uuid 39453393
  next;
1. ls_in_port_sec_ip (ovn-northd.c:2898): inport == "ds1-l6n7p.default" && eth.src == 0a:00:00:10:00:05 && ip4.src == {10.16.0.4}, priority 90, uuid 81bcd485
  next;
3. ls_in_pre_acl (ovn-northd.c:3269): ip, priority 100, uuid 7b4f4971
  reg0[0] = 1;
  next;
5. ls_in_pre_stateful (ovn-northd.c:3396): reg0[0] == 1, priority 100, uuid 36cdd577
  ct_next;

ct_next(ct_state=new|trk)
-----
6. ls_in_acl (ovn-northd.c:3759): ip && (!ct.est || (ct.est && ct_label.blocked == 1)), priority 1, uuid 7608af5b
  reg0[1] = 1;
  next;
10. ls_in_stateful (ovn-northd.c:3995): reg0[1] == 1, priority 100, uuid 2aba1b90
  ct_commit(ct_label=0/0x1);
  next;
16. ls_in_l2_lkup (ovn-northd.c:4470): eth.dst == 00:00:00:b8:ca:43, priority 50, uuid 5c9c3c9f
  output = "ovn-default-ovn-cluster";
  output;

...
```

|       |          |     |
|-------|----------|-----|
| trace | Underlay | Mac |
|-------|----------|-----|

```
kubectl ko trace default/virt-handler-7lvm1 8.8.8.8 82:7c:9f:83:8c:01 icmp
```

**diagnose {all|node|subnet|IPPorts} [nodename|subnetName|{proto1}-{IP1}-{Port1},{proto2}-{IP2}-{Port2}]**

```
kube-ovn-pinger
```

```
# kubectl ko diagnose all
switch c7cd17e8-ceee-4a91-9bb3-e5a313fe1ece (snat)
  port snat-ovn-cluster
    type: router
    router-port: ovn-cluster-snat
switch 20e0c6d0-023a-4756-aec5-200ec60f95d (join)
  port node-liumengxin-ovn3-192.168.137.178
    addresses: ["00:00:00:64:FF:A8 100.64.0.4"]
  port node-liumengxin-ovn1-192.168.137.176
    addresses: ["00:00:00:AF:98:62 100.64.0.2"]
  port join-ovn-cluster
    type: router
    router-port: ovn-cluster-join
switch 0191705c-f827-427b-9de3-3c3b7d971ba5 (central)
  port central-ovn-cluster
    type: router
    router-port: ovn-cluster-central
switch 2a45ff05-388d-4f85-9daf-e6fccd5833dc (ovn-default)
  port ovn-default-ovn-cluster
    type: router
    router-port: ovn-cluster-ovn-default
  port prometheus-k8s-1.monitoring
    addresses: ["00:00:00:AA:37:DF 10.16.0.23"]
router 212f73dd-d63d-4d72-864b-a537e9afbee1 (ovn-cluster)
  port ovn-cluster-snat
    mac: "00:00:00:7A:82:8F"
    networks: ["172.22.0.1/16"]
  port ovn-cluster-join
    mac: "00:00:00:F8:18:5A"
    networks: ["100.64.0.1/16"]
  port ovn-cluster-central
```

```

    mac: "00:00:00:4D:8C:F5"
    networks: ["192.101.0.1/16"]
  port ovn-cluster-ovn-default
    mac: "00:00:00:A3:F8:18"
    networks: ["10.16.0.1/16"]
Routing Policies
31000 ip4.dst == 10.16.0.0/16 allow
31000 ip4.dst == 100.64.0.0/16 allow
30000 ip4.dst == 192.168.137.177 reroute 100.64.0.3
30000 ip4.dst == 192.168.137.178 reroute 100.64.0.4
29000 ip4.src == $ovn.default.fake.6_ip4 reroute 100.64.0.22
29000 ip4.src == $ovn.default.fake.7_ip4 reroute 100.64.0.21
29000 ip4.src == $ovn.default.fake.8_ip4 reroute 100.64.0.23
20000 ip4.src == $ovn.default.liumengxin.ovn3.192.168.137.178_ip4 reroute 100.64.0.4
20000 ip4.src == $ovn.default.liumengxin.ovn1.192.168.137.176_ip4 && ip4.dst != $ovn.cluster.overlay.subnets.IPV4 reroute 100.
64.0.2
20000 ip4.src == $ovn.default.liumengxin.ovn2.192.168.137.177_ip4 && ip4.dst != $ovn.cluster.overlay.subnets.IPV4 reroute 100.
64.0.3
20000 ip4.src == $ovn.default.liumengxin.ovn3.192.168.137.178_ip4 && ip4.dst != $ovn.cluster.overlay.subnets.IPV4 reroute 100.
64.0.4
IPv4 Routes
Route Table <main>:
0.0.0.0/0 100.64.0.1 dst-ip
UUID LB PROTO VIP IPs
e9b9cf9d-793e-4431-9073-6dec96b75d71 cluster-tcp-load tcp 10.100.209.132:10660 192.168.137.176:10660
tcp 10.101.239.192:6641 192.168.137.177:6641
tcp 10.101.240.101:3000 10.16.0.7:3000
tcp 10.103.184.186:6642 192.168.137.177:6642
35d2b7a5-e3a7-485a-a4b7-b4970eb0e63b cluster-tcp-sess tcp 10.100.158.128:8080 10.16.0.10:8080,10.16.0.5:8080,10.16.63.30:8080
tcp 10.107.26.215:8080 10.16.0.19:8080,10.16.0.20:8080,10.16.0.21:8080
tcp 10.107.26.215:9093 10.16.0.19:9093,10.16.0.20:9093,10.16.0.21:9093
tcp 10.98.187.99:8080 10.16.0.22:8080,10.16.0.23:8080
tcp 10.98.187.99:9090 10.16.0.22:9090,10.16.0.23:9090
f43303e4-89aa-4d3e-a3dc-278a552fe27b cluster-udp-load udp 10.96.0.10:53 10.16.0.4:53,10.16.0.9:53
_uuid : 06776304-5a96-43ed-90c4-c4854c251699
addresses : []
external_ids : {vendor=kube-ovn}
name : node_liumengxin_ovn2_192.168.137.177_underlay_v6
_uuid : 62690625-87d5-491c-8675-9fd83b1f433c
addresses : []
external_ids : {vendor=kube-ovn}
name : node_liumengxin_ovn1_192.168.137.176_underlay_v6
_uuid : b03a9bae-94d5-4562-b34c-b5f6198e180b
addresses : ["10.16.0.0/16", "100.64.0.0/16", "172.22.0.0/16", "192.101.0.0/16"]
external_ids : {vendor=kube-ovn}
name : ovn.cluster.overlay.subnets.IPV4
_uuid : e1056f3a-24cc-4666-8a91-75ee6c3c2426
addresses : []
external_ids : {vendor=kube-ovn}
name : ovn.cluster.overlay.subnets.IPV6
_uuid : 3e5d5fff-e670-47b2-a2f5-a39f4698a8c5
addresses : []
external_ids : {vendor=kube-ovn}
name : node_liumengxin_ovn3_192.168.137.178_underlay_v6
_uuid : 2d85dbdc-d0db-4abe-b19e-cc806d32b492
action : drop
direction : from-lport
external_ids : {}
label : 0
log : false
match : "inport==@ovn.sg.kubeovn_deny_all && ip"
meter : []
name : []
options : {}
priority : 2003
severity : []
_uuid : de790cc8-f155-405f-bb32-5a51f30c545f
action : drop
direction : to-lport
external_ids : {}
label : 0
log : false
match : "outport==@ovn.sg.kubeovn_deny_all && ip"
meter : []
name : []
options : {}
priority : 2003
severity : []
Chassis "e15ed4d4-1780-4d50-b09e-ea8372ed48b8"
  hostname: liumengxin-ovn1-192.168.137.176
  Encap stt
    ip: "192.168.137.176"
    options: {csum="true"}
  Port_Binding node-liumengxin-ovn1-192.168.137.176
  Port_Binding perf-6vxkn.default
  Port_Binding kube-state-metrics-5d6885d89-4nf8h.monitoring
  Port_Binding alertmanager-main-0.monitoring
  Port_Binding kube-ovn-pinger-6ftdf.kube-system
  Port_Binding fake-kubelet-67c55dfd89-pv86k.kube-system

```

```

Port_Binding prometheus-k8s-0.monitoring
Chassis "eef07da1-f8ad-4775-b14d-bd6a3b4eb0d5"
hostname: liumengxin-ovn3-192.168.137.178
Encap stt
  ip: "192.168.137.178"
  options: {csum="true"}
Port_Binding kube-ovn-pinger-7twb4.kube-system
Port_Binding prometheus-adapter-86df476d87-r188g.monitoring
Port_Binding prometheus-k8s-1.monitoring
Port_Binding node-liumengxin-ovn3-192.168.137.178
Port_Binding perf-ff475.default
Port_Binding alertmanager-main-1.monitoring
Port_Binding blackbox-exporter-676d976865-tvsjd.monitoring
Chassis "efa253c9-494d-4719-83ae-b48ab0f11c03"
hostname: liumengxin-ovn2-192.168.137.177
Encap stt
  ip: "192.168.137.177"
  options: {csum="true"}
Port_Binding grafana-6c4c6b8fb7-pzd2c.monitoring
Port_Binding node-liumengxin-ovn2-192.168.137.177
Port_Binding alertmanager-main-2.monitoring
Port_Binding coredns-6789c94dd8-9jqsz.kube-system
Port_Binding coredns-6789c94dd8-25d4r.kube-system
Port_Binding prometheus-operator-7bbc99fc8b-wgjm4.monitoring
Port_Binding prometheus-adapter-86df476d87-gdxmc.monitoring
Port_Binding perf-fjnws.default
Port_Binding kube-ovn-pinger-vh2xg.kube-system
ds kube-proxy ready
kube-proxy ready
deployment ovn-central ready
deployment kube-ovn-controller ready
ds kube-ovn-cni ready
ds ovs-ovn ready
deployment coredns ready
ovn-nb leader check ok
ovn-sb leader check ok
ovn-northd leader check ok
### kube-ovn-controller recent log

### start to diagnose node liumengxin-ovn1-192.168.137.176
#### ovn-controller log:
2022-06-03T00:56:44.897Z|16722|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:06:44.912Z|16723|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:16:44.925Z|16724|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:26:44.936Z|16725|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:36:44.959Z|16726|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:46:44.974Z|16727|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:56:44.988Z|16728|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T02:06:45.001Z|16729|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T02:16:45.025Z|16730|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T02:26:45.040Z|16731|inc_proc_eng|INFO|User triggered force recompute.

#### ovs-vswnitchd log:
2022-06-02T23:03:00.137Z|00079|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,d1_src=00:00:00:f8:07:c8,d1_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:f9d1
  with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-02T23:23:31.840Z|00080|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,d1_src=00:00:00:f8:07:c8,d1_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:15b2
  with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-03T00:09:15.659Z|00081|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,d1_src=00:00:00:dc:e3:63,d1_dst=00:00:00:fa:1e:50,nw_src=10.16.63.30,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0 icmp_csum:e5a5
  with metadata skb_priority(0),tunnel(tun_id=0x150017000004,src=192.168.137.178,dst=192.168.137.176,ttl=64,tp_src=9239,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.63.30,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-03T00:02:33.832Z|00082|dpif(handler2)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,d1_src=00:00:00:f8:07:c8,d1_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:6b4a
  with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-03T00:02:33.832Z|00082|dpif(handler2)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,d1_src=00:00:00:f8:07:c8,d1_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:a819
  with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0

#### ovs-vsctl show results:
0d4c4675-c9cc-440a-8c1a-878e17f81b88
  Bridge br-int
    fail_mode: secure
    datapath_type: system
    Port a2c1a8a8b83a_h
      Interface a2c1a8a8b83a_h
    Port "4fa5c4cbb1a5_h"
      Interface "4fa5c4cbb1a5_h"
    Port ovn-eef07d-0
      Interface ovn-eef07d-0
        type: stt
        options: {csum="true", key=flow, remote_ip="192.168.137.178"}
    Port ovn0
      Interface ovn0

```

```

        type: internal
    Port "04d03360e9a0_h"
        Interface "04d03360e9a0_h"
    Port eeb4d9e51b5d_h
        Interface eeb4d9e51b5d_h
    Port mirror0
        Interface mirror0
        type: internal
    Port "8e5d887ccd80_h"
        Interface "8e5d887ccd80_h"
    Port ovn-efa253-0
        Interface ovn-efa253-0
        type: stt
        options: {csum="true", key=flow, remote_ip="192.168.137.177"}
    Port "17512d5be1f1_h"
        Interface "17512d5be1f1_h"
    Port br-int
        Interface br-int
        type: internal
    ovs_version: "2.17.2"

### pinger diagnose results:
I0603 10:35:04.349404 17619 pinger.go:19]
-----
Kube-OVN:
Version:      v1.15.21
Build:        2022-04-24_08:02:50
Commit:       git-73f9d15
Go Version:   go1.17.8
Arch:         amd64
-----
I0603 10:35:04.376797 17619 config.go:166] pinger config is &{KubeConfigFile: KubeClient:0xc000493380 Port:8080 DaemonSetNameSpace:kube-system
DaemonSetName:kube-ovn-pinger Interval:5 Mode:job ExitCode:0 InternalDNS:kubernetes.default ExternalDNS: NodeName:liumengxin-ovn1-192.168.137.176 HostIP:
192.168.137.176 PodName:kube-ovn-pinger-6ftdf PodIP:10.16.0.10 PodProtocols:[IPv4] ExternalAddress: NetworkMode:kube-ovn PollTimeout:2 PollInterval:15
SystemRunDir:/var/run/openvswitch DatabaseVswitchName:Open_vSwitch DatabaseVswitchSocketRemote:unix:/var/run/openvswitch/db.sock DatabaseVswitchFileDataPath:/
etc/openvswitch/conf.db DatabaseVswitchFileLogPath:/var/log/openvswitch/ovsdb-server.log DatabaseVswitchFilePidPath:/var/run/openvswitch/ovsdb-server.pid
DatabaseVswitchFileSystemIDPath:/etc/openvswitch/system-id.conf ServiceVswitchdFileLogPath:/var/log/openvswitch/ovs-vswitchd.log ServiceVswitchdFilePidPath:/
var/run/openvswitch/ovs-vswitchd.pid ServiceOvnControllerFileLogPath:/var/log/ovn/ovn-controller.log ServiceOvnControllerFilePidPath:/var/run/ovn/ovn-
controller.pid}
I0603 10:35:04.449166 17619 exporter.go:75] liumengxin-ovn1-192.168.137.176: exporter connect successfully
I0603 10:35:04.554011 17619 ovn.go:21] ovs-vswitchd and ovsdb are up
I0603 10:35:04.651293 17619 ovn.go:33] ovn_controller is up
I0603 10:35:04.651342 17619 ovn.go:39] start to check port binding
I0603 10:35:04.749613 17619 ovn.go:135] chassis id is 1d7f3d6c-ee5-4b3c-adca-2969d9cdfd80
I0603 10:35:04.763487 17619 ovn.go:49] port in sb is [node-liumengxin-ovn1-192.168.137.176 perf-6vxkn.default kube-state-metrics-5d6885d89-4nf8h.monitoring
alertmanager-main-0.monitoring kube-ovn-pinger-6ftdf.kube-system fake-kubelet-67c55dfd89-pv86k.kube-system prometheus-k8s-0.monitoring]
I0603 10:35:04.763583 17619 ovn.go:61] ovs and ovn-sb binding check passed
I0603 10:35:05.049309 17619 ping.go:259] start to check apiserver connectivity
I0603 10:35:05.053666 17619 ping.go:268] connect to apiserver success in 4.27ms
I0603 10:35:05.053786 17619 ping.go:129] start to check pod connectivity
I0603 10:35:05.249590 17619 ping.go:159] ping pod: kube-ovn-pinger-6ftdf 10.16.0.10, count: 3, loss count 0, average rtt 16.30ms
I0603 10:35:05.354135 17619 ping.go:159] ping pod: kube-ovn-pinger-7twb4 10.16.63.30, count: 3, loss count 0, average rtt 1.81ms
I0603 10:35:05.458460 17619 ping.go:159] ping pod: kube-ovn-pinger-vh2xg 10.16.0.5, count: 3, loss count 0, average rtt 1.92ms
I0603 10:35:05.458523 17619 ping.go:83] start to check node connectivity

```

|          |         |                 |           |                 |           |     |           |
|----------|---------|-----------------|-----------|-----------------|-----------|-----|-----------|
| diagnose | subnet  | subnet          | daemonset | kube-ovn-pinger | daemonset | pod | daemonset |
| diagnose | IPPorts | kube-ovn-pinger | pod       | IP Port         |           |     |           |

reload

Kube-OVN

```

# kubectl ko reload
pod "ovn-central-8684dd94bd-vzgcr" deleted
Waiting for deployment "ovn-central" rollout to finish: 0 of 1 updated replicas are available...
deployment "ovn-central" successfully rolled out
pod "ovs-ovn-bsnvz" deleted
pod "ovs-ovn-m9b98" deleted
pod "kube-ovn-controller-8459db5ff4-64c62" deleted
Waiting for deployment "kube-ovn-controller" rollout to finish: 0 of 1 updated replicas are available...
deployment "kube-ovn-controller" successfully rolled out
pod "kube-ovn-cni-2klmh" deleted
pod "kube-ovn-cni-t2jz4" deleted
Waiting for daemon set "kube-ovn-cni" rollout to finish: 0 of 2 updated pods are available...
Waiting for daemon set "kube-ovn-cni" rollout to finish: 1 of 2 updated pods are available...
daemon set "kube-ovn-cni" successfully rolled out
pod "kube-ovn-pinger-ln72z" deleted
pod "kube-ovn-pinger-w8lrk" deleted
Waiting for daemon set "kube-ovn-pinger" rollout to finish: 0 of 2 updated pods are available...
Waiting for daemon set "kube-ovn-pinger" rollout to finish: 1 of 2 updated pods are available...
daemon set "kube-ovn-pinger" successfully rolled out
pod "kube-ovn-monitor-7fb67d5488-7q6zb" deleted
Waiting for deployment "kube-ovn-monitor" rollout to finish: 0 of 1 updated replicas are available...
deployment "kube-ovn-monitor" successfully rolled out

```

log

kube-ovn      Kube-OVN OVN Open vSwitch   log   linux   debug

```
# kubectl ko log all
Collecting kube-ovn logging files
Collecting ovn logging files
Collecting openvswitch logging files
Collecting linux dmesg files
Collecting linux iptables-legacy files
Collecting linux iptables-nft files
Collecting linux route files
Collecting linux link files
Collecting linux neigh files
Collecting linux memory files
Collecting linux top files
Collecting linux sysctl files
Collecting linux netstat files
Collecting linux addr files
Collecting linux ipset files
Collecting linux tcp files
Collected files have been saved in the directory /root/kubectl-ko-log
```

```
# tree kubectl-ko-log/
kubectl-ko-log/
|-- kube-ovn-control-plane
|   |-- kube-ovn
|   |   |-- kube-ovn-cni.log
|   |   |-- kube-ovn-monitor.log
|   |   '-- kube-ovn-pinger.log
|   |-- linux
|   |   |-- addr.log
|   |   |-- dmesg.log
|   |   |-- ipset.log
|   |   |-- iptables-legacy.log
|   |   |-- iptables-nft.log
|   |   |-- link.log
|   |   |-- memory.log
|   |   |-- neigh.log
|   |   |-- netstat.log
|   |   |-- route.log
|   |   |-- sysctl.log
|   |   |-- tcp.log
|   |   '-- top.log
|   |-- openvswitch
|   |   |-- ovs-vswitchd.log
|   |   '-- ovssdb-server.log
|   '-- ovn
|       |-- ovn-controller.log
|       |-- ovn-northd.log
|       |-- ovssdb-server-nb.log
|       '-- ovssdb-server-sb.log
```

perf [image]

Kube-OVN

- 1.
- 2. Hostnetwork
- 3.
- 4. OVN-NB, OVN-SB, OVN-Northd leader

image      Pod      kubeovn/test:v1.12.0

```
# kubectl ko perf
===== Prepareing Performance Test Resources =====
pod/test-client created
pod/test-host-client created
pod/test-server created
pod/test-host-server created
service/test-server created
pod/test-client condition met
pod/test-host-client condition met
pod/test-host-server condition met
pod/test-server condition met
=====
===== Start Pod Network Unicast Performance Test =====
Size          TCP Latency    TCP Bandwidth  UDP Latency    UDP Lost Rate  UDP Bandwidth
64            82.8 us        97.7 Mbits/sec 67.6 us        (0%)           8.42 Mbits/sec
128           85.4 us        167 Mbits/sec  67.2 us        (0%)           17.2 Mbits/sec
```

```
512      85.8 us      440 Mbits/sec  68.7 us      (0%)      68.4 Mbits/sec
1k       85.1 us      567 Mbits/sec  68.7 us      (0%)      134 Mbits/sec
4k       138 us      826 Mbits/sec  78.1 us      (1.4%)    583 Mbits/sec
=====
===== Start Host Network Performance Test =====
Size      TCP Latency    TCP Bandwidth  UDP Latency    UDP Lost Rate  UDP Bandwidth
64        49.7 us      120 Mbits/sec  37.9 us      (0%)          18.6 Mbits/sec
128       49.7 us      200 Mbits/sec  38.1 us      (0%)          35.5 Mbits/sec
512       51.9 us      588 Mbits/sec  38.9 us      (0%)          142 Mbits/sec
1k        51.7 us      944 Mbits/sec  37.2 us      (0%)          279 Mbits/sec
4k        74.9 us      1.66 Gbits/sec 39.9 us      (0%)          1.20 Gbits/sec
=====
===== Start Service Network Performance Test =====
Size      TCP Latency    TCP Bandwidth  UDP Latency    UDP Lost Rate  UDP Bandwidth
64        111 us      96.3 Mbits/sec  88.4 us      (0%)          7.59 Mbits/sec
128       83.7 us      150 Mbits/sec  69.2 us      (0%)          16.9 Mbits/sec
512       87.4 us      374 Mbits/sec  75.8 us      (0%)          60.9 Mbits/sec
1k        88.2 us      521 Mbits/sec  73.1 us      (0%)          123 Mbits/sec
4k        148 us      813 Mbits/sec  77.6 us      (0.0044%)     451 Mbits/sec
=====
===== Start Pod Multicast Network Performance Test =====
Size      UDP Latency    UDP Lost Rate  UDP Bandwidth
64        0.014 ms      (0.17%)       5.80 Mbits/sec
128       0.012 ms      (0%)          11.4 Mbits/sec
512       0.016 ms      (0%)          46.1 Mbits/sec
1k        0.023 ms      (0.073%)      89.8 Mbits/sec
4k        0.035 ms      (1.3%)        126 Mbits/sec
=====
===== Start Host Multicast Network Performance =====
Size      UDP Latency    UDP Lost Rate  UDP Bandwidth
64        0.007 ms      (0%)          9.95 Mbits/sec
128       0.005 ms      (0%)          21.8 Mbits/sec
512       0.008 ms      (0%)          86.8 Mbits/sec
1k        0.013 ms      (0.045%)     168 Mbits/sec
4k        0.010 ms      (0.31%)     242 Mbits/sec
=====
===== Start Leader Recover Time Test =====
Delete ovn central nb pod
pod "ovn-central-5cb9c67d75-tlz9w" deleted
Waiting for ovn central nb pod running
===== OVN nb Recovery takes 3.305236803 s =====
Delete ovn central sb pod
pod "ovn-central-5cb9c67d75-szx4c" deleted
Waiting for ovn central sb pod running
===== OVN sb Recovery takes 3.462698535 s =====
Delete ovn central northd pod
pod "ovn-central-5cb9c67d75-zqmqv" deleted
Waiting for ovn central northd pod running
===== OVN northd Recovery takes 2.691291403 s =====
===== Remove Performance Test Resource =====
rm -f unicast-test-client.log
rm -f unicast-test-host-client.log
rm -f unicast-test-client.log
kubectl ko nbctl lb-del test-server
rm -f multicast-test-server.log
kubectl exec ovs-ovn-gxdrf -n kube-system -- ip maddr del 01:00:5e:00:00:64 dev eth0
kubectl exec ovs-ovn-h57bf -n kube-system -- ip maddr del 01:00:5e:00:00:64 dev eth0
rm -f multicast-test-host-server.log
pod "test-client" deleted
pod "test-host-client" deleted
pod "test-host-server" deleted
pod "test-server" deleted
service "test-server" deleted
=====
```

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## 6.2

Kubernetes

ovs-ovn

ovn-controller

ovn-central

Kubernetes

ovn-central

ovn-central

### 6.2.1

```
# kubectl drain kube-ovn-worker --ignore-daemonsets --force
node/kube-ovn-worker cordoned
WARNING: ignoring DaemonSet-managed Pods: kube-system/kube-ovn-cni-zt74b, kube-system/kube-ovn-pinger-5rxfs, kube-system/kube-proxy-jpmnm, kube-system/ovs-ovn-v2kll
evicting pod kube-system/coredns-64897985d-qsgpt
evicting pod local-path-storage/local-path-provisioner-5ddd94ff66-1lss6
evicting pod kube-system/kube-ovn-controller-8459db5ff4-94lxb
pod/kube-ovn-controller-8459db5ff4-94lxb evicted
pod/coredns-64897985d-qsgpt evicted
pod/local-path-provisioner-5ddd94ff66-1lss6 evicted
node/kube-ovn-worker drained
```

### 6.2.2 kubelet docker

ovs-ovn

ovn-central

```
systemctl stop kubelet
systemctl stop docker
```

#### Containerd

containerd CRI

Pod/

ovs-ovn

containerd

ovs-ovn

#### 1. kubelet

```
systemctl stop kubelet
```

#### 2. ovs-ovn Pod

```
crictl rm -f $(crictl ps | grep openvswitch | awk '{print $1}')
```

#### 3. containerd

```
systemctl stop containerd
```

### 6.2.3 Node

```
rm -rf /var/run/openvswitch
rm -rf /var/run/ovn
rm -rf /etc/origin/openvswitch/
rm -rf /etc/origin/ovn/
rm -rf /etc/cni/net.d/00-kube-ovn.conflist
rm -rf /etc/cni/net.d/01-kube-ovn.conflist
rm -rf /var/log/openvswitch
rm -rf /var/log/ovn
```

### 6.2.4 kubectl

```
kubectl delete no kube-ovn-01
```



## 6.2.5      ovn-sb

kube-ovn-worker

```
# kubectl ko sbctl show
Chassis "b0564934-5a0d-4804-a4c0-476c93596a17"
  hostname: kube-ovn-worker
  Encap geneve
    ip: "172.18.0.2"
    options: {csum="true"}
  Port_Binding kube-ovn-pinger-5rxfs.kube-system
Chassis "6a29de7e-d731-4eaf-bacd-2f239ee52b28"
  hostname: kube-ovn-control-plane
  Encap geneve
    ip: "172.18.0.3"
    options: {csum="true"}
  Port_Binding coredns-64897985d-nbfln.kube-system
  Port_Binding node-kube-ovn-control-plane
  Port_Binding local-path-provisioner-5ddd94ff66-h4tn9.local-path-storage
  Port_Binding kube-ovn-pinger-hf2p6.kube-system
  Port_Binding coredns-64897985d-fhwlw.kube-system
```

## 6.2.6      chassis

uuid      Chassis    id

```
# kubectl ko sbctl chassis-del b0564934-5a0d-4804-a4c0-476c93596a17
# kubectl ko sbctl show
Chassis "6a29de7e-d731-4eaf-bacd-2f239ee52b28"
  hostname: kube-ovn-control-plane
  Encap geneve
    ip: "172.18.0.3"
    options: {csum="true"}
  Port_Binding coredns-64897985d-nbfln.kube-system
  Port_Binding node-kube-ovn-control-plane
  Port_Binding local-path-provisioner-5ddd94ff66-h4tn9.local-path-storage
  Port_Binding kube-ovn-pinger-hf2p6.kube-system
  Port_Binding coredns-64897985d-fhwlw.kube-system
```



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## 6.2.7

## 6.3 ovn-central

ovn-central    ovn-nb    ovn-sb    etcd Raft    ovn-central

### 6.3.1 ovn-central

kube-ovn-control-plane2

ovn-central

```
# kubectl -n kube-system get pod -o wide | grep central
ovn-central-6bf58cbc97-2cdhg          1/1      Running   0           21m    172.18.0.3    kube-ovn-control-plane    <none>    <none>
ovn-central-6bf58cbc97-crmfp          1/1      Running   0           21m    172.18.0.5    kube-ovn-control-plane2   <none>    <none>
ovn-central-6bf58cbc97-lxmpl          1/1      Running   0           21m    172.18.0.4    kube-ovn-control-plane3   <none>    <none>
```

#### ovn-nb

ID

```
# kubectl ko nb status
1b9a
Name: OVN_Northbound
Cluster ID: 32ca (32ca07fb-739b-4257-b510-12fa18e7cce8)
Server ID: 1b9a (1b9a5d76-e69b-410c-8085-39943d0cd38c)
Address: tcp:[172.18.0.3]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2135194 ms ago, reason: timeout
Last Election won: 2135188 ms ago
Election timer: 5000
Log: [135, 135]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-d64b ->d64b <-4984 ->4984
Disconnections: 0
Servers:
  4984 (4984 at tcp:[172.18.0.4]:6643) next_index=135 match_index=134 last msg 1084 ms ago
  1b9a (1b9a at tcp:[172.18.0.3]:6643) (self) next_index=2 match_index=134
  d64b (d64b at tcp:[172.18.0.5]:6643) next_index=135 match_index=134 last msg 1084 ms ago
status: ok
```

kube-ovn-control-plane2

IP 172.18.0.5

ID d64b

ovn-nb

```
# kubectl ko nb kick d64b
started removal
```

```
# kubectl ko nb status
1b9a
Name: OVN_Northbound
Cluster ID: 32ca (32ca07fb-739b-4257-b510-12fa18e7cce8)
Server ID: 1b9a (1b9a5d76-e69b-410c-8085-39943d0cd38c)
Address: tcp:[172.18.0.3]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2297649 ms ago, reason: timeout
Last Election won: 2297643 ms ago
Election timer: 5000
Log: [136, 136]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-4984 ->4984
Disconnections: 2
Servers:
  4984 (4984 at tcp:[172.18.0.4]:6643) next_index=136 match_index=135 last msg 1270 ms ago
  1b9a (1b9a at tcp:[172.18.0.3]:6643) (self) next_index=2 match_index=135
status: ok
```

**ovn-sb**

ovn-sb

ID

```
kubectl ko sb status
3722
Name: OVN_Southbound
Cluster ID: d4bd (d4bd37a4-0400-499f-b4df-b4fd389780f0)
Server ID: 3722 (3722d5ae-2ced-4820-a6b2-8b744d11fb3e)
Address: tcp:[172.18.0.3]:6644
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2395317 ms ago, reason: timeout
Last Election won: 2395316 ms ago
Election timer: 5000
Log: [130, 130]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-e9f7 ->e9f7 <-6e84 ->6e84
Disconnections: 0
Servers:
  e9f7 (e9f7 at tcp:[172.18.0.5]:6644) next_index=130 match_index=129 last msg 1006 ms ago
  6e84 (6e84 at tcp:[172.18.0.4]:6644) next_index=130 match_index=129 last msg 1004 ms ago
  3722 (3722 at tcp:[172.18.0.3]:6644) (self) next_index=2 match_index=129
status: ok
```

kube-ovn-control-plane2      IP   172.18.0.5      ID   e9f7      ovn-sb

```
# kubectl ko sb kick e9f7
started removal
```

```
# kubectl ko sb status
3722
Name: OVN_Southbound
Cluster ID: d4bd (d4bd37a4-0400-499f-b4df-b4fd389780f0)
Server ID: 3722 (3722d5ae-2ced-4820-a6b2-8b744d11fb3e)
Address: tcp:[172.18.0.3]:6644
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2481636 ms ago, reason: timeout
Last Election won: 2481635 ms ago
Election timer: 5000
Log: [131, 131]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-6e84 ->6e84
Disconnections: 2
Servers:
  6e84 (6e84 at tcp:[172.18.0.4]:6644) next_index=131 match_index=130 last msg 642 ms ago
  3722 (3722 at tcp:[172.18.0.3]:6644) (self) next_index=2 match_index=130
status: ok
```

**ovn-central**

ovn-central

NODE\_IPS

```
kubectl label node kube-ovn-control-plane2 kube-ovn/role-
kubectl scale deployment -n kube-system ovn-central --replicas=2
kubectl set env deployment/ovn-central -n kube-system NODE_IPS="172.18.0.3,172.18.0.4"
kubectl rollout status deployment/ovn-central -n kube-system
```

**ovn-central**

OVS-OVN

```
# kubectl set env daemonset/ovs-ovn -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4"
daemonset.apps/ovs-ovn env updated
# kubectl delete pod -n kube-system -lapp=ovs
pod "ovs-ovn-4f6jc" deleted
```

```
pod "ovs-ovn-csn2w" deleted
pod "ovs-ovn-mpbmb" deleted
```

#### kube-ovn-controller

```
# kubectl set env deployment/kube-ovn-controller -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4"
deployment.apps/kube-ovn-controller env updated

# kubectl rollout status deployment/kube-ovn-controller -n kube-system
Waiting for deployment "kube-ovn-controller" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "kube-ovn-controller" rollout to finish: 2 of 3 updated replicas are available...
deployment "kube-ovn-controller" successfully rolled out
```

#### kube-ovn-control-plane2

```
rm -rf /etc/origin/ovn
```

### Kubernetes

## 6.3.2 ovn-central

#### Kubernetes

#### ovn-central

```
/etc/origin/ovn      ovnnb_db.db  ovnsb_db.db
```

```
rm -rf /etc/origin/ovn
```

### ovn-central

#### ovn-central

```
# kubectl ko nb status
1b9a
Name: OVN_Northbound
Cluster ID: 32ca (32ca07fb-739b-4257-b510-12fa18e7cce8)
Server ID: 1b9a (1b9a5d76-e69b-410c-8085-39943d0cd38c)
Address: tcp:[172.18.0.3]:6643
Status: cluster member
Role: leader
Term: 44
Leader: self
Vote: self

Last Election started 1855739 ms ago, reason: timeout
Last Election won: 1855729 ms ago
Election timer: 5000
Log: [147, 147]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: ->4984 <-4984
Disconnections: 0
Servers:
  4984 (4984 at tcp:[172.18.0.4]:6643) next_index=147 match_index=146 last msg 367 ms ago
  1b9a (1b9a at tcp:[172.18.0.3]:6643) (self) next_index=140 match_index=146
status: ok

# kubectl ko sb status
3722
Name: OVN_Southbound
Cluster ID: d4bd (d4bd37a4-0400-499f-b4df-b4fd389780f0)
Server ID: 3722 (3722d5ae-2ced-4820-a6b2-8b744d11fb3e)
Address: tcp:[172.18.0.3]:6644
Status: cluster member
Role: leader
Term: 33
Leader: self
Vote: self

Last Election started 1868589 ms ago, reason: timeout
Last Election won: 1868579 ms ago
Election timer: 5000
Log: [142, 142]
Entries not yet committed: 0
```

```
Entries not yet applied: 0
Connections: ->6e84 <-6e84
Disconnections: 0
Servers:
  6e84 (6e84 at tcp:[172.18.0.4]:6644) next_index=142 match_index=141 last msg 728 ms ago
  3722 (3722 at tcp:[172.18.0.3]:6644) (self) next_index=134 match_index=141
status: ok
```

ovn-central    NODE\_IPS

```
kubectl label node kube-ovn-control-plane2 kube-ovn/role=master
kubectl scale deployment -n kube-system ovn-central --replicas=3
kubectl set env deployment/ovn-central -n kube-system NODE_IPS="172.18.0.3,172.18.0.4,172.18.0.5"
kubectl rollout status deployment/ovn-central -n kube-system
```

ovn-central

ovs-ovn

```
# kubectl set env daemonset/ovs-ovn -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4,172.18.0.5"
daemonset.apps/ovs-ovn env updated
# kubectl delete pod -n kube-system -lapp=ovs
pod "ovs-ovn-4f6jc" deleted
pod "ovs-ovn-csn2w" deleted
pod "ovs-ovn-mpbmb" deleted
```

kube-ovn-controller

```
# kubectl set env deployment/kube-ovn-controller -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4,172.18.0.5"
deployment.apps/kube-ovn-controller env updated

# kubectl rollout status deployment/kube-ovn-controller -n kube-system
Waiting for deployment "kube-ovn-controller" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "kube-ovn-controller" rollout to finish: 2 of 3 updated replicas are available...
deployment "kube-ovn-controller" successfully rolled out
```



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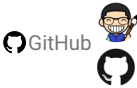
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6.3.3

## 6.4 OVN

### 6.4.1

kubectl      backup

```
# kubectl ko nb backup
tar: Removing leading '/' from member names
backup ovn-nb db to /root/ovnnb_db.060223191654183154.backup

# kubectl ko sb backup
tar: Removing leading '/' from member names
backup ovn-sb db to /root/ovnsb_db.060223191654183154.backup
```

### 6.4.2

/var/log/ovn/ovn-northd.log

```
* ovn-northd is not running
ovsdb-server: ovsdb error: error reading record 2739 from OVN_Northbound log: record 2739 advances commit index to 6308 but last log index is 6307
* Starting ovsdb-nb
```

OVN\_Northbound

OVN\_Southbound

OVN\_Northbound

ovn-nb

```
# kubectl ko nb status
9182
Name: OVN_Northbound
Cluster ID: e75f (e75fa340-49ed-45ab-990e-26cb865ebc85)
Server ID: 9182 (9182e8dd-b5b0-4dd8-8518-598cc1e374f3)
Address: tcp:[10.0.128.61]:6643
Status: cluster member
Role: leader
Term: 1454
Leader: self
Vote: self

Last Election started 1732603 ms ago, reason: timeout
Last Election won: 1732587 ms ago
Election timer: 1000
Log: [7332, 12512]
Entries not yet committed: 1
Entries not yet applied: 1
Connections: ->f080 <-f080 <-e631 ->e631
Disconnections: 1
Servers:
  f080 (f080 at tcp:[10.0.129.139]:6643) next_index=12512 match_index=12510 last msg 63 ms ago
  9182 (9182 at tcp:[10.0.128.61]:6643) (self) next_index=10394 match_index=12510
  e631 (e631 at tcp:[10.0.131.173]:6643) next_index=12512 match_index=0
```

```
kubectl ko nb kick e631
```

```
mv /etc/origin/ovn/ovnnb_db.db /tmp
```

ovn-central Pod

```
kubectl delete pod -n kube-system ovn-central-xxxx
```

## 6.4.3

leader

**ovn-central**

ovn-central      ovn-central

```
kubectl scale deployment -n kube-system ovn-central --replicas=0
```

ovsdb-tool cluster-to-standalone

ovn-central      NODE\_IPS      /etc/origin/ovn

```
docker run -it -v /etc/origin/ovn:/etc/ovn kubeovn/kube-ovn:v1.15.21 bash
cd /etc/ovn/
ovsdb-tool cluster-to-standalone ovnnb_db_standalone.db ovnnb_db.db
ovsdb-tool cluster-to-standalone ovnsb_db_standalone.db ovnsb_db.db
```

**ovn-central**

```
mv /etc/origin/ovn/ovnnb_db.db /tmp
mv /etc/origin/ovn/ovnsb_db.db /tmp
```

ovnnb\_db.db      ovnsb\_db.db      ovn-central      NODE\_IPS      /etc/origin/ovn/

```
mv /etc/origin/ovn/ovnnb_db_standalone.db /etc/origin/ovn/ovnnb_db.db
mv /etc/origin/ovn/ovnsb_db_standalone.db /etc/origin/ovn/ovnsb_db.db
```

ovn-central

```
kubectl scale deployment -n kube-system ovn-central --replicas=3
kubectl rollout status deployment/ovn-central -n kube-system
```



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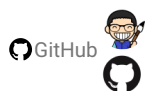
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## 6.4.4

## 6.5 CIDR

CIDR

CIDR

Pod

CIDR

Join

CIDR

Join

CIDR

### 6.5.1

kubectl edit      cidrBlock   gateway   excludeIps

```
kubectl edit subnet test-subnet
```

### 6.5.2 Namespace Pod

test Namespace

```
for pod in $(kubectl get pod --no-headers -n "$ns" --field-selector spec.restartPolicy=Always -o custom-columns=NAME:.metadata.name,HOST:spec.hostNetwork |
awk '{if ($2!="true") print $1}'); do
    kubectl delete pod "$pod" -n test --ignore-not-found
done
```

host

Pod

```
for ns in $(kubectl get ns --no-headers -o custom-columns=NAME:.metadata.name); do
  for pod in $(kubectl get pod --no-headers -n "$ns" --field-selector spec.restartPolicy=Always -o custom-columns=NAME:.metadata.name,HOST:spec.hostNetwork |
    awk '{if ($2!="true") print $1}'); do
    kubectl delete pod "$pod" -n "$ns" --ignore-not-found
  done
done
```

### 6.5.3

CIDR

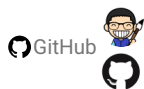
kube-ovn-controller Deployment

```
args:
- --default-cidr=10.17.0.0/16
- --default-gateway=10.17.0.1
- --default-exclude-ips=10.17.0.1
```



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### 6.5.4



## 6.6 Join CIDR

Join CIDR

Join CIDR Pod

### 6.6.1 Join

```
kubectl patch subnet join --type='json' -p '[{"op": "replace", "path": "/metadata/finalizers", "value": []}]'
kubectl delete subnet join
```

### 6.6.2

```
kubectl annotate node ovn.kubernetes.io/allocated=false --all --overwrite
```

### 6.6.3 Join

kube-ovn-controller Join

```
kubectl edit deployment -n kube-system kube-ovn-controller
```

```
args:
- --node-switch-cidr=100.51.0.0/16
```

kube-ovn-controller join

```
kubectl delete pod -n kube-system -lapp=kube-ovn-controller
```

Join

| # kubectl get subnet |          |             |          |               |         |       |         |             |        |             |        |             |
|----------------------|----------|-------------|----------|---------------|---------|-------|---------|-------------|--------|-------------|--------|-------------|
| NAME                 | PROVIDER | VPC         | PROTOCOL | CIDR          | PRIVATE | NAT   | DEFAULT | GATEWAYTYPE | V4USED | V4AVAILABLE | V6USED | V6AVAILABLE |
| EXCLUDEIPS           |          |             |          |               |         |       |         |             |        |             |        |             |
| join                 | ovn      | ovn-cluster | IPv4     | 100.51.0.0/16 | false   | false | false   | distributed | 2      | 65531       | 0      | 0           |
| ["100.51.0.1"]       |          |             |          |               |         |       |         |             |        |             |        |             |
| ovn-default          | ovn      | ovn-cluster | IPv4     | 10.17.0.0/16  | false   | true  | true    | distributed | 5      | 65528       | 0      | 0           |
| ["10.17.0.1"]        |          |             |          |               |         |       |         |             |        |             |        |             |

### 6.6.4 ovn0

ovn0 kube-ovn-cni

```
kubectl delete pod -n kube-system -l app=kube-ovn-cni
```

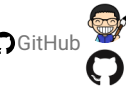
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### 6.6.5

# 6.7

---

kube-ovn.yaml1

```
vi kube-ovn.yaml1
# ...
- name: kube-ovn-controller
  image: "docker.io/kubeovn/kube-ovn:v1.15.21"
  imagePullPolicy: IfNotPresent
  args:
    - /kube-ovn/start-controller.sh
    - --v=3
# ...
#
```

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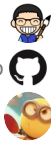
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## 6.7.1

---

6.8

Kube-OVN

6.8.1

1.
  - Pod Pod
  - Pod
  - Pod Service
  - Pod
  - Pod
  - kube-ovn-pinger
2.
  - Pod
  - kube-ovn-cni CNI
  - ovs-ovn OVS
  - kube-ovn-controller
  - ovn-central OVN
  - dmesg
  - netstat -s
3. CPU, IO
- kubectl ko logs

6.8.2 Pod IP

Pod Running kubectl describe Pod duplicate IPv4 address <ip> found on logical switch port <port>

1. Pod IP Pod IP IP IP Pod
2. kube-ovn-controller IP
3. kube-ovn-controller IP
4. kubectl ko nbctl show OVN IP
5. OVN Kubernetes IP kubectl ko nbctl del-port <port>

6.8.3 Pod ping gateway failed

Pod Running kubectl describe Pod network <ip> with gateway <gw ip> is not ready for interface eth0 after 30 checks

1. kubectl ko sbctl show Pod
2. ovn-central ovs-ovn ovn-central ovs-ovn

3. Pod
4. Underlay Underlay

6.8.4 Pod

VPC Pod

1. kubectl ko trace OVN ACL
2. ACL ACL
3. ACL Subnet stats
4. Subnet Spec
5. kube-ovn-controller

6.8.5 Pod IP CIDR

Pod IP CIDR

1. /etc/cni/net.d/ Kube-OVN CNI
2.
3. kubelet Pod

6.8.6 Debug Pod

kubectl debug Pod ContainerCreating Pod Event network not ready no address allocated

debug Pod yaml yaml Annotation

```
ovn.kubernetes.io/ip_address
ovn.kubernetes.io/mac_address
ovn.kubernetes.io/allocated
ovn.kubernetes.io/routed
```

debug Pod yaml debug Pod

6.8.7 ARM

ARM Offload

netstat

```
# netstat -us
IcmpMsg:
  InType0: 22
  InType3: 24
  InType8: 117852
  OutType0: 117852
```

```
OutType3: 29
OutType8: 22
Udp:
  3040636 packets received
  0 packets to unknown port received.
  4 packet receive errors
  602 packets sent
  0 receive buffer errors
  0 send buffer errors
  InCsumErrors: 4
UdpLite:
IpExt:
  InBcastPkts: 10244
  InOctets: 4446320361
  OutOctets: 1496815600
  InBcastOctets: 3095950
  InNoECTPkts: 7683903
```

InCsumErrors

tx offload TCP

```
ethtool -K eth0 tx off
```

CentOS 7 4.19.90-25.16.v2101

6.8.8 Pod Service

Pod Service dmesg

```
netlink: Unknown conntrack attr (type=6, max=5)
openvswitch: netlink: Flow actions may not be safe on all matching packets.
```

OVS NAT

- 1. OVS
- 2. Overlay kube-ovn-controller --enable-lb=false OVN LB kube-proxy Service

6.8.9 ovn-central

v1.11.x 1 Pod OVN NB SB Kube-OVN ovsdb-server/compact

ovn-central compact

```
- name: ENABLE_COMPACT
  value: "false"
```

 PDF

 Slack

 Support

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6.8.10

---

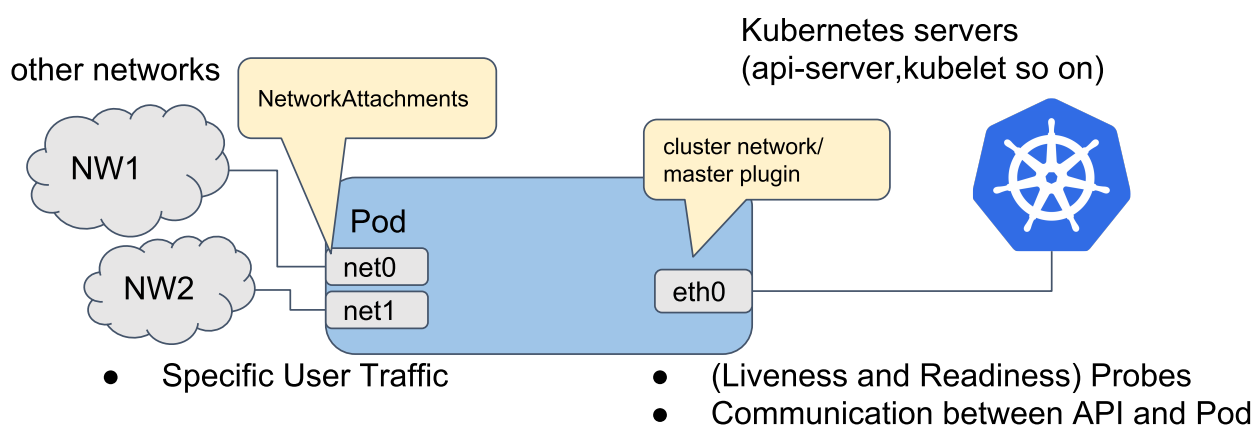
## 7.

## 7.1

|          |     |          |      |             |      |          |    |
|----------|-----|----------|------|-------------|------|----------|----|
| Kube-OVN | CNI | macvlan  | vlan | host-device | IPAM | Kube-OVN | IP |
| Kube-OVN |     | Kube-OVN |      |             |      |          |    |

## 7.1.1

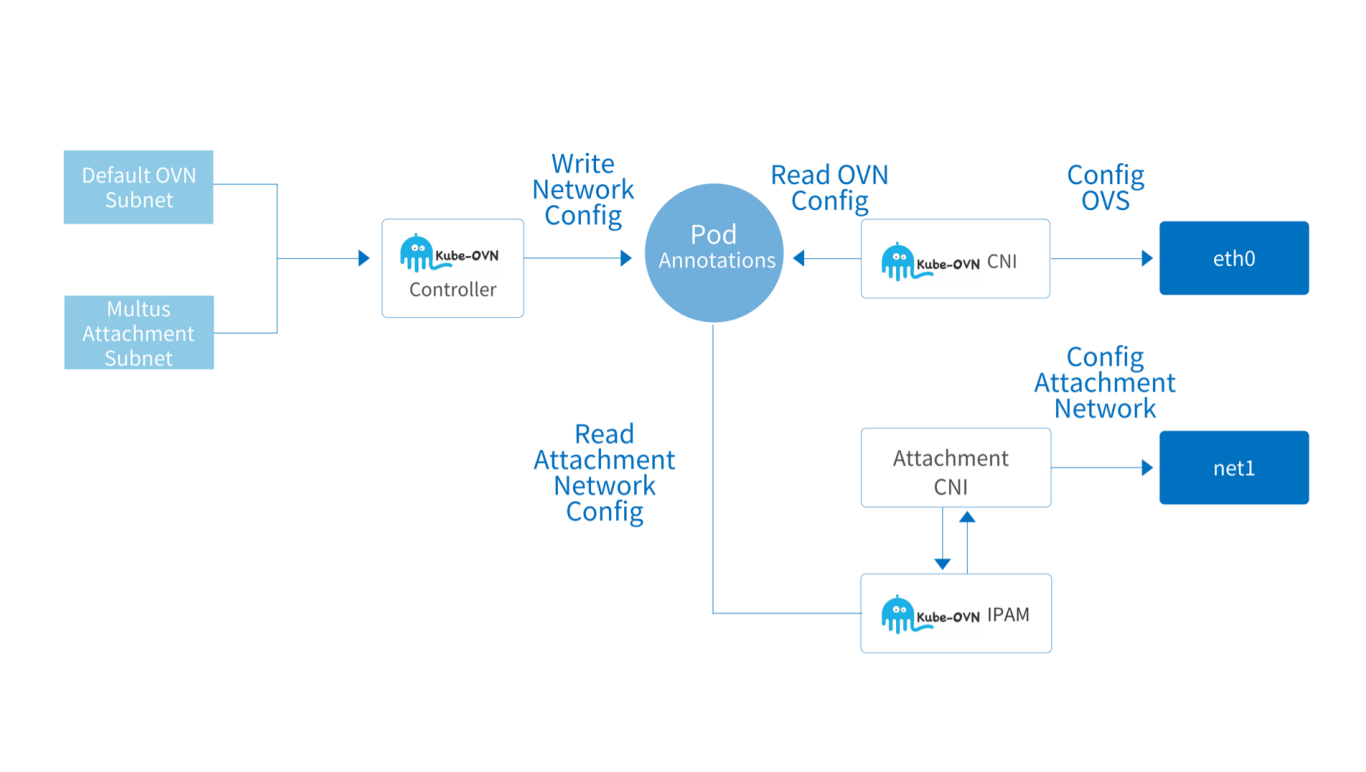
Multus CNI      pod      pod      eth0 net0 net1 eth0      kubernetes      kubernetes      kubernetes api-server kubelet  
net0 net1      CNI      vlan/vxlan/ptp



IPAM

Multus CNI      Pod      IP      Kube-OVN      Subnet IP CRD      IP      IP

Kube-OVN IPAM



|     | Kube-OVN            | IP           | eth0 | OVN          | net1           | CNI    | net1 | multus-cni | NetworkAttachmentDefinition |
|-----|---------------------|--------------|------|--------------|----------------|--------|------|------------|-----------------------------|
| Pod | kube-ovn-controller | Pod          | Pod  | Pod          | annotation     | Subnet | IP   | Pod        | Pod annotation              |
|     | CNI                 | kube-ovn-cni | ipam | kube-ovn-cni | Pod annotation |        | CNI  |            | CNI                         |

7.1.2

NetworkAttachmentDefinition spec multus defaultConfDir CNI

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: macvlan-conf-2
```

kube-ovn-controller NetworkAttachmentDefinition provider spec kube-ovn-controller provider spec Kube-OVN IPAM

7.1.3

Kube-OVN Multus

[Kube-OVN](#) [Multus how to use](#) [Kube-OVN](#) [Multus-CNI](#)

CNI IPAM

[Kube-OVN](#) [CNI](#)

NETWORKATTACHMENTDEFINITION

[macvlan](#) [ipam](#) [kube-ovn](#)

```
# macvlan
sudo modprobe macvlan
```



```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: macvlan
  namespace: default
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "macvlan",
    "master": "eth0",
    "mode": "bridge",
    "ipam": {
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "macvlan.default"
    }
  }'
```

- spec.config.ipam.type: kube-ovn kube-ovn
- server\_socket: Kube-OVN socket /run/openvswitch/kube-ovn-daemon.sock
- provider: NetworkAttachmentDefinition <name>.<namespace> Kube-OVN Subnet
- master:


**Info**

provider Underlay ProviderNetwork

#### KUBE-OVN SUBNET

Kube-OVN Subnet
 cidrBlock
 exclude\_ips
 provider
 NetworkAttachmentDefinition
 <name>.<namespace>
 macvlan
 Subnet

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: macvlan
spec:
  protocol: IPv4
  provider: macvlan.default
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
    - 172.17.0.0..172.17.0.10
```

gateway
 private
 nat
 provider
 ovn
 attachment network

#### Pod

Pod
 annotation k8s.v1.cni.cncf.io/networks
 NetworkAttachmentDefinition
 <namespace>/<name>

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/macvlan
spec:
  containers:
    - name: samplepod
      command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
      image: docker.io/library/alpine:edge
```

#### IP Pod

IP Pod
 <networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/ip\_address
 annotation

```
apiVersion: v1
kind: Pod
metadata:
  name: static-ip
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/macvlan
    ovn.kubernetes.io/ip_address: 10.16.0.15
    ovn.kubernetes.io/mac_address: 00:00:00:53:6B:B6
```

```
macvlan.default.kubernetes.io/ip_address: 172.17.0.100
macvlan.default.kubernetes.io/mac_address: 00:00:00:53:6B:BB
spec:
  containers:
  - name: static-ip
    image: docker.io/library/nginx:alpine
```

IP

ippool `<networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/ip_pool` annotation

```
apiVersion: apps/v1
kind: Deployment
metadata:
  namespace: default
  name: static-workload
  labels:
    app: static-workload
spec:
  replicas: 2
  selector:
    matchLabels:
      app: static-workload
  template:
    metadata:
      labels:
        app: static-workload
      annotations:
        k8s.v1.cni.cncf.io/networks: default/macvlan
        ovn.kubernetes.io/ip_pool: 10.16.0.15,10.16.0.16,10.16.0.17
        macvlan.default.kubernetes.io/ip_pool: 172.17.0.200,172.17.0.201,172.17.0.202
    spec:
      containers:
      - name: static-workload
        image: docker.io/library/nginx:alpine
```

macvlan Pod

macvlan Pod Pod annotation default-route

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod-route
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{"name": "macvlan", "namespace": "default", "default-route": ["172.17.0.1"]}]'
spec:
  containers:
  - name: samplepod-route
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

macvlan Pod

macvlan Pod annotation `v1.multus-cni.io/default-network` NetworkAttachmentDefinition `<namespace>/<name>`

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod-macvlan
  namespace: default
  annotations:
    v1.multus-cni.io/default-network: default/macvlan
spec:
  containers:
  - name: samplepod-macvlan
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

KUBE-OVN SUBNET PROVIDER OVN

provider ovn subnet IP provider ovn Kube-OVN Subnet cidrBlock exclude\_ips Subnet

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: macvlan
spec:
  protocol: IPv4
  provider: ovn
```

```
cidrBlock: 172.17.0.0/16
gateway: 172.17.0.1
excludeIps:
- 172.17.0.0..172.17.0.10
```

Pod

provider      ovn   subnet   IP   Pod   annotation k8s.v1.cni.cncf.io/networks

<networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/logical\_switch

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/macvlan
    macvlan.default.kubernetes.io/logical_switch: macvlan
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

- k8s.v1.cni.cncf.io/networks:      NetworkAttachmentDefinition   <namespace>/<name>
- macvlan.default.kubernetes.io/logical\_switch:

- <networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/logical\_switch      provider
- ovn      subnet   ipam      IP   Pod      IP      macvlan   Pod
- macvlan   Pod

## Kube-OVN

### Kube-OVN

#### NETWORKATTACHMENTDEFINITION

provider      ovn

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: attachnet
  namespace: default
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "kube-ovn",
    "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
    "provider": "attachnet.default.ovn"
  }'
```

- spec.config.type:      kube-ovn      CNI      Kube-OVN
- server\_socket: Kube-OVN      socket      /run/openvswitch/kube-ovn-daemon.sock
- provider:      NetworkAttachmentDefinition   <name>.<namespace>.ovn      Kube-OVN      Subnet      ovn

#### KUBE-OVN SUBNET

Kube-OVN      provider      NetworkAttachmentDefinition   <name>.<namespace>.ovn      ovn      Kube-OVN      Subnet

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attachnet
spec:
  protocol: IPv4
  provider: attachnet.default.ovn
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
  - 172.17.0.0..172.17.0.10
```

Pod

Pod      annotation k8s.v1.cni.cncf.io/networks      NetworkAttachmentDefinition      <namespace>/<name>

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

Kube-OVN      Pod      <networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/routes annotation

```
apiVersion: v1
kind: Pod
metadata:
  name: custom-routes-attach
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
    attachnet.default.ovn.kubernetes.io/routes: |
      [{
        "dst": "192.168.0.101/24",
        "gw": "172.17.0.254"
      }, {
        "gw": "172.17.0.254"
      }]
spec:
  containers:
  - name: custom-routes-attach
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

dst

Deployment DaemonSet StatefulSet      Annotation      .spec.template.metadata.annotations

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: custom-routes-attach
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
      annotations:
        k8s.v1.cni.cncf.io/networks: default/attachnet
        attachnet.default.ovn.kubernetes.io/routes: |
          [{
            "dst": "192.168.0.101/24",
            "gw": "172.17.0.254"
          }, {
            "gw": "172.17.0.254"
          }]
    spec:
      containers:
      - name: nginx
        image: docker.io/library/nginx:alpine
```

KUBE-OVN SUBNET PROVIDER OVN

provider      ovn      subnet      IP      provider      ovn      Kube-OVN Subnet      cidrBlock      exclude\_ips      Subnet

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attachnet
spec:
  protocol: IPv4
  provider: ovn
  cidrBlock: 172.17.0.0/16
```

```
gateway: 172.17.0.1
excludeIps:
- 172.17.0.0..172.17.0.10
```

Pod

```
provider    ovn  subnet    IP  Pod    annotation k8s.v1.cni.cncf.io/networks
```

```
<networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/logical_switch
```

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
    attachnet.default.ovn.kubernetes.io/logical_switch: attachnet
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

- `k8s.v1.cni.cncf.io/networks:` `NetworkAttachmentDefinition` `<namespace>/<name>`
- `attachnet.default.ovn.kubernetes.io/logical_switch:`

- |   |                                                                                       |     |                                                                               |     |    |         |     |            |     |
|---|---------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------|-----|----|---------|-----|------------|-----|
| • | <networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/logical_switch |     |                                                                               |     |    |         |     | provider   |     |
| • | Kube-OVN                                                                              | Pod | IP                                                                            | Pod | IP | macvlan | Pod | Kube-OVN   | Pod |
| • | Kube-OVN                                                                              | Pod | <networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/routes |     |    |         |     | annotation |     |



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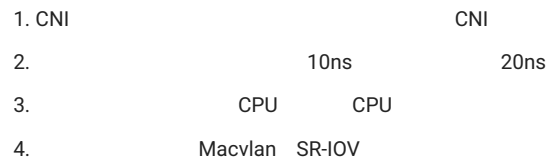
### 7.1.4

7.2

Kube-OVN

Kube-OVN

7.2.1



7.2.2

Overlay

- Kubernetes: 1.22.0
- OS: CentOS 7
- Kube-OVN: 1.8.0 *Overlay*
- CPU: Intel(R) Xeon(R) E-2278G
- Network: 2\*10Gbps, xmit\_hash\_policy=layer3+4

```
qperf -t 60 <server ip> -ub -oo msg_size:1 -vu tcp_lat tcp_bw udp_lat udp_bw 1 tcp/udp
```

| Type               | tcp_lat (us) | udp_lat (us) | tcp_bw (Mb/s) | udp_bw(Mb/s) |
|--------------------|--------------|--------------|---------------|--------------|
| Kube-OVN Default   | 25.7         | 22.9         | 27.1          | 1.59         |
| Kube-OVN Optimized | 13.9         | 12.9         | 27.6          | 5.57         |
| HOST Network       | 13.1         | 12.4         | 28.2          | 6.02         |

Overlay Underlay

Kube-OVN Overlay Underlay

Environment:

- Kubernetes: 1.22.0
- OS: CentOS 7
- Kube-OVN: 1.8.0
- CPU: AMD EPYC 7402P 24-Core Processor
- Network: Intel Corporation Ethernet Controller XXV710 for 25GbE SFP28

```
qperf -t 60 <server ip> -ub -oo msg_size:1 -vu tcp_lat tcp_bw udp_lat udp_bw
```

| Type              | tcp_lat (us) | udp_lat (us) | tcp_bw (Mb/s) | udp_bw(Mb/s) |
|-------------------|--------------|--------------|---------------|--------------|
| Kube-OVN Overlay  | 15.2         | 14.6         | 23.6          | 2.65         |
| Kube-OVN Underlay | 14.3         | 13.8         | 24.2          | 3.46         |
| HOST Network      | 16.6         | 15.4         | 24.8          | 2.64         |

```
qperf -t 60 <server ip> -ub -oo msg_size:1K -vu tcp_lat tcp_bw udp_lat udp_bw
```

| Type              | tcp_lat (us) | udp_lat (us) | tcp_bw (Gb/s) | udp_bw(Gb/s) |
|-------------------|--------------|--------------|---------------|--------------|
| Kube-OVN Overlay  | 16.5         | 15.8         | 10.2          | 2.77         |
| Kube-OVN Underlay | 15.9         | 14.5         | 9.6           | 3.22         |
| HOST Network      | 18.1         | 16.6         | 9.32          | 2.66         |

```
qperf -t 60 <server ip> -ub -oo msg_size:4K -vu tcp_lat tcp_bw udp_lat udp_bw
```

| Type              | tcp_lat (us) | udp_lat (us) | tcp_bw (Gb/s) | udp_bw(Gb/s) |
|-------------------|--------------|--------------|---------------|--------------|
| Kube-OVN Overlay  | 34.7         | 41.6         | 16.0          | 9.23         |
| Kube-OVN Underlay | 32.6         | 44           | 15.1          | 6.71         |
| HOST Network      | 35.9         | 45.9         | 14.6          | 5.59         |



7.2.3

CPU

CPU

CPU

```
cpupower frequency-set -g performance
```

```
# ethtool -g eno1
Ring parameters for eno1:
Pre-set maximums:
RX:                4096
RX Mini:           0
RX Jumbo:          0
TX:                4096
Current hardware settings:
RX:                255
RX Mini:           0
RX Jumbo:          0
TX:                255
```

```
ethtool -G eno1 rx 4096
ethtool -G eno1 tx 4096
```

tuned

tuned      profile

```
tuned-adm profile network-latency
```

```
tuned-adm profile network-throughput
```

irqbalance      CPU      CPU

OVN LB

OVN   L2 LB      conntrack      recirculate      CPU      20%      CPU      Overlay      kube-proxy      Service  
Pod-to-Pod      kube-ovn-controller

```
command:
- /kube-ovn/start-controller.sh
args:
...
- --enable-lb=false
...
```

Underlay      kube-proxy      iptables      ipvs      LB      Service

conntrack

OVN LB      Service      Service      NetworkPolicy      Subnet A      Pod      Subnet B      Service  
kube-ovn-controller      --skip-conntrack-dst-cidrs      conntrack

```
--skip-conntrack-dst-cidrs="10.17.0.0/16,169.254.169.245/32"
```

FastPath

network ns      netfilter      20%      CPU      netfilter      FastPath      netfilter      CPU

netfilter      iptables      ipvs      nftables

FastPath

insmod kube\_ovn\_fastpath.ko      FastPath      dmesg

```
# dmesg
...
[619631.323788] init_module,kube_ovn_fastpath_local_out
[619631.323798] init_module,kube_ovn_fastpath_post_routing
[619631.323800] init_module,kube_ovn_fastpath_pre_routing
[619631.323801] init_module,kube_ovn_fastpath_local_in
...
```

OVS

OVS   flow      10%      CPU      x86 CPU      popcnt      sse4.2      flow      CPU  
5%

FastPath



CPU

```
cat /proc/cpuinfo | grep popcnt
cat /proc/cpuinfo | grep sse4_2
```

CENTOS

```
yum install -y gcc kernel-devel-$(uname -r) python3 autoconf automake libtool rpm-build openssl-devel
```

OVS                  RPM    :

```
git clone -b branch-3.5 --depth=1 https://github.com/openvswitch/ovs.git
cd ovs
curl -s https://github.com/kubeovn/ovs/commit/2d2c83c26d4217446918f39d5cd5838e9ac27b32.patch | git apply
./boot.sh
./configure --with-linux=/lib/modules/$(uname -r)/build CFLAGS="-g -O2 -mpopcnt -msse4.2"
make rpm-fedora-kmod
cd rpm/rpmbuild/RPMS/x86_64/
```

RPM

```
rpm -i openvswitch-kmod-3.5.1-1.el7.x86_64.rpm
```

Kube-OVN          OVS

UBUNTU

```
apt install -y autoconf automake libtool gcc build-essential libssl-dev
```

OVS

```
apt install -y autoconf automake libtool gcc build-essential libssl-dev

git clone -b branch-3.5 --depth=1 https://github.com/openvswitch/ovs.git
cd ovs
curl -s https://github.com/kubeovn/ovs/commit/2d2c83c26d4217446918f39d5cd5838e9ac27b32.patch | git apply
./boot.sh
./configure --prefix=/usr/ --localstatedir=/var --enable-ssl --with-linux=/lib/modules/$(uname -r)/build
make -j `nproc`
make install
make modules_install

cat > /etc/depmod.d/openvswitch.conf << EOF
override openvswitch * extra
override vport-* * extra
EOF

depmod -a
cp debian/openvswitch-switch.init /etc/init.d/openvswitch-switch
/etc/init.d/openvswitch-switch force-reload-kmod
```

Kube-OVN          OVS

STT

 **Warning**

OpenVswitch    3.6    [STT](#)   [Tunnel](#)

|     |     |        |       |     |     |     |     |  |     |         |     |     |
|-----|-----|--------|-------|-----|-----|-----|-----|--|-----|---------|-----|-----|
|     |     | Geneve | Vxlan | UDP |     | UDP | TCP |  | TCP | offload |     | TCP |
|     | CPU | TCP    |       |     |     |     |     |  |     |         |     |     |
| STT |     | TCP    |       |     | TCP |     | TCP |  | TCP |         | TCP |     |
| STT |     |        | OVS   | OVS |     |     |     |  |     |         |     |     |

STT

```
kubect1 set env daemonset/ovs-ovn -n kube-system TUNNEL_TYPE=stt
kubect1 delete pod -n kube-system -lapp=ovs
```

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7.2.4

---

## 7.3 FastPath

Profile **Netfilter**

20%

CPU

FastPath

**Netfilter**

CPU

FastPath

### 7.3.1

```
git clone --depth=1 https://github.com/kubeovn/kube-ovn.git
```

### 7.3.2

RPM CentOS RHEL

```
yum install -y kernel-devel-$(uname -r) gcc elfutils-libelf-devel
```

DEB Ubuntu Debian

```
apt install -y linux-headers-$(uname -r) build-essential
```

### 7.3.3

3.x

```
cd kube-ovn/fastpath/3.x
make all
```

4.x ~ 6.x

```
cd kube-ovn/fastpath/4.x-6.x
make all
```

### 7.3.4

```
kube_ovn_fastpath.ko
```

```
insmod kube_ovn_fastpath.ko
```

```
dmesg
```

```
# dmesg
[619631.323788] init_module,kube_ovn_fastpath_local_out
[619631.323798] init_module,kube_ovn_fastpath_post_routing
[619631.323800] init_module,kube_ovn_fastpath_pre_routing
[619631.323801] init_module,kube_ovn_fastpath_local_in
```

```
rmmod kube_ovn_fastpath.ko
```



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7.3.5

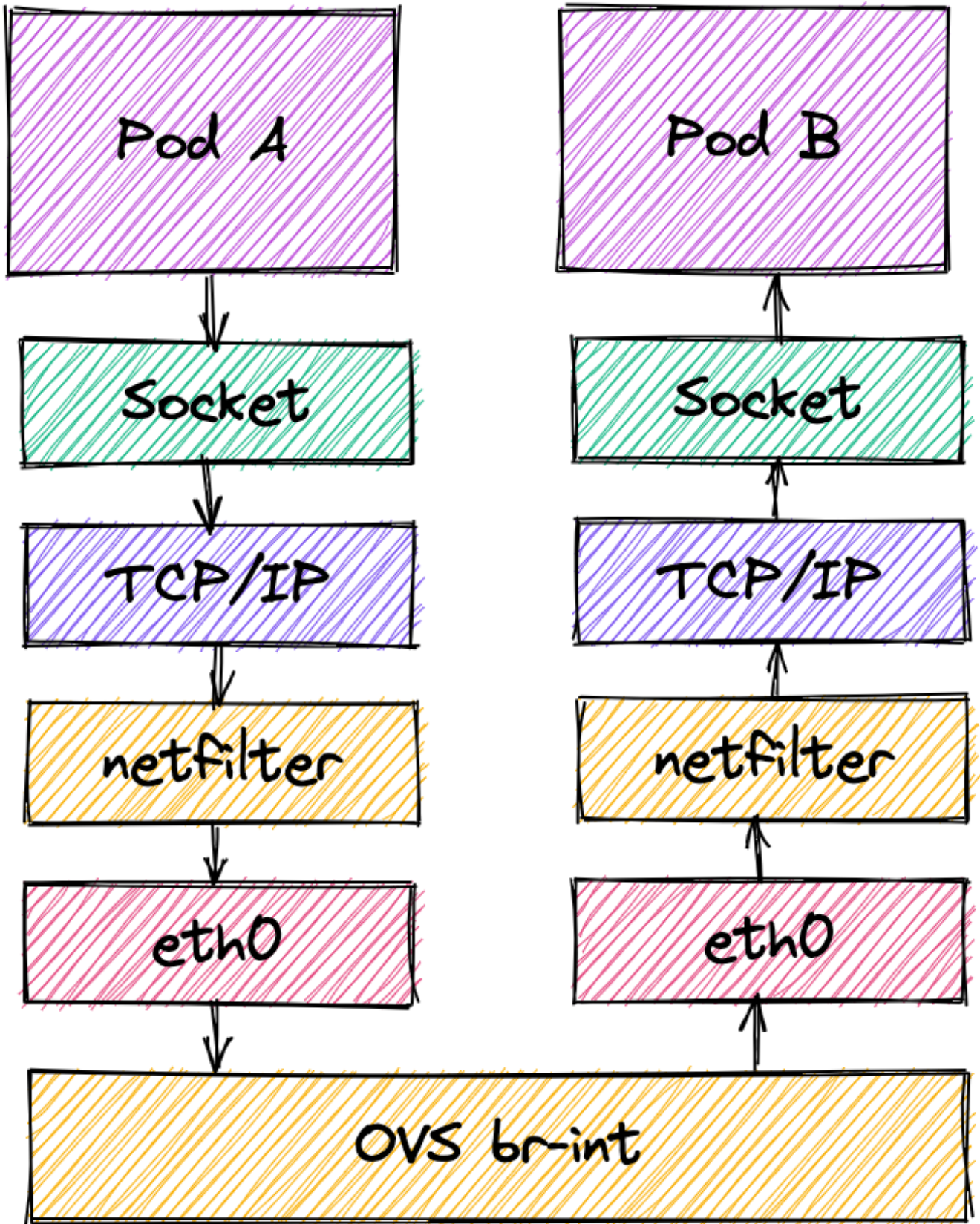
---

7.4 eBPF TCP

|    |     |     |       |                                    |     |      |        |        |
|----|-----|-----|-------|------------------------------------|-----|------|--------|--------|
| 5G | Pod | TCP | Intel | <a href="#">istio-tcpip-bypass</a> | Pod | eBPF | TCP/IP | socket |
|----|-----|-----|-------|------------------------------------|-----|------|--------|--------|

7.4.1

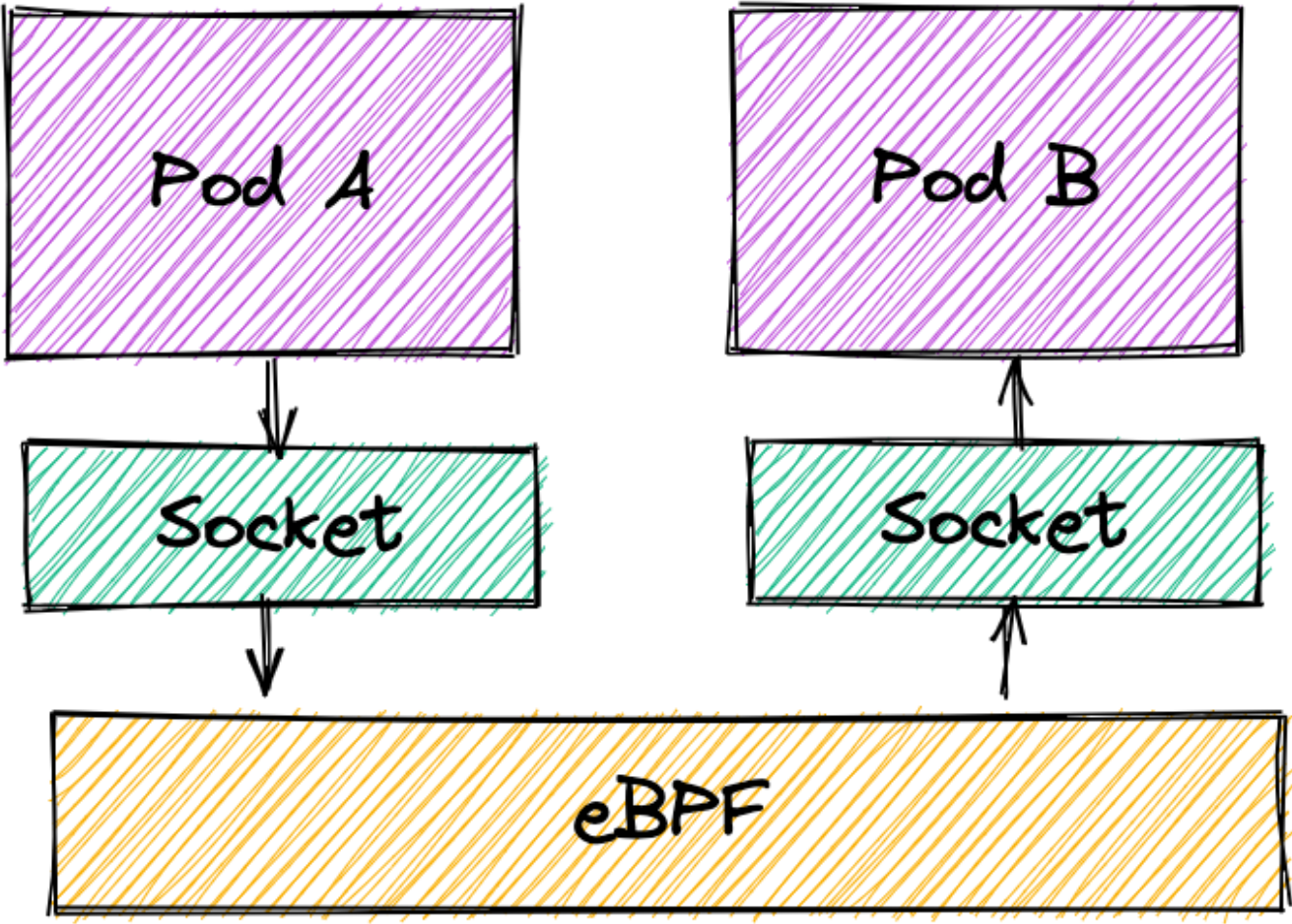
|     |     |                   |     |
|-----|-----|-------------------|-----|
| Pod | TCP | TCP/IP, netfilter | OVS |
|-----|-----|-------------------|-----|



istio-tcpip-bypass

TCP

socket



TCP                      Service Mesh                      Service Mesh

Tanzu Service Mesh Acceleration using eBPF

7.4.2

eBPF                      Ubuntu 20.04    Linux 5.4.0-74-generic

7.4.3

Pod

nodeSelector

```
# kubectl create deployment perf --image=kubeovn/perf:dev --replicas=2
deployment.apps/perf created
# kubectl get pod -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE   NOMINATED NODE   READINESS GATES
perf-7697bc6ddf-b2cpv               1/1     Running   0           28s   100.64.0.3    sealos <none>         <none>
perf-7697bc6ddf-p2xpt               1/1     Running   0           28s   100.64.0.2    sealos <none>         <none>
```

Pod    qperf server

Pod    qperf client

```
# kubectl exec -it perf-7697bc6ddf-b2cpv sh
/ # qperf

# kubectl exec -it perf-7697bc6ddf-p2xpt sh
/ # qperf -t 60 100.64.0.3 -ub -oo msg_size:1:16K:*4 -vu tcp_lat tcp_bw
```

istio-tcpip-bypass

```
kubectl apply -f https://raw.githubusercontent.com/intel/istio-tcpip-bypass/main/bypass-tcpip-daemonset.yaml
```

perf client

```
# kubectl exec -it perf-7697bc6ddf-p2xpt sh
/ # qperf -t 60 100.64.0.3 -ub -oo msg_size:1:16K:*4 -vu tcp_lat tcp_bw
```

7.4.4

| TCP                | 40% ~ 60%         | 1024                 | 40% ~ 80%          |                      |
|--------------------|-------------------|----------------------|--------------------|----------------------|
| Packet Size (byte) | eBPF tcp_lat (us) | Default tcp_lat (us) | eBPF tcp_bw (Mb/s) | Default tcp_bw(Mb/s) |
| 1                  | 20.2              | 44.5                 | 1.36               | 4.27                 |
| 4                  | 20.2              | 48.7                 | 5.48               | 16.7                 |
| 16                 | 19.6              | 41.6                 | 21.7               | 63.5                 |
| 64                 | 18.8              | 41.3                 | 96.8               | 201                  |
| 256                | 19.2              | 36                   | 395                | 539                  |
| 1024               | 18.3              | 42.4                 | 1360               | 846                  |
| 4096               | 16.5              | 62.6                 | 4460               | 2430                 |
| 16384              | 20.2              | 58.8                 | 9600               | 6900                 |

512

eBPF

TCP

eBPF

eBPF TCP

7.4.5

1. [istio-tcpip-bypass](#)

2. [Deep Dive TCP/IP Bypass with eBPF in Service Mesh](#)

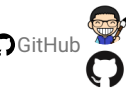
3. [Tanzu Service Mesh Acceleration using eBPF](#)
-  PDF

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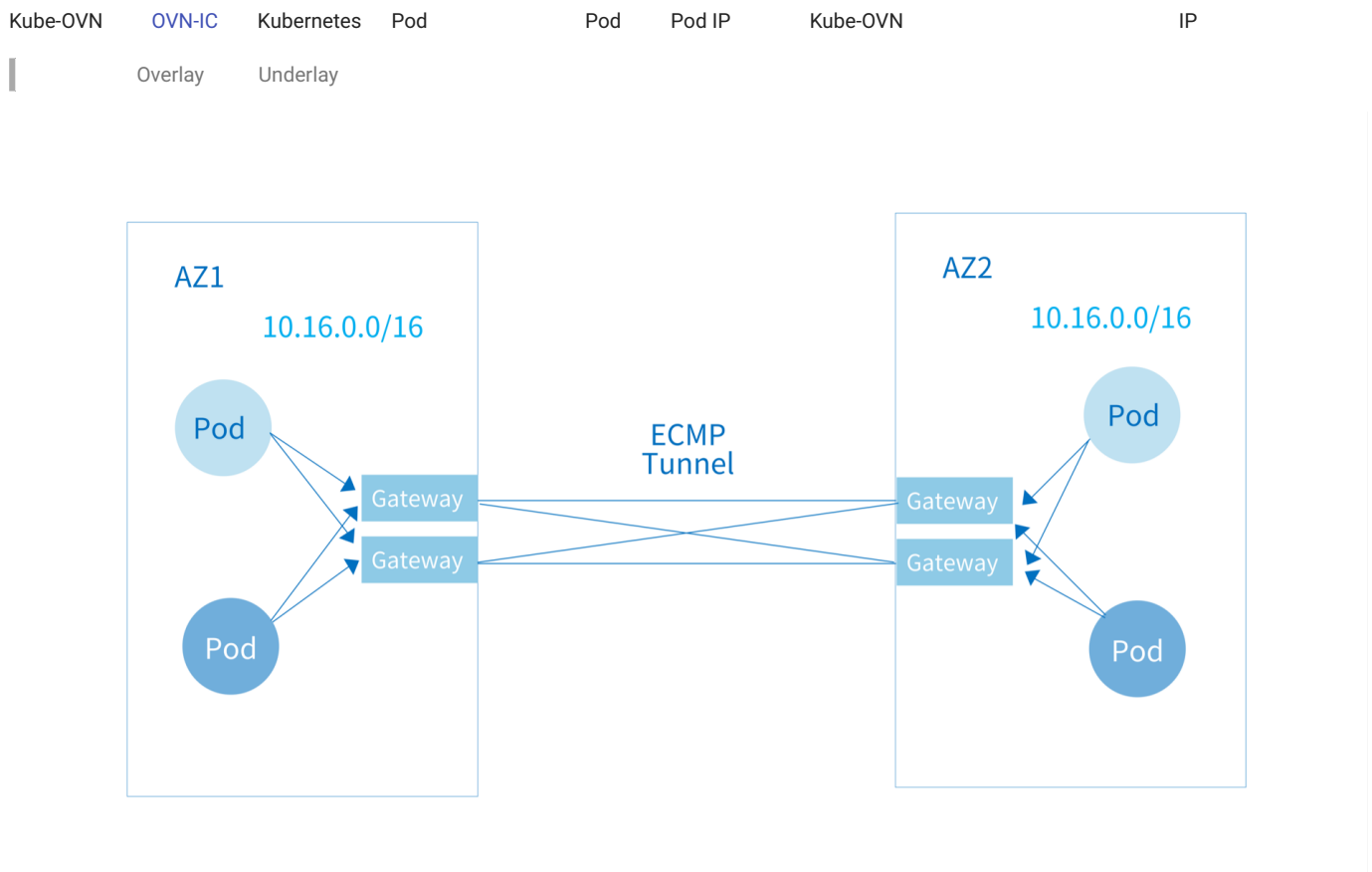
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7.4.6



## 7.5 OVN-IC



 **Limitation**

| OVN-IC | Pod IP | Service DNS | NetworkPolicy | Istio |
|--------|--------|-------------|---------------|-------|
|--------|--------|-------------|---------------|-------|

### 7.5.1

```
1.1.11.16 install.sh
```

ENABLE\_IC=true

deployment ovn-ic-controller

2. CIDR

3. kube-ovn-controller IP

4. IP

5. VPC VPC

### 7.5.2 OVN-IC

1

1 Kube-OVN v1.11.16

|   |   |   |   |            |        |        |   |        |
|---|---|---|---|------------|--------|--------|---|--------|
| " | " | " | " | Deployment | master | master | 1 | master |
|---|---|---|---|------------|--------|--------|---|--------|

```
install-ovn-ic.sh
```

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.15/dist/images/install-ic-server.sh
```

TS\_NUM ECMP Path

```
sed 's/VERSION=.*//VERSION=v1.15.21/' dist/images/install-ic-server.sh | TS_NUM=3 bash
```

```
deployment.apps/ovn-ic-server created
Waiting for deployment spec update to be observed...
Waiting for deployment "ovn-ic-server" rollout to finish: 0 out of 3 new replicas have been updated...
Waiting for deployment "ovn-ic-server" rollout to finish: 0 of 3 updated replicas are available...
Waiting for deployment "ovn-ic-server" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "ovn-ic-server" rollout to finish: 2 of 3 updated replicas are available...
deployment "ovn-ic-server" successfully rolled out
OVN IC Server installed Successfully
```

```
kubect1 ko icsbctl show
```

```
kubect1 ko icsbctl show
availability-zone az0
gateway 059b5c54-c540-4d77-b009-02d65f181a02
  hostname: kube-ovn-worker
  type: geneve
  ip: 172.18.0.3
  port ts-az0
    transit switch: ts
    address: ["00:00:00:B4:8E:BE 169.254.100.97/24"]
gateway 74ee4b9a-ba48-4a07-861e-1a8e4b9f905f
  hostname: kube-ovn-worker2
  type: geneve
  ip: 172.18.0.2
  port ts1-az0
    transit switch: ts1
    address: ["00:00:00:19:2E:F7 169.254.101.90/24"]
gateway 7e2428b6-344c-4dd5-a0d5-972c1ccec581
  hostname: kube-ovn-control-plane
  type: geneve
  ip: 172.18.0.4
  port ts2-az0
    transit switch: ts2
    address: ["00:00:00:EA:32:BA 169.254.102.103/24"]
availability-zone az1
gateway 034da7cb-3826-4318-81ce-6a877a9bf285
  hostname: kube-ovn1-worker
  type: geneve
  ip: 172.18.0.6
  port ts-az1
    transit switch: ts
    address: ["00:00:00:25:3A:B9 169.254.100.51/24"]
gateway 2531a683-283e-4fb8-a619-bdbcb33539b8
  hostname: kube-ovn1-worker2
  type: geneve
  ip: 172.18.0.5
  port ts1-az1
    transit switch: ts1
    address: ["00:00:00:52:87:F4 169.254.101.118/24"]
gateway b0efb0be-e5a7-4323-ad4b-317637a757c4
  hostname: kube-ovn1-control-plane
  type: geneve
  ip: 172.18.0.8
  port ts2-az1
    transit switch: ts2
    address: ["00:00:00:F6:93:1A 169.254.102.17/24"]
```

## 2

kube-ovn-controller IP OVN-IC

docker OVN-IC

```
docker run --name=ovn-ic-db -d --env "ENABLE_OVN_LEADER_CHECK=false" --network=host --privileged -v /etc/ovn:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/log/ovn:/var/log/ovn kubeovn/kube-ovn:v1.15.21 bash start-ic-db.sh
```

containerd docker

```
ctr -n k8s.io run -d --env "ENABLE_OVN_LEADER_CHECK=false" --net-host --privileged --mount="type=bind,src=/etc/ovn/,dst=/etc/ovn,options=rbind:rw" --mount="type=bind,src=/var/run/ovn,dst=/var/run/ovn,options=rbind:rw" --mount="type=bind,src=/var/log/ovn,dst=/var/log/ovn,options=rbind:rw" docker.io/kubeovn/kube-ovn:v1.15.21 ovn-ic-db bash start-ic-db.sh
```

## 7.5.3

VPC Subnet CIDR OVN-IC Subnet CIDR

kube-system Namespace ovn-ic-config ConfigMap

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "true"
```

- enable-ic:
- az-name:
- ic-db-host: OVN-IC
- ic-nb-port: OVN-IC 6645
- ic-sb-port: OVN-IC 6646
- gw-nodes:
- auto-route:

ovn-ic-config ConfigMap ConfigMap ConfigMap

ovn-ic ts

```
# ovn-ic-sbctl show
availability-zone az1
gateway deee03e0-af16-4f45-91e9-b50c3960f809
  hostname: az1-gw
  type: geneve
  ip: 192.168.42.145
  port ts-az1
  transit switch: ts
  address: ["00:00:00:50:AC:8C 169.254.100.45/24"]
availability-zone az2
gateway e94cc831-8143-40e3-a478-90352773327b
  hostname: az2-gw
  type: geneve
  ip: 192.168.42.149
  port ts-az2
  transit switch: ts
  address: ["00:00:00:07:4A:59 169.254.100.63/24"]
```

```
# kubectl ko nbctl lr-route-list ovn-cluster
IPv4 Routes
10.42.1.1 169.254.100.45 dst-ip (learned)
10.42.1.3 100.64.0.2 dst-ip
10.16.0.2 100.64.0.2 src-ip
10.16.0.3 100.64.0.2 src-ip
10.16.0.4 100.64.0.2 src-ip
10.16.0.6 100.64.0.2 src-ip
10.17.0.0/16 169.254.100.45 dst-ip (learned)
100.65.0.0/16 169.254.100.45 dst-ip (learned)
```

1 Pod ping 2 Pod IP

Subnet disableInterConnection

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
```

```
name: no-advertise
spec:
  cidrBlock: 10.199.0.0/16
  disableInterConnection: true
```

7.5.4

CIDR

```
kube-system Namespace    ovn-ic-config ConfigMap    auto-route    false
```

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "false"
```

```
[root@az1 ~]# kubectl ko nbctl show
switch a391d3a1-14a0-4841-9836-4bd930c447fb (ts)
  port ts-az1
    type: router
    router-port: az1-ts
  port ts-az2
    type: remote
    addresses: ["00:00:00:4B:E2:9F 169.254.100.31/24"]

[root@az2 ~]# kubectl ko nbctl show
switch da6138b8-de81-4908-abf9-b2224ec4edf3 (ts)
  port ts-az2
    type: router
    router-port: az2-ts
  port ts-az1
    type: remote
    addresses: ["00:00:00:FB:2A:F7 169.254.100.79/24"]
```

```
az1      az2      169.254.100.31 az2      az1      169.254.100.79
```

```
az1      CIDR  10.16.0.0/24  az2      CIDR  10.17.0.0/24
```

```
az1      az2
```

```
kubectl ko nbctl lr-route-add ovn-cluster 10.17.0.0/24 169.254.100.31
```

```
az2      az1
```

```
kubectl ko nbctl lr-route-add ovn-cluster 10.16.0.0/24 169.254.100.79
```

7.5.5 OVN-IC

1

1 Kube-OVN v1.11.16

1

2

```
OVN-IC      Raft      3
```

```
OVN-IC      leader
```

```
docker
```

```
docker run --name=ovn-ic-db -d --env "ENABLE_OVN_LEADER_CHECK=false" --network=host --privileged -v /etc/ovn:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/log/ovn:/var/log/ovn -e LOCAL_IP="192.168.65.3" -e NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1" kubeovn/kube-ovn:v1.15.21 bash start-ic-db.sh
```

containerd

```
ctr -n k8s.io run -d --env "ENABLE_OVN_LEADER_CHECK=false" --net-host --privileged --mount="type=bind,src=/etc/ovn,dst=/etc/ovn,options=rbind:rw" --mount="type=bind,src=/var/run/ovn,dst=/var/run/ovn,options=rbind:rw" --mount="type=bind,src=/var/log/ovn,dst=/var/log/ovn,options=rbind:rw" --env="NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1" --env="LOCAL_IP="192.168.65.3" docker.io/kubeovn/kube-ovn:v1.15.21 ovn-ic-db bash start-ic-db.sh
```

- LOCAL\_IP IP
- NODE\_IPS OVN-IC IP
- OVN-IC follower

docker

```
docker run --name=ovn-ic-db -d --network=host --privileged -v /etc/ovn:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/log/ovn:/var/log/ovn -e LOCAL_IP="192.168.65.2" -e NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1" -e LEADER_IP="192.168.65.3" kubeovn/kube-ovn:v1.15.21 bash start-ic-db.sh
```

containerd

```
ctr -n k8s.io run -d --net-host --privileged --mount="type=bind,src=/etc/ovn,dst=/etc/ovn,options=rbind:rw" --mount="type=bind,src=/var/run/ovn,dst=/var/run/ovn,options=rbind:rw" --mount="type=bind,src=/var/log/ovn,dst=/var/log/ovn,options=rbind:rw" --env="NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1" --env="LOCAL_IP="192.168.65.2" --env="LEADER_IP="192.168.65.3" docker.io/kubeovn/kube-ovn:v1.15.21 ovn-ic-db bash start-ic-db.sh
```

- LOCAL\_IP IP
- NODE\_IPS OVN-IC IP
- LEADER\_IP: OVN-IC leader IP
- ovn-ic-config OVN-IC

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3,192.168.65.2,192.168.65.1"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "true"
```

## 7.5.6 ECMP

1

ECMP ECMP path 3 ECMP path

```
kubectrl edit deployment ovn-ic-server -n kube-system
```

'TS\_NUM' TS\_NUM ECMP Path

## 7.5.7

ovn-ic-config Configmap

```
kubectrl -n kube-system delete cm ovn-ic-config
```

ts

```
kubectrl ko nbctl ls-del ts
```

7.5.8 az-name

kubectl edit ovn-ic-config configmap az-name ovn-cni pod 10

```
ovn-appctl -t ovn-controller inc-engine/recompute
```

7.5.9

ovn-ic-config Configmap

```
kubectl -n kube-system delete cm ovn-ic-config
```

ts

```
kubectl ko nbctl ls-del ts
```

OVN-IC

docker

```
docker stop ovn-ic-db
docker rm ovn-ic-db
```

containerd

```
ctr -n k8s.io task kill ovn-ic-db
ctr -n k8s.io containers rm ovn-ic-db
```

deployment ovn-ic-server

```
kubectl delete deployment ovn-ic-server -n kube-system
```

master DB

```
rm -f /etc/origin/ovn/ovn_ic_nb_db.db
rm -f /etc/origin/ovn/ovn_ic_sb_db.db
```

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7.5.10

## 7.6 Submariner

|            |            |          |          |          |            |
|------------|------------|----------|----------|----------|------------|
| Submariner | Kubernetes | Pod      | Service  | Kube-OVN |            |
| OVN-IC     | Submariner | Kube-OVN | Kube-OVN | Service  | Submariner |

### 7.6.1

- Service CIDR CIDR

### 7.6.2 Submariner

#### Note

Submariner v0.22.0 nftables Kube-OVN Submariner

```
kubectl create namespace submariner-operator
kubectl create configmap submariner-global --namespace=submariner-operator --from-literal=use-nftables=false
```

```
subctl
```

```
curl -Ls https://get.submariner.io | bash
export PATH=$PATH:~/local/bin
echo export PATH=$PATH:~/local/bin >> ~/.profile
```

```
kubeconfig submariner-broker
```

```
subctl deploy-broker
```

```
cluster0 CIDR 10.16.0.0/16 join CIDR 100.64.0.0/16 cluster1 CIDR 11.16.0.0/16 join CIDR 100.68.0.0/16
```

```
kubeconfig cluster0 broker :
```

```
subctl join broker-info.subm --clusterid cluster0 --clustercidr 100.64.0.0/16,10.16.0.0/16 --natt=false --cable-driver vxlan --health-check=false
kubectl label nodes cluster0 submariner.io/gateway=true
```

```
kubeconfig cluster1 broker :
```

```
subctl join broker-info.subm --clusterid cluster1 --clustercidr 100.68.0.0/16,11.16.0.0/16 --natt=false --cable-driver vxlan --health-check=false
kubectl label nodes cluster1 submariner.io/gateway=true
```

```
join gateway, routeagent pod , submariner-operator clusterrole :
```

```
- apiGroups:
- "apps"
resources:
- daemonsets
verbs:
- create
- get
- list
- watch
- update
```

```
subnet ovn-default centralized submariner gateway subnet
```

```
Pod IP
```

```
subctl
```

```
subctl show all
subctl diagnose all
```

Submariner Submariner

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7.6.3

---



## 7.7 Overlay

| Underlay |         | Pod    | IP |
|----------|---------|--------|----|
| VPC      | Overlay | Pod IP | IP |

### 7.7.1

- `ip_forward`
- `iptables` `forward` `Drop`
- `ct` `INVALID`

### 7.7.2

|                          |                    |                  |        |
|--------------------------|--------------------|------------------|--------|
| <code>natOutgoing</code> | <code>false</code> | <code>nat</code> | Pod IP |
|--------------------------|--------------------|------------------|--------|

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: routed
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
    - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: distributed
  natOutgoing: false
```

|     |            |
|-----|------------|
| Pod | Kubernetes |
|-----|------------|

```
ip route add 10.166.0.0/16 via 192.168.2.10 dev eth0
```

|               |              |            |
|---------------|--------------|------------|
| 10.166.0.0/16 | 192.168.2.10 | Kubernetes |
|---------------|--------------|------------|

|        |                          |                          |                             |
|--------|--------------------------|--------------------------|-----------------------------|
| IP     | Keepalived               | VIP                      | VIP                         |
| Subnet | <code>gatewayType</code> | <code>centralized</code> | <code>gatewayNode</code> IP |

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: routed
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
    - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: centralized
  gatewayNode: "node1"
  natOutgoing: false
```

|                  |         |
|------------------|---------|
| <code>nat</code> | VPC NAT |
|------------------|---------|

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7.7.3

---

## 7.8 Overlay

| Overlay                                                            | tunnel |
|--------------------------------------------------------------------|--------|
| <ul style="list-style-type: none"><li></li><li></li><li></li></ul> |        |

### 7.8.1

| Kube-OVN                                                                     | Kubernetes Node IP                     |
|------------------------------------------------------------------------------|----------------------------------------|
| <pre>IFACE=eth1</pre>                                                        |                                        |
| <pre>ens[a-z0-9]*,eth[a-z0-9]*</pre>                                         |                                        |
| <pre>kube-ovn-cni DaemonSet</pre>                                            |                                        |
| <pre>args: - --iface=eth1</pre>                                              |                                        |
| <pre>annotation ovn.kubernetes.io/tunnel_interface</pre>                     | <pre>annotation iface annotation</pre> |
| <pre>kubect1 annotate node no1 ovn.kubernetes.io/tunnel_interface=ethx</pre> |                                        |

### 7.8.2

| Kube-OVN                                                                                                                                   | IP          | tunnel                       |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------|
| <ul style="list-style-type: none"><li>IP</li><li></li><li>Overlay</li></ul>                                                                |             |                              |
| 1. Node                                                                                                                                    | IP          |                              |
| 2. Subnet                                                                                                                                  | nodeNetwork |                              |
| 3. kube-ovn-daemon                                                                                                                         | IP          | OVS                          |
| 4. Pod                                                                                                                                     | nodeNetwork | Pod OVS IP                   |
| <pre>ovn.kubernetes.io/node_networks</pre>                                                                                                 |             | <pre>JSON key value IP</pre> |
| <pre>kubect1 annotate node &lt;node-name&gt; ovn.kubernetes.io/node_networks='{ "storage": "192.168.100.10", "app": "172.16.0.10" }'</pre> |             |                              |
| <ul style="list-style-type: none"><li>storage IP 192.168.100.10 IP</li><li>app IP 172.16.0.10 IP</li></ul>                                 |             | IP                           |

spec.nodeNetwork

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: storage-subnet
spec:
  protocol: IPv4
  cidrBlock: 10.100.0.0/16
  gateway: 10.100.0.1
  nodeNetwork: storage
```

|                |     |         |        |
|----------------|-----|---------|--------|
| storage-subnet | Pod | storage | tunnel |
| nodeNetwork    | IP  | IFACE   | IP     |

OVS IP

OVS IP

```
ovs-vsctl get open . external-ids:ovn-encap-ip
```

IP

```
"192.168.1.10,192.168.100.10,172.16.0.10"
```

IP

```
ovs-vsctl get open . external-ids:ovn-encap-ip-default
```

POD IP

Pod Pod OVS IP

```
ovs-vsctl --columns=external_ids find interface external-ids:iface-id="<pod-name>.<namespace>"
```

nodeNetwork encap-ip

```
external_ids : {encap-ip="192.168.100.10", iface-id="test-pod.default", ...}
```

1.

- eth0 IP 192.168.1.10 192.168.1.11
- eth1 IP 10.10.10.10 10.10.10.11

```
kubect1 annotate node node1 ovn.kubernetes.io/node_networks='{"storage": "10.10.10.10"}'
kubect1 annotate node node2 ovn.kubernetes.io/node_networks='{"storage": "10.10.10.11"}'
```

2.

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: storage-net
spec:
  protocol: IPv4
  cidrBlock: 10.200.0.0/16
  gateway: 10.200.0.1
  nodeNetwork: storage
```

```
namespaces:
- storage-namespace
```

3. POD

```
apiVersion: v1
kind: Pod
metadata:
  name: storage-pod
  namespace: storage-namespace
spec:
  containers:
  - name: app
    image: docker.io/library/nginx:alpine
```

Pod eth1

- 1.
- 2. IP IP
- 3. Overlay Underlay Underlay Underlay
- 4. nodeNetwork IP
- 5. Pod nodeNetwork Pod
- 6. ovn.kubernetes.io/node\_networks Pod

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## 7.9 BGP

| Kube-OVN | Pods   | Subnets    | Services        | EIPs | IP | BGP  | IP | kube-ovn-speaker |
|----------|--------|------------|-----------------|------|----|------|----|------------------|
| Pod      | Subnet | annotation |                 |      |    |      |    |                  |
| EIP      | BGP    |            | VPC NAT Gateway |      |    | EIPs |    |                  |

### 7.9.1 kube-ovn-speaker

kube-ovn-speaker [GoBGP](#)

kube-ovn-speaker

```
kubect1 label nodes speaker-node-1 ovn.kubernetes.io/bgp=true
kubect1 label nodes speaker-node-2 ovn.kubernetes.io/bgp=true
```

kube-ovn-speaker ECMP

yaml:

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.15/yamls/speaker.yaml
```

yaml

```
- --neighbor-address=10.32.32.254
- --neighbor-ipv6-address=2409:AB00:AB00:2000::AFB:8AFE
- --neighbor-as=65030
- --cluster-as=65000
```

```
- --neighbor-address=10.32.32.252,10.32.32.253
- --neighbor-ipv6-address=2409:AB00:AB00:2000::AFB:8AFC,2409:AB00:AB00:2000::AFB:8AFD
- --neighbor-as=65030
- --cluster-as=65000
```

- neighbor-address: BGP Peer
- neighbor-as: BGP Peer AS
- cluster-as: AS

yaml:

```
kubect1 apply -f speaker.yaml
```

### 7.9.2 Pod/Subnet

BGP Subnet natOutgoing false Pod IP

annotation

```
kubect1 annotate pod sample ovn.kubernetes.io/bgp=true
kubect1 annotate subnet ovn-default ovn.kubernetes.io/bgp=true
```

annotation

```
kubect1 annotate pod sample ovn.kubernetes.io/bgp-
kubect1 annotate subnet ovn-default ovn.kubernetes.io/bgp-
```

BGP

7.9.3 ClusterIP Service

Service

ClusterIP

kube-ovn-speaker

--announce-cluster-ip

true

BGP

annotation

kubectl annotate service sample ovn.kubernetes.io/bgp=true

annotation

kubectl annotate service sample ovn.kubernetes.io/bgp-

7.9.4 EIPs

EIPs

VPC NAT Gateway

BGP

BGP Sidecar

VPC NAT Gateway

BGP

BGP Speaker Sidecar

NetworkAttachmentDefinition

NAD

VPC

Subnet

Sidecar

Kubernetes API

EIPs

VPC

CoreDNS

NAD

NetworkAttachmentDefinition

Subnet

provider

{nadName}.{nadNamespace}.ovn

apiVersion: "k8s.cni.cncf.io/v1"  
kind: NetworkAttachmentDefinition  
metadata:  
 name: api-ovn-nad  
 namespace: default  
spec:  
 config: '{  
 "cniVersion": "0.3.0",  
 "type": "kube-ovn",  
 "server\_socket": "/run/openvswitch/kube-ovn-daemon.sock",  
 "provider": "api-ovn-nad.default.ovn"  
 }'  
---  
apiVersion: kubeovn.io/v1  
kind: Subnet  
metadata:  
 name: vpc-apiserver-subnet  
spec:  
 protocol: IPv4  
 cidrBlock: 100.100.100.0/24  
 provider: api-ovn-nad.default.ovn

ovn-vpc-nat-config

ConfigMap

apiNadProvider

BGP Speaker

:

apiVersion: v1  
kind: ConfigMap  
metadata:  
 name: ovn-vpc-nat-config  
 namespace: kube-system  
data:  
 apiNadProvider: api-ovn-nad.default.ovn # What NetworkAttachmentDefinition provider to use so that the sidecar  
# can access the K8S API, as it can't by default due to VPC segmentation  
 bgpSpeakerImage: docker.io/kubeovn/kube-ovn:v1.13.0 # Sets the BGP speaker image used  
 image: docker.io/kubeovn/vpc-nat-gateway:v1.13.0

ovn-default

provider

provider: api-ovn-nad.default.ovn

VPC NAT Gateway

BGP

kind: VpcNatGateway  
apiVersion: kubeovn.io/v1  
metadata:  
 name: vpc-natgw  
spec:  
 vpc: vpc1  
 subnet: net1  
 lanIp: 10.0.1.10  
 bgpSpeaker:  
 enabled: true  
 asn: 65500  
 remoteAsn: 65000  
 neighbors:  
 - 100.127.4.161  
 - fd:01::1  
 enableGracefulRestart: true # Optional

```
routerId: 1.1.1.1      # Optional
holdTime: 1m          # Optional
password: "password123" # Optional
extraArgs:             # Optional, passed directly to the BGP speaker
  - -v5                # Enables verbose debugging of the BGP speaker sidecar
selector:
  - "kubernetes.io/os: linux"
externalSubnets:
  - ovn-vpc-external-network # Network on which we'll speak BGP and receive/send traffic to the outside world
                                # BGP neighbors need to be on that network
```

BGP      EIP

```
kubectl annotate eip sample ovn.kubernetes.io/bgp=true
```

7.9.5

kube-ovn-speaker :

• Cluster:

speaker

Pod IPs/Subnet CIDRs

IP

CIDR

Pod

speaker

Pod

• Local:

Pod IPs

Pod

Cluster

Pod

:

Local

kube-ovn-speaker

Pod

speaker

Cluster

Pod/Subnet

annotation

ovn.kubernetes.io/bgp

• ovn.kubernetes.io/bgp=cluster

ovn.kubernetes.io/bgp=yes

Cluster

• ovn.kubernetes.io/bgp=local

Local

Service

kube-proxy

ClusterIP

Service

Cluster

7.9.6 BGP

kube-ovn-speaker      BGP

• announce-cluster-ip:

Service

false

• auth-password:

BGP peer

• holdtime:

BGP

90

• graceful-restart:

BGP Graceful Restart

• graceful-restart-time:

BGP Graceful restart time

RFC4724 3

• graceful-restart-deferral-time:

BGP Graceful restart deferral time

RFC4724 4.1

• passivemode:

Speaker

passive

peer

• ebgp-multihop:

ebgp ttl

1

7.9.7 BGP routes debug

```
# show peer neighbor
gobgp neighbor

# show announced routes to one peer
gobgp neighbor 10.32.32.254 adj-out
```

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7.9.8

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7.10 MetalLB Kube-OVN Underlay

MetalLB      Kubernetes      Kubernetes      Service

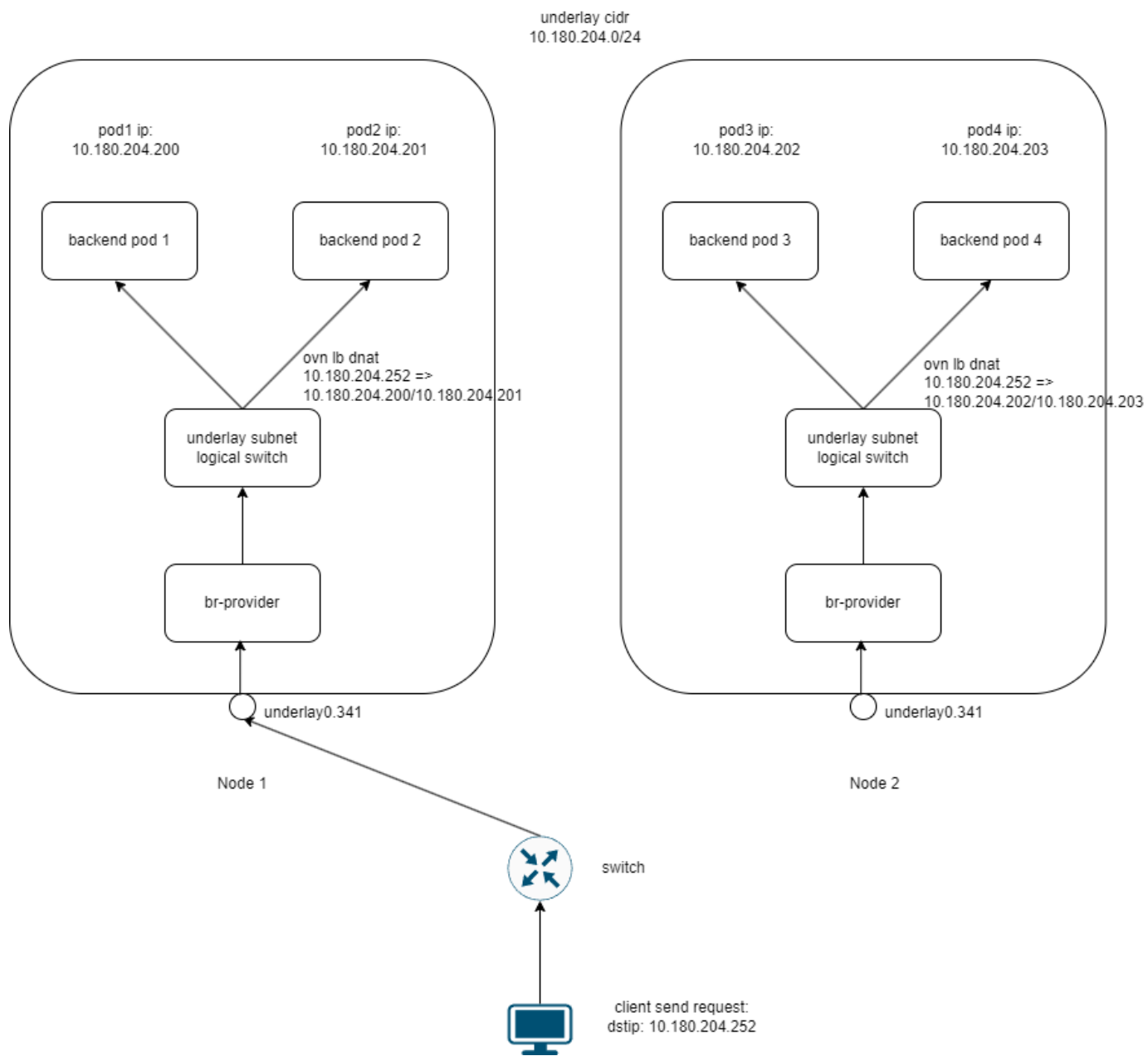
MetalLB   Kube-OVN   Underlay

7.10.1

Kube-OVN    1.14.0    MetalLB   Underlay

- MetalLB   IP
- Pod   MetalLB VIP    Underlay
- IP                SNAT

7.10.2



### 1 MetalLB VIP Kube-OVN Underlay

#### MetalLB Kube-OVN Underlay

1. VIP 10.180.204.252 IP MetalLB L2 Node1 metallb VIP
2. VIP underlay0.341
3. br-provider Underlay
4. br-provider OpenFlow OVN
5. underlay subnet OVN ovn lb dnat
6. OVN Pod  
10.180.204.0/24 VIP Pod IP

## 7.10.3

- Kube-OVN `--enable-ovn-lb-prefer-local=true`
- Underlay `enableExternalLBAddress=true`
- Underlay `excludeIps` MetalLB IP

## 7.10.4

### 1. Kube-OVN

Kube-OVN Kube-OVN `--enable-ovn-lb-prefer-local=true` `--ls-ct-skip-dst-lport-ips=false`

```
# kube-ovn-controller Deployment
kubectl edit deployment -n kube-system kube-ovn-controller
```

```
--enable-ovn-lb-prefer-local=true
--ls-ct-skip-dst-lport-ips=false
```

### 2. Underlay

Underlay LoadBalancer excludeIps MetalLB IP

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: underlay-subnet
spec:
  protocol: IPv4
  provider: ovn
  cidrBlock: 10.180.204.0/24 #
  gateway: 10.180.204.1
  excludeIps:
    - 10.180.204.250
    - 10.180.204.251
    - 10.180.204.252 # MetalLB VIP 10.180.204.252
  natOutgoing: false
  enableExternalLBAddress: true # subnet cidr ip metallb vip
```

### 3. MetalLB

MetalLB MetalLB

```
kubectl apply -f https://raw.githubusercontent.com/metallb/metallb/v0.13.7/config/manifests/metallb-native.yaml
```

MetalLB L2

```
apiVersion: metallb.io/v1beta1
kind: IPAddressPool
metadata:
```

```

name: underlay-pool
namespace: metallb-system
spec:
  addresses:
    - 10.180.204.250-10.180.204.254 # VIP 10.180.204.252
---
apiVersion: metallb.io/v1beta1
kind: L2Advertisement
metadata:
  name: l2-advert
  namespace: metallb-system
spec:
  ipAddressPools:
    - underlay-pool

```

#### 4. LoadBalancer Service

LoadBalancer      Service      Underlay      Pod

```

apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: nginx
  name: deploy-169402624
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      annotations:
        ovn.kubernetes.io/logical_switch: underlay-subnet
      labels:
        app: nginx
    spec:
      containers:
        - args:
            - netexec
            - --http-port
            - "80"
          image: kubeovn/agnhost:2.47
          imagePullPolicy: IfNotPresent
          name: nginx
---
apiVersion: v1
kind: Service
metadata:
  name: nginx-lb
spec:
  externalTrafficPolicy: Local
  ipFamilies:
    - IPv4
  ipFamilyPolicy: PreferDualStack
  ports:
    - port: 80
      protocol: TCP
      targetPort: 80
  selector:
    app: nginx
  type: LoadBalancer

```

### 7.10.5

#### 1. Service MetalLB IP

```
kubectl get svc nginx-lb
```

EXTERNAL-IP      IP      10.180.204.252

#### 1. Service IP

```
curl http://10.180.204.252
```

#### 1. Pod

Service endpoints Pod

```

# Service endpoints
kubectl get endpoints nginx-lb

```

```
# Pod
kubectl get pods -l app=nginx -o wide
```

1. IP

| nginx Pod | IP | IP | SNAT | IP |
|-----------|----|----|------|----|
|-----------|----|----|------|----|

```
kubectl exec -it $(kubectl get pods -l app=nginx -o name | head -n1) -- cat /var/log/nginx/access.log
```

7.10.6

IP

|         |          |      |          |            |    |
|---------|----------|------|----------|------------|----|
| MetalLB | Underlay | CIDR | Underlay | excludeIps | IP |
|---------|----------|------|----------|------------|----|

|         |                   |               |      |      |     |             |
|---------|-------------------|---------------|------|------|-----|-------------|
| MetalLB | Kube-OVN Underlay | underlay0.341 | VLAN | VLAN | ARP | MetalLB VIP |
|---------|-------------------|---------------|------|------|-----|-------------|

|            |                                   |           |                              |
|------------|-----------------------------------|-----------|------------------------------|
| - Kube-OVN | --enable-ovn-lb-prefer-local=true | - Service | externalTrafficPolicy: Local |
|------------|-----------------------------------|-----------|------------------------------|

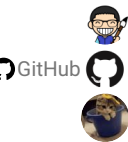
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7.10.7

### 7.11 Cilium

## Cilium

eBPF

## Kube-OVN

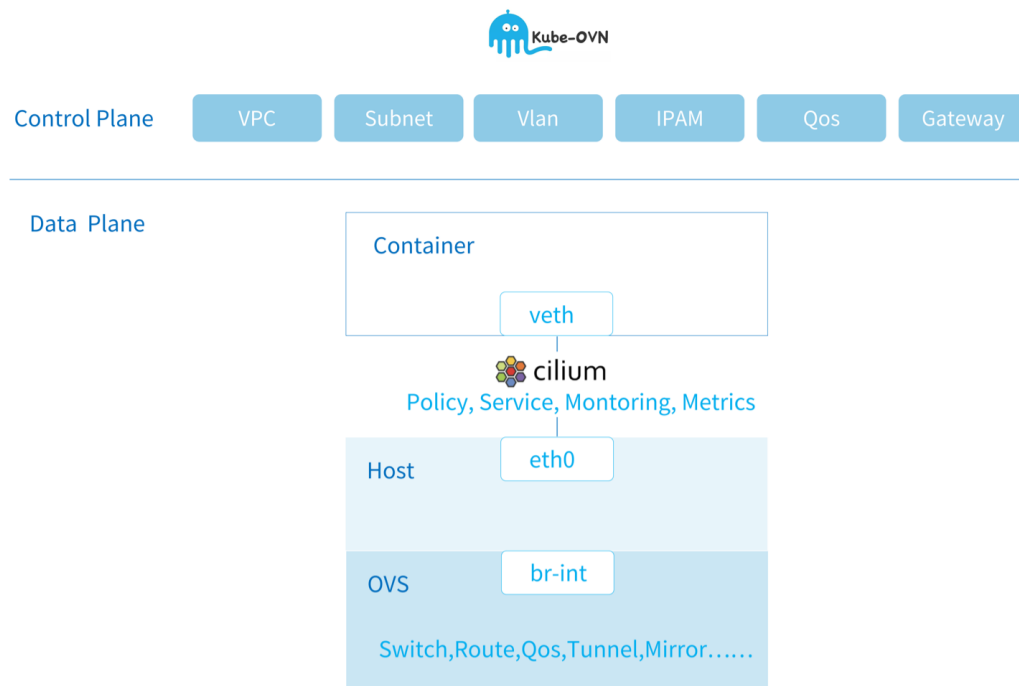
## CNI Chaining

## Kube-OVN

eBPF

## Cilium Kube-OVN

- Hubble



### 7.11.1

## 1. Linux

4.19

eBPF

## 2. Helm

Cilium

Helm

## Installing Helm

### 7.11.2 Kube-OVN

Cilium

## Kube-OVN

networkpolicy

CNI

```
install.sh
```

```
ENABLE_NP=false
CNI_CONFIG_PRIORITY=10
```

kube-ovn-controller

networkpolicy

```
args:
- --enable-np=false
```

```
kube-ovn-cni      CNI
```

```
args:
- --cni-conf-name=10-kube-ovn.conf
```

Kube-OVN

Cilium

```
mv /etc/cni/net.d/01-kube-ovn.conflist /etc/cni/net.d/10-kube-ovn.conflist
```

### 7.11.3 Cilium

chaining.yaml

Cilium generic-veth

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: cni-configuration
  namespace: kube-system
data:
  cni-config: |-
    {
      "name": "generic-veth",
      "cniVersion": "0.3.1",
      "plugins": [
        {
          "type": "kube-ovn",
          "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
          "ipam": {
            "type": "kube-ovn",
            "server_socket": "/run/openvswitch/kube-ovn-daemon.sock"
          }
        },
        {
          "type": "portmap",
          "snat": true,
          "capabilities": {"portMappings": true}
        },
        {
          "type": "cilium-cni"
        }
      ]
    }
```

```
kubectl apply -f chaining.yaml
```

Helm Cilium

```
helm repo add cilium https://helm.cilium.io/
helm install cilium cilium/cilium --version 1.11.6 \
  --namespace kube-system \
  --set cni.chainingMode=generic-veth \
  --set cni.customConf=true \
  --set cni.configMap=cni-configuration \
  --set tunnel=disabled \
  --set enableIPv4Masquerade=false \
  --set devices="eth+ ovn0 genev_sys_6081 vxlan_sys_4789" \
  --set enableIdentityMark=false
```

Cilium

```
# cilium status
/---\
|--\  /---\ Cilium:      OK
\--\  /---\ Operator:    OK
/---\  /---\ Hubble:     disabled
\--\  /---\ ClusterMesh: disabled
      \--\

DaemonSet      cilium           Desired: 2, Ready: 2/2, Available: 2/2
Deployment      cilium-operator   Desired: 2, Ready: 2/2, Available: 2/2
Containers:     cilium           Running: 2
                 cilium-operator Running: 2
Cluster Pods:   8/11 managed by Cilium
Image versions  cilium           quay.io/cilium/cilium:v1.10.5@sha256:0612218e28288db360c63677c09fafa2d17edda4f13867bcabf87056046b33bb: 2
                 cilium-operator  quay.io/cilium/operator-generic:v1.10.5@sha256:2d2f730f219d489ff0702923bf24c0002cd93eb4b47ba344375566202f56d972: 2
```



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7.11.4

---



## 7.12 Cilium NetworkPolicy

|          |        |              |
|----------|--------|--------------|
| Kube-OVN | Cilium | Cilium       |
| Cilium   | Cilium | Cilium L3 L4 |

### 7.12.1

#### Pod

namespace test    yaml    test namespace    label app=test    Pod    Pod

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: test
  name: test
  namespace: test
spec:
  replicas: 1
  selector:
    matchLabels:
      app: test
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      labels:
        app: test
    spec:
      containers:
        - image: docker.io/library/nginx:alpine
          imagePullPolicy: IfNotPresent
          name: nginx
```

yaml    default namespace    label app=dynamic    Pod    Pod

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: dynamic
  name: dynamic
  namespace: default
spec:
  replicas: 2
  selector:
    matchLabels:
      app: dynamic
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: dynamic
    spec:
      containers:
        - image: docker.io/library/nginx:alpine
          imagePullPolicy: IfNotPresent
          name: nginx
```

Pod    Label :

```
# kubectl get pod -o wide --show-labels
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE                                NOMINATED NODE   READINESS GATES   LABELS
dynamic-7d8d7874f5-9v5c4            1/1     Running   0           28h   10.16.0.35    kube-ovn-worker                    <none>            <none>              app=dynamic, pod-
dynamic-7d8d7874f5-8z2n             1/1     Running   0           28h   10.16.0.36    kube-ovn-control-plane              <none>            <none>              app=dynamic, pod-
dynamic-7d8d7874f5-6dsg6            1/1     Running   0           7h20m 10.16.0.2     kube-ovn-control-plane              <none>            <none>              app=dynamic, pod-
dynamic-7d8d7874f5-6dsg6            1/1     Running   0           7h20m 10.16.0.2     kube-ovn-control-plane              <none>            <none>              app=dynamic, pod-
dynamic-7d8d7874f5-6dsg6            1/1     Running   0           7h20m 10.16.0.2     kube-ovn-control-plane              <none>            <none>              app=dynamic, pod-
```

|                                                            |     |         |   |       |            |                 |        |        |                   |
|------------------------------------------------------------|-----|---------|---|-------|------------|-----------------|--------|--------|-------------------|
| dynamic-7d8d7874f5-tjgtp<br>template-hash=7d8d7874f5       | 1/1 | Running | 0 | 7h46m | 10.16.0.42 | kube-ovn-worker | <none> | <none> | app=dynamic, pod- |
| label=test1-77b6764857-swq4k<br>template-hash=77b6764857   | 1/1 | Running | 0 | 3h43m | 10.16.0.12 | kube-ovn-worker | <none> | <none> | app=test1, pod-   |
| # Pod<br>test-54c98bc466-mft5s<br>template-hash=54c98bc466 | 1/1 | Running | 0 | 8h    | 10.16.0.41 | kube-ovn-worker | <none> | <none> | app=test, pod-    |

### L3

yaml CiliumNetworkPolicy :

```
apiVersion: "cilium.io/v2"
kind: CiliumNetworkPolicy
metadata:
  name: "l3-rule"
  namespace: test
spec:
  endpointSelector:
    matchLabels:
      app: test
  ingress:
    - fromEndpoints:
      - matchLabels:
          app: dynamic
```

default namespace Pod Pod test namespace Pod

default namespace :

```
# kubectl exec -it dynamic-7d8d7874f5-9v5c4 -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
```

test namespace Pod :

```
# kubectl exec -it -n test dynamic-7d8d7874f5-6dsg6 -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes
64 bytes from 10.16.0.41: seq=0 ttl=64 time=2.558 ms
64 bytes from 10.16.0.41: seq=1 ttl=64 time=0.223 ms
64 bytes from 10.16.0.41: seq=2 ttl=64 time=0.304 ms

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.223/1.028/2.558 ms
```

Cilium CiliumNetworkPolicy Namespace Cilium

Namespace Pod Namespace Pod

Namespace Namespace

CiliumNetworkPolicy namespace :

```
ingress:
- fromEndpoints:
- matchLabels:
    app: dynamic
    k8s:io.kubernetes.pod.namespace: default # Namespace Pod
```

CiliumNetworkPolicy :

```
# kubectl get cnp -n test -o yaml l3-rule
apiVersion: cilium.io/v2
kind: CiliumNetworkPolicy
metadata:
  name: l3-rule
  namespace: test
spec:
  endpointSelector:
    matchLabels:
      app: test
  ingress:
    - fromEndpoints:
      - matchLabels:
```

```

    app: dynamic
  - matchLabels:
    app: dynamic
    k8s:io.kubernetes.pod.namespace: default

```

default namespace Pod Pod :

```

# kubectl exec -it dynamic-7d8d7874f5-9v5c4 -n test -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes
64 bytes from 10.16.0.41: seq=0 ttl=64 time=2.383 ms
64 bytes from 10.16.0.41: seq=1 ttl=64 time=0.115 ms
64 bytes from 10.16.0.41: seq=2 ttl=64 time=0.142 ms

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.115/0.880/2.383 ms

```

| Kubernetes | networkpolicy | Cilium | Namespace | Namespace | Namespace | Pod       | Pod       | Pod |
|------------|---------------|--------|-----------|-----------|-----------|-----------|-----------|-----|
| Kube-OVN   | Kube-OVN      | k8s    | Namespace | Pod       | Pod       | Namespace | Namespace | Pod |

## L4

yaml L4 :

```

apiVersion: "cilium.io/v2"
kind: CiliumNetworkPolicy
metadata:
  name: "l4-rule"
  namespace: test
spec:
  endpointSelector:
    matchLabels:
      app: test
  ingress:
    - fromEndpoints:
      - matchLabels:
          app: dynamic
    toPorts:
      - ports:
          - port: "80"
            protocol: TCP

```

Namespace Pod

```

# kubectl exec -it -n test dynamic-7d8d7874f5-6dsg6 -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
bash-5.0#
bash-5.0# curl 10.16.0.41:80
<html>
<head>
  <title>Hello World!</title>
  <link href="//fonts.googleapis.com/css?family=Open+Sans:400,700" rel="stylesheet" type="text/css">
  <style>
    body {
      background-color: white;
      text-align: center;
      padding: 50px;
      font-family: "Open Sans", "Helvetica Neue", Helvetica, Arial, sans-serif;
    }
    #logo {
      margin-bottom: 40px;
    }
  </style>
</head>
<body>
  <h1>Hello World!</h1>
  <h3>Links found</h3>
  <h3>I am on test-54c98bc466-mft5s</h3>
  <h3>Cookie =</h3>
  <b>KUBERNETES</b> listening in 443 available at tcp://10.96.0.1:443<br />
  <h3>my name is hanhouchao!</h3>
  <h3> RequestURI='/' </h3>
</body>
</html>

```

Namespace

Pod

```
# kubectl exec -it -n test label-test1-77b6764857-swq4k -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
bash-5.0#
bash-5.0# curl -v 10.16.0.41:80 --connect-timeout 10
* Trying 10.16.0.41:80...
* After 10000ms connect time, move on!
* connect to 10.16.0.41 port 80 failed: Operation timed out
* Connection timeout after 10001 ms
* Closing connection 0
curl: (28) Connection timeout after 10001 ms
```

Namespace

L3

L4

ping

TCP

ICMP

L4

L7

chaining

L7

Cilium

[Generic Veth Chaining](#)

[issue 12454](#)

 PDF

 Slack

 Support

 2026 7 4

 2022 8 2



GitHub



+1

7.12.2

## 7.13 Cilium

|          |        |        |
|----------|--------|--------|
| Kube-OVN | Cilium | Cilium |
| Cilium   | Hubble | Hubble |

### 7.13.1 Hubble

|        |        |        |
|--------|--------|--------|
| Cilium | Hubble | Hubble |
| helm   | Hubble |        |

```
helm upgrade cilium cilium/cilium --version 1.11.6 \
  --namespace kube-system \
  --reuse-values \
  --set hubble.relay.enabled=true \
  --set hubble.ui.enabled=true
```

|        |               |        |
|--------|---------------|--------|
| Hubble | cilium status | Hubble |
|--------|---------------|--------|

```
# cilium status
/---\
/---\ Cilium:      OK
\---\ Operator:   OK
/---\ Hubble:     OK
\---\ ClusterMesh: disabled

Deployment      hubble-relay      Desired: 1, Ready: 1/1, Available: 1/1
Deployment      cilium-operator    Desired: 2, Ready: 2/2, Available: 2/2
DaemonSet      cilium             Desired: 2, Ready: 2/2, Available: 2/2
Deployment      hubble-ui          Desired: 1, Ready: 1/1, Available: 1/1
Containers:     cilium             Running: 2
                hubble-ui          Running: 1
                hubble-relay       Running: 1
                cilium-operator    Running: 2

Cluster Pods:   16/17 managed by Cilium
Image versions  hubble-relay       quay.io/cilium/hubble-relay:v1.11.6@sha256:fd9034a2d04d5b973f1e8ed44f230ea195b89c37955ff32e34e5aa68f3ed675a: 1
                  cilium-operator    quay.io/cilium/operator-generic:v1.11.6@sha256:9f6063c7bcaede801a39315ec7c166309f6a6783e98665f6693939cf1701bc17: 2
                  cilium             quay.io/cilium/cilium:v1.11.6@sha256:f7f93c26739b6641a3fa3d76b1e1605b15989f25d06625260099e01c8243f54c: 2
                  hubble-ui          quay.io/cilium/hubble-ui:v0.9.0@sha256:0ef04e9a29212925da6bdfd0ba5b581765e41a01f1cc30563cef9b30b457fea0: 1
                  hubble-ui          quay.io/cilium/hubble-ui-backend:v0.9.0@sha256:000df6b76719f607a9edefb9af94dfd1811a6f1b6a8a9c537cba90bf12df474b: 1

apple@bogon cilium %
```

|        |              |
|--------|--------------|
| Hubble | Hubble CLI : |
|--------|--------------|

```
curl -L --fail --remote-name-all https://github.com/cilium/hubble/releases/download/v0.10.0/hubble-linux-amd64.tar.gz
sudo tar xzvfC hubble-linux-amd64.tar.gz /usr/local/bin
```

### 7.13.2

#### Cilium

|                          |        |             |           |             |
|--------------------------|--------|-------------|-----------|-------------|
| cilium connectivity test | Cilium | cilium-test | Namespace | cilium-test |
| cilium-test namespace    |        |             |           |             |

```
# kubectl get all -n cilium-test
NAME                                READY   STATUS    RESTARTS   AGE
pod/client-7df6cfbf7b-z5t2j         1/1     Running   0           21s
pod/client2-547996d7d8-nvgxg        1/1     Running   0           21s
pod/echo-other-node-d79544ccf-h14gg 2/2     Running   0           21s
pod/echo-same-node-5d466d5444-ml7tc 2/2     Running   0           21s

NAME                                TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
service/echo-other-node             NodePort      10.109.58.126    <none>            8080:32269/TCP   21s
service/echo-same-node              NodePort      10.108.70.32     <none>            8080:32490/TCP   21s

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/client              1/1     1             1           21s
deployment.apps/client2            1/1     1             1           21s
deployment.apps/echo-other-node     1/1     1             1           21s
deployment.apps/echo-same-node      1/1     1             1           21s

NAME                                DESIRED   CURRENT   READY   AGE
```

```

replicaset.apps/client-7df6cfbf7b      1      1      1      21s
replicaset.apps/client2-547996d7d8     1      1      1      21s
replicaset.apps/echo-other-node-d79544ccf 1      1      1      21s
replicaset.apps/echo-same-node-5d466d5444 1      1      1      21s

```

### 7.13.3

Cilium                      kube-system namespace      Cilium    pod            hubble observe

```

# kubectl get pod -n kube-system -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP              NODE               NOMINATED NODE   READINESS GATES
cilium-d6h56                        1/1     Running   0           2d20h  172.18.0.2      kube-ovn-worker    <none>           <none>
cilium-operator-5887f78bbb-c7sb2    1/1     Running   0           2d20h  172.18.0.2      kube-ovn-worker    <none>           <none>
cilium-operator-5887f78bbb-wj8gt    1/1     Running   0           2d20h  172.18.0.3      kube-ovn-control-plane <none>           <none>
cilium-tq5xb                        1/1     Running   0           2d20h  172.18.0.3      kube-ovn-control-plane <none>           <none>
kube-ovn-pinger-71gk8               1/1     Running   0           21h    10.16.0.19      kube-ovn-control-plane <none>           <none>
kube-ovn-pinger-msvcn               1/1     Running   0           21h    10.16.0.18      kube-ovn-worker     <none>           <none>

# kubectl exec -it -n kube-system cilium-d6h56 -- bash
root@kube-ovn-worker:/home/cilium# hubble observe --from-namespace kube-system
Jul 29 03:24:25.551: kube-system/kube-ovn-pinger-msvcn:35576 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 29 03:24:25.561: kube-system/kube-ovn-pinger-msvcn:35576 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: RST)
Jul 29 03:24:25.561: kube-system/kube-ovn-pinger-msvcn:35576 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, RST)
Jul 29 03:24:25.572: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: SYN)
Jul 29 03:24:25.572: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.651: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 29 03:24:25.661: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: RST)
Jul 29 03:24:25.661: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, RST)
Jul 29 03:24:25.761: kube-system/kube-ovn-pinger-msvcn:52004 -> 172.18.0.3:6443 to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 29 03:24:25.779: kube-system/kube-ovn-pinger-msvcn -> kube-system/kube-ovn-pinger-71gk8 to-stack FORWARDED (ICMPv4 EchoRequest)
Jul 29 03:24:25.779: kube-system/kube-ovn-pinger-msvcn <- kube-system/kube-ovn-pinger-71gk8 to-endpoint FORWARDED (ICMPv4 EchoReply)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 <- kube-system/hubble-relay-959988db5-zc5vv:4245 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 <- kube-system/hubble-relay-959988db5-zc5vv:80 to-endpoint FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 -> kube-system/hubble-relay-959988db5-zc5vv:4245 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 -> kube-system/hubble-relay-959988db5-zc5vv:4245 to-endpoint FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.975: kube-system/kube-ovn-pinger-71gk8 -> kube-system/kube-ovn-pinger-msvcn to-endpoint FORWARDED (ICMPv4 EchoRequest)
Jul 29 03:24:25.975: kube-system/kube-ovn-pinger-71gk8 <- kube-system/kube-ovn-pinger-msvcn to-stack FORWARDED (ICMPv4 EchoReply)
Jul 29 03:24:25.979: kube-system/kube-ovn-pinger-msvcn -> 172.18.0.3 to-stack FORWARDED (ICMPv4 EchoRequest)
Jul 29 03:24:26.037: kube-system/coredns-6d4b75cb6d-lbgjg:36430 -> 172.18.0.3:6443 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:26.282: kube-system/kube-ovn-pinger-msvcn -> 172.18.0.2 to-stack FORWARDED (ICMPv4 EchoRequest)

```

Hubble Relay    Hubble

Hubble API                      Hubble Service                      kubectl port-forward deployment/hubble-relay -n kube-system 4245:4245

kubectl port-forward

hubble status

```

# hubble status
Healthcheck (via localhost:4245): Ok
Current/Max Flows: 8,190/8,190 (100.00%)
Flows/s: 22.86
Connected Nodes: 2/2

```

hubble observe

cilium-test

```

apple@bogon ovn-test % hubble observe
Jul 28 08:00:07.033: cilium-test/client-7df6cfbf7b-qn7q6:56906 (ID:15432) -> kube-system/coredns-6d4b75cb6d-b444j:53 (ID:11307) to-endpoint FORWARDED (UDP)
Jul 28 08:00:07.033: cilium-test/client-7df6cfbf7b-qn7q6:56906 (ID:15432) <- kube-system/coredns-6d4b75cb6d-b444j:53 (ID:11307) to-stack FORWARDED (UDP)
Jul 28 08:00:07.095: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.095: 100.64.0.3:43037 (world) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.096: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.096: 100.64.0.3:43037 (world) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.096: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.097: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.249: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.249: 100.64.0.2:33419 (host) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.250: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.251: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.251: 100.64.0.2:33419 (host) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.251: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.323: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) <- 172.18.0.3:31514 (kube-apiserver) to-endpoint FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.323: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) -> 172.18.0.3:31514 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) -> cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.324: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) -> 172.18.0.3:31514 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) -> cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) <- cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) <- 172.18.0.3:31514 (kube-apiserver) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) -> 172.18.0.3:31514 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) -> cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.340: kube-system/hubble-relay-959988db5-zc5vv:46938 (ID:53347) -> 172.18.0.2:4244 (host) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.340: kube-system/hubble-relay-959988db5-zc5vv:46938 (ID:53347) -> 172.18.0.3:4244 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.340: kube-system/hubble-relay-959988db5-zc5vv:46938 (ID:53347) <- 172.18.0.2:4244 (host) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.341: kube-system/hubble-relay-959988db5-zc5vv:46938 (ID:53347) <- 172.18.0.3:4244 (kube-apiserver) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.409: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.409: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.409: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.411: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.411: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
apple@bogon ovn-test %

```

hubble observe

hubble help observe

Hubble CLI

## 7.13.4 UI

cilium status

Hubble UI

Hubble

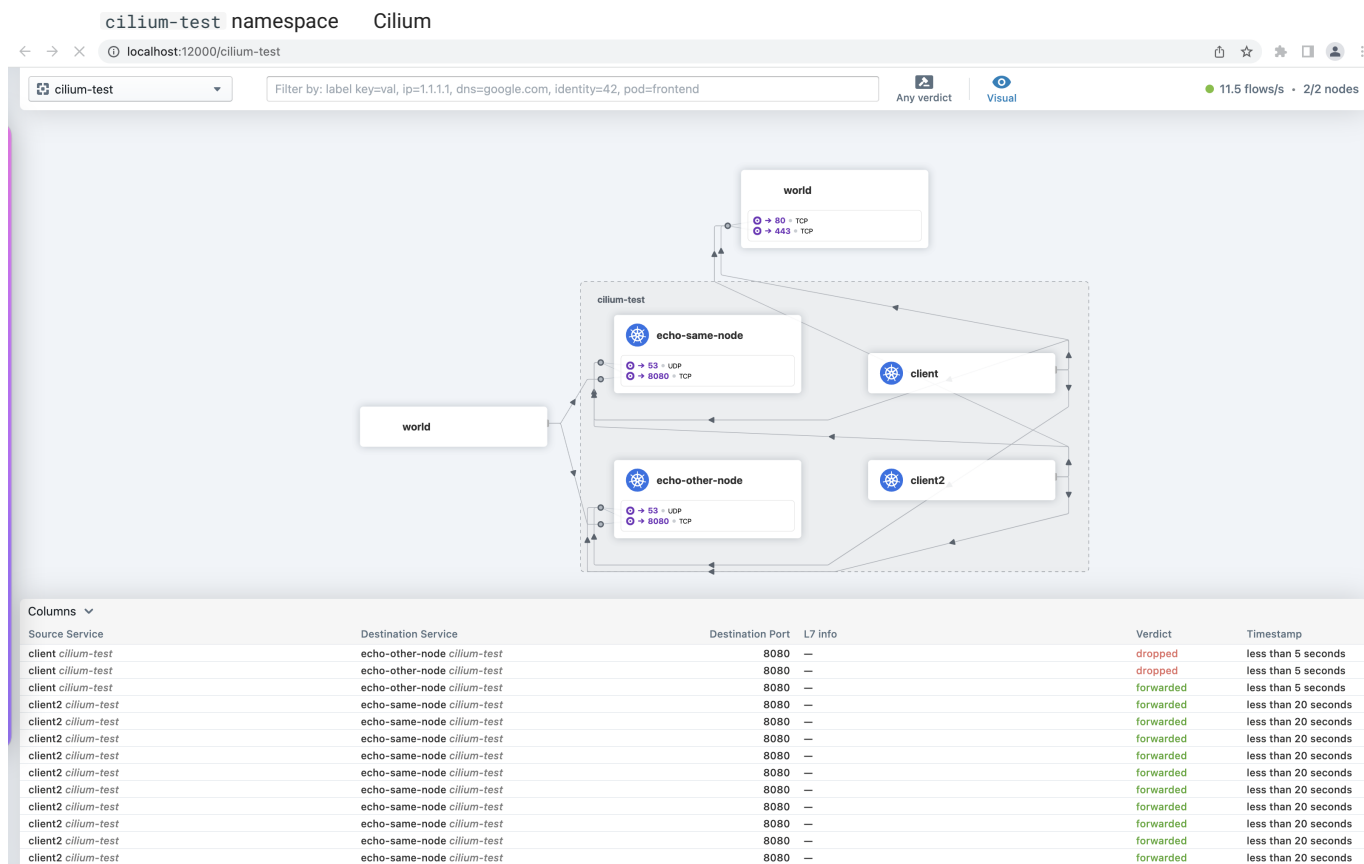
UI

cilium hubble ui

hubble-ui service

Hubble UI

http://localhost:12000 UI



## 7.13.5 Hubble

Hubble Pod Hubble

hubble.metrics.enabled :

```
helm upgrade cilium cilium/cilium --version 1.11.6 \
--namespace kube-system \
--reuse-values \
--set hubble.relay.enabled=true \
--set hubble.ui.enabled=true \
--set hubble.metrics.enabled="{dns,drop,tcp,flow,icmp,http}"
```

kube-system namespace hubble-metrics Endpoints Hubble :

```
# curl 172.18.0.2:9091/metrics
# HELP hubble_drop_total Number of drops
# TYPE hubble_drop_total counter
hubble_drop_total{protocol="ICMPv6",reason="Unsupported L3 protocol"} 2
# HELP hubble_flows_processed_total Total number of flows processed
# TYPE hubble_flows_processed_total counter
hubble_flows_processed_total{protocol="ICMPv4",subtype="to-endpoint",type="Trace",verdict="FORWARDED"} 335
hubble_flows_processed_total{protocol="ICMPv4",subtype="to-stack",type="Trace",verdict="FORWARDED"} 335
hubble_flows_processed_total{protocol="ICMPv6",subtype="",type="Drop",verdict="DROPPED"} 2
hubble_flows_processed_total{protocol="TCP",subtype="to-endpoint",type="Trace",verdict="FORWARDED"} 8282
hubble_flows_processed_total{protocol="TCP",subtype="to-stack",type="Trace",verdict="FORWARDED"} 6767
hubble_flows_processed_total{protocol="UDP",subtype="to-endpoint",type="Trace",verdict="FORWARDED"} 1642
hubble_flows_processed_total{protocol="UDP",subtype="to-stack",type="Trace",verdict="FORWARDED"} 1642
# HELP hubble_icmp_total Number of ICMP messages
# TYPE hubble_icmp_total counter
hubble_icmp_total{family="IPv4",type="EchoReply"} 335
hubble_icmp_total{family="IPv4",type="EchoRequest"} 335
hubble_icmp_total{family="IPv4",type="RouterSolicitation"} 2
# HELP hubble_tcp_flags_total TCP flag occurrences
# TYPE hubble_tcp_flags_total counter
hubble_tcp_flags_total{family="IPv4",flag="FIN"} 2043
hubble_tcp_flags_total{family="IPv4",flag="RST"} 301
hubble_tcp_flags_total{family="IPv4",flag="SYN"} 1169
hubble_tcp_flags_total{family="IPv4",flag="SYN-ACK"} 1169
```



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7.13.6

---

# 7.14

## Kube-OVN

### 7.14.1

```
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: external
spec:
  cidrBlock: 172.31.0.0/16
  gatewayType: centralized
  natOutgoing: false
  externalEgressGateway: 192.168.0.1
  policyRoutingTableID: 1000
  policyRoutingPriority: 1500
```

- natOutgoing: false
- externalEgressGateway
- policyRoutingTableID TableID
- policyRoutingPriority

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### 7.14.2

7.15 VIP IP

[illegible]

- Allowed-Address-Pairs VIP
- Switch LB rule VIP
- Pod      VIP      IP

### 7.15.1 1. Allowed-Address-Pairs VIP

|              | IP                    | Pod        |          |        |  |
|--------------|-----------------------|------------|----------|--------|--|
| • Kubernetes | Kubernetes            | Kubernetes | Underlay | Subnet |  |
| • LB         | Subnet                | IP         | Pod      |        |  |
| VIP          | Allowed-Address-Pairs | IP         | IP       |        |  |
| • Keepalived | IP                    |            |          |        |  |

## 1.1 VIP

```
IP    IP                                yaml
```

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: vip-dynamic-01
spec:
  subnet: ovn-default
  type: ""
```

- subnet: Subnet IP
- type: ipam ip switch\_lb\_vip vip switch lb vip ip

VIP

```
# kubectl get vip
```

| NAME           | V4IP       | PV4IP | MAC               | PMAC | V6IP | PV6IP | SUBNET      | READY |
|----------------|------------|-------|-------------------|------|------|-------|-------------|-------|
| vip-dynamic-01 | 10.16.0.12 |       | 00:00:00:F0:DB:25 |      |      |       | ovn-default | true  |

VIP 10.16.0.12 IP

## 1.2 VIP

```
VIP IP yaml
```

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: static-vip01
spec:
  subnet: ovn-default
  v4ip: "10.16.0.121"
```

- `subnet:` Subnet IP
- `v4ip:` IP subnet CIDR

VIP

```
# kubectl get vip
NAME          V4IP      PV4IP  MAC          PMAC  V6IP  PV6IP  SUBNET    READY
static-vip01  10.16.0.121  10.16.0.121  00:00:00:F0:DB:26  00:00:00:F0:DB:26  00:00:00:F0:DB:26  00:00:00:F0:DB:26  ovn-default  true
```

VIP IP

1.3 Pod VIP AAP

Pod annotation VIP AAP labels VIP

Pod annotation VIP ovn.kubernetes.io/aaps: vip-aap,vip-aap2,vip-aap3

AAP Pod AAP Pod VIP subnet Port

1.3.1 VIP AAP

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: vip-aap
spec:
  subnet: ovn-default
  namespace: default
  selector:
    - "app: aap1"
```

VIP

- namespace: AAP VIP AAP
- selector: AAP VIP Pod Kubernetes NodeSelector

VIP Port

```
# kubectl ko nbctl show ovn-default
switch e32e1d3b-c539-45f4-ab19-be4e33a061f6 (ovn-default)
  port aap-vip
    type: virtual
```

```
apiVersion: v1
kind: Pod
metadata:
  name: busybox
  annotations:
    ovn.kubernetes.io/aaps: vip-aap
  labels:
    app: aap1
spec:
  containers:
    - name: busybox
      image: busybox
      command: ["sleep", "3600"]
      securityContext:
        capabilities:
          add:
            - NET_ADMIN
```

AAP

```
# kubectl ko nbctl list logical_switch_port aap-vip
_uuid      : cd930750-0533-4f06-a6c0-217ddac73272
addresses  : []
dhcpv4_options : []
dhcpv6_options : []
dynamic_addresses : []
enabled     : []
external_ids : {ls=ovn-default, vendor=kube-ovn}
ha_chassis_group : []
mirror_rules : []
name       : aap-vip
options    : {virtual-ip="10.16.0.100", virtual-parents="busybox.default"}
parent_name : []
port_security : []
tag        : []
tag_request : []
type       : virtual
up         : false
```

virtual-ip VIP IP virtual-parents AAP Pod Port

Pod

```
# kubectl exec -it busybox -- ip addr add 10.16.0.100/16 dev eth0
# kubectl exec -it busybox01 -- ip addr show eth0
35: eth0@if36: <BROADCAST,MULTICAST,UP,LOWER_UP,M-DOWN> mtu 1400 qdisc noqueue
    link/ether 00:00:00:e2:ab:0c brd ff:ff:ff:ff:ff:ff
    inet 10.16.0.7/16 brd 10.16.255.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet 10.16.0.100/16 scope global secondary eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fee2:ab0c/64 scope link
        valid_lft forever preferred_lft forever
```

|     |    |     |    |        |     |    |
|-----|----|-----|----|--------|-----|----|
| Pod | IP | VIP | IP | subnet | Pod | IP |
|-----|----|-----|----|--------|-----|----|

7.15.2.2. Switch LB rule vip

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: slr-01
spec:
  subnet: ovn-default
  type: switch_lb_vip
```

- subnet: Subnet IP
- type: ipam ip switch\_lb\_vip vip switch lb vip ip

7.15.3.3. Pod VIP IP

v1.12

IP

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: pod-use-vip
spec:
  subnet: ovn-default
  type: ""
```

|            |     |     |
|------------|-----|-----|
| annotation | VIP | Pod |
|------------|-----|-----|

```
apiVersion: v1
kind: Pod
metadata:
  name: static-ip
  annotations:
    ovn.kubernetes.io/vip: pod-use-vip # vip
  namespace: default
spec:
  containers:
    - name: static-ip
      image: docker.io/library/nginx:alpine
```

3.1 StatefulSet Kubevirt VM VIP

|             |     |                     |                             |
|-------------|-----|---------------------|-----------------------------|
| StatefulSet | VM  | Pod                 | VIP                         |
| VM          | VIP | kube-ovn-controller | keep-vm-ip true Kubevirt VM |

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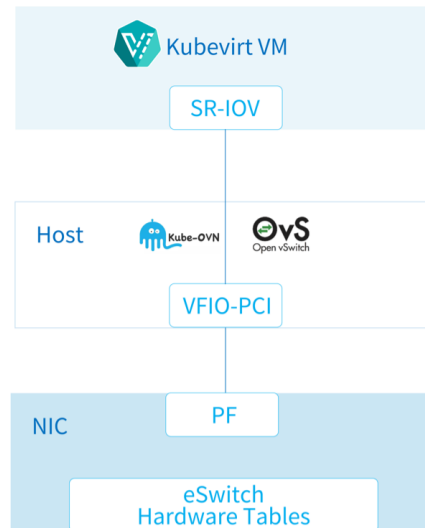
GitHub

7.15.4

---

## 7.16 Mellanox Offload

Kube-OVN Processing (ASAP<sup>2</sup>)      OVS OVS      CPU eSwitch      CPU OVS      Mellanox CPU      Accelerated Switching And Packet



### Note

2022

### 7.16.1

- Mellanox CX5/CX6/CX7/BlueField      ASAP<sup>2</sup>
- CentOS 8 Stream      Linux 5.7
- dp\_hash      hash      OVN LB
- bond

### 7.16.2 SR-IOV Device Plugin

Mellanox      offload      SR-IOV Device Plugin      [sriov-network-operator](#)

#### SR-IOV Device Plugin

SR-IOV

ID      84:00.0      84.00.1

```
# lspci -nn | grep ConnectX-5
84:00.0 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
84:00.1 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
```

## ID

```
# ls -l /sys/class/net/ | grep 84:00.0
lrwxrwxrwx 1 root root 0 Feb 4 16:16 enp132s0f0np0 -> ../../devices/pci0000:80/0000:80:08.0/0000:84:00.0/net/enp132s0f0np0
# ls -l /sys/class/net/ | grep 84:00.1
lrwxrwxrwx 1 root root 0 Feb 4 16:16 enp132s0f1np1 -> ../../devices/pci0000:80/0000:80:08.0/0000:84:00.1/net/enp132s0f1np1
```

## bond

```
enp132s0f0np0 enp132s0f1np1 bond1
```

```
# ip link show enp132s0f0np0 | grep bond
160: enp132s0f0np0: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master bond1 state UP mode DEFAULT group default qlen 1000
# ip link show enp132s0f1np1 | grep bond
169: enp132s0f1np1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master bond1 state UP mode DEFAULT group default qlen 1000
```

## bond VF

```
ifenslave -d bond1 enp132s0f0np0
ifenslave -d bond1 enp132s0f1np1
echo 0 > /sys/class/net/enp132s0f0np0/device/sriov_numvfs
echo 0 > /sys/class/net/enp132s0f1np1/device/sriov_numvfs
ip link set enp132s0f0np0 down
ip link set enp132s0f1np1 down
```

## OVS

## SMFS DMFS

- SMFS (software-managed flow steering)
- DMFS (device-managed flow steering)

## sysfs devlink API

```
# sysfs
echo <smfs|dmfs> > /sys/class/net/enp132s0f0np0/compat/devlink/steering_mode
echo <smfs|dmfs> > /sys/class/net/enp132s0f1np1/compat/devlink/steering_mode
# devlink
devlink dev param set pci/84:00.0 name flow_steering_mode value smfs cmode runtime
devlink dev param set pci/84:00.1 name flow_steering_mode value smfs cmode runtime
```

## VF

```
# cat /sys/class/net/enp132s0f0np0/device/sriov_totalvfs
127
# cat /sys/class/net/enp132s0f1np1/device/sriov_totalvfs
127
```

## VF

```
# echo '4' > /sys/class/net/enp132s0f0np0/device/sriov_numvfs
# echo '4' > /sys/class/net/enp132s0f1np1/device/sriov_numvfs
# ip link show enp132s0f0np0
160: enp132s0f0np0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000
    link/ether 08:c0:eb:74:c3:4a brd ff:ff:ff:ff:ff:ff
    vf 0 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 1 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 2 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 3 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
# ip link show enp132s0f1np1
169: enp132s0f1np1: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000
    link/ether 08:c0:eb:74:c3:4b brd ff:ff:ff:ff:ff:ff
    vf 0 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 1 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 2 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 3 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
# ip link set enp132s0f0np0 up
# ip link set enp132s0f1np1 up
```

## VF

## ID

```
# lspci -nn | grep ConnectX-5 | grep Virtual
84:00.2 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.3 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.4 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
```



```
84:00.5 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.6 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.7 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:01.0 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:01.1 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
```

VF

```
echo 0000:84:00.2 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.3 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.4 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.5 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.6 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.7 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:01.0 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:01.1 > /sys/bus/pci/drivers/mlx5_core/unbind
```

eSwitch

```
devlink dev eswitch set pci/0000:84:00.0 mode switchdev
devlink dev eswitch set pci/0000:84:00.1 mode switchdev
ethtool -K enp132s0f0np0 hw-tc-offload on
ethtool -K enp132s0f1np1 hw-tc-offload on
```

SR-IOV VF



- Active-backup
- XOR
- LACP

LACP

```
modprobe bonding mode=802.3ad
ip link set enp132s0f0np0 master bond1
ip link set enp132s0f1np1 master bond1
ip link set enp132s0f0np0 up
ip link set enp132s0f1np1 up
ip link set bond1 up
```

bond

VF

```
echo 0000:84:00.2 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.3 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.4 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.5 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.6 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.7 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:01.0 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:01.1 > /sys/bus/pci/drivers/mlx5_core/bind
```

NetworkManager

NetworkManager

```
systemctl stop NetworkManager
systemctl disable NetworkManager
```

DEVICE PLUGIN

VF Pod VF SR-IOV Device Plugin

SR-IOV Configmap

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
```

```
{
  "resourceList": [{
    "resourcePrefix": "mellanox.com",
    "resourceName": "cx5_sriov_switchdev",
    "selectors": {
      "vendors": ["15b3"],
      "devices": ["1018"],
      "drivers": ["mlx5_core"]
    }
  }]
}
```

SR-IOV Device Plugin      ConfigMap      ConfigMap    name   sriovdp-config

- selectors: VF
- vendors:
- devices:
- drivers:

selectors      pciAddresses    acpiIndexes      VF      [SR-IOV ConfigMap](#)

[SR-IOV](#)

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/sriov-network-device-plugin/v3.6.2/deployments/sriovdp-daemonset.yaml
```

SR-IOV      Kubernetes Node

```
kubectl describe node kube-ovn-01 | grep mellanox

mellanox.com/cx5_sriov_switchdev: 8
mellanox.com/cx5_sriov_switchdev: 8
mellanox.com/cx5_sriov_switchdev 0      0
```

**sriov-network-operator**    **SR-IOV**    **Device Plugin**

[node-feature-discovery](#)

```
kubectl apply -k https://github.com/kubernetes-sigs/node-feature-discovery/deployment/overlays/default?ref=v0.11.3
```

offload      annotation:

```
kubectl label nodes [offloadNicNode] feature.node.kubernetes.io/network-sriov.capable=true
```

Operator

```
git clone --depth=1 https://github.com/kubeovn/sriov-network-operator.git
kubectl apply -k sriov-network-operator/deploy
```

Operator

```
# kubectl get -n kube-system all | grep sriov
NAME                                READY    STATUS    RESTARTS    AGE
pod/sriov-network-config-daemon-bf9nt 1/1      Running   0            8s
pod/sriov-network-operator-54d7545f65-296gb 1/1      Running   0            10s

NAME                                DESIRED    CURRENT    READY    UP-TO-DATE    AVAILABLE    NODE
SELECTOR
daemonset.apps/sriov-network-config-daemon 1          1          1        1              1            beta.kubernetes.io/os=linux,feature.node.kubernetes.io/network-sriov.capable=true 8s

NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
deployment.apps/sriov-network-operator 1/1      1              1            10s

NAME                                DESIRED    CURRENT    READY    AGE
replicaset.apps/sriov-network-operator-54d7545f65 1          1          1        10s
```

SriovNetworkNodeState      node1      Mellanox

```
# kubectl get sriovnetworknodestates.sriovnetwork.openshift.io -n kube-system node1 -o yaml
apiVersion: sriovnetwork.openshift.io/v1
kind: SriovNetworkNodeState
spec: ...
```

```
status:
  interfaces:
  - deviceID: "1017"
    driver: mlx5_core
    mtu: 1500
    pciAddress: "0000:5f:00.0"
    totalVfs: 8
    vendor: "15b3"
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:bb
    name: ens41f0np0
  - deviceID: "1017"
    driver: mlx5_core
    mtu: 1500
    pciAddress: "0000:5f:00.1"
    totalVfs: 8
    vendor: "15b3"
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:bb
    name: ens41f1np1
```

SriovNetworkNodePolicy

nicSelector

```
apiVersion: srionetwork.openshift.io/v1
kind: SrionetworkNodePolicy
metadata:
  name: policy
  namespace: kube-system
spec:
  nodeSelector:
    feature.node.kubernetes.io/network-srionetwork.capable: "true"
  eSwitchMode: switchdev
  numVfs: 3
  nicSelector:
    pfNames:
    - ens41f0np0
    - ens41f1np1
  resourceName: cx_srionetwork_switchdev
```

SrionetworkNodeState

status

```
# kubectl get srionetworknodestates.srionetwork.openshift.io -n kube-system node1 -o yaml

...
spec:
  interfaces:
  - eSwitchMode: switchdev
    name: ens41f0np0
    numVfs: 3
    pciAddress: 0000:5f:00.0
    vfGroups:
    - policyName: policy
      vfRange: 0-2
      resourceName: cx_srionetwork_switchdev
  - eSwitchMode: switchdev
    name: ens41f1np1
    numVfs: 3
    pciAddress: 0000:5f:00.1
    vfGroups:
    - policyName: policy
      vfRange: 0-2
      resourceName: cx_srionetwork_switchdev
status:
  interfaces:
  - Vfs:
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.2
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.3
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.4
      vendor: "15b3"
    deviceID: "1017"
    driver: mlx5_core
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:ab
    mtu: 1500
    name: ens41f0np0
    numVfs: 3
    pciAddress: 0000:5f:00.0
    totalVfs: 3
    vendor: "15b3"
  - Vfs:
```

```
- deviceID: 1018
  driver: mlx5_core
  pciAddress: 0000:5f:00.5
  vendor: "15b3"
- deviceID: 1018
  driver: mlx5_core
  pciAddress: 0000:5f:00.6
  vendor: "15b3"
- deviceID: 1018
  driver: mlx5_core
  pciAddress: 0000:5f:00.7
  vendor: "15b3"
deviceID: "1017"
driver: mlx5_core
linkSeed: 25000Mb/s
linkType: ETH
mac: 08:c0:eb:f4:85:bb
mtu: 1500
name: ens41f1np1
numVfs: 3
pciAddress: 0000:5f:00.1
totalvfs: 3
vendor: "15b3"
```

VF

```
# lspci -nn | grep ConnectX
5f:00.0 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
5f:00.1 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
5f:00.2 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.3 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.4 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.5 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.6 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.7 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
```

PF

```
# cat /sys/class/net/ens41f0np0/compat/devlink/mode
switchdev
```

7.16.3 Multus-CNI

SR-IOV Device Plugin      ID      Multus-CNI      Kube-OVN      Multus-CNI

Multus-CNI

kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/v4.0.2/deployments/multus-daemonset-thick.yml

multus    Thin    Thick                      SR-IOV            Thick

NetworkAttachmentDefinition

apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
 name: sriov
 namespace: default
 annotations:
 k8s.v1.cni.cncf.io/resourceName: mellanox.com/cx5\_sriov\_switchdev
spec:
 config: '{
 "cniVersion": "0.3.1",
 "name": "kube-ovn",
 "plugins":[
 {
 "type": "kube-ovn",
 "server\_socket": "/run/openvswitch/kube-ovn-daemon.sock",
 "provider": "sriov.default.ovn"
 },
 {
 "type": "portmap",
 "capabilities": {
 "portMappings": true
 }
 }
 ]
 }'

• provider:      NetworkAttachmentDefinition    {name}.{namespace}.ovn

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7.16.4 Overlay

Kube-OVN

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.15/dist/images/install.sh
```

| IFACE                                                                                                                          |        |               | IP   |       |       |      |       |               |               |
|--------------------------------------------------------------------------------------------------------------------------------|--------|---------------|------|-------|-------|------|-------|---------------|---------------|
| <pre>ENABLE_MIRROR=\${ENABLE_MIRROR:-false} HW_OFFLOAD=\${HW_OFFLOAD:-true} ENABLE_LB=\${ENABLE_LB:-false} IFACE="bond1"</pre> |        |               |      |       |       |      |       |               |               |
| #                                                                                                                              | SR-IOV | Device Plugin | bond | IFACE | bond1 | bond | IFACE | enp132s0f0np0 | enp132s0f1np1 |

Kube-OVN

```
bash install.sh
```

VF Pod

yaml

VF

Pod:

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-overlay
  annotations:
    v1.multus-cni.io/default-network: default/sriov
    sriov.default.ovn.kubernetes.io/logical_switch: ovn-default
spec:
  containers:
  - name: nginx-overlay
    image: docker.io/library/nginx:alpine
  resources:
    requests:
      mellanox.com/cx5_sriov_switchdev: '1'
    limits:
      mellanox.com/cx5_sriov_switchdev: '1'
```

- v1.multus-cni.io/default-network: NetworkAttachmentDefinition {namespace}/{name}
- sriov.default.ovn.kubernetes.io/logical\_switch: Pod Pod

7.16.5 Underlay

Kube-OVN

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.15/dist/images/install.sh
```

| IFACE                                                                                                                     |          | IP    |     |
|---------------------------------------------------------------------------------------------------------------------------|----------|-------|-----|
| <pre>ENABLE_MIRROR=\${ENABLE_MIRROR:-false} HW_OFFLOAD=\${HW_OFFLOAD:-true} ENABLE_LB=\${ENABLE_LB:-false} IFACE=""</pre> |          |       |     |
| #                                                                                                                         | Underlay | IFACE | PF  |
|                                                                                                                           |          | IFACE | K8s |
|                                                                                                                           |          |       | PF  |

Kube-OVN

```
bash install.sh
```

VF Pod

yaml

VF

Pod:

```

apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: underlay-offload
spec:
  defaultInterface: bond1

---
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan0
spec:
  id: 0
  provider: underlay-offload

---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vlan0
spec:
  protocol: IPv4
  provider: ovn
  cidrBlock: 10.10.204.0/24
  gateway: 10.10.204.254
  vlan: vlan0
  excludeIps:
    - 10.10.204.1..10.10.204.100

---
apiVersion: v1
kind: Pod
metadata:
  name: nginx-underlay
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{
      "name": "sriov",
      "namespace": "default",
      "default-route": ["10.10.204.254"]
    }]'
    sriov.default.ovn.kubernetes.io/logical_switch: vlan0
spec:
  containers:
    - name: nginx-underlay
      image: docker.io/library/nginx:alpine
  resources:
    requests:
      mellanox.com/cx5_sriov_switchdev: '1'
    limits:
      mellanox.com/cx5_sriov_switchdev: '1'

```

• v1.multus-cni.io/default-network: NetworkAttachmentDefinition {namespace}/{name}

multus VF Pod VF Pod VF Pod multus

yaml VF Pod:

```

apiVersion: v1
kind: Pod
metadata:
  name: nginx-underlay-noVF
  annotations:
    ovn.kubernetes.io/logical_switch: vlan0
spec:
  containers:
    - name: nginx-underlay-noVF
      image: docker.io/library/nginx:alpine

```

VF Pod ovs-kernel e-switch

## 7.16.6

Pod ovs-ovn

```

# ovs-appctl dpctl/dump-flows -m type=offloaded
ufid:91cc45de-e7e9-4935-8f82-1890430b0f66, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/0),in_port(5b45c61b307e_h),packet_type(ns=0/0,id=0/0),eth(src=00:00:00:c5:6d:4e,dst=00:00:00:e7:16:ce),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no), packets:941539, bytes:62142230, used:0.260s, offloaded:yes, dp:tc, actions:54235e5753b8_h
ufid:e00768d7-e652-4d79-8182-3291d852b791, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/0),in_port(54235e5753b8_h),packet_type(ns=0/0,id=0/0),eth(src=00:00:00:e7:16:ce,dst=00:00:00:c5:6d:4e),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no), packets:82386659, bytes:115944854173, used:0.260s, offloaded:yes, dp:tc, actions:5b45c61b307e_h

```

offloaded:yes, dp:tc

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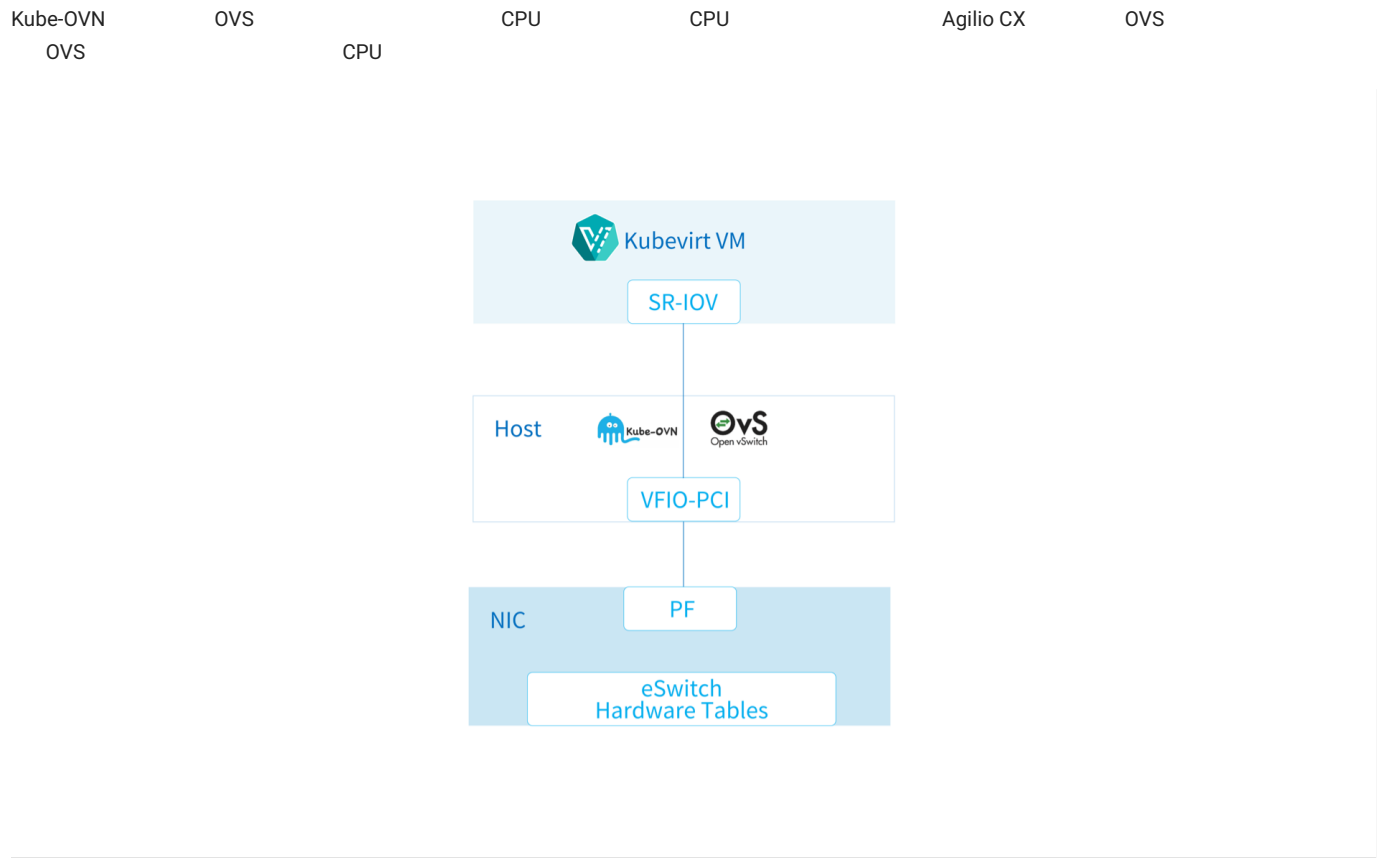
  
  
  


GitHub

7.16.7

---

7.17 Offload



**Note**

2022

7.17.1

- Agilio CX
- CentOS 8 Stream    Linux 5.7
- dp\_hash    hash        OVN LB

7.17.2 SR-IOV

Agilio Open vSwitch TC User Guide

```
#!/bin/bash
DEVICE=${1}
DEFAULT_ASSY=scan
ASSY=${2:-${DEFAULT_ASSY}}
APP=${3:-flower}

if [ "x${DEVICE}" = "x" -o ! -e /sys/class/net/${DEVICE} ]; then
    echo Syntax: ${0} device [ASSY] [APP]
    echo
    echo This script associates the TC Offload firmware
    echo with a Netronome SmartNIC.
    echo
    echo device: is the network device associated with the SmartNIC
```



```
echo ASSY: defaults to ${DEFAULT_ASSY}
echo APP: defaults to flower. flower-next is supported if updated
echo     firmware has been installed.
exit 1
fi

# It is recommended that the assembly be determined by inspection
# The following code determines the value via the debug interface
if [ "${ASSY}x" = "scanx" ]; then
    ethtool -W ${DEVICE} 0
    DEBUG=$(ethtool -w ${DEVICE} data /dev/stdout | strings)
    SERIAL=$(echo "${DEBUG}" | grep "^SN:")
    ASSY=$(echo ${SERIAL} | grep -oE AMDA[0-9]{4})
fi

PCIADDR=$(basename $(readlink -e /sys/class/net/${DEVICE}/device))
FWDIR="/lib/firmware/netronome"

# AMDA0081 and AMDA0097 uses the same firmware
if [ "${ASSY}" = "AMDA0081" ]; then
    if [ ! -e ${FWDIR}/${APP}/nic_AMDA0081.nffw ]; then
        ln -sf nic_AMDA0097.nffw ${FWDIR}/${APP}/nic_AMDA0081.nffw
    fi
fi

FW="${FWDIR}/pci-${PCIADDR}.nffw"
ln -sf "${APP}/nic-${ASSY}.nffw" "${FW}"

# insert distro-specific initramfs section here...
```

```
./agilio-tc-fw-select.sh ens47np0 scan
rmmod nfp
modprobe nfp
```

VF      VF

```
# cat /sys/class/net/ens3/device/sriov_totalvfs
65

# echo 4 > /sys/class/net/ens47/device/sriov_numvfs
```

7.17.3 SR-IOV Device Plugin

VF          Pod      VF          SR-IOV Device Plugin

SR-IOV      Configmap

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
      "resourceList": [{
        "resourcePrefix": "corigine.com",
        "resourceName": "agilio_sriov",
        "selectors": {
          "vendors": ["19ee"],
          "devices": ["6003"],
          "drivers": ["nfp_netvf"]
        }
      }]
    }
```

SR-IOV          :

```
kubectl apply -f https://raw.githubusercontent.com/intel/sriov-network-device-plugin/master/deployments/k8s-v1.16/sriovdp-daemonset.yaml
```

SR-IOV          Kubernetes Node

```
kubectl describe no containerserver | grep corigine

corigine.com/agilio_sriov: 4
corigine.com/agilio_sriov: 4
corigine.com/agilio_sriov 0          0
```

7.17.4 Multus-CNI

|                                                                                                                                       |    |            |          |            |
|---------------------------------------------------------------------------------------------------------------------------------------|----|------------|----------|------------|
| SR-IOV Device Plugin                                                                                                                  | ID | Multus-CNI | Kube-OVN | Multus-CNI |
| Multus-CNI                                                                                                                            |    |            |          |            |
| <pre>kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/master/deployments/multus-daemonset.yml</pre> |    |            |          |            |

NetworkAttachmentDefinition

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: default
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/resourceName: corigine.com/agilio_srlov
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins":[
      {
        "type":"kube-ovn",
        "server_socket":"/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "default.default.ovn"
      },
      {
        "type":"portmap",
        "capabilities":{"
          "portMappings":true
        }
      }
    ]
  }'
```

• provider: NetworkAttachmentDefinition {name}.{namespace}.ovn

7.17.5 Kube-OVN

|                                                                                                        |  |
|--------------------------------------------------------------------------------------------------------|--|
| <pre>wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.15/dist/images/install.sh</pre> |  |
|--------------------------------------------------------------------------------------------------------|--|

IFACE IP

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE="ensp01"
```

Kube-OVN

```
bash install.sh
```

7.17.6 VF Pod

|      |    |      |
|------|----|------|
| yaml | VF | Pod: |
|------|----|------|

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: default
  annotations:
    v1.multus-cni.io/default-network: default/default
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          corigine.com/agilio_srlov: '1'
```

```
limits:
  corigine.com/agilio_sriov: '1'
```

• v1.multus-cni.io/default-network: NetworkAttachmentDefinition {namespace}/{name}

Pod ovs-ovn

```
# ovs-appctl dpctl/dump-flows -m type=offloaded
ufid:91cc45de-e7e9-4935-8f82-1890430b0f66, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/0),in_port(5b45c61b307e_h),packet_type(ns=0/0,id=0/0),eth(src=00:00:00:c5:6d:4e,dst=00:00:00:e7:16:ce),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no), packets:941539, bytes:62142230, used:0.260s, offloaded:yes, dp:tc, actions:54235e5753b8_h
ufid:e00768d7-e652-4d79-8182-3291d852b791, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/0),in_port(54235e5753b8_h),packet_type(ns=0/0,id=0/0),eth(src=00:00:00:e7:16:ce,dst=00:00:00:c5:6d:4e),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no), packets:82386659, bytes:115944854173, used:0.260s, offloaded:yes, dp:tc, actions:5b45c61b307e_h
```

offloaded:yes, dp:tc

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## 7.18 Offload

|          |     |     |     |     |  |           |     |
|----------|-----|-----|-----|-----|--|-----------|-----|
| Kube-OVN | OVS |     | CPU | CPU |  | metaScale | OVS |
| OVS      |     | CPU |     |     |  |           |     |

### Note

1. 2024
2. 1.11 Kube-OVN

### 7.18.1

- metaScale
- MCR
- BIOS SR-IOV VT-d

### 7.18.2

#### hw-offload Kube-OVN

1.

```
wget https://github.com/yunsilicon/kube-ovn/blob/release-1.11/dist/images/install.sh
```

1.

```
/opt/ovs-config/ovs-dpdk-config
```

```
# specify log level for ovs dpdk, the value is info or dbg, default is info
VLOG=info
# specify nic offload, the value is true or false, default is true
HW_OFFLOAD=true
# specify cpu mask for ovs dpdk, not specified by default
CPU_MASK=0x02
# specify socket memory, not specified by default
SOCKET_MEM="2048,2048"
# specify encap IP
ENCAP_IP=6.6.6.200/24
# specify pci device
DPDK_DEV=0000:b3:00.0
# specify mtu, default is 1500
PF_MTU=1500
# specify bond name if bond enabled, not specified by default
BR_PHY_BOND_NAME=bond0
```

1. Kube-OVN

```
bash install.sh
```

#### SR-IOV

1. metaScale ID b3:00.0:

```
[root@k8s-master ~]# lspci -d 1f67:
b3:00.0 Ethernet controller: Device 1f67:1111 (rev 02)
b3:00.1 Ethernet controller: Device 1f67:1111 (rev 02)
```

1. ID p3p1

```
ls -l /sys/class/net/ | grep b3:00:0
lrwxrwxrwx 1 root root 0 May  7 16:30 p3p1 -> ../../devices/pci0000:b2:000:b2:00.0/0000:b3:00.0/net/p3p1
```

## 1. VF

```
cat /sys/class/net/p3p1/device/sriov_totalvfs
512
```

## 1. VF

```
echo '10' > /sys/class/net/p3p1/device/sriov_numvfs
```

## 1. VF

```
lspci -d 1f67:
b3:00.0 Ethernet controller: Device 1f67:1111 (rev 02)
b3:00.1 Ethernet controller: Device 1f67:1111 (rev 02)
b3:00.2 Ethernet controller: Device 1f67:1112
b3:00.3 Ethernet controller: Device 1f67:1112
b3:00.4 Ethernet controller: Device 1f67:1112
b3:00.5 Ethernet controller: Device 1f67:1112
b3:00.6 Ethernet controller: Device 1f67:1112
b3:00.7 Ethernet controller: Device 1f67:1112
b3:01.0 Ethernet controller: Device 1f67:1112
b3:01.1 Ethernet controller: Device 1f67:1112
b3:01.2 Ethernet controller: Device 1f67:1112
b3:01.3 Ethernet controller: Device 1f67:1112
```

## 1. switchdev

```
devlink dev eswitch set pci/0000:b3:00.0 mode switchdev
```

## SR-IOV Device Plugin

## 1. SR-IOV ConfigMap

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
      "resourceList": [{
        "resourceName": "xsc_sriov",
        "resourcePrefix": "yunsilicon.com",
        "selectors": {
          "vendors": ["1f67"],
          "devices": ["1012", "1112"]
        }
      }]
    }
```

## 1. SR-IOV Device Plugin DevicePlugin

## 2. SR-IOV

```
# kubectl describe node <node name> | grep yunsilicon.com/xsc_sriov
yunsilicon.com/xsc_sriov: 10
yunsilicon.com/xsc_sriov: 10
yunsilicon.com/xsc_sriov 0 0
```

## Multus-CNI

## 1. Multus-CNI Multus-CNI

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/master/deployments/multus-daemonset.yml
```

## 1. NetworkAttachmentDefinition

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
```

```

metadata:
  name: sriov-net1
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/resourceName: yunsilicon.com/xsc_sriov
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins": [
      {
        "type": "kube-ovn",
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "sriov-net1.default.ovn"
      },
      {
        "type": "portmap",
        "capabilities": {
          "portMappings": true
        }
      }
    ]
  }'

```

## SR-IOV Pod

```

apiVersion: v1
kind: Pod
metadata:
  name: nginx
  annotations:
    v1.multus-cni.io/default-network: default/sriov-net1
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          yunsilicon.com/xsc_sriov: '1'
        limits:
          yunsilicon.com/xsc_sriov: '1'

```

## Offload

Pod `ovs-ovn`

```

ovs-appctl dpctl/dump-flows type=offloaded
flow-dump from pmd on cpu core: 9
ct_state(-new+est-rel+rpl+trk),ct_mark(0/0x3),recirc_id(0x2d277),in_port(15),packet_type(ns=0,id=0),eth(src=00:00:00:9d:fb:1a,dst=00:00:00:ce:cf:b9),eth_type(0x0800),ipv4(dst=10.16.0.14,frag=no), packets:6, bytes:588, used:7.276s, actions:ct(zone=4,nat),recirc(0x2d278)
ct_state(-new+est-rel+rpl+trk),ct_mark(0/0x3),recirc_id(0x2d275),in_port(8),packet_type(ns=0,id=0),eth(src=00:00:00:ce:cf:b9,dst=00:00:00:9d:fb:1a),eth_type(0x0800),ipv4(dst=10.16.0.18,frag=no), packets:5, bytes:490, used:7.434s, actions:ct(zone=6,nat),recirc(0x2d276)
ct_state(-new+est-rel+rpl+trk),ct_mark(0/0x1),recirc_id(0x2d276),in_port(8),packet_type(ns=0,id=0),eth(src=00:00:00:ce:cf:b9,dst=00:00:00:9d:fb:1a/01:00:00:00:00:00),eth_type(0x0800),ipv4(frag=no), packets:5, bytes:490, used:7.434s, actions:15
recirc_id(0),in_port(15),packet_type(ns=0,id=0),eth(src=00:00:00:9d:fb:1a/01:00:00:00:00:00,dst=00:00:00:ce:cf:b9),eth_type(0x0800),ipv4(dst=10.16.0.14/255.192.0.0,frag=no), packets:6, bytes:588, used:7.277s, actions:ct(zone=6,nat),recirc(0x2d277)
recirc_id(0),in_port(8),packet_type(ns=0,id=0),eth(src=00:00:00:ce:cf:b9/01:00:00:00:00:00,dst=00:00:00:9d:fb:1a),eth_type(0x0800),ipv4(dst=10.16.0.18/255.192.0.0,frag=no), packets:6, bytes:588, used:7.434s, actions:ct(zone=4,nat),recirc(0x2d275)
ct_state(-new+est-rel+rpl+trk),ct_mark(0/0x1),recirc_id(0x2d278),in_port(15),packet_type(ns=0,id=0),eth(dst=00:00:00:ce:cf:b9/01:00:00:00:00:00),eth_type(0x0800),ipv4(frag=no), packets:6, bytes:588, used:7.277s, actions:8

```

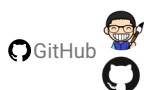

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7.18.3

7.19 Offload

 **Note**

2024

7.19.1

- 2200E
- HADOS
- BIOS SR-IOV

7.19.2

SR-IOV

1. 2200E vendor ID 1f47 ID 00:0a.0 00:0b.0 2200E

```
lspci | grep 1f47
00:0a.0 Ethernet controller: Device 1f47:1001 (rev 10)
00:0b.0 Ethernet controller: Device 1f47:1001 (rev 10)
```

1. VF

```
cat /sys/bus/pci/devices/0000\:00\:0a.0/sriov_totalvfs
256
```

1. VF VF

```
echo 7 > /sys/bus/pci/devices/0000\:00\:0a.0/sriov_numvfs
```

1. VF

```
lspci | grep 1f47
00:0a.0 Ethernet controller: Device 1f47:1001 (rev 10)
00:0a.1 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.2 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.3 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.4 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.5 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.6 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.7 Ethernet controller: Device 1f47:110f (rev 10)
00:0b.0 Ethernet controller: Device 1f47:1001 (rev 10)
```

SR-IOV Device Plugin

1. SR-IOV Configmap SR-IOV Device Plugin VF Pod

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
      "resourceList": [{
        "resourceName": "sriov_dpu",
        "resourcePrefix": "yusur.tech",
        "selectors": {
          "vendors": ["1f47"],
          "devices": ["110f"]
        }
      }]
    }
```

```
]
}
```

1. SR-IOV Device Plugin

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/sriov-network-device-plugin/v3.6.2/deployments/sriovdp-daemonset.yaml
```

1. SR-IOV kubernetes Node

```
kubectl describe node node1 | grep yusur
yusur.tech/sriov_dpu: 7
yusur.tech/sriov_dpu: 7
yusur.tech/sriov_dpu: 0 0
```

7.19.3 Multus-CNI

Multus-CNI Kube-OCN SRIOV Device ID

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/v4.0.2/deployments/multus-daemonset-thick.yml
```

NetworkAttachmentDefinition

```
apiVersion:
  "k8s.cni.cncf.io/v1"
kind:
  NetworkAttachmentDefinition
metadata:
  name: test
  namespace: kube-system
  annotations:
    k8s.v1.cni.cncf.io/resourceName: yusur.tech/sriov_dpu
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins": [
      {
        "type": "kube-ovn",
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "test.kube-system.ovn"
      },
      {
        "type": "portmap",
        "capabilities": {
          "portMappings": true
        }
      }
    ]
  }
}
```

• provider: NetworkAttachmentDefinition {name}.{namespace}.ovn

7.19.4 Kube-OVN

1.

```
wget https://github.com/kubeovn/kube-ovn/blob/release-1.12/dist/images/install.sh
```

1. IFACE IP

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE="p0"
```

1. kube-ovn

```
bash install.sh
```

| VF   | pod    |
|------|--------|
| yaml | VF Pod |



```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: default
  annotations:
    v1.multus-cni.io/default-network: kube-system/test
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          yusur.tech/sriov_dpu: '1'
        limits:
          yusur.tech/sriov_dpu: '1'
```

• v1.multus-cni.io/default-network: NetworkAttachmentDefinition {namespace}/{name}

Offload

Pod ovs-ovn

```
# ovs-appctl dpctl/dump-flows -m type=offloaded
ufid:67c2e10f-92d4-4574-be70-d072815ff166, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0),recirc_id(0),dp_hash(0/0),in_port(d85b161b6840_h),packet_type(ns=0/0,id=0/0),eth(src=0a:c9:1c:70:01:09,dst=8a:18:a4:22:b7:7d),eth_type(0x0800),ipv4(src=10.0.1.10,dst=10.0.1.6,proto=6,tos=0/0x3,ttl=0/0,frag=no),tcp(src=60774,dst=9001), packets:75021, bytes:109521630, offload_packets:75019, offload_bytes:109521498, used: 3.990s, offloaded:yes, dp:tc, actions:set(tunnel(tun_id=0x5,dst=192.168.201.12,ttl=64,tp_dst=6081,geneve({class=0x102,type=0x80,len=4,0xa0006})), flags(csum|key))), genev_sys_6081
ufid:7940666e-a0bd-42a5-8116-1e84e81bb338, skb_priority(0/0),tunnel(tun_id=0x5,src=192.168.201.12,dst=192.168.201.11,ttl=0/0,tp_dst=6081,geneve({class=0x102,type=0x80,len=4,0x6000a})), flags(+key)),skb_mark(0/0),ct_state(0/0),ct_zone(0/0),ct_mark(0/0),ct_label(0/0),recirc_id(0),dp_hash(0/0),in_port(genev_sys_6081),packet_type(ns=0/0,id=0/0),eth(src=8a:18:a4:22:b7:7d,dst=0a:c9:1c:70:01:09),eth_type(0x0800),ipv4(src=10.0.1.6,dst=10.0.1.10,proto=6,tos=0/0,ttl=0/0,frag=no),tcp(src=9001,dst=60774), packets:6946, bytes:459664, offload_packets:6944, offload_bytes:459532, used: 4.170s, dp:tc, offloaded:yes, actions:d85b161b6840_h
```

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7.19.5

## 7.20 DPDK

Kube-OVN   OVS-DPDK   KubeVirt   DPDK

KubeVirt   OVS-DPDK   patchVhostuser implementation   KubeVirt   KVM Device Plugin   OVS-DPDK

### 7.20.1

- DPDK
- Hugepages

### 7.20.2 DPDK

driverctl   DPDK

```
driverctl set-override 0000:00:0b.0 uio_pci_generic
```

### 7.20.3

OVS-DPDK   Kube-OVN

```
kubectl label nodes <node> ovn.kubernetes.io/ovs_dp_type="userspace"
```

OVS-DPDK   /opt/ovs-config   ovs-dpdk-config

```
ENCAP_IP=192.168.122.193/24
DPDK_DEV=0000:00:0b.0
```

- ENCAP\_IP:
- DPDK\_DEV:   PCI ID

### 7.20.4 Kube-OVN

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.15/dist/images/install.sh
```

DPDK

```
bash install.sh --with-hybrid-dpdk
```

### 7.20.5

vhostuser   OVS-DPDK

KVM Device Plugin   KVM Device Plugin

```
kubectl apply -f https://raw.githubusercontent.com/kubevirt/kubernetes-device-plugins/master/manifests/kvm-ds.yml
```

NetworkAttachmentDefinition

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-dpdk
  namespace: default
spec:
  config: >-
    {
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
```

```

    "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
    "provider": "ovn-dpdk.default.ovn",
    "vhost_user_socket_volume_name": "vhostuser-sockets",
    "vhost_user_socket_name": "sock"
}

```

## Dockerfile VM

```

FROM quay.io/kubevirt/virt-launcher:v0.46.1

# wget http://cloud.centos.org/centos/7/images/CentOS-7-x86_64-GenericCloud.qcow2
COPY CentOS-7-x86_64-GenericCloud.qcow2 /var/lib/libvirt/images/CentOS-7-x86_64-GenericCloud.qcow2

```

```

apiVersion: v1
kind: ConfigMap
metadata:
  name: vm-config
data:
  start.sh: |
    chmod u+w /etc/libvirt/qemu.conf
    echo "hugetlbfs_mount = \"/dev/hugepages\" >> /etc/libvirt/qemu.conf
    virtlogd &
    libvirtd &

    mkdir /var/lock

    sleep 5

    virsh define /root/vm/vm.xml
    virsh start vm

  tail -f /dev/null
  vm.xml: |
    <domain type='kvm'>
      <name>vm</name>
      <uuid>4a9b3f53-fa2a-47f3-a757-dd87720d9d1d</uuid>
      <memory unit='KiB'>2097152</memory>
      <currentMemory unit='KiB'>2097152</currentMemory>
      <memoryBacking>
        <hugepages>
          <page size='2' unit='M' nodeset='0' />
        </hugepages>
      </memoryBacking>
      <vcpu placement='static'>2</vcpu>
      <cputune>
        <shares>4096</shares>
        <vcpupin vcpu='0' cpuset='4' />
        <vcpupin vcpu='1' cpuset='5' />
        <emulatorpin cpuset='1,3' />
      </cputune>
      <os>
        <type arch='x86_64' machine='pc'>hvm</type>
        <boot dev='hd' />
      </os>
      <features>
        <acpi />
        <apic />
      </features>
      <cpu mode='host-model'>
        <model fallback='allow' />
        <topology sockets='1' cores='2' threads='1' />
        <numa>
          <cell id='0' cpus='0-1' memory='2097152' unit='KiB' memAccess='shared' />
        </numa>
      </cpu>
      <on_reboot>restart</on_reboot>
      <devices>
        <emulator>usr/libexec/qemu-kvm</emulator>
        <disk type='file' device='disk'>
          <driver name='qemu' type='qcow2' cache='none' />
          <source file='/var/lib/libvirt/images/CentOS-7-x86_64-GenericCloud.qcow2' />
          <target dev='vda' bus='virtio' />
        </disk>

        <interface type='vhostuser'>
          <mac address='00:00:00:0A:30:89' />
          <source type='unix' path='/var/run/vm/sock' mode='server' />
          <model type='virtio' />
          <driver queues='2'>
            <host mrg_rxbuf='off' />
          </driver>
        </interface>
        <serial type='pty'>
          <target type='isa-serial' port='0'>
            <model name='isa-serial' />
          </target>
        </serial>
        <console type='pty'>
          <target type='serial' port='0' />
        </console>
      </devices>
    </domain>

```

```

    </console>
    <channel type='unix'>
      <source mode='bind' path='/var/lib/libvirt/qemu/channel/target/domain-1-vm/org.qemu.guest_agent.0' />
      <target type='virtio' name='org.qemu.guest_agent.0' state='connected' />
      <alias name='channel0' />
      <address type='virtio-serial' controller='0' bus='0' port='1' />
    </channel>

  </devices>
</domain>
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: vm-deployment
  labels:
    app: vm
spec:
  replicas: 1
  selector:
    matchLabels:
      app: vm
  template:
    metadata:
      labels:
        app: vm
    annotations:
      k8s.v1.cnf.cncf.io/networks: default/ovn-dpdk
      ovn-dpdk.default.ovn.kubernetes.io/ip_address: 10.16.0.96
      ovn-dpdk.default.ovn.kubernetes.io/mac_address: 00:00:00:0A:30:89
  spec:
    nodeSelector:
      ovn.kubernetes.io/ovs_dp_type: userspace
    securityContext:
      runAsUser: 0
    volumes:
      - name: vhostuser-sockets
        emptyDir: {}
      - name: xml
        configMap:
          name: vm-config
      - name: hugepage
        emptyDir:
          medium: HugePages-2Mi
      - name: libvirt-runtime
        emptyDir: {}
    containers:
      - name: vm
        image: vm-vhostuser:latest
        command: ["bash", "/root/vm/start.sh"]
        securityContext:
          capabilities:
            add:
              - NET_BIND_SERVICE
              - SYS_NICE
              - NET_RAW
              - NET_ADMIN
          privileged: false
          runAsUser: 0
        resources:
          limits:
            cpu: '2'
            devices.kubevirt.io/kvm: '1'
            memory: '8784969729'
            hugepages-2Mi: 2Gi
          requests:
            cpu: 666m
            devices.kubevirt.io/kvm: '1'
            ephemeral-storage: 50M
            memory: '4490002433'
        volumeMounts:
          - name: vhostuser-sockets
            mountPath: /var/run/vm
          - name: xml
            mountPath: /root/vm/
          - mountPath: /dev/hugepages
            name: hugepage
          - name: libvirt-runtime
            mountPath: /var/run/libvirt

```

## Pod

```

# virsh set-user-password vm root 12345
Password set successfully for root in vm

# virsh console vm
Connected to domain 'vm'
Escape character is ^] (Ctrl + ])

CentOS Linux 7 (Core)
Kernel 3.10.0-1127.el7.x86_64 on an x86_64

```

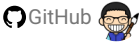
```
localhost login: root
Password:
Last login: Fri Feb 25 09:52:54 on ttyS0
```

```
ip link set eth0 mtu 1400
ip addr add 10.16.0.96/16 dev eth0
ip ro add default via 10.16.0.1
ping 114.114.114.114
```



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7.20.6

---

## 7.21 OpenStack

|     |           |            |  |                   |     |          |
|-----|-----------|------------|--|-------------------|-----|----------|
|     | OpenStack | Kubernetes |  | OpenStack Neutron | OVN | Kube-OVN |
| OVN | OpenStack | Kubernetes |  |                   |     |          |

### 7.21.1

|        |            |           |            |
|--------|------------|-----------|------------|
| OVN-IC | Kubernetes | OpenStack | Kubernetes |
|--------|------------|-----------|------------|

- OpenStack    Kubernetes    CIDR
- 
- IP
- Kubernetes    OpenStack    VPC

#### OVN-IC

OVN-IC

```
docker run --name=ovn-ic-db -d --network=host -v /etc/ovn:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/log/ovn:/var/log/ovn kubeovn/kube-ovn:v1.15.21 bash start-ic-db.sh
```

#### Kubernetes

kube-system Namespace    ovn-ic-config ConfigMap

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "true"
```

- enable-ic:
- az-name:
- ic-db-host: OVN-IC
- ic-nb-port: OVN-IC      6645
- ic-sb-port: OVN-IC      6646
- gw-nodes:
- auto-route:

#### OpenStack

Kubernetes

```
# openstack router create router0
# openstack router list
+-----+-----+-----+-----+-----+
| ID | Name | Status | State | Project |
+-----+-----+-----+-----+-----+
| d5b38655-249a-4192-8046-71aa4d2b4af1 | router0 | ACTIVE | UP | 98a29ab7388347e7b5ff8bdd181ba4f9 |
+-----+-----+-----+-----+-----+
```

## OpenStack OVN

```
ovn-nbctl set NB_Global . name=op-az
```

OVN-IC

OVN-IC

```
/usr/share/ovn/scripts/ovn-ctl --ovn-ic-nb-db=tcp:192.168.65.3:6645 \
--ovn-ic-sb-db=tcp:192.168.65.3:6646 \
--ovn-northd-nb-db=unix:/run/ovn/ovnnb_db.sock \
--ovn-northd-sb-db=unix:/run/ovn/ovnsb_db.sock \
start_ic
```

- `ovn-ic-nb-db` `ovn-ic-sb-db` : OVN-IC
- `ovn-northd-nb-db` `ovn-northd-sb-db` : OVN

```
ovs-vsctl set open_vswitch . external_ids:ovn-is-interconn=true
```

## OpenStack OVN

ts router0

```
ovn-nbctl lrp-add router0 lrp-router0-ts 00:02:ef:11:39:4f 169.254.100.73/24
ovn-nbctl lsp-add ts lsp-ts-router0 -- lsp-set-addresses lsp-ts-router0 router \
-- lsp-set-type lsp-ts-router0 router \
-- lsp-set-options lsp-ts-router0 router-port=lrp-router0-ts
ovn-nbctl lrp-set-gateway-chassis lrp-router0-ts {gateway chassis} 1000
ovn-nbctl set NB_Global . options:ic-route-adv=true options:ic-route-learn=true
```

## Kubernetes

```
# ovn-nbctl lr-route-list router0
IPv4 Routes
      10.0.0.22          169.254.100.34 dst-ip (learned)
      10.16.0.0/16       169.254.100.34 dst-ip (learned)
```

router0

Kubernetes Pod

## 7.21.2 OVN

OpenStack Kubernetes OVN VPC Subnet

Kube-OVN OVN OpenStack Neutron OVN OpenStack networking-ovn Neutron

## Neutron

Neutron /etc/neutron/plugins/ml2/ml2\_conf.ini

```
[ovn]
...
ovn_nb_connection = tcp:[192.168.137.176]:6641,tcp:[192.168.137.177]:6641,tcp:[192.168.137.178]:6641
ovn_sb_connection = tcp:[192.168.137.176]:6642,tcp:[192.168.137.177]:6642,tcp:[192.168.137.178]:6642
ovn_l3_scheduler = OVN_L3_SCHEDULER
```

- `ovn_nb_connection` `ovn_sb_connection` : Kube-OVN `ovn-central`

## OVS

```
ovs-vsctl set open . external-ids:ovn-remote=tcp:[192.168.137.176]:6642,tcp:[192.168.137.177]:6642,tcp:[192.168.137.178]:6642
ovs-vsctl set open . external-ids:ovn-encap-type=geneve
ovs-vsctl set open . external-ids:ovn-encap-ip=192.168.137.200
```

- `external-ids:ovn-remote` : Kube-OVN `ovn-central`
- `ovn-encap-ip` : IP

## Kubernetes OpenStack

Kubernetes OpenStack OpenStack Pod



## Note

```
kube-ovn-controller args --enable-external-vpc=true
```

## OpenStack

```
# openstack router list
+-----+-----+-----+-----+-----+
| ID | Name | Status | State | Project |
+-----+-----+-----+-----+-----+
| 22040ed5-0598-4f77-bffd-e7fd4db47e93 | router0 | ACTIVE | UP | 62381a21d569404aa236a5dd8712449c |
+-----+-----+-----+-----+-----+

# openstack network list
+-----+-----+-----+
| ID | Name | Subnets |
+-----+-----+-----+
| cd59e36a-37db-4c27-b709-d35379a7920f | provider | 01d73d9f-fdaa-426c-9b60-aa34abbfacae |
+-----+-----+-----+

# openstack subnet list
+-----+-----+-----+-----+
| ID | Name | Network | Subnet |
+-----+-----+-----+-----+
| 01d73d9f-fdaa-426c-9b60-aa34abbfacae | provider-v4 | cd59e36a-37db-4c27-b709-d35379a7920f | 192.168.1.0/24 |
+-----+-----+-----+-----+

# openstack server list
+-----+-----+-----+-----+-----+-----+
| ID | Name | Status | Networks | Image | Flavor |
+-----+-----+-----+-----+-----+-----+
| 8433d622-a8d6-41a7-8b31-49abfd64f639 | provider-instance | ACTIVE | provider=192.168.1.61 | ubuntu | m1 |
+-----+-----+-----+-----+-----+-----+
```

## Kubernetes VPC

```
# kubectl get vpc
NAME                                STANDBY SUBNETS
neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93 true ["neutron-cd59e36a-37db-4c27-b709-d35379a7920f"]
ovn-cluster                         true ["join", "ovn-default"]
```

neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93 OpenStack VPC

Kube-OVN VPC Subnet Pod

VPC, Subnet Namespace net2 Pod:

```
apiVersion: v1
kind: Namespace
metadata:
  name: net2
---
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  creationTimestamp: "2021-06-20T13:34:11Z"
  generation: 2
  labels:
    ovn.kubernetes.io/vpc_external: "true"
  name: neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93
  resourceVersion: "583728"
  uid: 18d4c654-f511-4def-a3a0-a6434d237c1e
spec:
  namespaces:
    - net2
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net2
spec:
  vpc: neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93
  namespaces:
    - net2
  cidrBlock: 12.0.1.0/24
  natOutgoing: false
---
apiVersion: v1
kind: Pod
metadata:
  name: ubuntu
```

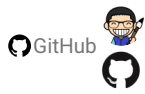


```
namespace: net2
spec:
  containers:
  - image: docker.io/kubeovn/kube-ovn:v1.8.0
    command:
    - "sleep"
    - "604800"
    imagePullPolicy: IfNotPresent
    name: ubuntu
  restartPolicy: Always
```



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7.21.3

---

7.22 IPsec

|         |         |      |
|---------|---------|------|
| v1.13.0 | UDP 500 | 4500 |
|---------|---------|------|

7.22.1

|                       |                           |                     |                     |         |              |       |
|-----------------------|---------------------------|---------------------|---------------------|---------|--------------|-------|
| kube-ovn-cni<br>ipsec | certificatesigningrequest | kube-ovn-controller | kube-ovn-controller | approve | kube-ovn-cni | ipsec |
|-----------------------|---------------------------|---------------------|---------------------|---------|--------------|-------|

7.22.2 IPsec

kube-ovn-controller

kube-ovn-cni

args --enable-ovn-ipsec=false

--enable-ovn-ipsec=true

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7.22.3

7.23 OVN

|          |            |
|----------|------------|
| Pod      | GRE/ERSPAN |
| Kube-OVN | v1.12      |

7.23.1 Multus-CNI

|            |        |
|------------|--------|
| Multus-CNI | Multus |
|------------|--------|

7.23.2

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: attachnet
  namespace: default
spec:
  config: |
    {
      "cniVersion": "0.3.1",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "attachnet.default.ovn"
    }

provider <NAME>.<NAMESPACE>.ovn
```

7.23.3 Underlay

|          |         |          |
|----------|---------|----------|
| MTU      | LSP/Pod | Underlay |
| Underlay |         |          |

```
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: net1
spec:
  defaultInterface: eth1
---
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan1
spec:
  id: 0
  provider: net1
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 172.19.0.0/16
  excludeIps:
    - 172.19.0.2..172.19.0.20
  gateway: 172.19.0.1
  vlan: vlan1
  provider: attachnet.default.ovn

provider provider
```

7.23.4 Pod

```
Pod

apiVersion: v1
kind: Pod
```

```

metadata:
  name: pod1
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
spec:
  containers:
    - name: bash
      image: docker.io/kubeovn/kube-ovn:v1.15.21
      args:
        - bash
        - -c
        - sleep infinity
  securityContext:
    privileged: true

```

#### Pod IP

```

$ kubectl get ips | grep pod1
pod1.default          10.16.0.12   00:00:00:FF:34:24   kube-ovn-worker   ovn-default
pod1.default.attachnet.default.ovn  172.19.0.21  00:00:00:A0:30:68   kube-ovn-worker   subnet1

```

IP 172.19.0.21

## 7.23.5 OVN

### OVN

```

kubectl ko nbctl mirror-add mirror1 gre 99 from-lport 172.19.0.21
kubectl ko nbctl lsp-attach-mirror coredns-787d4945fb-gpnkb.kube-system mirror1

```

coredns-787d4945fb-gpnkb.kube-system OVN LSP <POD\_NAME>.<POD\_NAMESPACE>

### OVN

```

ovn-nbctl mirror-add <NAME> <TYPE> <INDEX> <FILTER> <IP>

NAME - add a mirror with given name
TYPE - specify TYPE 'gre' or 'erspan'
INDEX - specify the tunnel INDEX value
        (indicates key if GRE, erspan_idx if ERSPAN)
FILTER - specify FILTER for mirroring selection
        ('to-lport' / 'from-lport')
IP - specify Sink / Destination i.e. Remote IP

ovn-nbctl mirror-del [NAME]          remove mirrors
ovn-nbctl mirror-list                print mirrors

ovn-nbctl lsp-attach-mirror PORT MIRROR attach source PORT to MIRROR
ovn-nbctl lsp-detach-mirror PORT MIRROR detach source PORT from MIRROR

```

## 7.23.6 Pod

### Pod

```

root@pod1:/kube-ovn# ip link add mirror1 type gretap local 172.19.0.21 key 99 dev net1
root@pod1:/kube-ovn# ip link set mirror1 up

```

### Pod

```

root@pod1:/kube-ovn# tcpdump -i mirror1 -nnve
tcpdump: listening on mirror1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
05:13:30.328008 00:00:00:a3:f5:e2 > 00:00:00:97:0f:6e, ethertype ARP (0x0806), length 42: Ethernet (len 6), IPv4 (len 4), Request who-has 10.16.0.7 tell 10.16.0.4, length 28
05:13:30.559167 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 212: (tos 0x0, ttl 64, id 57364, offset 0, flags [DF], proto UDP (17), length 198)
10.16.0.4.53 > 10.16.0.6.50472: 34511 NXDomain*- 0/1/1 (170)
05:13:30.559343 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 212: (tos 0x0, ttl 64, id 57365, offset 0, flags [DF], proto UDP (17), length 198)
10.16.0.4.53 > 10.16.0.6.45177: 1659 NXDomain*- 0/1/1 (170)
05:13:30.560625 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 200: (tos 0x0, ttl 64, id 57367, offset 0, flags [DF], proto UDP (17), length 186)
10.16.0.4.53 > 10.16.0.6.43848: 2636*- 0/1/1 (158)
05:13:30.562774 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 191: (tos 0x0, ttl 64, id 57368, offset 0, flags [DF], proto UDP (17), length 177)
10.16.0.4.53 > 10.16.0.6.37755: 48737 NXDomain*- 0/1/1 (149)
05:13:30.563523 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 187: (tos 0x0, ttl 64, id 57369, offset 0, flags [DF], proto UDP (17), length 173)
10.16.0.4.53 > 10.16.0.6.53887: 45519 NXDomain*- 0/1/1 (145)
05:13:30.564940 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 201: (tos 0x0, ttl 64, id 57370, offset 0, flags [DF], proto UDP (17),

```

```
length 187)
10.16.0.4.53 > 10.16.0.6.40846: 25745 NXDomain*- 0/1/1 (159)
05:13:30.565140 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 201: (tos 0x0, ttl 64, id 57371, offset 0, flags [DF], proto UDP (17),
length 187)
10.16.0.4.53 > 10.16.0.6.45214: 61875 NXDomain*- 0/1/1 (159)
05:13:30.566023 00:00:00:a3:f5:e2 > 00:00:00:55:e4:4e, ethertype IPv4 (0x0800), length 80: (tos 0x0, ttl 64, id 45937, offset 0, flags [DF], proto UDP (17),
length 66)
10.16.0.4.44116 > 172.18.0.1.53: 16025+ [1au] AAAA? kube-ovn.io. (38)
```

7.23.7

1.      ERSPAN              OVN              Linux              4.14      ERSPAN              IPv6              Linux              4.16
2.                              OVN

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7.23.8

7.24 DNS Kube-OVN

NodeLocal DNSCache DaemonSet DNS DNS Kube-OVN

7.24.1 DNS

Kubernetes DNS

Kubernetes Nodelocaldns

```
#!/bin/bash

localdns=169.254.20.10
domain=cluster.local
kubedns=10.96.0.10

wget https://raw.githubusercontent.com/kubernetes/kubernetes/master/cluster/addons/dns/nodelocaldns/nodelocaldns.yaml
sed -i "s/___PILLAR__LOCAL__DNS___/$localdns/g; s/___PILLAR__DNS__DOMAIN___/$domain/g; s/___PILLAR__DNS__SERVER___/g; s/___PILLAR__CLUSTER__DNS___/$kubedns/g"
nodelocaldns.yaml

kubectl apply -f nodelocaldns.yaml
```

kubelet /var/lib/kubelet/config.yaml clusterDNS DNS IP 169.254.20.10 kubelet

Kube-OVN DNS

Kubernetes Nodelocal DNSCache Kube-OVN

UNDERLAY SUBNET U2O

Underlay Subnet DNS U2O kubectl edit subnet {your subnet} spec.u2oInterconnection = true , Overlay Subnet

KUBE-OVN-CONTROLLER DNS IP

```
kubectl edit deployment kube-ovn-controller -n kube-system
```

```
spec.template.spec.containers.args --node-local-dns-ip=169.254.20.10
```

POD

Pod /etc/resolv.conf nameserver DNS IP Pod nameserver DNS ClusterIP u2o Pod Pod

7.24.2 DNS

Pod Pod DNS 169.254.20.10

```
# kubectl exec -it pod1 -- nslookup github.com
Server: 169.254.20.10
Address: 169.254.20.10:53

Name: github.com
Address: 20.205.243.166
```

DNS ovn0 DNS DNS

```
# tcpdump -i any port 53

06:20:00.441889 659246098c56_h P ifindex 17 00:00:00:73:f1:06 ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1291+ A? baidu.com. (27)
06:20:00.441889 ovn0 In ifindex 7 00:00:00:50:32:cd ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1291+ A? baidu.com. (27)
06:20:00.441950 659246098c56_h P ifindex 17 00:00:00:73:f1:06 ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1611+ AAAA? baidu.com. (27)
06:20:00.441950 ovn0 In ifindex 7 00:00:00:50:32:cd ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1611+ AAAA? baidu.com. (27)
06:20:00.442203 ovn0 Out ifindex 7 00:00:00:52:99:d8 ethertype IPv4 (0x0800), length 145: 169.254.20.10.53 > 10.16.0.2.40230: 1611* 0/1/0 (97)
```

```
06:20:00.442219 659246098c56_h Out ifindex 17 00:00:00:ea:b3:5e ethertype IPv4 (0x0800), length 145: 169.254.20.10.53 > 10.16.0.2.40230: 1611* 0/1/0 (97)
06:20:00.442273 ovn0 Out ifindex 7 00:00:00:52:99:d8 ethertype IPv4 (0x0800), length 125: 169.254.20.10.53 > 10.16.0.2.40230: 1291* 2/0/0 A 39.156.66.10, A
110.242.68.66 (77)
06:20:00.442278 659246098c56_h Out ifindex 17 00:00:00:ea:b3:5e ethertype IPv4 (0x0800), length 125: 169.254.20.10.53 > 10.16.0.2.40230: 1291* 2/0/0 A 39.
156.66.10, A 110.242.68.66 (77)
```

7.24.3



NetworkPolicy   NetworkPolicy   DNS IP   169.254.20.10   CIDR   NetworkPolicy   DNS   Pod

NetworkPolicy

Pod   DNS   NetworkPolicy

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-local-dns-and-node-cidr
  namespace: default #
spec:
  podSelector: {} # Pod
  policyTypes:
    - Ingress
    - Egress
  egress:
    # DNS
    - to:
      - ipBlock:
          cidr: 169.254.20.10/32
        # CIDR CIDR
    - to:
      - ipBlock:
          cidr: 10.0.0.0/8 # CIDR
  ingress:
    # DNS
    - from:
      - ipBlock:
          cidr: 169.254.20.10/32
        # CIDR CIDR
    - from:
      - ipBlock:
          cidr: 10.0.0.0/8 # CIDR
```

- 169.254.20.10/32   DNS   IP
- 10.0.0.0/8   CIDR

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7.24.4

## 7.25 VPC NAT

### 7.25.1

|     |         |             |        |      |      |    |  |     |      |
|-----|---------|-------------|--------|------|------|----|--|-----|------|
| VPC | Overlay | natOutgoing | Subnet | Pod  | SNAT | IP |  | Pod | SNAT |
| NAT |         | CIDR        | IP     | SNAT |      |    |  |     |      |

### 7.25.2

subnet.SpecnatOutgoingnatOutgoingPolicyRules

```
spec:
  natOutgoing: true
  natOutgoingPolicyRules:
    - action: forward
      match:
        srcIPs: 10.0.11.0/30,10.0.11.254
    - action: nat
      match:
        srcIPs: 10.0.11.128/26
        dstIPs: 114.114.114.114,8.8.8.8
```

NAT

1. IP 10.0.11.0/30 10.0.11.254 SNAT
2. IP 10.0.11.128/26 IP 114.114.114.114 8.8.8.8 SNAT

actionmatchaction, actionforwardnatforwardSNAT, natSNATnatOutgoingPolicyRules

SNAT

matchsrcIPsdstIPsIPIP match.srcIPsmatch.dstIPsCIDRIP

matchnatOutgoingPolicyRulesaction

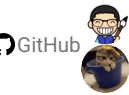
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### 7.25.3



# 8.

## 8.1

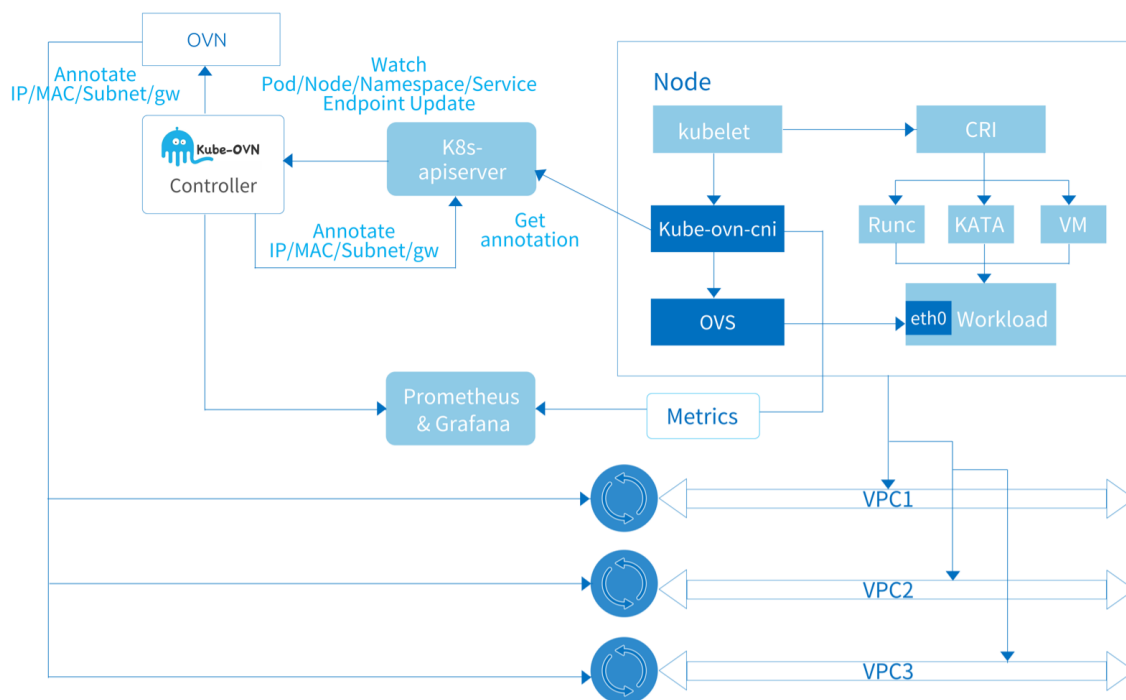
Kube-OVN

Kube-OVN    Kubernetes    OVN    SDN    Kube-OVN    OVN    Kubernetes    CNI Service    Networkpolicy  
SDN    VPC    QoS    ACL  
Kube-OVN    Cilium    Submariner    Prometheus    KubeVirt

### 8.1.1

Kube-OVN

- OVN/OVS
- Agent
- 



OVN/OVS

OVN/OVS    Kube-OVN    OVN/OVS    SDN    Kube-OVN    [ovn-architecture\(7\)](#)  
OVN    Kube-OVN    OVN    Kubernetes  
OVN/OVS    Kubernetes

OVN-CENTRAL

ovn-central Deployment OVN ovn-nb, ovn-sb, ovn-northd

- ovn-nb API kube-ovn-controller ovn-nb
- ovn-sb ovn-nb
- ovn-northd ovn-nb ovn-sb

ovn-central Raft

OVS-OVN

ovs-ovn DaemonSet Pod openvswitch, ovssdb, ovn-controller ovn-central Agent

Agent

Kube-OVN OVN Kubernetes

KUBE-OVN-CONTROLLER

Deployment Kubernetes OVN Kube-OVN kube-ovn-controller OVN

Pod Service Endpoint Node NetworkPolicy VPC Subnet Vlan ProviderNetwork

Pod kube-ovn-controller Pod IPAM ovn-central ACL kube-ovn-controller

CIDR Pod annotation annotation kube-ovn-cni

KUBE-OVN-CNI

DaemonSet CNI OVS

DaemonSet kube-ovn kubelet kube-ovn-cni CNI kube-ovn-cni /opt/cni/bin

kube-ovn-cni

- 1. ovn-controller vswitchd
- 2. CNI add/del
  - a. veth OVS
  - b. OVS
  - c. iptables/ipset/route
- 3. QoS.
- 4. ovn0
- 5. Vlan/Underlay/EIP
- 6.

Kube-OVN

KUBE-OVN-SPEAKER

DaemonSet Pod IP

BGP

KUBE-OVN-PINGER

DaemonSet OVS Kube-OVN

KUBE-OVN-MONITOR

Deployment OVN Kube-OVN

KUBECTL-KO

kubectl

kubectl

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8.1.2

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## 8.2 What's Next

---

This document lists the features merged into the master branch for the next minor release.

### 8.2.1 Post-v1.15.0

---

- NetworkPolicy now supports provider-scoped policies for multi-network pods using the `ovn.kubernetes.io/policy-for` annotation. [#6223](#)
- Support static IP/MAC for multiple interfaces on the same logical switch. [#6060](#)
- MetalLB underlay integration now supports IPv6 and dual-stack. [#6159](#)
- KubeVirt live-migration multi-chassis options now apply to all VM NICs, not just the primary one. [#6241](#)
- Add human-readable descriptions to all Kube-OVN CRD fields for better `kubectl explain` output. [#6133](#) [#6147](#)
- Support BGP and EVPN (L3VPN) for VPC Egress Gateway, running FRR in the egress gateway Pod. [#6224](#)
- Support per-pod NIC granular DHCP control via annotations, overriding subnet-level DHCP settings. [#6475](#)
- Extend SecurityGroup API with tiers, larger priority range (1-16384), localAddress and port matches. [#6330](#)
- Use BigInt for SubnetStatus IP count fields to support large address ranges. [#6403](#)
- Improve hairpin logic to support FIP, SNAT from any CIDR within the VPC and LoadBalancers. [#6445](#)
- VPC NAT Gateway
  - Support user-defined annotations on NAT gateway Pod template. [#6256](#)
  - Allow SNAT EIP to FIP EIP traffic. [#6329](#)
  - Allow any EIP to share external subnet gateway in one native VLAN. [#6426](#)
- VPC Egress Gateway
  - Support custom resources and bandwidth limits. [#6407](#)
  - Accept integer Mbps values and `M`, `Mi`, `G`, or `Gi` bit-rate quantities for per-replica bandwidth limits. [#7124](#)
- Interconnection
  - Add vendor ID to transit switches to avoid conflicts with other OVN controllers. [#6186](#)
- Reliability
  - OpenFlow synchronization: detect and recover from stale or missing OVS flows automatically. [#6117](#)
  - OVN DB: back up Raft header before rejoining cluster to improve recovery. [#6106](#)
- Performance
  - Strip `managedFields` from informer cache to reduce memory usage. [#6119](#)
  - Add field selectors to informer factory to reduce API server load. [#6091](#)
- Security
  - Replace wildcard RBAC verbs with explicit verb lists. [#6233](#)
  - Specify ephemeral storage limits for containers. [#6259](#)
- Helm Chart
  - Make DaemonSet update strategy configurable via `values.yaml`. [#6136](#)
  - Introduce `extraEnv` variable for all components. [#6142](#)
  - Add `affinity` and `nodeSelector` support for ovs-ovn and ovs-ovn-dpdk DaemonSets. [#6308](#)
  - Add `external-gateway-config-ns` option for controller. [#6211](#)
  - Introduce ServiceMonitor and labels to services. [#6340](#)

### 8.2.2 Post-v1.14.0

---

- ACL log supports ratelimiting. [#5938](#)

- Subnet with centralized gateway now supports nodeSelectors. [#5956](#)
- Overlay encapsulation NIC selection. [#5946](#)
- Performance: skip conntrack for specific dst CIDRs. [#5821](#)
- NetworkPolicy supports `lax` mode which only deny traffic type of TCP, UDP and SCTP. That means ARP, ICMP and DHCP traffic are always allowed. [#5745](#)
- Remove internal-port type interface code. [#5794](#)
- IPPool
- Multiple IPPools now can bind to the same Namespace. [#5731](#)
- Pods in a bound namespace will only get IPs from the bound pool(s), not other ranges in the subnet. [#5731](#)
- IPPool will create an AddressSet that can be work with VPC Policy Route and ACL. [#5920](#)
- `AdminNetworkPolicy` now supports specify egress peers using FQDNs. [#5703](#)
- Using ARP for IPv4 network ready check: now you don't need ACL allow rules for gateway to make Pod running. [#5716](#)
- Non-primary CNI mode: you can run Kube-OVN as the secondary only network, without annoying unused annotations and logical switch port allocations. [#5618](#)
- VPC NAT Gateway:
  - No default EIP mode: the secondary interface can initialize without a default EIP to avoid the waste. [#5605](#)
  - Custom routes: you can control the route rules within the vpc-nat-gateway Pods to control traffic paths. [#5608](#)
  - Gratuitous ARP: VPC NAT Gateway automatically sends gratuitous ARP packets during initialization to accelerate network convergence. [#5607](#)
  - Healthchecks for static endpoints in `SwitchLBRules`: SLR with both selector or endpoints key can support healthchecks. [#5435](#)
- Underlay
  - Node Selectors for `ProviderNetwork`: instead of adding/removing nodes to the `ProviderNetwork` one by one, you can use node selectors to simplify the workflow. [#5518](#)
  - Different `NetworkProvider`s can now share the same VLAN. [#5471](#)
  - Auto create VLAN sub-interfaces. [#5966](#)
  - Auto move VLAN sub-interfaces to OVS bridges. [#5949](#)
  - Adding `pod_name` and `pod_namespace` labels to interface metrics. [#5463](#)
- IPsec
  - Support `cert-manager` to issue certificates. [#5365](#)
  - Request new certificate if current certificate is not trusted. [#5710](#)
- kubectl-ko
  - Collect IPsec and xfrm information. [#5472](#)
  - Replace `Endpoint` with `EndpointSlice`. [#5425](#)
  - NetworkAttachment caching: reduce APIServer load in large-scale deployments with Multus. [#5386](#)
  - Upgrade `ovs` to 3.5 and `ovn` to 25.03. [#5537](#)



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8.2.3

---

8.3

Kube-OVN

8.3.1

Kube-OVN

1. default
- Pod IP
- CIDR
- 10.16.0.0/16
- 10.16.0.1
2. join
- Node Pod
- ,
- CIDR
- 100.64.0.0/16
- 100.64.0.1

```
POD_CIDR="10.16.0.0/16"
POD_GATEWAY="10.16.0.1"
JOIN_CIDR="100.64.0.0/16"
EXCLUDE_IPS=""
```

EXCLUDE\_IP      POD\_CIDR      192.168.10.20..192.168.10.30

Overlay      Service CIDR

Join

8.3.2 Service

kube-proxy      iptables      Kube-OVN      Kube-OVN      Service      CIDR

```
SVC_CIDR="10.96.0.0/12"
```

kube-ovn-controller Deployment

```
args:
- --service-cluster-ip-range=10.96.0.0/12
```

8.3.3 Overlay

Kube-OVN      Kubernetes Node IP

```
IFACE=eth1
```

ens[a-z0-9]\*,eth[a-z0-9]\*

kube-ovn-cni DaemonSet

```
args:
- --iface=eth1
```

annotation ovn.kubernetes.io/tunnel\_interface      annotation      iface      annotation

```
kubect1 annotate node no1 ovn.kubernetes.io/tunnel_interface=ethx
```

Overlay

8.3.4 MTU

|                                   |          |              |           |         |     |     |           |          |     |
|-----------------------------------|----------|--------------|-----------|---------|-----|-----|-----------|----------|-----|
| Overlay                           | Kube-OVN | MTU          | MTU       | Overlay | Pod | MTU | MTU - 100 | Underlay | Pod |
| MTU                               |          |              |           |         |     |     |           |          |     |
| Overlay                           | MTU      | kube-ovn-cni | DaemonSet |         |     |     |           |          |     |
|                                   |          |              |           |         |     |     |           |          |     |
| <pre>args:<br/>- --mtu=1333</pre> |          |              |           |         |     |     |           |          |     |

8.3.5

|                                 |           |         |  |  |  |  |  |  |  |
|---------------------------------|-----------|---------|--|--|--|--|--|--|--|
| Kube-OVN                        | mirror0   | tcpdump |  |  |  |  |  |  |  |
| ENABLE_MIRROR=true              |           |         |  |  |  |  |  |  |  |
| kube-ovn-cni                    | DaemonSet | :       |  |  |  |  |  |  |  |
| args:<br>- --enable-mirror=true |           |         |  |  |  |  |  |  |  |

8.3.6 LB

|                                          |                         |                       |                   |       |           |         |    |
|------------------------------------------|-------------------------|-----------------------|-------------------|-------|-----------|---------|----|
| Underlay                                 | kube-proxy<br>ClusterIP | Service<br>kube-proxy | OVN<br>kube-proxy | L2 LB | ClusterIP | Service | LB |
| <pre>ENABLE_LB=false</pre>               |                         |                       |                   |       |           |         |    |
| kube-ovn-controller Deployment           |                         |                       |                   |       |           |         |    |
| <pre>args:<br/>- --enable-lb=false</pre> |                         |                       |                   |       |           |         |    |

LB

|                  |            |               |          |               |               |         |                     |  |  |
|------------------|------------|---------------|----------|---------------|---------------|---------|---------------------|--|--|
| Kube-OVN v1.12.0 | subnet crd | spec          | enableLb | Kube-OVN      | LB            | LB      | kube-ovn-controller |  |  |
| Deployment       | enable-lb  | load-balancer | enableLb | load-balancer | load-balancer | v1.12.0 | enableLb            |  |  |

8.3.7 NetworkPolicy

|                                |     |     |               |               |              |      |               |          |
|--------------------------------|-----|-----|---------------|---------------|--------------|------|---------------|----------|
| Kube-OVN                       | OVN | ACL | NetworkPolicy | NetworkPolicy | Cilium Chain | eBPF | NetworkPolicy | Kube-OVN |
| NetworkPolicy                  |     |     |               |               |              |      |               |          |
| ENABLE_NP=false                |     |     |               |               |              |      |               |          |
| kube-ovn-controller Deployment |     |     |               |               |              |      |               |          |
| args:<br>- --enable-np=false   |     |     |               |               |              |      |               |          |

NetworkPolicy

8.3.8 EIP SNAT

|     |      |                     |  |  |  |  |  |  |  |
|-----|------|---------------------|--|--|--|--|--|--|--|
| EIP | SNAT | kube-ovn-controller |  |  |  |  |  |  |  |
|-----|------|---------------------|--|--|--|--|--|--|--|



```
ENABLE_EIP_SNAT=false
```

kube-ovn-controller Deployment

```
args:
- --enable-eip-snat=false
```

EIP SNAT EIP SNAT

8.3.9 Load Balancer Service

VPC Load Balancer Service LoadBalancer Service

```
ENABLE_LB_SVC=true
```

kube-ovn-controller Deployment

```
args:
- --enable-lb-svc=true
```

8.3.10 ECMP

ECMP ECMP kube-ovn-controller Deployment :

```
args:
- --enable-ecmp=true
```

Kube-OVN v1.12.0 subnet crd spec enableEcmp ECMP ECMP kube-ovn-controller  
Deployment enable-ecmp v1.12.0

8.3.11 Kubevirt VM

Kubevirt VM kube-ovn-controller StatefulSet Pod IP VM

1.10.6 kube-ovn-controller Deployment

```
args:
- --keep-vm-ip=false
```

8.3.12 CNI

Kube-OVN /opt/cni/bin CNI /etc/cni/net.d CNI 01-kube-ovn.conflist CNI

```
CNI_CONF_DIR="/etc/cni/net.d"
CNI_BIN_DIR="/opt/cni/bin"
CNI_CONFIG_PRIORITY="01"
```

kube-ovn-cni DaemonSet Volume

```
volumes:
- name: cni-conf
  hostPath:
    path: "/etc/cni/net.d"
- name: cni-bin
  hostPath:
    path: "/opt/cni/bin"
...
args:
- --cni-conf-name=01-kube-ovn.conflist
```

8.3.13

Kube-OVN

Overlay

Geneve

Vxlan

STT

TUNNEL\_TYPE="vxlan"

ovs-ovn DaemonSet

env:  
- name: TUNNEL\_TYPE  
  value: "vxlan"

STT

ovs

8.3.14 SSL

OVN DB

API

SSL

:

ENABLE\_SSL=true

SSL

8.3.15 ip

kube-ovn-controller/kube-ovn-cni/kube-ovn-monitor

ip

0.0.0.0

ip

ENABLE\_BIND\_LOCAL\_IP=true

kube-ovn-monitor

pod ip

# netstat -tunlp |grep kube-ovn  
tcp 0 0 172.18.0.5:10661 0.0.0.0:\* LISTEN 2612/./kube-ovn-mon

deployment

daemonSet

env:  
- name: ENABLE\_BIND\_LOCAL\_IP  
  value: "false"

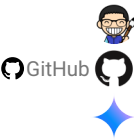
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8.3.16

8.4

|          |       |       |       |         |     |       |     |     |
|----------|-------|-------|-------|---------|-----|-------|-----|-----|
| Kube-OVN | Minor | Patch | Minor | OVN/OVS | API | Patch | Bug | API |
|----------|-------|-------|-------|---------|-----|-------|-----|-----|

8.4.1

|                      |                      |              |              |     |
|----------------------|----------------------|--------------|--------------|-----|
| Kube-OVN<br>backport | master, release-1.12 | release-1.11 | release-1.12 | Bug |
| release-1.11         | backport             | Bug          |              |     |

8.4.2

|       |       |     |     |
|-------|-------|-----|-----|
| Minor | Patch | Bug | Bug |
|-------|-------|-----|-----|

8.4.3 Patch

|       |                     |                 |          |                      |
|-------|---------------------|-----------------|----------|----------------------|
| Patch | hack/release.sh     |                 |          |                      |
| 1.    | Build               |                 |          |                      |
| 2.    | tag                 | Docker Hub      |          |                      |
| 3.    | tag                 | Github          |          |                      |
| 4.    |                     |                 |          |                      |
| 5.    |                     |                 |          |                      |
| 6.    | Release Note PR     |                 |          |                      |
| 7.    | Release Note ( )    |                 |          |                      |
| 8.    | Merge github action | Release Note PR |          |                      |
| 9.    | Github Release      |                 |          |                      |
| 10.   | Github Release      | Release         | v1.12.12 | Release Note Release |

8.4.4 Minor

|       |                                                              |                            |         |                      |
|-------|--------------------------------------------------------------|----------------------------|---------|----------------------|
| Minor |                                                              |                            |         |                      |
| 1.    | Github                                                       | release-1.13 ( )           |         |                      |
| 2.    | VERSION, dist/images/install.sh, charts/kube-ovn/values.yaml | charts/kube-ovn/Chart.yaml | Minor   | v1.14.0 ( )          |
| 3.    | tag                                                          | Docker Hub ( )             |         |                      |
| 4.    | tag                                                          | Github ( )                 |         |                      |
| 5.    | v1.13                                                        | mkdocs.yml                 | version | branch ( )           |
| 6.    | Release Note PR                                              |                            |         |                      |
| 7.    | Release Note ( )                                             |                            |         |                      |
| 8.    | Merge github action                                          | Release Note PR            |         |                      |
| 9.    | Github Release                                               |                            |         |                      |
| 10.   | Github Release                                               | Release                    | v1.13.0 | Release Note Release |
| 11.   | VERSION                                                      | Patch                      | v1.13.1 |                      |

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🐙GitHub



8.4.5

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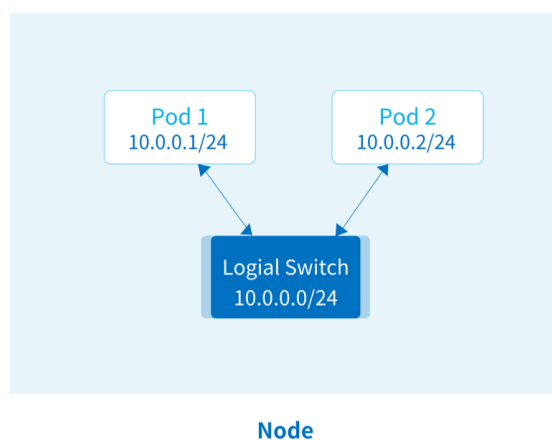
## 8.5 Underlay

---

Underlay

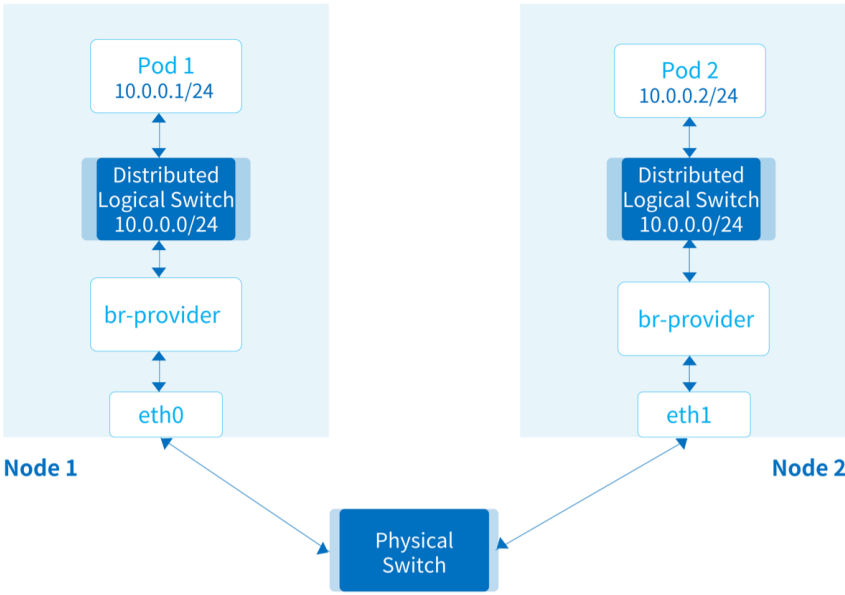
### 8.5.1

---

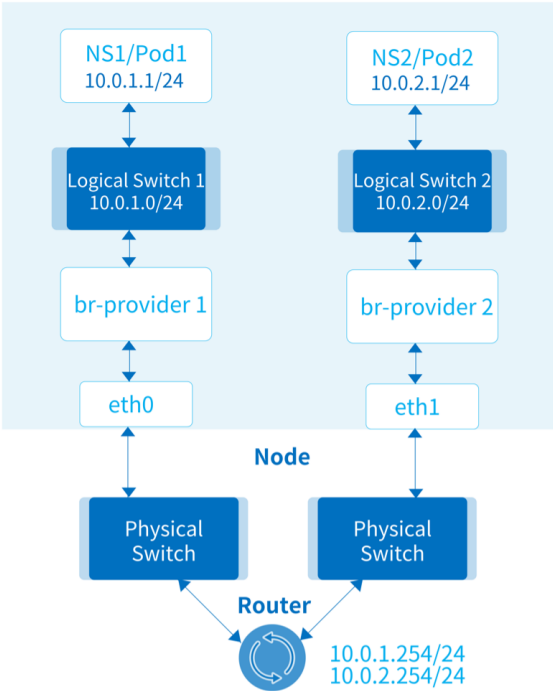


### 8.5.2

---

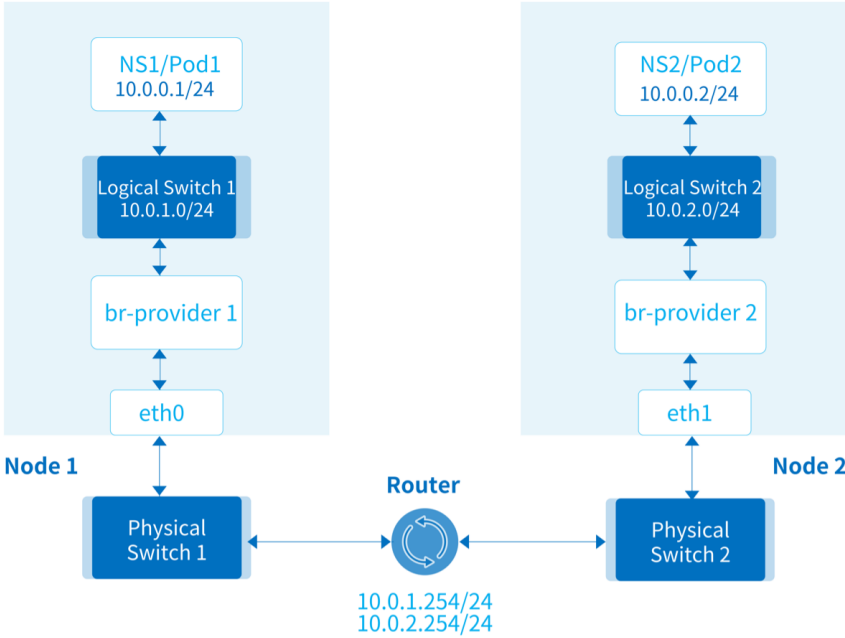


8.5.3

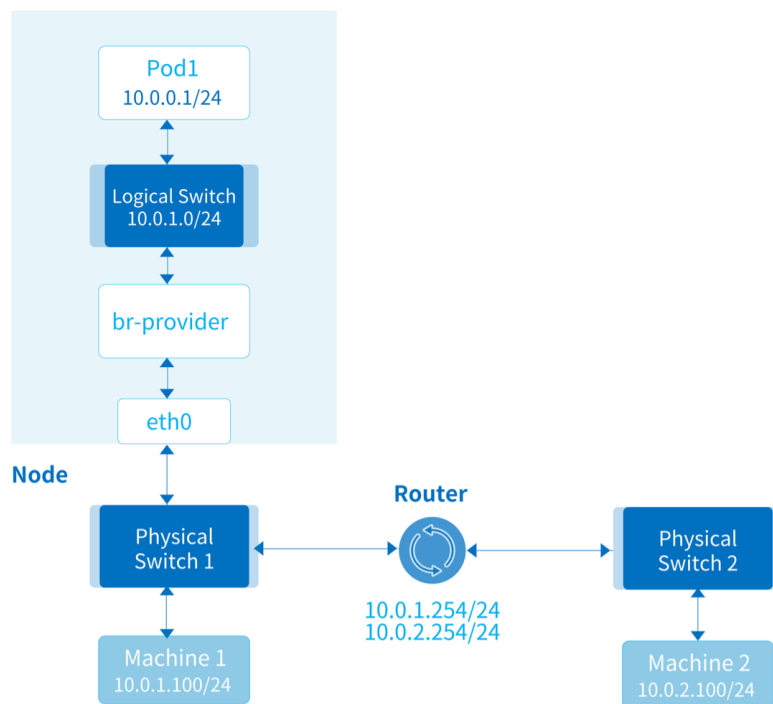


br-provider-1 br-provider-2 OVS Provider Network

8.5.4

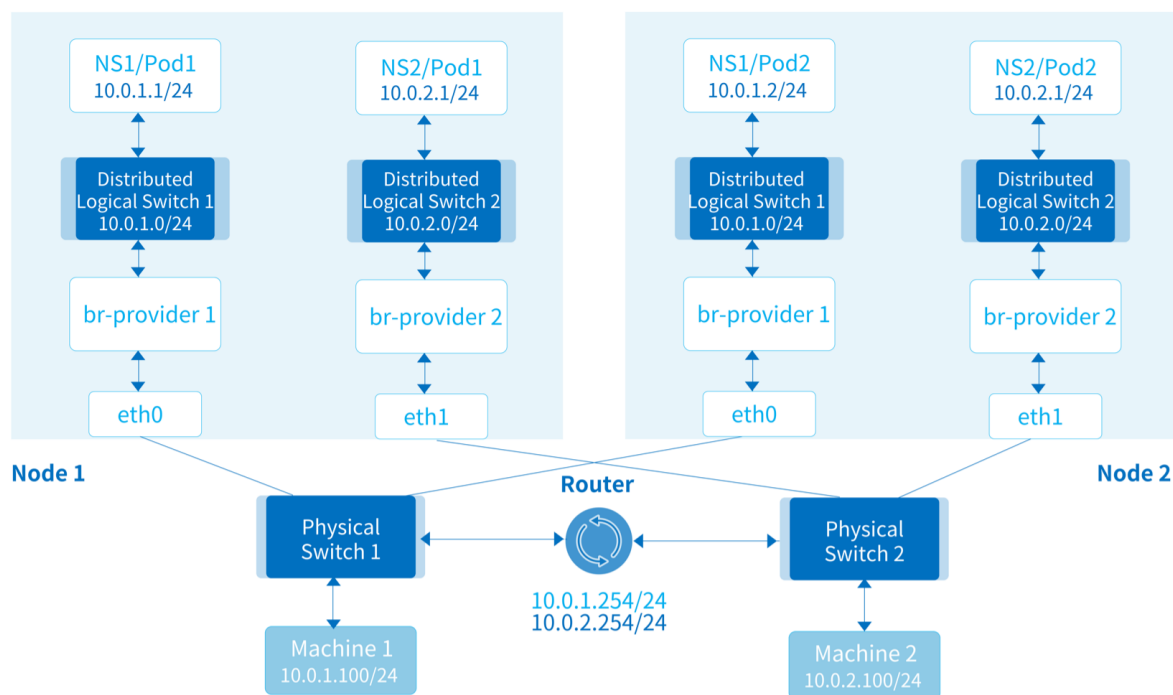


8.5.5



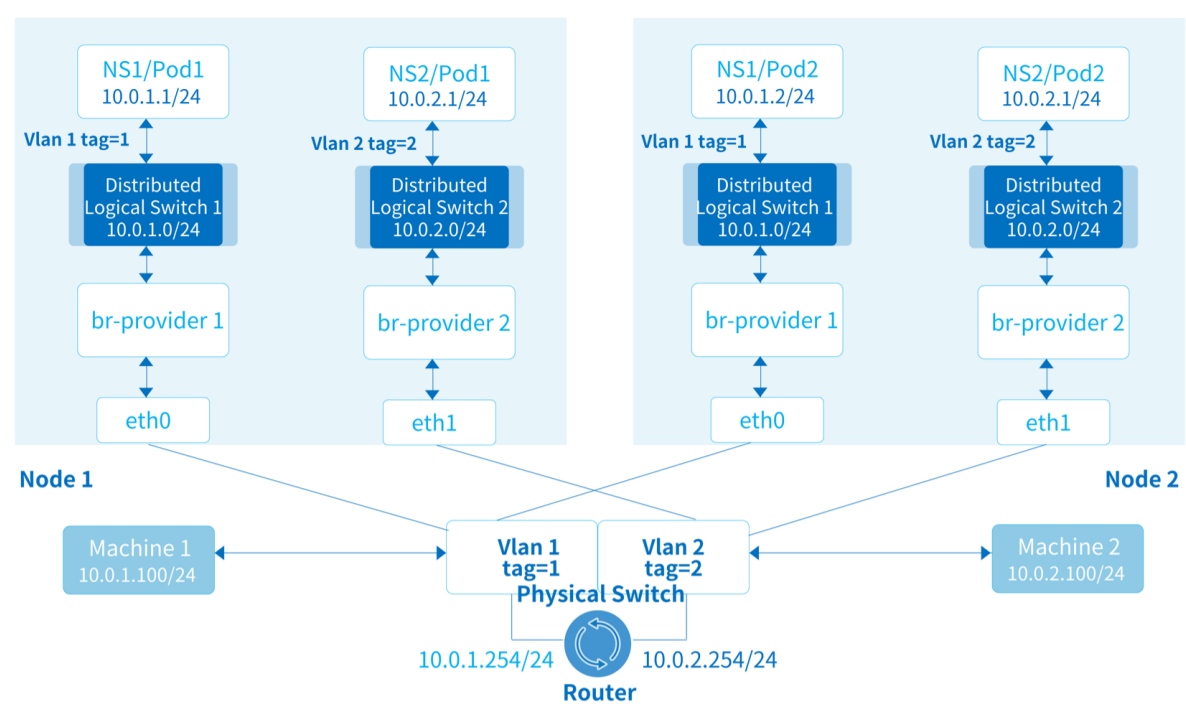
Pod

## 8.5.6 Vlan Tag





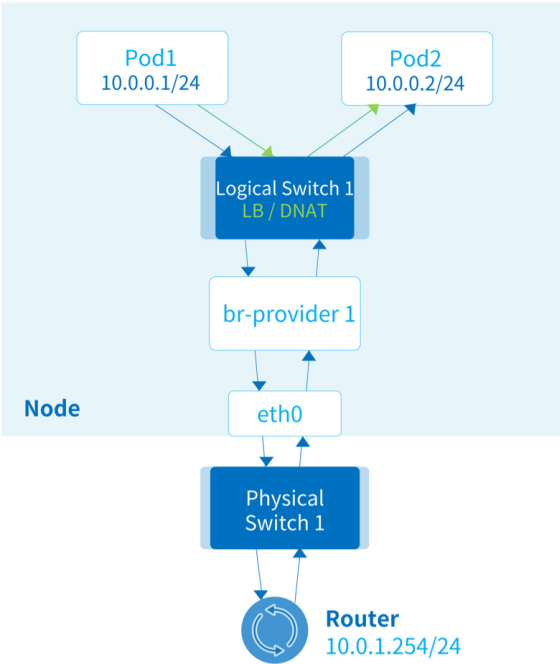
8.5.7 VLAN



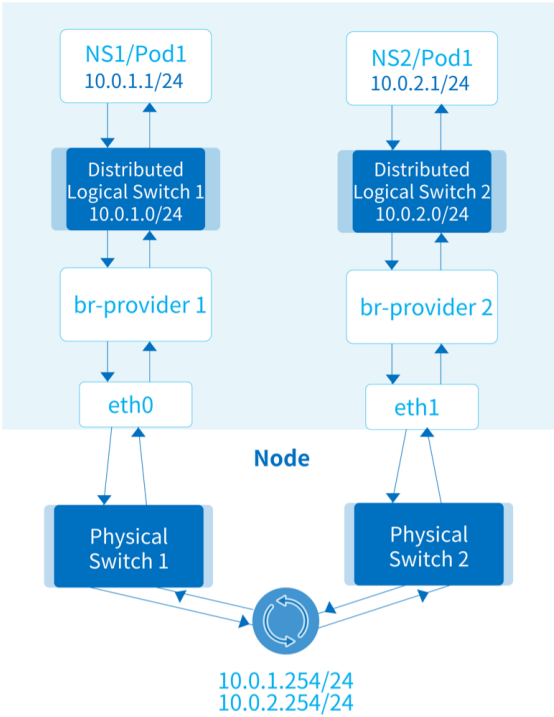
8.5.8 Pod Service IP

| Kube-OVN | Kubernetes Service<br>DNAT | Service<br>IP | Pod<br>Endpoint | Service IP<br>IP | Pod<br>MAC | Service IP<br>MAC | MAC | MAC |
|----------|----------------------------|---------------|-----------------|------------------|------------|-------------------|-----|-----|
|----------|----------------------------|---------------|-----------------|------------------|------------|-------------------|-----|-----|

Service                  Pod



Service                  Pod



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8.5.9

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## 8.6 Iptables

Kube-OVN   ipset   iptables   VPC   Overlay   NAT

ipset

| IPv4/IPv6                                               |          |                        |                  |        |
|---------------------------------------------------------|----------|------------------------|------------------|--------|
| ovn40services/ovn60services                             | hash:net | Service                |                  |        |
| ovn40subnets/ovn60subnets                               | hash:net | Overlay                | NodeLocal DNS IP |        |
| ovn40subnets-nat/ovn60subnets-nat                       | hash:net | NatOutgoing            | Overlay          |        |
| ovn40subnets-distributed-gw/ovn60subnets-distributed-gw | hash:net | Overlay                |                  |        |
| ovn40other-node/ovn60other-node                         | hash:net | IP                     |                  |        |
| ovn40local-pod-ip-nat/ovn60local-pod-ip-nat             | hash:ip  |                        |                  |        |
| ovn40subnets-nat-policy                                 | hash:net | natOutgoingPolicyRules |                  |        |
| ovn40natpr-418e79269dc5-dst                             | hash:net | natOutgoingPolicyRules | rule             | dstIPs |
| ovn40natpr-418e79269dc5-src                             | hash:net | natOutgoingPolicyRules | rule             | srcIPs |

iptables    IPv4

|        |                    |                                                                                                                                 |                           |                                                                                              |
|--------|--------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------|
| filter | INPUT              | -m set --match-set<br>ovn40services src -j<br>ACCEPT                                                                            | k8s Service Pod           | --                                                                                           |
| filter | INPUT              | -m set --match-set<br>ovn40services dst -j<br>ACCEPT                                                                            |                           | --                                                                                           |
| filter | INPUT              | -m set --match-set<br>ovn40subnets src -j<br>ACCEPT                                                                             |                           | --                                                                                           |
| filter | INPUT              | -m set --match-set<br>ovn40subnets dst -j<br>ACCEPT                                                                             |                           | --                                                                                           |
| filter | FORWARD            | -m set --match-set<br>ovn40services src -j<br>ACCEPT                                                                            |                           | --                                                                                           |
| filter | FORWARD            | -m set --match-set<br>ovn40services dst -j<br>ACCEPT                                                                            |                           | --                                                                                           |
| filter | FORWARD            | -m set --match-set<br>ovn40subnets src -j<br>ACCEPT                                                                             |                           | --                                                                                           |
| filter | FORWARD            | -m set --match-set<br>ovn40subnets dst -j<br>ACCEPT                                                                             |                           | --                                                                                           |
| filter | FORWARD            | -s 10.16.0.0/16 -m<br>comment --comment "ovn-<br>subnet-gateway,ovn-<br>default"                                                | subnet                    | 10.16.0.0/16 subnet cidr comment<br>ovn-subnet-gateway iptables<br>subnet ovn-default subnet |
| filter | FORWARD            | -d 10.16.0.0/16 -m<br>comment --comment "ovn-<br>subnet-gateway,ovn-<br>default"                                                | subnet                    |                                                                                              |
| filter | OUTPUT             | -p udp -m udp --dport 6081<br>-j MARK --set-xmark 0x0                                                                           | SNAT                      | <a href="#">UDP: bad checksum on VXLAN interface</a>                                         |
| nat    | PREROUTING         | -m comment --comment<br>"kube-ovn prerouting rules"<br>-j OVN-PREROUTING                                                        | OVN-PREROUTING            | --                                                                                           |
| nat    | POSTROUTING        | -m comment --comment<br>"kube-ovn postrouting<br>rules" -j OVN-<br>POSTROUTING                                                  | OVN-POSTROUTING           | --                                                                                           |
| nat    | OVN-<br>PREROUTING | -i ovn0 -m set --match-set<br>ovn40subnets src -m set --<br>match-set ovn40services<br>dst -j MARK --set-xmark<br>0x4000/0x4000 | Pod Service<br>masquerade | LB                                                                                           |
| nat    | OVN-<br>PREROUTING | -p tcp -m addrtype --dst-<br>type LOCAL -m set --match-                                                                         |                           | kube-proxy ipvs                                                                              |

|     |                 |                                                                                                                           |                                                            |                                      |                                  |
|-----|-----------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|--------------------------------------|----------------------------------|
|     |                 | set KUBE-NODE-PORT-LOCAL-TCP dst -j MARK --set-xmark 0x80000/0x80000                                                      | ExternalTrafficPolicy<br>Local Service TCP                 |                                      |                                  |
| nat | OVN-PREROUTING  | -p udp -m addrtype --dst-type LOCAL -m set --match-set KUBE-NODE-PORT-LOCAL-UDP dst -j MARK --set-xmark 0x80000/0x80000   | ExternalTrafficPolicy<br>Local Service UDP                 |                                      |                                  |
| nat | OVN-POSTROUTING | -m set --match-set ovn40services src -m set --match-set ovn40subnets dst -m mark --mark 0x4000/0x4000 -j SNAT --to-source | Service IP<br>Overlay Pod IP                               | kube-proxy                           | ipvs                             |
| nat | OVN-POSTROUTING | -m mark --mark 0x4000/0x4000 -j MASQUERADE                                                                                | SNAT                                                       | --                                   |                                  |
| nat | OVN-POSTROUTING | -m set --match-set ovn40subnets src -m set --match-set ovn40subnets dst -j MASQUERADE                                     | Pod Service<br>SNAT                                        | --                                   |                                  |
| nat | OVN-POSTROUTING | -m mark --mark 0x80000/0x80000 -m set --match-set ovn40subnets-distributed-gw dst -j RETURN                               | ExternalTrafficPolicy<br>Local Service<br>Endpoint<br>SNAT | --                                   |                                  |
| nat | OVN-POSTROUTING | -m mark --mark 0x80000/0x80000 -j MASQUERADE                                                                              | ExternalTrafficPolicy<br>Local Service<br>Endpoint<br>SNAT | --                                   |                                  |
| nat | OVN-POSTROUTING | -p tcp -m tcp --tcp-flags SYN NONE -m conntrack --ctstate NEW -j RETURN                                                   | Pod IP                                                     | SNAT                                 | --                               |
| nat | OVN-POSTROUTING | -s 10.16.0.0/16 -m set ! --match-set ovn40subnets dst -j SNAT --to-source 192.168.0.101                                   | Pod<br>NatOutgoing<br>SNAT                                 | IP                                   | 10.16.0.0/16<br>IP 192.168.0.101 |
| nat | OVN-POSTROUTING | -m set --match-set ovn40subnets-nat src -m set ! --match-set ovn40subnets dst -j MASQUERADE                               | Pod<br>NatOutgoing                                         | SNAT                                 | --                               |
| nat | OVN-POSTROUTING | -m set --match-set ovn40subnets-nat-policy src -m set ! --match-set ovn40subnets dst -j OVN-NAT-POLICY                    | Pod<br>natOutgoingPolicyRules<br>SNAT                      | natOutgoingPolicyRules<br>NAT-POLICY | OVN                              |
| nat |                 |                                                                                                                           |                                                            |                                      |                                  |

|        |                                      |                                                                                                                                                            |                   |                                                                                                                                                                                     |               |
|--------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
|        | OVN-<br>POSTROUTING                  | -m mark --mark<br>0x90001/0x90001 -j<br>MASQUERADE --random-<br>fully                                                                                      |                   | OVN-NAT-POLICY<br>0x90001/0x90001                                                                                                                                                   | tag<br>SNAT   |
| nat    | OVN-<br>POSTROUTING                  | -m mark --mark<br>0x90002/0x90002 -j<br>RETURN                                                                                                             |                   | OVN-NAT-POLICY<br>0x90002/0x90002                                                                                                                                                   | , tag<br>SNAT |
| nat    | OVN-NAT-POLICY                       | -s 10.0.11.0/24 -m<br>comment --comment<br>natPolicySubnet-net1 -j<br>OVN-NAT-PSUBNET-<br>aa98851157c5                                                     |                   | 10.0.11.0/24 net1 CIDR<br>OVN-NAT-<br>PSUBNET-aa98851157c5<br>natOutgoingPolicyRules                                                                                                |               |
| nat    | OVN-NAT-<br>PSUBNET-<br>xxxxxxxxxxxx | -m set --match-set<br>ovn40natpr-418e79269dc5-<br>src src -m set --match-set<br>ovn40natpr-418e79269dc5-<br>dst dst -j MARK --set-xmark<br>0x90002/0x90002 |                   | 418e79269dc5 natOutgoingPolicyRules<br>ID<br>status.natOutgoingPolicyRules[index].Rule<br>srcIPs ovn40natpr-418e79269dc5-<br>src dstIPs ovn40natpr-418e79269dc5-<br>dst tag 0x90002 |               |
| mangle | OVN-OUTPUT                           | -d 10.241.39.2/32 -p tcp -m<br>tcp --dport 80 -j MARK --set-<br>xmark 0x90003/0x90003                                                                      | kubelet<br>tproxy |                                                                                                                                                                                     |               |
| mangle | OVN-<br>PREROUTING                   | -d 10.241.39.2/32 -p tcp -m<br>tcp --dport 80 -j TPROXY --<br>on-port 8102 --on-ip<br>172.18.0.3 --tproxy-mark<br>0x90004/0x90004                          | kubelet<br>tproxy |                                                                                                                                                                                     |               |

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## 8.7

### 8.7.1

|          |        |   |        |       |    |            |        |               |
|----------|--------|---|--------|-------|----|------------|--------|---------------|
| Kube-OVN | Github | / | Github | Issue | PR | Maintainer | Review | Github Action |
|----------|--------|---|--------|-------|----|------------|--------|---------------|

### 8.7.2

Kube-OVN

Go

Go Modules

G0111MODULE="on"

golangci-lint

local-installation

Kube-OVN

Docker buildx

Docker

buildx:

docker buildx create --use

### 8.7.3

Kube-OVN

git clone https://github.com/kubeovn/kube-ovn.git

cd kube-ovn

make release

ARM

make release-arm

### 8.7.4 base

OVS/OVN

base

base

Dockerfile

dist/images/Dockerfile.base

# build x86 base image

make base-amd64

# build arm base image

make base-arm64

### 8.7.5 E2E

Kube-OVN

:

• KIND

Kubernetes

go install sigs.k8s.io/kind@latest

• jinjanator

:

pip install jinjanator

• Ginkgo

go install github.com/onsi/ginkgo/v2/ginkgo;

go get github.com/onsi/gomega/...

E2E

make kind-init

make kind-install

make e2e

Underlay E2E

```
make kind-init
make kind-install-underlay
make e2e-underlay-single-nic
```

ovn vpc nat gw eip, fip, snat, dnat

```
make kind-init
make kind-install
make ovn-vpc-nat-gw-conformance-e2e
```

iptables vpc nat gw eip, fip, snat, dnat

```
make kind-init
make kind-install-vpc-nat-gw
make iptables-vpc-nat-gw-conformance-e2e
```

loadbalancer service

```
make kind-init
make kind-install-lb-svc
make kube-ovn-lb-svc-conformance-e2e
```

```
make kind-clean
```

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8.8 OVS/OVN

| OVN/OVS                      | SDN                   | Kubernetes   | Kube-OVN              | Kube-OVN | OVN/OVS |
|------------------------------|-----------------------|--------------|-----------------------|----------|---------|
| OVN/OVS                      | Kube-OVN              |              |                       |          |         |
| • <a href="#">4228eab1d7</a> | vswitchd ofport_usage |              |                       |          |         |
| • <a href="#">54056ea65d</a> | timer                 |              |                       |          |         |
| • <a href="#">6b4dcb311f</a> | fdb                   |              |                       |          |         |
| • <a href="#">f627b7721e</a> | hairpin               | fdb          |                       |          |         |
| • <a href="#">3f3e3a436f</a> | ovsdb-tool            | join-cluster | Server ID             |          |         |
| • <a href="#">a6cb8215a8</a> | QoS                   |              |                       |          |         |
| • <a href="#">d4d76ddb2e</a> | ovsdb-tool            | fix-cluster  |                       |          |         |
| • <a href="#">ffd2328d4a</a> | netdev                | CPU          |                       |          |         |
| • <a href="#">d088c5d8c2</a> | ovs-router            | kube-ipvs0   |                       |          |         |
| • <a href="#">1b31f07dc6</a> |                       |              |                       |          |         |
| • <a href="#">54b7678229</a> | ovs-sandbox           | docker run   |                       |          |         |
| • <a href="#">9ee66bd91b</a> |                       |              |                       |          |         |
| • <a href="#">e889d46924</a> | Underlay              | resubmit     |                       |          |         |
| • <a href="#">f9e97031b5</a> | ovn-controller        | Kube-OVN     | localnet              | GARP     |         |
| • <a href="#">78cade0187</a> | conntrack             |              |                       |          |         |
| • <a href="#">85aa6263ad</a> | northd                | DNS IP       | conntrack             |          |         |
| • <a href="#">34dc3e3fcf</a> | lflow                 | lport        | conntrack             |          |         |
| • <a href="#">a297b840c2</a> |                       | DNAT         | lsp                   |          |         |
| • <a href="#">03e35ed9c5</a> | ovn-controller        |              |                       |          |         |
| • <a href="#">e7d3ba53cd</a> | ACL                   | DNS IP       | conntrack             |          |         |
| • <a href="#">9286e1fd57</a> |                       |              |                       |          |         |
| • <a href="#">e5916eb53a</a> |                       | lr-lb DNAT   |                       |          |         |
| • <a href="#">e4e6ea9c5f</a> | BFD LRP               |              |                       |          |         |
| • <a href="#">e76880e792</a> | northd                | nb           | version_compatibility |          |         |
| • <a href="#">477695a010</a> | northd                | localnet     | lrp                   | arp/nd   |         |
| • <a href="#">20626ea909</a> | LB                    | ACL          |                       |          |         |
| • <a href="#">a2d9ff3ccd</a> | Deb                   |              |                       |          |         |

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8.8.1

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8.9

|                                                  |         |        |       |     |     |
|--------------------------------------------------|---------|--------|-------|-----|-----|
| Kube-OVN                                         | OVN/OVS | Geneve | Vxlan | STT | OVN |
| <a href="#">OVN Architecture Design Decision</a> |         |        |       |     |     |

8.9.1 Geneve

|          |              |          |         |          |       |
|----------|--------------|----------|---------|----------|-------|
| Geneve   | Kube-OVN     | OVN      | Offload | Geneve   | 24bit |
| datapath |              | datapath | 32768   |          |       |
| Mellanox | OVS          | Geneve   | 5.4     | backport |       |
| UDP      | TCP over UDP | TCP      | CPU     |          |       |

8.9.2 Vxlan

|          |              |         |     |          |      |          |
|----------|--------------|---------|-----|----------|------|----------|
| Vxlan    | OVN          | Offload | OVN | datapath | 4096 | datapath |
| datapath | 4096         | inport  | ACL |          |      |          |
| Mellanox | OVS          | Vxlan   |     |          |      |          |
| UDP      | TCP over UDP | TCP     | CPU |          |      |          |

8.9.3 STT

 **Warning**

OpenVswitch 3.6 [STT](#) [Tunnel](#)

|     |     |     |     |     |     |          |
|-----|-----|-----|-----|-----|-----|----------|
| STT | OVN | TCP | TCP | TCP | OVN | datapath |
|     | OVS |     |     |     |     |          |
|     | OVS |     |     |     |     |          |

8.9.4

- [VXLAN vs GENEVE: Understand The Difference](#)
- [OVN FAQ](#)
- [What is Geneve](#)

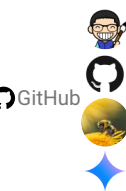
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8.9.5

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## 8.10 Kube-OVN

### Kube-OVN

#### 8.10.1 ovn-monitor

##### OVN

|       |                                                   |                                        |              |                |     |
|-------|---------------------------------------------------|----------------------------------------|--------------|----------------|-----|
|       |                                                   |                                        |              |                |     |
| Gauge | kube_ovn_ovn_status                               | OVN                                    | (2) follower | (1) leader,    | (0) |
| Gauge | kube_ovn_failed_req_count                         | OVN                                    |              |                |     |
| Gauge | kube_ovn_log_file_size_bytes                      | OVN                                    |              |                |     |
| Gauge | kube_ovn_db_file_size_bytes                       | OVN                                    |              |                |     |
| Gauge | kube_ovn_chassis_info                             | OVN chassis                            | (1)          | (0)            |     |
| Gauge | kube_ovn_db_status                                | OVN                                    | , (1)        | (0)            |     |
| Gauge | kube_ovn_logical_switch_info                      | OVN logical switch                     | (1)          | logical switch |     |
| Gauge | kube_ovn_logical_switch_external_id               | OVN logical switch external_id         | (1)          | external-id    |     |
| Gauge | kube_ovn_logical_switch_port_binding              | OVN logical switch logical switch port | (1)          |                |     |
| Gauge | kube_ovn_logical_switch_tunnel_key                | OVN logical switch tunnel key          |              |                |     |
| Gauge | kube_ovn_logical_switch_ports_num                 | OVN logical switch logical port        |              |                |     |
| Gauge | kube_ovn_logical_switch_port_info                 | OVN logical switch port                | (1)          |                |     |
| Gauge | kube_ovn_logical_switch_port_tunnel_key           | OVN logical switch port tunnel key     |              |                |     |
| Gauge | kube_ovn_cluster_enabled                          | (1) OVN                                | (0) OVN      |                |     |
| Gauge | kube_ovn_cluster_role                             |                                        | (1)          |                |     |
| Gauge | kube_ovn_cluster_status                           |                                        | (1)          |                |     |
| Gauge | kube_ovn_cluster_term                             | RAFT term                              |              |                |     |
| Gauge | kube_ovn_cluster_leader_self                      |                                        | leader (1)   | (0)            |     |
| Gauge | kube_ovn_cluster_vote_self                        |                                        | leader (1)   | (0)            |     |
| Gauge | kube_ovn_cluster_election_timer                   | election timer                         |              |                |     |
| Gauge | kube_ovn_cluster_log_not_committed                | commit RAFT                            |              |                |     |
| Gauge | kube_ovn_cluster_log_not_applied                  | apply RAFT                             |              |                |     |
| Gauge | kube_ovn_cluster_log_index_start                  | RAFT                                   |              |                |     |
| Gauge | kube_ovn_cluster_log_index_next                   | RAFT                                   |              |                |     |
| Gauge | kube_ovn_cluster_inbound_connections_total        |                                        |              |                |     |
| Gauge | kube_ovn_cluster_outbound_connections_total       |                                        |              |                |     |
| Gauge | kube_ovn_cluster_inbound_connections_error_total  |                                        |              |                |     |
| Gauge | kube_ovn_cluster_outbound_connections_error_total |                                        |              |                |     |

## 8.10.2 ovs-monitor

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ovsdb vswitchd



|       |                         |               |                       |     |
|-------|-------------------------|---------------|-----------------------|-----|
|       |                         |               |                       |     |
| Gauge | ovs_status              | OVS           | (1)                   | (0) |
| Gauge | ovs_info                | OVS           | (1)                   |     |
| Gauge | failed_req_count        | OVS           |                       |     |
| Gauge | log_file_size           | OVS           |                       |     |
| Gauge | db_file_size            | OVS           |                       |     |
| Gauge | datapath                | Datapath      | (1)                   |     |
| Gauge | dp_total                | OVS           | datapath              |     |
| Gauge | dp_if                   | Datapath      | (1)                   |     |
| Gauge | dp_if_total             | datapath      | port                  |     |
| Gauge | dp_flows_total          | Datapath      | flow                  |     |
| Gauge | dp_flows_lookup_hit     | Datapath      | flow                  |     |
| Gauge | dp_flows_lookup_missed  | Datapath      | flow                  |     |
| Gauge | dp_flows_lookup_lost    | Datapath      | userspace             |     |
| Gauge | dp_masks_hit            | Datapath      | mask                  |     |
| Gauge | dp_masks_total          | Datapath      | mask                  |     |
| Gauge | dp_masks_hit_ratio      | Datapath      | mask                  |     |
| Gauge | interface               | OVS           | (1)                   |     |
| Gauge | interface_admin_state   |               | (0) down, (1) up, (2) |     |
| Gauge | interface_link_state    |               | (0) down, (1) up, (2) |     |
| Gauge | interface_mac_in_use    | OVS Interface | MAC                   |     |
| Gauge | interface_mtu           | OVS Interface | MTU                   |     |
| Gauge | interface_of_port       | OVS Interface | OpenFlow Port ID      |     |
| Gauge | interface_if_index      | OVS Interface | Index                 |     |
| Gauge | interface_tx_packets    | OVS Interface |                       |     |
| Gauge | interface_tx_bytes      | OVS Interface |                       |     |
| Gauge | interface_rx_packets    | OVS Interface |                       |     |
| Gauge | interface_rx_bytes      | OVS Interface |                       |     |
| Gauge | interface_rx_crc_err    | OVS Interface |                       |     |
| Gauge | interface_rx_dropped    | OVS Interface |                       |     |
| Gauge | interface_rx_errors     | OVS Interface |                       |     |
| Gauge | interface_rx_frame_err  | OVS Interface |                       |     |
| Gauge | interface_rx_missed_err | OVS Interface | miss                  |     |
| Gauge | interface_rx_over_err   | OVS Interface | overrun               |     |
| Gauge | interface_tx_dropped    | OVS Interface |                       |     |

|       |                      |               |
|-------|----------------------|---------------|
|       |                      |               |
| Gauge | interface_tx_errors  | OVS Interface |
| Gauge | interface_collisions | OVS interface |

### 8.10.3 kube-ovn-pinger

|           |                                  |                    |               |
|-----------|----------------------------------|--------------------|---------------|
|           |                                  |                    |               |
| Gauge     | pinger_ovs_up                    | OVS                |               |
| Gauge     | pinger_ovs_down                  | OVS                |               |
| Gauge     | pinger_ovn_controller_up         | ovn-controller     |               |
| Gauge     | pinger_ovn_controller_down       | ovn-controller     |               |
| Gauge     | pinger_inconsistent_port_binding | OVN-SB portbinding | OVS interface |
| Gauge     | pinger_apiserver_healthy         | kube-ovn-pinger    | apiserver     |
| Gauge     | pinger_apiserver_unhealthy       | kube-ovn-pinger    | apiserver     |
| Histogram | pinger_apiserver_latency_ms      | kube-ovn-pinger    | apiserver     |
| Gauge     | pinger_internal_dns_healthy      | kube-ovn-pinger    |               |
| Gauge     | pinger_internal_dns_unhealthy    | kube-ovn-pinger    |               |
| Histogram | pinger_internal_dns_latency_ms   | kube-ovn-pinger    |               |
| Gauge     | pinger_external_dns_health       | kube-ovn-pinger    |               |
| Gauge     | pinger_external_dns_unhealthy    | kube-ovn-pinger    |               |
| Histogram | pinger_external_dns_latency_ms   | kube-ovn-pinger    |               |
| Histogram | pinger_pod_ping_latency_ms       | kube-ovn-pinger    | ping Pod      |
| Gauge     | pinger_pod_ping_lost_total       | kube-ovn-pinger    | ping Pod      |
| Gauge     | pinger_pod_ping_count_total      | kube-ovn-pinger    | ping Pod      |
| Histogram | pinger_node_ping_latency_ms      | kube-ovn-pinger    | ping Node     |
| Gauge     | pinger_node_ping_lost_total      | kube-ovn-pinger    | ping Node     |
| Gauge     | pinger_node_ping_count_total     | kube-ovn-pinger    | ping Node     |
| Histogram | pinger_external_ping_latency_ms  | kube-ovn-pinger    | ping          |
| Gauge     | pinger_external_lost_total       | kube-ovn-pinger    | ping          |

## 8.10.4 kube-ovn-controller

### kube-ovn-controller

|           |                                         |                  |
|-----------|-----------------------------------------|------------------|
|           |                                         |                  |
| Histogram | rest_client_request_latency_seconds     | apiserver        |
| Counter   | rest_client_requests_total              | apiserver        |
| Counter   | lists_total                             | API list         |
| Summary   | list_duration_seconds                   | API list         |
| Summary   | items_per_list                          | API list         |
| Counter   | watches_total                           | API watch        |
| Counter   | short_watches_total                     | API watch        |
| Summary   | watch_duration_seconds                  | API watch        |
| Summary   | items_per_watch                         | API watch        |
| Gauge     | last_resource_version                   | resource version |
| Histogram | ovs_client_request_latency_milliseconds | OVN              |
| Gauge     | subnet_available_ip_count               | IP               |
| Gauge     | subnet_used_ip_count                    | IP               |

## 8.10.5 kube-ovn-cni

### kube-ovn-cni

|           |                                         |                  |
|-----------|-----------------------------------------|------------------|
|           |                                         |                  |
| Histogram | cni_op_latency_seconds                  | CNI              |
| Counter   | cni_wait_address_seconds_total          | CNI              |
| Counter   | cni_wait_connectivity_seconds_total     | CNI              |
| Counter   | cni_wait_route_seconds_total            | CNI              |
| Histogram | rest_client_request_latency_seconds     | apiserver        |
| Counter   | rest_client_requests_total              | apiserver        |
| Counter   | lists_total                             | API list         |
| Summary   | list_duration_seconds                   | API list         |
| Summary   | items_per_list                          | API list         |
| Counter   | watches_total                           | API watch        |
| Counter   | short_watches_total                     | API watch        |
| Summary   | watch_duration_seconds                  | API watch        |
| Summary   | items_per_watch                         | API watch        |
| Gauge     | last_resource_version                   | resource version |
| Histogram | ovs_client_request_latency_milliseconds | OVN              |

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8.10.6

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## 8.11 Kube-OVN

Kube-OVN      Kube-OVN      CRD      CRD

### 8.11.1 Condition

|                    |        |                    |
|--------------------|--------|--------------------|
|                    |        |                    |
| type               | String |                    |
| status             | String | True False Unknown |
| reason             | String |                    |
| message            | String |                    |
| observedGeneration | Int64  |                    |
| lastUpdateTime     | Time   |                    |
| lastTransitionTime | Time   |                    |

CRD      Status      Condition

### 8.11.2

#### Subnet

SUBNET

|            |              |            |               |
|------------|--------------|------------|---------------|
|            |              |            |               |
| apiVersion | String       | Kubernetes | kubeovn.io/v1 |
| kind       | String       | Kubernetes | Subnet        |
| metadata   | ObjectMeta   | Kubernetes |               |
| spec       | SubnetSpec   | Subnet     |               |
| status     | SubnetStatus | Subnet     |               |

SubnetSpec

|                        |                         |                        |                                        |  |  |
|------------------------|-------------------------|------------------------|----------------------------------------|--|--|
|                        |                         |                        |                                        |  |  |
| default                | Bool                    |                        |                                        |  |  |
| vpc                    | String                  | VPC                    | ovn-cluster                            |  |  |
| protocol               | String                  | IP                     | IPv4 IPv6 Dual                         |  |  |
| namespaces             | []String                | namespace              |                                        |  |  |
| cidrBlock              | String                  | 10.16.0.0/16           |                                        |  |  |
| gateway                | String                  | CIDRBlock              |                                        |  |  |
| excludelpls            | []String                |                        |                                        |  |  |
| provider               | String                  | OVN Subnet             | NetworkAttachmentDefinition . Kube-OVN |  |  |
| gatewayType            | String                  | Overlay                | distributed centralized                |  |  |
| gatewayNode            | String                  | centralized            |                                        |  |  |
| natOutgoing            | Bool                    | NAT                    | externalEgressGateway                  |  |  |
| externalEgressGateway  | String                  |                        | natOutgoing                            |  |  |
| policyRoutingPriority  | Uint32                  |                        |                                        |  |  |
| policyRoutingTableID   | Uint32                  | TableID                |                                        |  |  |
| mtu                    | Uint32                  | MTU                    |                                        |  |  |
| private                | Bool                    |                        |                                        |  |  |
| allowSubnets           | []String                |                        |                                        |  |  |
| vlan                   | String                  | Vlan                   |                                        |  |  |
| vips                   | []String                | virtual lsp virtual-ip |                                        |  |  |
| logicalGateway         | Bool                    |                        |                                        |  |  |
| disableGatewayCheck    | Bool                    | Pod                    |                                        |  |  |
| disableInterConnection | Bool                    |                        |                                        |  |  |
| enableDHCP             | Bool                    | lsp dhcp               |                                        |  |  |
| dhcpV4Options          | String                  | lsp dhcpv4_options     | DHCP_Options                           |  |  |
| dhcpV6Options          | String                  | lsp dhcpv6_options     | DHCP_Options                           |  |  |
| enableIPv6RA           | Bool                    | lrp                    | ipv6_ra_configs                        |  |  |
| ipv6RAConfigs          | String                  | lrp                    | ipv6_ra_configs                        |  |  |
| acls                   | []Acl                   | logical-switch         | acls                                   |  |  |
| allowEWTraffic         | Bool                    |                        |                                        |  |  |
| natOutgoingPolicyRules | []NatOutgoingPolicyRule | NAT                    |                                        |  |  |
| u2oInterconnectionIP   | String                  | Underlay/Overlay       | IP                                     |  |  |
| u2oInterconnection     | Bool                    | Overlay/Underlay       |                                        |  |  |
| enableLb               | *Bool                   | logical-switch         | load-balancer                          |  |  |
| enableEcmp             | Bool                    | ECMP                   |                                        |  |  |

|                         |                 |
|-------------------------|-----------------|
|                         |                 |
| enableMulticastSnoop    | Bool            |
| enableExternalLBAddress | Bool            |
| routeTable              | String          |
| namespaceSelectors      | []LabelSelector |

Acl

|           |        |     |               |                                   |
|-----------|--------|-----|---------------|-----------------------------------|
|           |        |     |               |                                   |
| direction | String | Acl | from-lport    | to-lport                          |
| priority  | Int    | Acl | 0             | 32767                             |
| match     | String | Acl |               |                                   |
| action    | String | Acl | allow-related | allow-stateless allow drop reject |

NatOutgoingPolicyRule

|        |                        |
|--------|------------------------|
|        |                        |
| match  | NatOutGoingPolicyMatch |
| action | String                 |

NatOutGoingPolicyMatch

|        |        |    |
|--------|--------|----|
|        |        |    |
| srcIPs | String | IP |
| dstIPs | String | IP |



SubnetStatus

| conditions             | []SubnetCondition             |                    | Condition    |
|------------------------|-------------------------------|--------------------|--------------|
| v4availableIPs         | Float64                       | IPv4 IP            |              |
| v4availableIPrange     | String                        | IPv4               |              |
| v4usingIPs             | Float64                       | IPv4 IP            |              |
| v4usingIPrange         | String                        | IPv4               |              |
| v6availableIPs         | Float64                       | IPv6 IP            |              |
| v6availableIPrange     | String                        | IPv6               |              |
| v6usingIPs             | Float64                       | IPv6 IP            |              |
| v6usingIPrange         | String                        | IPv6               |              |
| activateGateway        | String                        |                    |              |
| dhcpV4OptionsUUID      | String                        | lsp dhcpv4_options | DHCP_Options |
| dhcpV6OptionsUUID      | String                        | lsp dhcpv6_options | DHCP_Options |
| u2oInterconnectionIP   | String                        | Overlay/Underlay   | IP           |
| u2oInterconnectionMAC  | String                        | Overlay/Underlay   | MAC          |
| u2oInterconnectionVPC  | String                        | Overlay/Underlay   | VPC          |
| natOutgoingPolicyRules | []NatOutgoingPolicyRuleStatus | NAT                |              |
| mcastQuerierIP         | String                        | IP                 |              |
| mcastQuerierMAC        | String                        | MAC                |              |

IP

IP

| apiVersion | String     | Kubernetes | kubeovn.io/v1 |
|------------|------------|------------|---------------|
| kind       | String     | Kubernetes | IP            |
| metadata   | ObjectMeta | Kubernetes |               |
| spec       | IPSpec     | IP         |               |

IPSpec

|               |          |                                |
|---------------|----------|--------------------------------|
|               |          |                                |
| podName       | String   | Pod                            |
| namespace     | String   | Pod Namespace                  |
| subnet        | String   | IP Subnet                      |
| attachSubnets | []String | IP                             |
| nodeName      | String   | Pod                            |
| ipAddress     | String   | IP v4IP v6IP                   |
| v4IpAddress   | String   | IPv4 IP                        |
| v6IpAddress   | String   | IPv6 IP                        |
| attachIps     | []String | IP IP                          |
| macAddress    | String   | Pod MAC                        |
| attachMacs    | []String | IP MAC                         |
| containerID   | String   | Pod Container ID               |
| podType       | String   | Pod StatefulSet VirtualMachine |

Vpc

VPC

|            |            |            |               |
|------------|------------|------------|---------------|
|            |            |            |               |
| apiVersion | String     | Kubernetes | kubeovn.io/v1 |
| kind       | String     | Kubernetes | Vpc           |
| metadata   | ObjectMeta | Kubernetes |               |
| spec       | VpcSpec    | Vpc        |               |
| status     | VpcStatus  | Vpc        |               |

VpcSpec

|                      |               |         |
|----------------------|---------------|---------|
|                      |               |         |
| defaultSubnet        | String        |         |
| namespaces           | []String      | Vpc     |
| staticRoutes         | []StaticRoute |         |
| policyRoutes         | []PolicyRoute |         |
| vpcPeerings          | []VpcPeering  | VPC     |
| enableExternal       | Bool          |         |
| extraExternalSubnets | []String      |         |
| enableBfd            | Bool          | BFD ( ) |
| bfdPort              | BFDPort       | BFD     |

StaticRoute

|            |        |        |  |
|------------|--------|--------|--|
|            |        |        |  |
| policy     | String |        |  |
| cidr       | String |        |  |
| nextHopIP  | String | IP     |  |
| ecmpMode   | String | ECMP   |  |
| bfdId      | String | BFD ID |  |
| routeTable | String |        |  |

PolicyRoute

|           |        |       |                |
|-----------|--------|-------|----------------|
|           |        |       |                |
| priority  | Int    |       |                |
| match     | String |       |                |
| action    | String | allow | drop reroute   |
| nextHopIP | String | IP    | action reroute |

VpcPeering

|                |        |     |
|----------------|--------|-----|
|                |        |     |
| remoteVpc      | String | VPC |
| localConnectIP | String | IP  |

BFDPort

|              |               |         |    |
|--------------|---------------|---------|----|
|              |               |         |    |
| enabled      | Bool          | BFD     |    |
| ip           | String        | BFD     | IP |
| nodeSelector | LabelSelector | BFD LRP |    |

VpcStatus

| conditions              | []VpcCondition | Vpc  | Condition |
|-------------------------|----------------|------|-----------|
| standby                 | Bool           | VPC  |           |
| default                 | Bool           | VPC  |           |
| defaultLogicalSwitch    | String         |      |           |
| router                  | String         |      |           |
| tcpLoadBalancer         | String         | TCP  |           |
| udpLoadBalancer         | String         | UDP  |           |
| sctpLoadBalancer        | String         | SCTP |           |
| tcpSessionLoadBalancer  | String         | TCP  |           |
| udpSessionLoadBalancer  | String         | UDP  |           |
| sctpSessionLoadBalancer | String         | SCTP |           |
| subnets                 | []String       | VPC  |           |
| vpcPeerings             | []String       | VPC  |           |
| enableExternal          | Bool           |      |           |
| extraExternalSubnets    | []String       |      |           |
| enableBfd               | Bool           | BFD  |           |

8.11.3 Underlay

Vlan

| apiVersion | String     | Kubernetes | kubeovn.io/v1 |
|------------|------------|------------|---------------|
| kind       | String     | Kubernetes | Vlan          |
| metadata   | ObjectMeta | Kubernetes |               |
| spec       | VlanSpec   | Vlan       |               |
| status     | VlanStatus | Vlan       |               |

VLANSPEC

| id       | Int    | Vlan tag | 0~4096          |
|----------|--------|----------|-----------------|
| provider | String | Vlan     | ProviderNetwork |

VLANSTATUS

|            |                 |      |           |
|------------|-----------------|------|-----------|
|            |                 |      |           |
| subnets    | []String        | Vlan |           |
| conflict   | Bool            |      |           |
| conditions | []VlanCondition | Vlan | Condition |

ProviderNetwork

|            |                       |                 |                 |
|------------|-----------------------|-----------------|-----------------|
|            |                       |                 |                 |
| apiVersion | String                | Kubernetes      | kubeovn.io/v1   |
| kind       | String                | Kubernetes      | ProviderNetwork |
| metadata   | ObjectMeta            | Kubernetes      |                 |
| spec       | ProviderNetworkSpec   | ProviderNetwork |                 |
| status     | ProviderNetworkStatus | ProviderNetwork |                 |

PROVIDERNETWORKSPEC

|                  |                   |     |             |                  |                           |
|------------------|-------------------|-----|-------------|------------------|---------------------------|
|                  |                   |     |             |                  |                           |
| defaultInterface | String            |     |             |                  |                           |
| customInterfaces | []CustomInterface |     |             |                  |                           |
| nodeSelector     | LabelSelector     | OVS | matchLabels | matchExpressions | nodeSelector excludeNodes |
| excludeNodes     | []String          |     |             |                  |                           |
| exchangeLinkName | Bool              | OVS |             |                  |                           |

CustomInterface

|           |          |          |
|-----------|----------|----------|
|           |          |          |
| interface | String   | Underlay |
| nodes     | []String |          |

PROVIDERNETWORKSTATUS

|               |                            |                 |           |
|---------------|----------------------------|-----------------|-----------|
|               |                            |                 |           |
| ready         | Bool                       |                 |           |
| readyNodes    | []String                   |                 |           |
| notReadyNodes | []String                   |                 |           |
| vlan          | []String                   | Vlan            |           |
| conditions    | []ProviderNetworkCondition | ProviderNetwork | Condition |

## 8.11.4

**SecurityGroup**

|            |                     |               |               |
|------------|---------------------|---------------|---------------|
|            |                     |               |               |
| apiVersion | String              | Kubernetes    | kubeovn.io/v1 |
| kind       | String              | Kubernetes    | SecurityGroup |
| metadata   | ObjectMeta          | Kubernetes    |               |
| spec       | SecurityGroupSpec   | SecurityGroup |               |
| status     | SecurityGroupStatus | SecurityGroup |               |

**SECURITYGROUPSPEC**

|                       |                     |
|-----------------------|---------------------|
|                       |                     |
| ingressRules          | []SecurityGroupRule |
| egressRules           | []SecurityGroupRule |
| allowSameGroupTraffic | Bool                |

**SecurityGroupRule**

|                     |              |    |         |               |
|---------------------|--------------|----|---------|---------------|
|                     |              |    |         |               |
| ipVersion           | String       | IP | ipv4    | ipv6          |
| protocol            | SgProtocol   |    | all     | icmp tcp udp  |
| priority            | Int          |    | 1-200   |               |
| remoteType          | SgRemoteType |    | address | securityGroup |
| remoteAddress       | String       |    |         |               |
| remoteSecurityGroup | String       |    |         |               |
| portRangeMin        | Int          |    | 1       |               |
| portRangeMax        | Int          |    | 65535   |               |
| policy              | SgPolicy     |    | allow   | drop          |

**SECURITYGROUPSTATUS**

|                        |        |     |
|------------------------|--------|-----|
|                        |        |     |
| portGroup              | String |     |
| allowSameGroupTraffic  | Bool   |     |
| ingressMd5             | String | MD5 |
| egressMd5              | String | MD5 |
| ingressLastSyncSuccess | Bool   |     |
| egressLastSyncSuccess  | Bool   |     |

8.11.5IP

Vip

|            |            |            |               |
|------------|------------|------------|---------------|
|            |            |            |               |
| apiVersion | String     | Kubernetes | kubeovn.io/v1 |
| kind       | String     | Kubernetes | Vip           |
| metadata   | ObjectMeta | Kubernetes |               |
| spec       | VipSpec    | Vip        |               |
| status     | VipStatus  | Vip        |               |

VIPSPEC

|               |          |      |
|---------------|----------|------|
|               |          |      |
| namespace     | String   | VIP  |
| subnet        | String   | VIP  |
| type          | String   | VIP  |
| v4ip          | String   | IPv4 |
| v6ip          | String   | IPv6 |
| macAddress    | String   | MAC  |
| selector      | []String |      |
| attachSubnets | []String |      |

VIPSTATUS

|            |                |      |           |
|------------|----------------|------|-----------|
|            |                |      |           |
| conditions | []VipCondition | VIP  | Condition |
| type       | String         | VIP  |           |
| v4ip       | String         | IPv4 |           |
| v6ip       | String         | IPv6 |           |
| mac        | String         | MAC  |           |

SwitchLBRule

|            |                    |              |               |
|------------|--------------------|--------------|---------------|
|            |                    |              |               |
| apiVersion | String             | Kubernetes   | kubeovn.io/v1 |
| kind       | String             | Kubernetes   | SwitchLBRule  |
| metadata   | ObjectMeta         | Kubernetes   |               |
| spec       | SwitchLBRuleSpec   | SwitchLBRule |               |
| status     | SwitchLBRuleStatus | SwitchLBRule |               |

SWITCHLBRULESPEC

|                 |                    |    |
|-----------------|--------------------|----|
|                 |                    |    |
| vip             | String             | IP |
| namespace       | String             |    |
| selector        | []String           |    |
| endpoints       | []String           |    |
| sessionAffinity | String             |    |
| ports           | []SwitchLBRulePort |    |

SwitchLBRulePort

|            |        |
|------------|--------|
|            |        |
| name       | String |
| port       | Int32  |
| targetPort | Int32  |
| protocol   | String |

SWITCHLBRULESTATUS

|            |                         |              |           |
|------------|-------------------------|--------------|-----------|
|            |                         |              |           |
| conditions | []SwitchLBRuleCondition | SwitchLBRule | Condition |
| ports      | String                  | SwitchLBRule |           |
| service    | String                  | SwitchLBRule | service   |

8.11.6 QoS IP

QoSPolicy

|            |               |            |               |
|------------|---------------|------------|---------------|
|            |               |            |               |
| apiVersion | String        | Kubernetes | kubeovn.io/v1 |
| kind       | String        | Kubernetes | QoSPolicy     |
| metadata   | ObjectMeta    | Kubernetes |               |
| spec       | QoSPolicySpec | QoSPolicy  |               |

QOSPOLICYSPEC

|                     |                              |
|---------------------|------------------------------|
|                     |                              |
| bandwidthLimitRules | QoSPolicyBandwidthLimitRules |
| shared              | Bool                         |
| bindingType         | QoSPolicyBindingType         |



IPPool

|            |              |            |               |
|------------|--------------|------------|---------------|
|            |              |            |               |
| apiVersion | String       | Kubernetes | kubeovn.io/v1 |
| kind       | String       | Kubernetes | IPPool        |
| metadata   | ObjectMeta   | Kubernetes |               |
| spec       | IPPoolSpec   | IPPool     |               |
| status     | IPPoolStatus | IPPool     |               |

IPPOOLSPEC

|            |          |    |
|------------|----------|----|
|            |          |    |
| subnet     | String   |    |
| namespaces | []String |    |
| ips        | []String | IP |

IPPOOLSTATUS

|                    |                   |      |           |
|--------------------|-------------------|------|-----------|
|                    |                   |      |           |
| v4AvailableIPs     | BigInt            | IPv4 | IP        |
| v4AvailableIPRange | String            | IPv4 | IP        |
| v4UsingIPs         | BigInt            | IPv4 | IP        |
| v4UsingIPRange     | String            | IPv4 | IP        |
| v6AvailableIPs     | BigInt            | IPv6 | IP        |
| v6AvailableIPRange | String            | IPv6 | IP        |
| v6UsingIPs         | BigInt            | IPv6 | IP        |
| v6UsingIPRange     | String            | IPv6 | IP        |
| conditions         | []IPPoolCondition | IP   | Condition |

8.11.7 NAT IP

IptablesEIP

|            |                   |             |               |
|------------|-------------------|-------------|---------------|
|            |                   |             |               |
| apiVersion | String            | Kubernetes  | kubeovn.io/v1 |
| kind       | String            | Kubernetes  | IptablesEIP   |
| metadata   | ObjectMeta        | Kubernetes  |               |
| spec       | IptablesEIPSpec   | IptablesEIP |               |
| status     | IptablesEIPStatus | IptablesEIP |               |

IPTABLESEIPSPEC

|                |        |      |
|----------------|--------|------|
|                |        |      |
| v4ip           | String | IPv4 |
| v6ip           | String | IPv6 |
| macAddress     | String | MAC  |
| natGwDp        | String | NAT  |
| qosPolicy      | String | QoS  |
| externalSubnet | String |      |

IPTABLESEIPSTATUS

|            |                        |                 |               |      |
|------------|------------------------|-----------------|---------------|------|
|            |                        |                 |               |      |
| ready      | Bool                   | IptablesEIP     |               |      |
| ip         | String                 | IptablesEIP     | IP            | IPv4 |
| redo       | String                 | IptablesEIP CRD |               |      |
| nat        | String                 | IptablesEIP     | fip snat dnat |      |
| qosPolicy  | String                 | QoS             |               |      |
| conditions | []IptablesEIPCondition | IptablesEIP     | Condition     |      |

OvnEip

|            |              |            |               |
|------------|--------------|------------|---------------|
|            |              |            |               |
| apiVersion | String       | Kubernetes | kubeovn.io/v1 |
| kind       | String       | Kubernetes | OvnEip        |
| metadata   | ObjectMeta   | Kubernetes |               |
| spec       | OvnEipSpec   | OvnEip     |               |
| status     | OvnEipStatus | OvnEip     |               |

OVNEIPSPEC

|                |        |             |
|----------------|--------|-------------|
|                |        |             |
| externalSubnet | String |             |
| v4Ip           | String | IPv4        |
| v6Ip           | String | IPv6        |
| macAddress     | String | MAC         |
| type           | String | lrp lsp nat |

IptablesFIPRule

|            |                     |                 |                 |
|------------|---------------------|-----------------|-----------------|
|            |                     |                 |                 |
| apiVersion | String              | Kubernetes      | kubeovn.io/v1   |
| kind       | String              | Kubernetes      | IptablesFIPRule |
| metadata   | ObjectMeta          | Kubernetes      |                 |
| spec       | IptablesFIPRuleSpec | IptablesFIPRule |                 |

IPTABLESFIPRULESPEC

|            |        |    |
|------------|--------|----|
|            |        |    |
| eip        | String | IP |
| internalIP | String | IP |

OvnFip

|            |              |            |               |
|------------|--------------|------------|---------------|
|            |              |            |               |
| apiVersion | String       | Kubernetes | kubeovn.io/v1 |
| kind       | String       | Kubernetes | OvnFip        |
| metadata   | ObjectMeta   | Kubernetes |               |
| spec       | OvnFipSpec   | OvnFip     |               |
| status     | OvnFipStatus | OvnFip     |               |

OVNFIPSPEC

|        |        |                         |
|--------|--------|-------------------------|
|        |        |                         |
| ovnEip | String | OVN EIP                 |
| ipType | String | IP vip ip               |
| ipName | String | IP                      |
| vpc    | String | VPC                     |
| v4Ip   | String | IPv4                    |
| v6Ip   | String | IPv6                    |
| type   | String | distributed centralized |

IptablesDnatRule

|            |                      |                  |                  |
|------------|----------------------|------------------|------------------|
|            |                      |                  |                  |
| apiVersion | String               | Kubernetes       | kubeovn.io/v1    |
| kind       | String               | Kubernetes       | IptablesDnatRule |
| metadata   | ObjectMeta           | Kubernetes       |                  |
| spec       | IptablesDnatRuleSpec | IptablesDnatRule |                  |

IPTABLESDNATRULESPEC

|              |        |    |
|--------------|--------|----|
|              |        |    |
| eip          | String | IP |
| externalPort | String |    |
| protocol     | String |    |
| internalIP   | String | IP |
| internalPort | String |    |

OvnDnatRule

|            |                   |             |               |
|------------|-------------------|-------------|---------------|
|            |                   |             |               |
| apiVersion | String            | Kubernetes  | kubeovn.io/v1 |
| kind       | String            | Kubernetes  | OvnDnatRule   |
| metadata   | ObjectMeta        | Kubernetes  |               |
| spec       | OvnDnatRuleSpec   | OvnDnatRule |               |
| status     | OvnDnatRuleStatus | OvnDnatRule |               |

OVNDNATRULESPEC

|              |        |           |
|--------------|--------|-----------|
|              |        |           |
| ovnEip       | String | OVN EIP   |
| ipType       | String | IP vip ip |
| ipName       | String | IP        |
| internalPort | String |           |
| externalPort | String |           |
| protocol     | String |           |
| vpc          | String | VPC       |
| v4Ip         | String | IPv4      |
| v6Ip         | String | IPv6      |

OVNDNATRULESTATUS

|              |                        |          |           |
|--------------|------------------------|----------|-----------|
|              |                        |          |           |
| vpc          | String                 | VPC      |           |
| v4Eip        | String                 | IPv4 EIP |           |
| v6Eip        | String                 | IPv6 EIP |           |
| externalPort | String                 |          |           |
| v4Ip         | String                 | IPv4     |           |
| v6Ip         | String                 | IPv6     |           |
| internalPort | String                 |          |           |
| protocol     | String                 |          |           |
| ipName       | String                 | IP       |           |
| ready        | Bool                   | DNAT     |           |
| conditions   | []OvnDnatRuleCondition | OVN DNAT | Condition |

IptablesSnatRule

|            |                      |                  |                  |
|------------|----------------------|------------------|------------------|
|            |                      |                  |                  |
| apiVersion | String               | Kubernetes       | kubeovn.io/v1    |
| kind       | String               | Kubernetes       | IptablesSnatRule |
| metadata   | ObjectMeta           | Kubernetes       |                  |
| spec       | IptablesSnatRuleSpec | IptablesSnatRule |                  |

IPTABLESSNATRULESPEC

|              |        |      |
|--------------|--------|------|
|              |        |      |
| eip          | String | IP   |
| internalCIDR | String | CIDR |

OvnSnatRule

|            |                   |             |               |
|------------|-------------------|-------------|---------------|
|            |                   |             |               |
| apiVersion | String            | Kubernetes  | kubeovn.io/v1 |
| kind       | String            | Kubernetes  | OvnSnatRule   |
| metadata   | ObjectMeta        | Kubernetes  |               |
| spec       | OvnSnatRuleSpec   | OvnSnatRule |               |
| status     | OvnSnatRuleStatus | OvnSnatRule |               |

OVNSNATRULESPEC

|           |        |           |
|-----------|--------|-----------|
|           |        |           |
| ovnEip    | String | OVN EIP   |
| vpcSubnet | String | VPC       |
| ipName    | String | IP        |
| vpc       | String | VPC       |
| v4IpCidr  | String | IPv4 CIDR |
| v6IpCidr  | String | IPv6 CIDR |

8.11.8 VPC

VpcNatGateway

|            |                     |               |               |
|------------|---------------------|---------------|---------------|
|            |                     |               |               |
| apiVersion | String              | Kubernetes    | kubeovn.io/v1 |
| kind       | String              | Kubernetes    | VpcNatGateway |
| metadata   | ObjectMeta          | Kubernetes    |               |
| spec       | VpcNatGatewaySpec   | VpcNatGateway |               |
| status     | VpcNatGatewayStatus | VpcNatGateway |               |

VPCNATGATEWAYSPEC

|                 |               |                     |     |     |
|-----------------|---------------|---------------------|-----|-----|
|                 |               |                     |     |     |
| vpc             | String        | VPC                 | Pod | VPC |
| subnet          | String        | VPC                 | Pod |     |
| externalSubnets | []String      |                     |     |     |
| lanIp           | String        | VPC                 | Pod | IP  |
| selector        | []String      | Kubernetes Selector |     |     |
| tolerations     | []Toleration  | Kubernetes          |     |     |
| affinity        | Affinity      | Kubernetes          |     |     |
| qosPolicy       | String        | QoS                 |     |     |
| bgpSpeaker      | VpcBgpSpeaker | BGP speaker         |     |     |
| routes          | []Route       |                     |     |     |

**VpcBgpSpeaker**

|                       |          |             |
|-----------------------|----------|-------------|
|                       |          |             |
| enabled               | Bool     | BGP speaker |
| asn                   | Uint32   |             |
| remoteAsn             | Uint32   |             |
| neighbors             | []String | BGP         |
| holdTime              | Duration | BGP         |
| routerId              | String   | BGP ID      |
| password              | String   | BGP         |
| enableGracefulRestart | Bool     |             |
| extraArgs             | []String |             |

**Route**

|           |        |      |
|-----------|--------|------|
|           |        |      |
| cidr      | String | CIDR |
| nextHopIP | String | IP   |

**VPCNATGATEWAYSTATUS**

|                 |              |                     |
|-----------------|--------------|---------------------|
|                 |              |                     |
| qosPolicy       | String       | QoS                 |
| externalSubnets | []String     |                     |
| selector        | []String     | Kubernetes Selector |
| tolerations     | []Toleration | Kubernetes          |
| affinity        | Affinity     | Kubernetes          |

**VpcEgressGateway**

|            |                        |                  |                  |
|------------|------------------------|------------------|------------------|
|            |                        |                  |                  |
| apiVersion | String                 | Kubernetes       | kubeovn.io/v1    |
| kind       | String                 | Kubernetes       | VpcEgressGateway |
| metadata   | ObjectMeta             | Kubernetes       |                  |
| spec       | VpcEgressGatewaySpec   | VpcEgressGateway |                  |
| status     | VpcEgressGatewayStatus | VpcEgressGateway |                  |

VPCEGRESSGATEWAYSPEC

|                |                |     |      |                   |
|----------------|----------------|-----|------|-------------------|
|                |                |     |      |                   |
| vpc            | String         | VPC |      |                   |
| replicas       | Int32          |     |      |                   |
| prefix         | String         |     |      |                   |
| image          | String         |     |      |                   |
| internalSubnet | String         |     |      |                   |
| externalSubnet | String         |     |      |                   |
| internalIPs    | []String       | IP  |      |                   |
| externalIPs    | []String       | IP  |      |                   |
| trafficPolicy  | String         |     |      |                   |
| bandwidth      | BandwidthLimit | /   | Mbps | bit-rate quantity |

BandwidthLimit

|         |       |        |     |                                                                                   |
|---------|-------|--------|-----|-----------------------------------------------------------------------------------|
|         |       |        |     |                                                                                   |
| ingress | Int64 | String | VPC | Mbps quantity <span>M</span> <span>Mi</span> <span>G</span> <span>Gi</span> bit/s |
| egress  | Int64 | String | VPC | Mbps quantity <span>M</span> <span>Mi</span> <span>G</span> <span>Gi</span> bit/s |

VpcDns

|            |              |            |               |
|------------|--------------|------------|---------------|
|            |              |            |               |
| apiVersion | String       | Kubernetes | kubeovn.io/v1 |
| kind       | String       | Kubernetes | VpcDns        |
| metadata   | ObjectMeta   | Kubernetes |               |
| spec       | VpcDNSSpec   | VpcDns     |               |
| status     | VpcDNSStatus | VpcDns     |               |

VPCDNSSPEC

|          |            |
|----------|------------|
|          |            |
| replicas | Int32      |
| vpc      | String VPC |
| subnet   | String     |

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8.11.9

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# 8.12 Annotation

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| Kube-OVN | Pod | Node Annotation | Annotation |
|----------|-----|-----------------|------------|
|          |     |                 | Annotation |

## 8.12.1 Pod Annotation

| Key                                   | Value                | Description                                   |
|---------------------------------------|----------------------|-----------------------------------------------|
| ovn.kubernetes.io/allocated           | true or false        | Pod                                           |
| ovn.kubernetes.io/routed              | true or false        | Pod OVN                                       |
| ovn.kubernetes.io/routes              | String               | Pod                                           |
| ovn.kubernetes.io/mac_address         | String               | Pod Mac Pod Annotation Mac                    |
| ovn.kubernetes.io/ip_address          | String               | Pod IP Pod Annotation IP                      |
| ovn.kubernetes.io/cidr                | String               | Pod CIDR                                      |
| ovn.kubernetes.io/gateway             | String               | Pod Gateway                                   |
| ovn.kubernetes.io/ip_pool             | IP                   | Pod Workload IP                               |
| ovn.kubernetes.io/bgp                 | true, cluster, local | BGP Pod                                       |
| ovn.kubernetes.io/snat                | String               | Pod SNAT                                      |
| ovn.kubernetes.io/eip                 | String               | Pod EIP                                       |
| ovn.kubernetes.io/vip                 | String               | Pod VIP Annotation VIP                        |
| ovn.kubernetes.io/aaps                | String               | Pod AAPs (Additional Allowed Addresses Pairs) |
| ovn.kubernetes.io/virtualmachine      | String               | Pod VirtualMachineInstance                    |
| ovn.kubernetes.io/activation_strategy | String               | Pod                                           |
| ovn.kubernetes.io/logical_router      | String               | Pod VPC                                       |
| ovn.kubernetes.io/layer2_forward      | true or false        | Pod OVN LSP unknown                           |
| ovn.kubernetes.io/port_security       | true or false        | Pod Port Security                             |
| ovn.kubernetes.io/logical_switch      | String               | Pod                                           |
| ovn.kubernetes.io/vlan_id             | Int                  | Pod Vlan ID                                   |
| ovn.kubernetes.io/ingress_rate        | Int                  | Pod Mbits/s                                   |
| ovn.kubernetes.io/egress_rate         | Int                  | Pod Mbits/s                                   |
| ovn.kubernetes.io/security_groups     | String               | Pod Security Group                            |
| ovn.kubernetes.io/default_route       | true or false        |                                               |
| ovn.kubernetes.io/provider_network    | String               | Pod ProviderNetwork                           |
| ovn.kubernetes.io/mirror              | true or false        | Pod                                           |
| ovn.kubernetes.io/north_gateway       | String               | Pod                                           |
| ovn.kubernetes.io/latency             | Int                  | Pod ms                                        |
| ovn.kubernetes.io/limit               | Int                  | Pod qdisc                                     |
| ovn.kubernetes.io/loss                | Float                | Pod                                           |
| ovn.kubernetes.io/jitter              | Int                  | Pod ms                                        |
| ovn.kubernetes.io/generate-hash       | true or false        | Pod                                           |
| ovn.kubernetes.io/attachmentprovider  | String               | Pod                                           |

### 8.12.2 Node Annotation

| Key                                | Value         | Description                      |
|------------------------------------|---------------|----------------------------------|
| ovn.kubernetes.io/allocated        | true or false | ovn0 join                        |
| ovn.kubernetes.io/mac_address      | String        | Node ovn0 Mac                    |
| ovn.kubernetes.io/ip_address       | String        | Node ovn0 IP                     |
| ovn.kubernetes.io/cidr             | String        | Node ovn0 join CIDR              |
| ovn.kubernetes.io/gateway          | String        | Node ovn0 join Gateway           |
| ovn.kubernetes.io/chassis          | String        | Node OVN-SouthBoundDB Chassis ID |
| ovn.kubernetes.io/port_name        | String        | Node ovn0 OVN-NorthboundDB LSP   |
| ovn.kubernetes.io/logical_switch   | String        | Node ovn0                        |
| ovn.kubernetes.io/tunnel_interface | String        |                                  |

### 8.12.3 Namespace Annotation

| Key                           | Value      | Description          |
|-------------------------------|------------|----------------------|
| ovn.kubernetes.io/cidr        | CIDR       | Namespace CIDR       |
| ovn.kubernetes.io/exclude_ips | excludeIPs | Namespace excludeIPs |

### 8.12.4 Subnet Annotation

| Key                   | Value                | Description |
|-----------------------|----------------------|-------------|
| ovn.kubernetes.io/bgp | true, cluster, local | BGP         |

### 8.12.5 Service Annotation

| Key                                               | Value         | Description          |
|---------------------------------------------------|---------------|----------------------|
| ovn.kubernetes.io/bgp                             | true or false | BGP Service          |
| ovn.kubernetes.io/switch_lb_vip                   | String        | Service Kube-OVN VIP |
| ovn.kubernetes.io/vpc                             | String        | Service VPC          |
| ovn.kubernetes.io/service_external_ip_from_subnet | true or false | Service IP           |
| ovn.kubernetes.io/service_health_check            | true or false | Service              |
| ovn.kubernetes.io/lb_svc_img                      | String        |                      |

### 8.12.6 Networkpolicy Annotation

| Key                                  | Value             | Description     |
|--------------------------------------|-------------------|-----------------|
| ovn.kubernetes.io/enable_log         | true or false     | NetworkPolicy   |
| ovn.kubernetes.io/log_acl_actions    | "allow,drop,pass" | Action ACL      |
| ovn.kubernetes.io/acl_log_meter_rate | Int               | NetworkPolicy / |

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8.12.7

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8.13

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8.13.1

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| Bad | Good |
|-----|------|
|     |      |

| Bad                  | Good                 |
|----------------------|----------------------|
| Kube-OVN1.10Kube-OVN | Kube-OVN1.10Kube-OVN |



| Bad            | Good           |
|----------------|----------------|
| wget 127.0.0.1 | wget 127.0.0.1 |

8.13.2

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yamlyaml

| Bad                                                                           | Good                                                                          |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| apiVersion: kubeovn.io/v1<br>kind: Subnet<br>metadata:<br>name: attach-subnet | apiVersion: kubeovn.io/v1<br>kind: Subnet<br>metadata:<br>name: attach-subnet |

bash

| Bad            | Good           |
|----------------|----------------|
| wget 127.0.0.1 | wget 127.0.0.1 |

#

| Bad                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Good                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>oilbeater@macdeMac-3 ~ ping 114.114.114.114 -c 3 PING 114.114.114.114 (114.114.114.114): 56 data bytes 64 bytes from 114.114.114.114: icmp_seq=0 ttl=83 time=10.429 ms 64 bytes from 114.114.114.114: icmp_seq=1 ttl=79 time=11.360 ms 64 bytes from 114.114.114.114: icmp_seq=2 ttl=76 time=10.794 ms  --- 114.114.114.114 ping statistics --- 3 packets transmitted, 3 packets received, 0.0% packet loss round-trip min/avg/max/stddev = 10.429/10.861/11.360/0.383 ms</pre> | <pre># ping 114.114.114.114 -c 3 PING 114.114.114.114 (114.114.114.114): 56 data bytes 64 bytes from 114.114.114.114: icmp_seq=0 ttl=83 time=10.429 ms 64 bytes from 114.114.114.114: icmp_seq=1 ttl=79 time=11.360 ms 64 bytes from 114.114.114.114: icmp_seq=2 ttl=76 time=10.794 ms  --- 114.114.114.114 ping statistics --- 3 packets transmitted, 3 packets received, 0.0% packet loss round-trip min/avg/max/stddev = 10.429/10.861/11.360/0.383 ms</pre> |

#

| Bad                                                                                    | Good                                                                               |
|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <pre># mv /etc/origin/ovn/ovnnb_db.db /tmp # mv /etc/origin/ovn/ovnsb_db.db /tmp</pre> | <pre>mv /etc/origin/ovn/ovnnb_db.db /tmp mv /etc/origin/ovn/ovnsb_db.db /tmp</pre> |

8.13.3

md

| Bad                                              | Good                         |
|--------------------------------------------------|------------------------------|
| <pre>[ ](http://kubeovn.github.io/prepare)</pre> | <pre>[ ](./prepare.md)</pre> |

| Bad                                            | Good                                                               |
|------------------------------------------------|--------------------------------------------------------------------|
| <pre>[Kubernetes ](http://kubernetes.io)</pre> | <pre>[Kubernetes ](http://kubernetes.io){: target="_blank" }</pre> |

8.13.4

| Bad                                   | Good                                  |
|---------------------------------------|---------------------------------------|
| <pre>'''bash wget 127.0.0.1 '''</pre> | <pre>'''bash wget 127.0.0.1 '''</pre> |

| Bad                                   | Good                                  |
|---------------------------------------|---------------------------------------|
| <pre>'''bash wget 127.0.0.1 '''</pre> | <pre>'''bash wget 127.0.0.1 '''</pre> |

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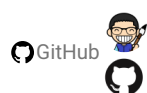


8.13.5

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9.

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